MightyXDDoS™ - DDoS Mitigator

- High-Performance Security Bridge based on FD.io VPP and eBPF XDP
- L2 Mitigator for AI LLM Inference Server

MightyLink Lab.
(www.mightyconn. com)
Doc. Revision: 1.02

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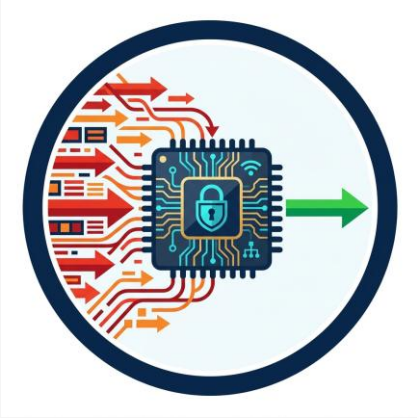
1. DDoS Attack & Mitigator
2. Base Technology – DPDK, FD.io VPP, eBPF XDP
3. MightyXDDoS
4. XDDoS User Interface

Appendix

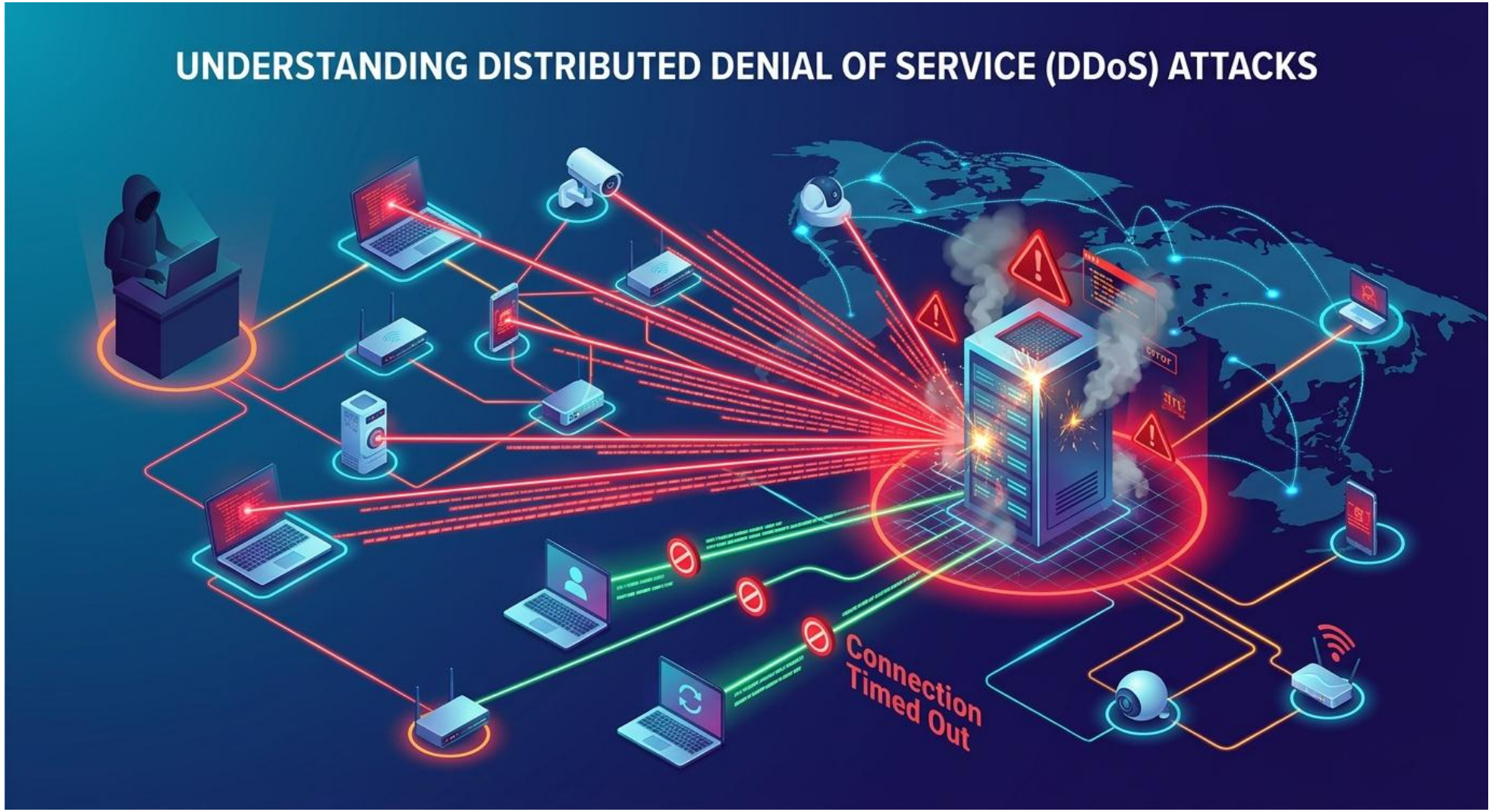


1. DDoS Attack & Mitigator

DDoS Attack and DDoS Mitigator

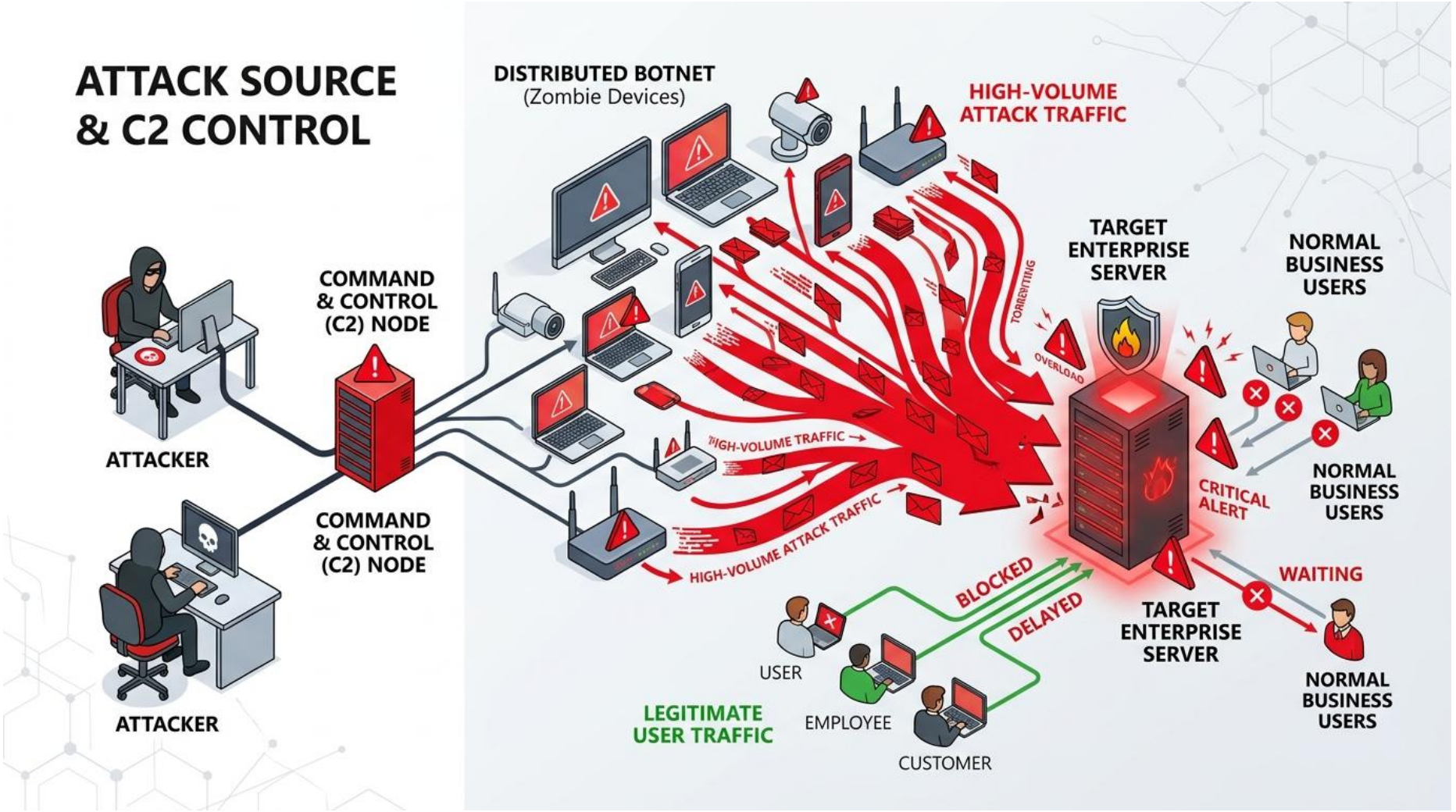


1. DDoS Attack & Mitigator(1) – Attack(1)



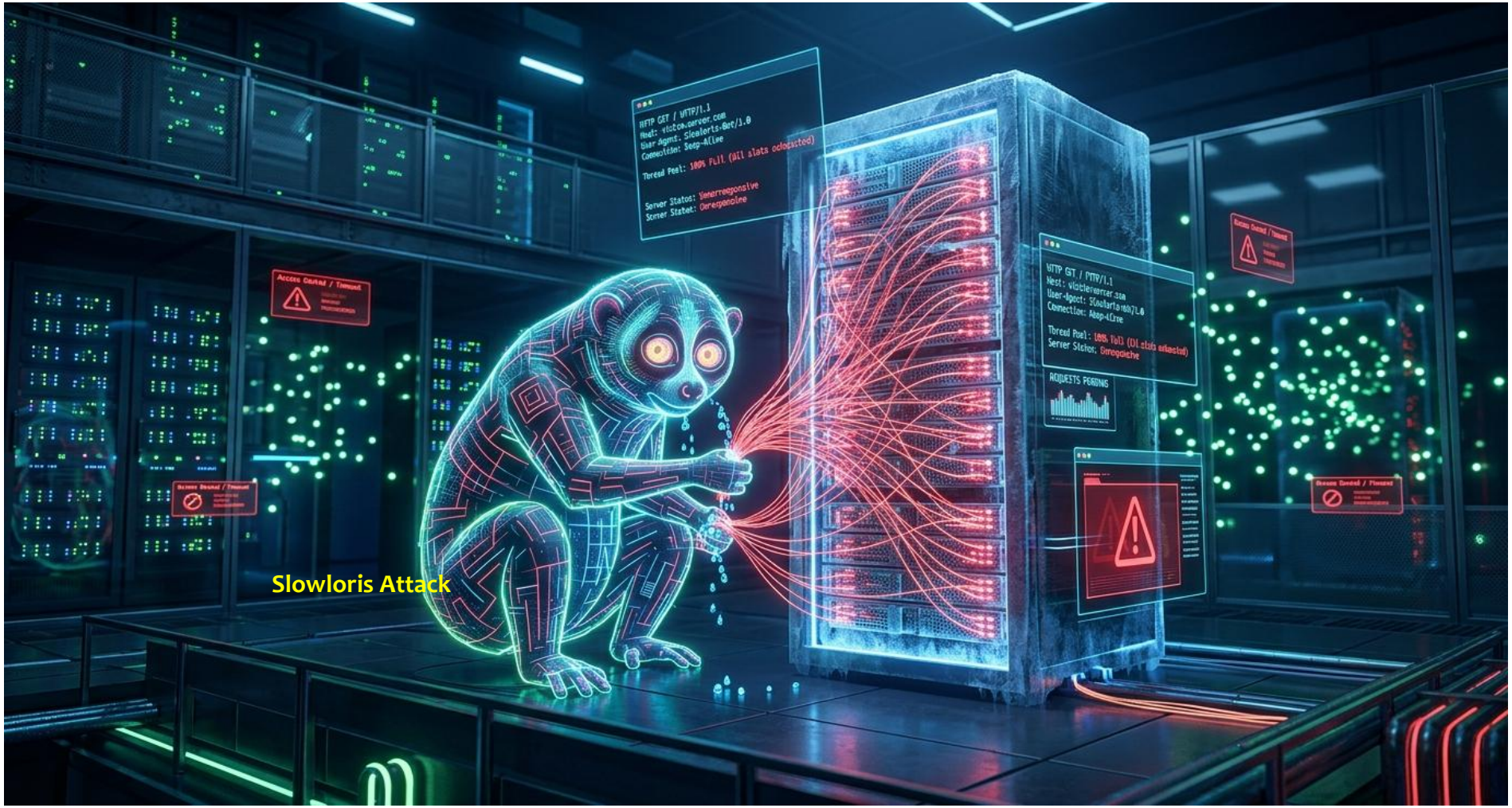
Generated by Gemini

1. DDoS Attack & Mitigator(1) – Attack(2)



Generated by Gemini

1. DDoS Attack & Mitigator(1) – Attack(3)



Generated by Gemini

1. DDoS Attack & Mitigator(1) – Attack(4)

Types of DDoS Attacks

Volumetric attacks(L3/L4)

- Flooding the network with a large volume of data to consume available bandwidth..

  Ex) UDP Flood, ICMP Flood(Ping Flood), DNS amplification attack

- **Protocol attacks(L3/L4)**
  - Exhausting server or firewall resources by exploiting vulnerabilities in communication protocols..

    Ex) SYN Flood, Ping of Death, Smurf Attack, Fragmentation Packet Attack

- **Application layer attacks(L7)**
  - Mimics the behavior of legitimate users to target specific applications and evade detection..

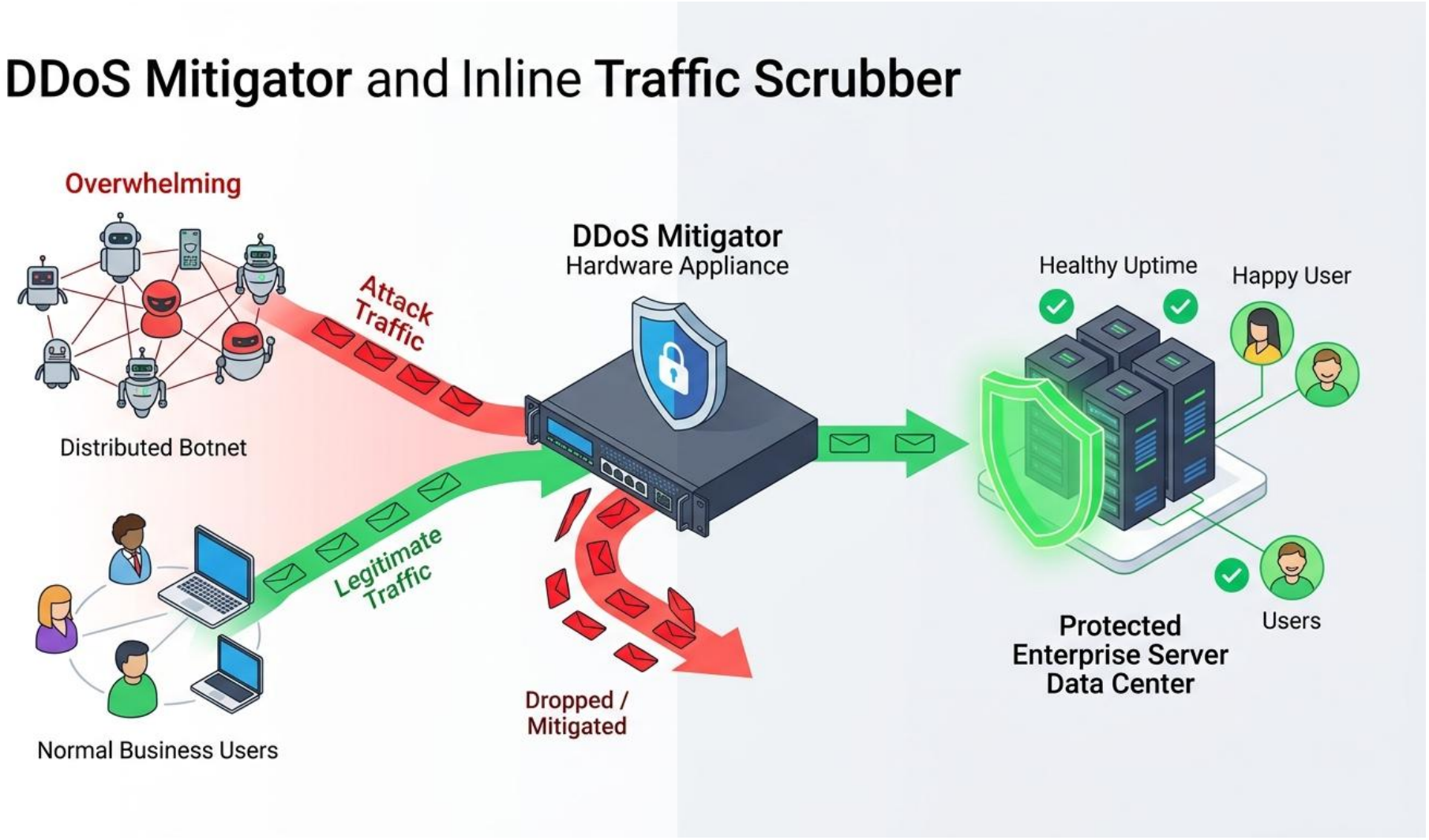
    Ex) HTTP GET/POST Flood, Slowloris Attack, XML-RPC Flood, WordPress Administrator Login Flood

1. DDoS Attack & Mitigator(2) – Mitigator(1)



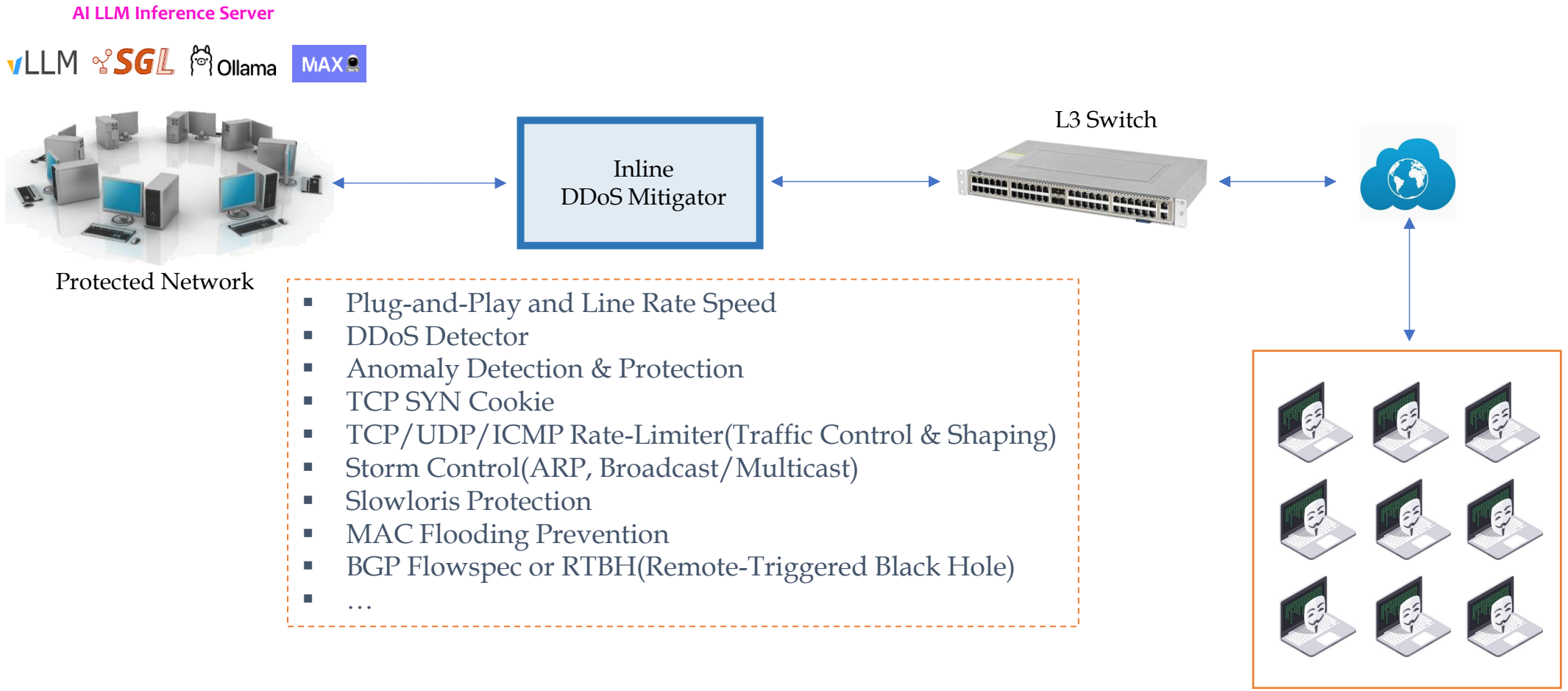
Generated by Gemini

1. DDoS Attack & Mitigator(2) – Mitigator(2)

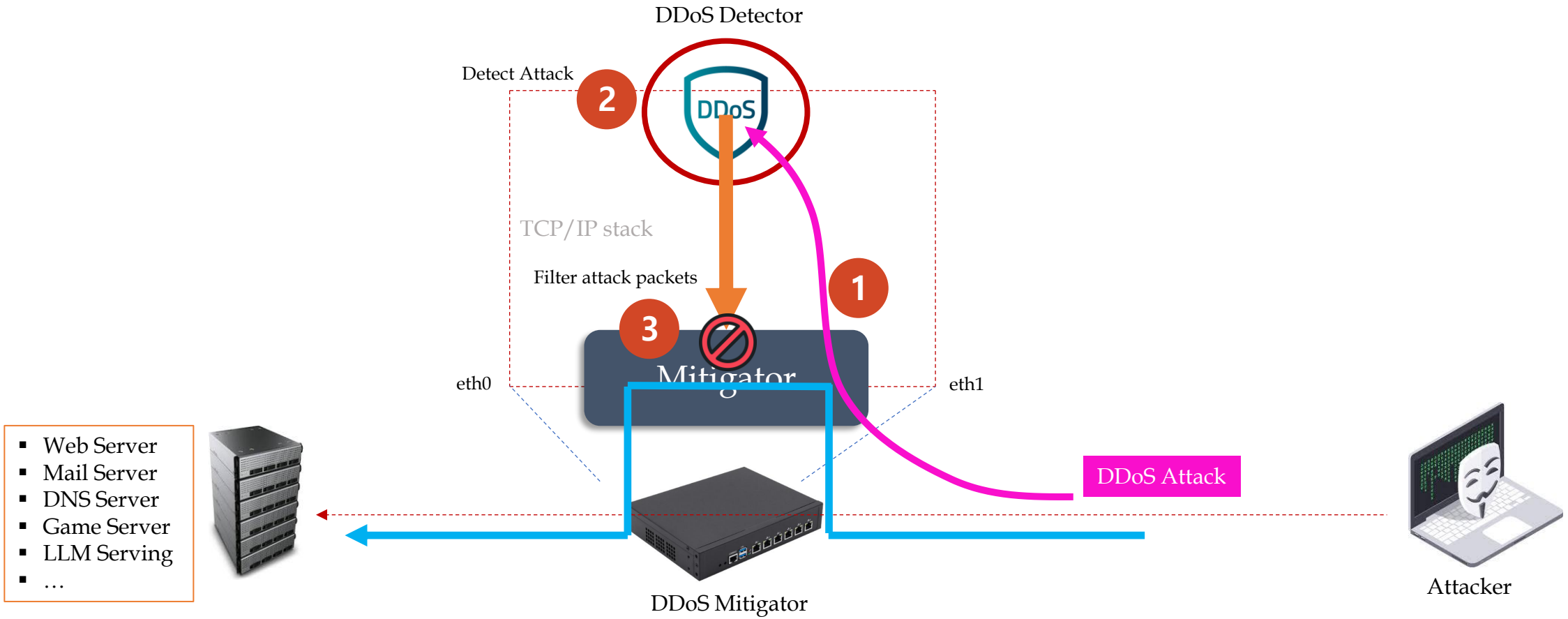


Generated by Gemini

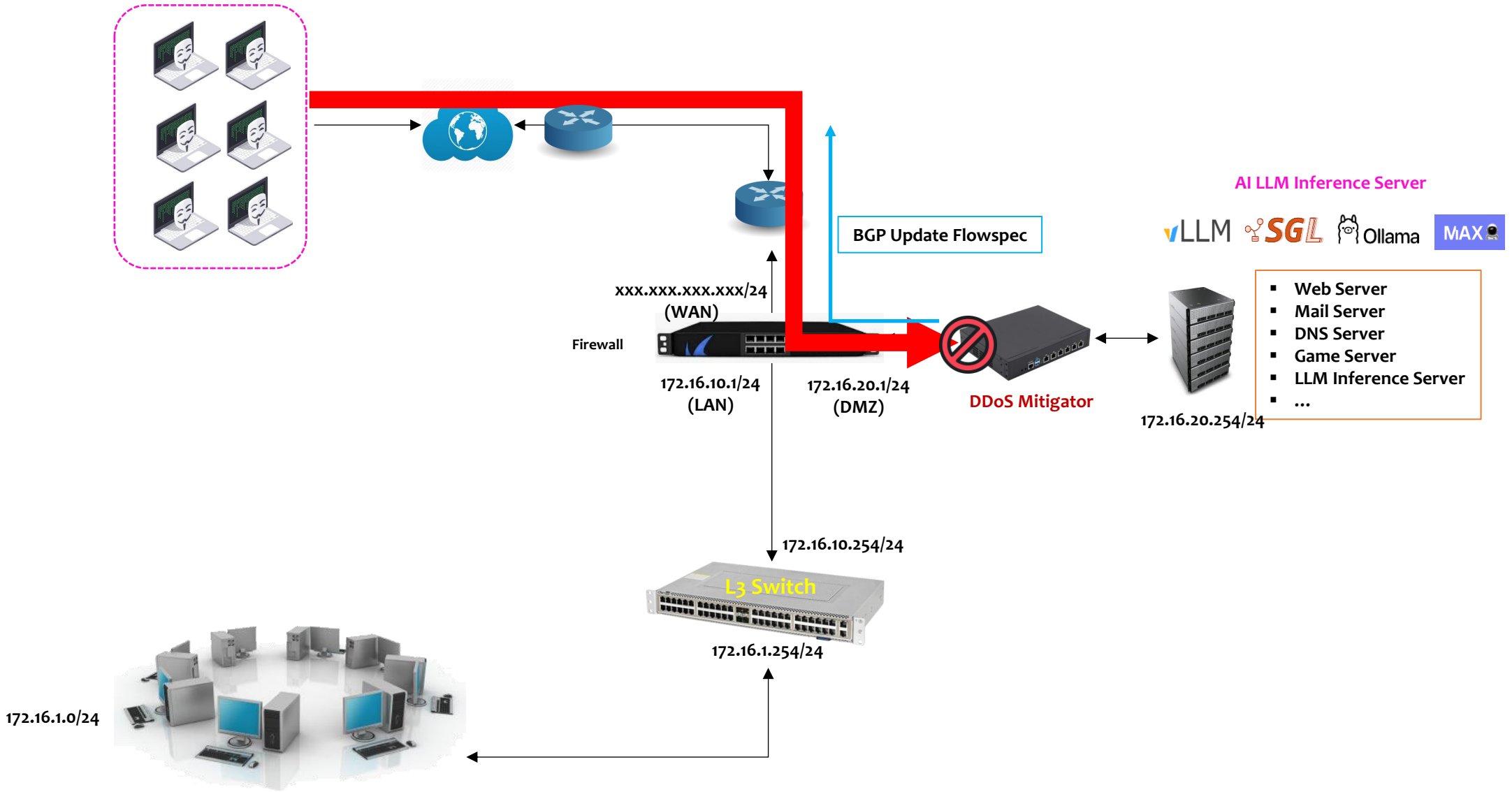
1. DDoS Attack & Mitigator(2) – Mitigator(3)



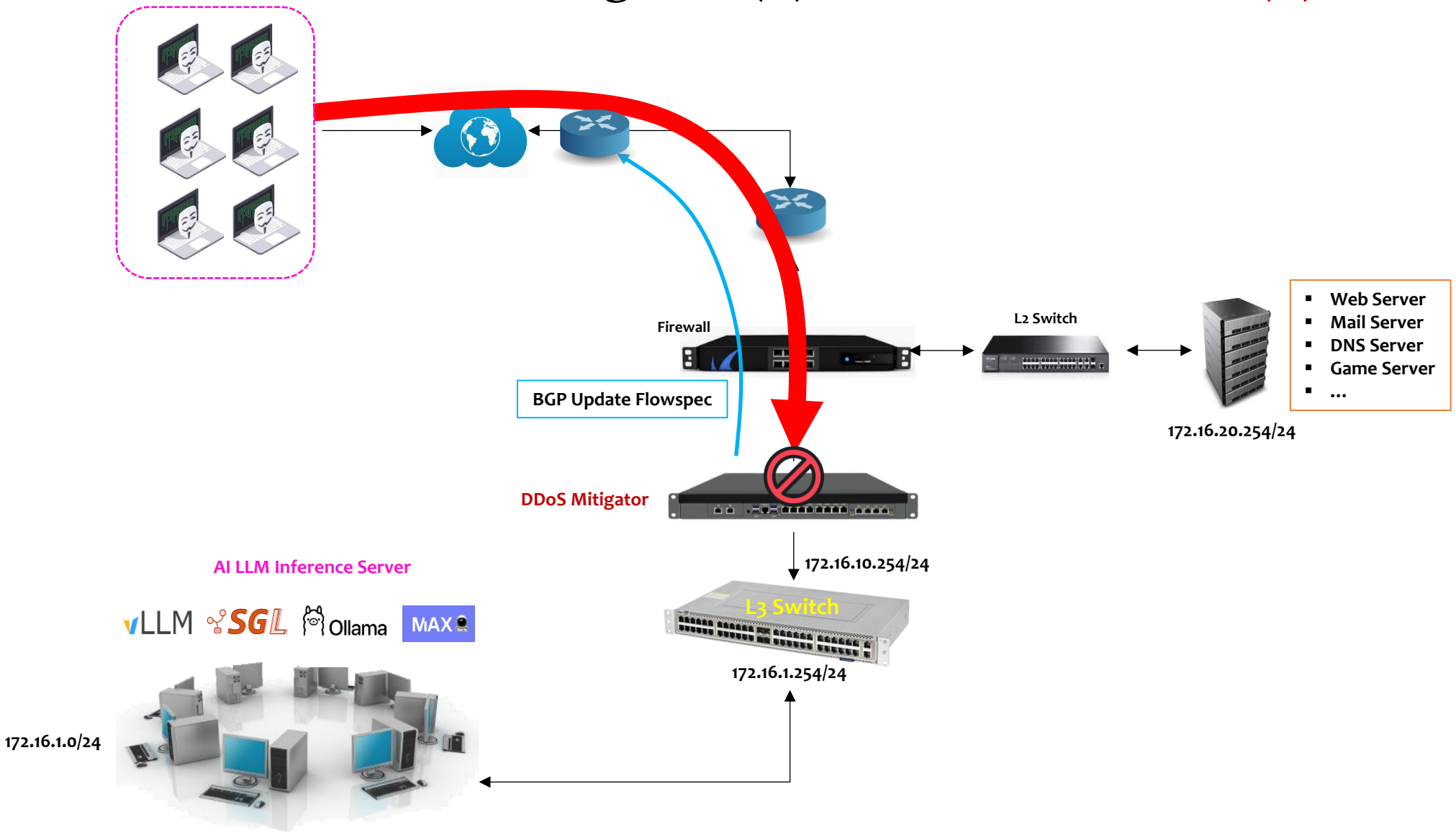
1. DDoS Attack & Mitigator(2) – Mitigator(4)



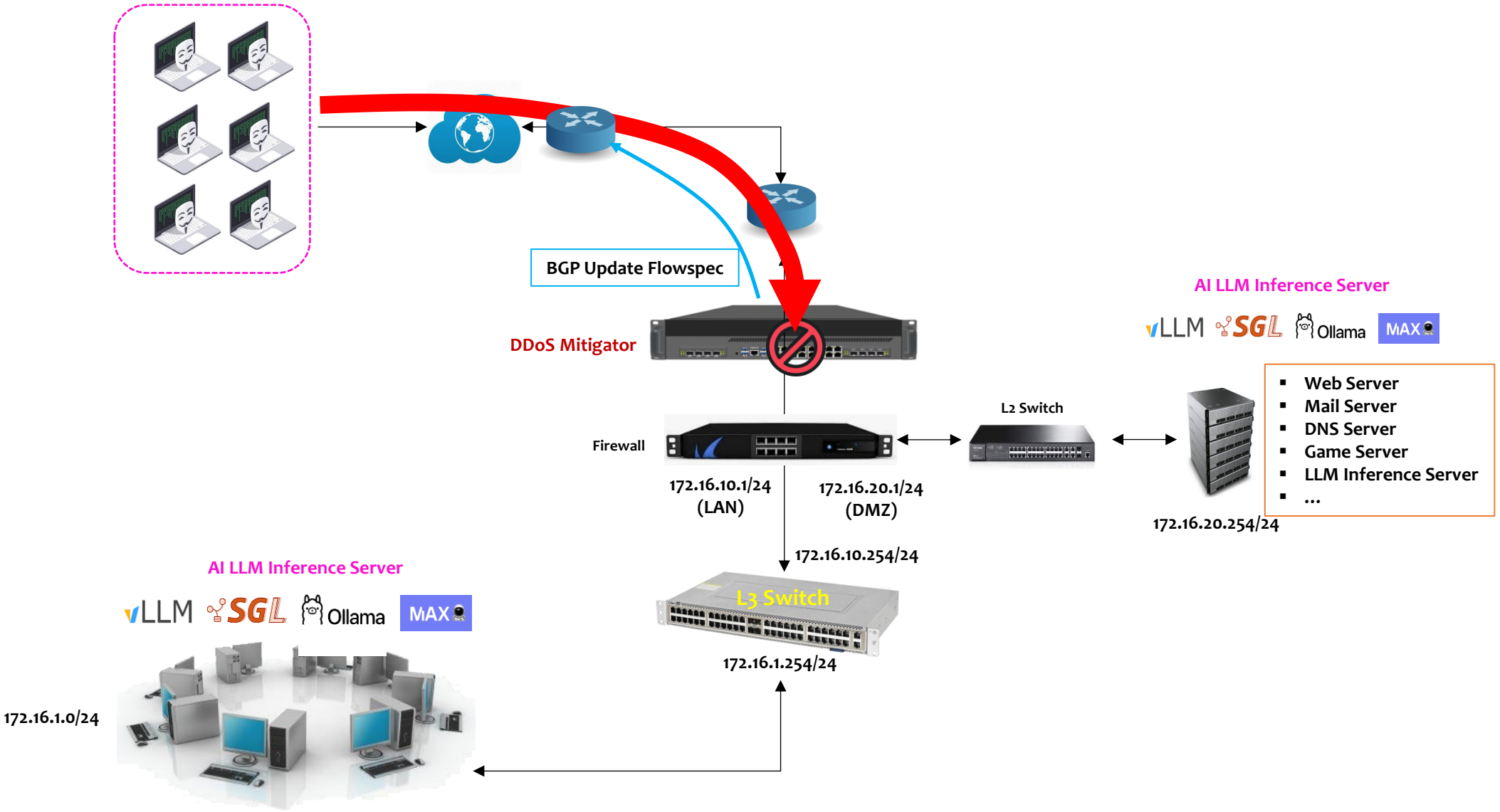
1. DDoS Attack & Mitigator(3) – Network 구성(1)



1. DDoS Attack & Mitigator(3) – Network 구성(2)



1. DDoS Attack & Mitigator(3) – Network 구성(3)



2. Base Technology

DPDK, FD.io VPP and XDP

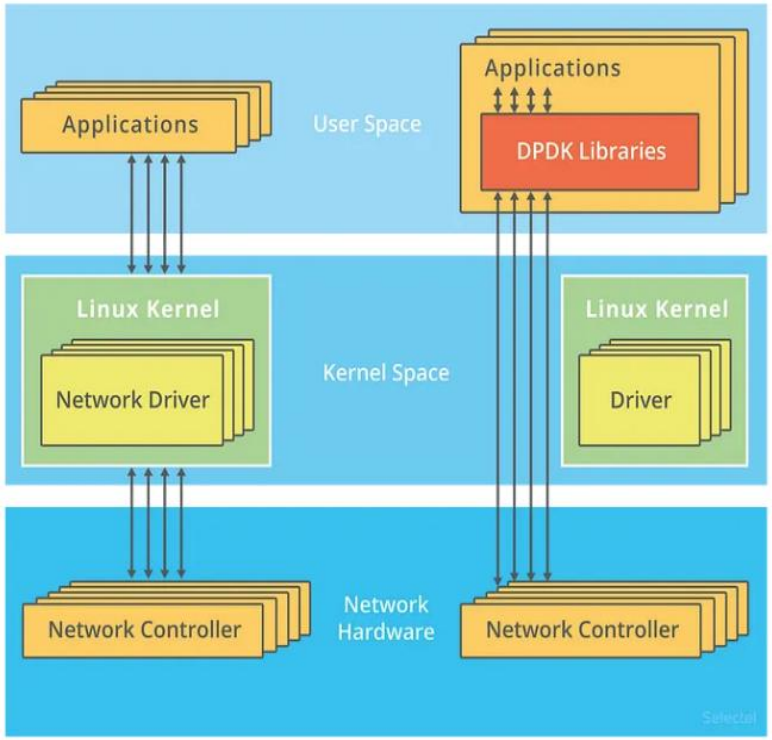
2.1 DPDK

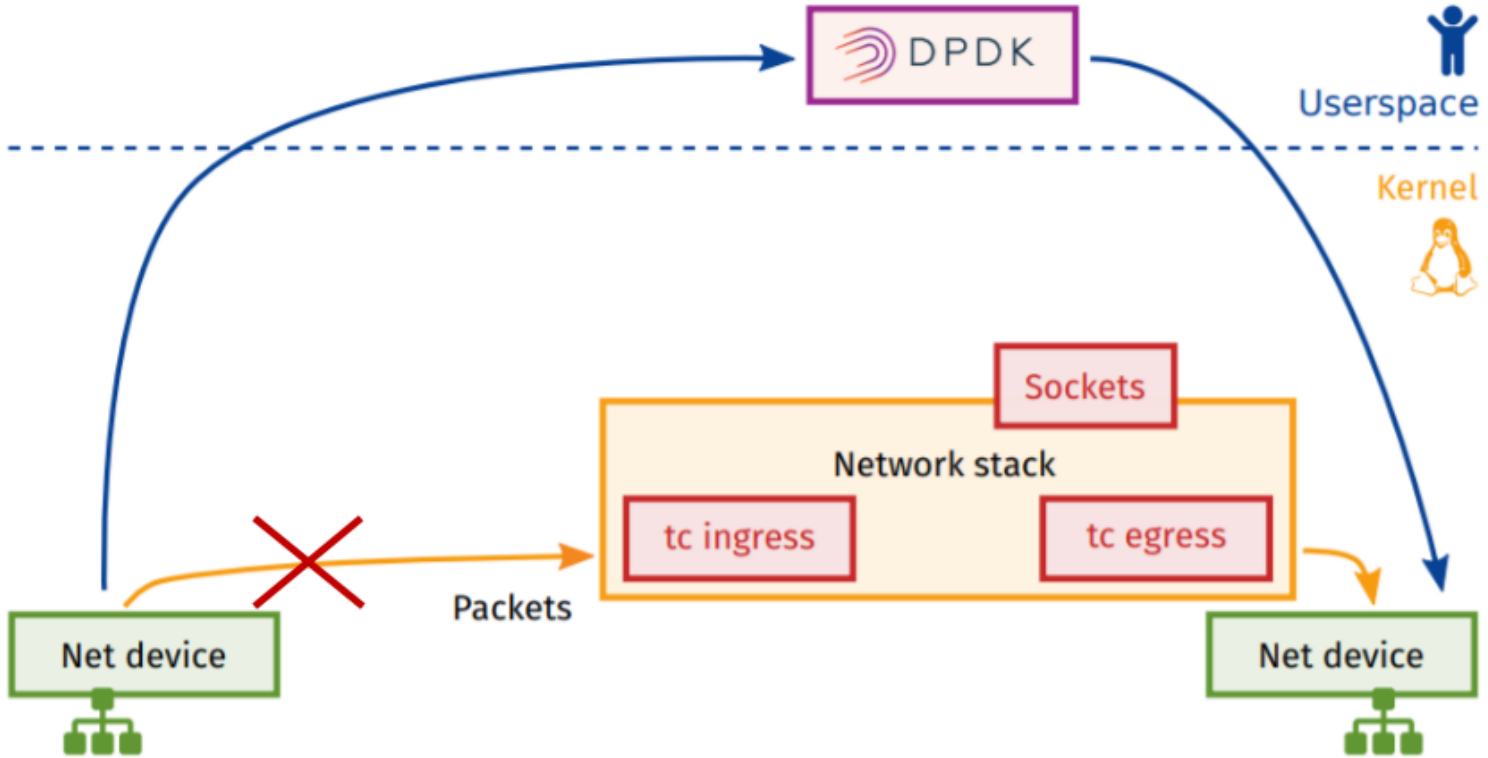
Data Plane Development Kit



https://www.dpdk.org/

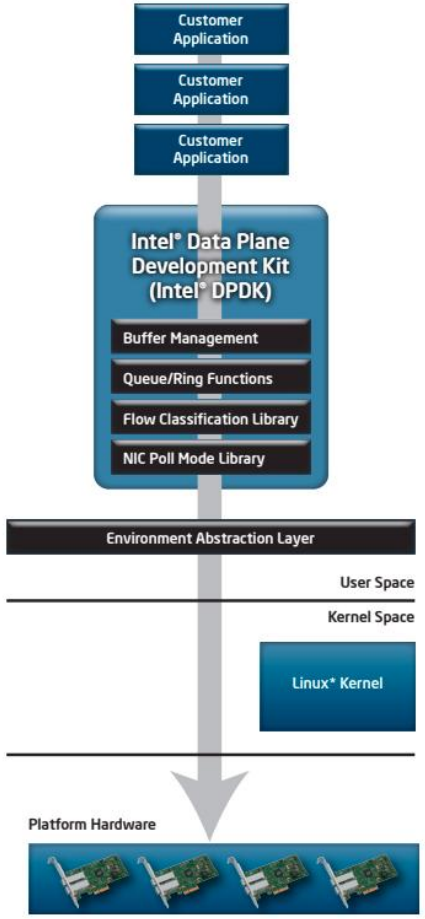
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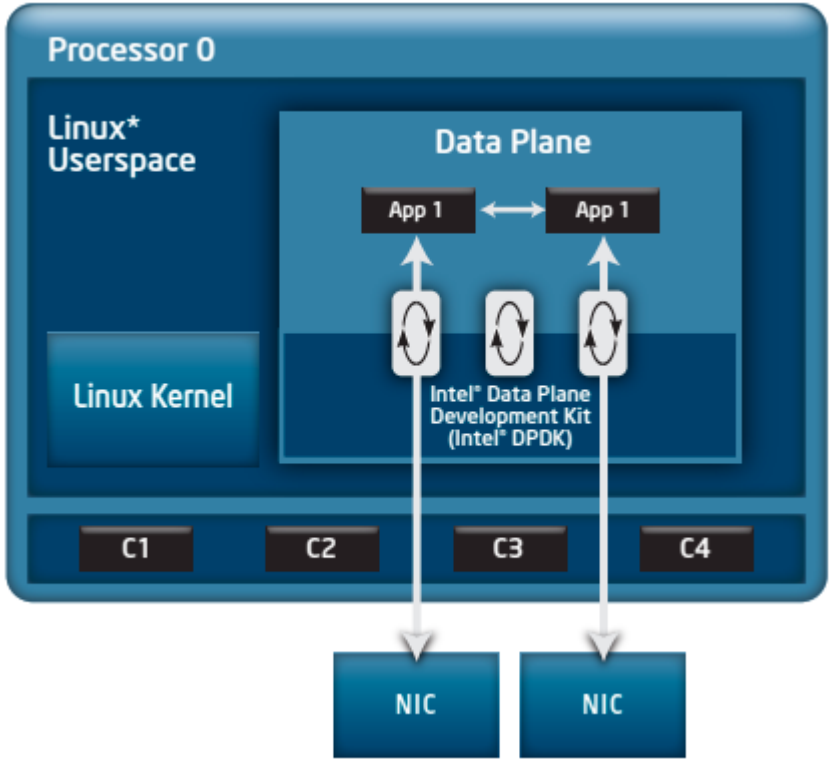




Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

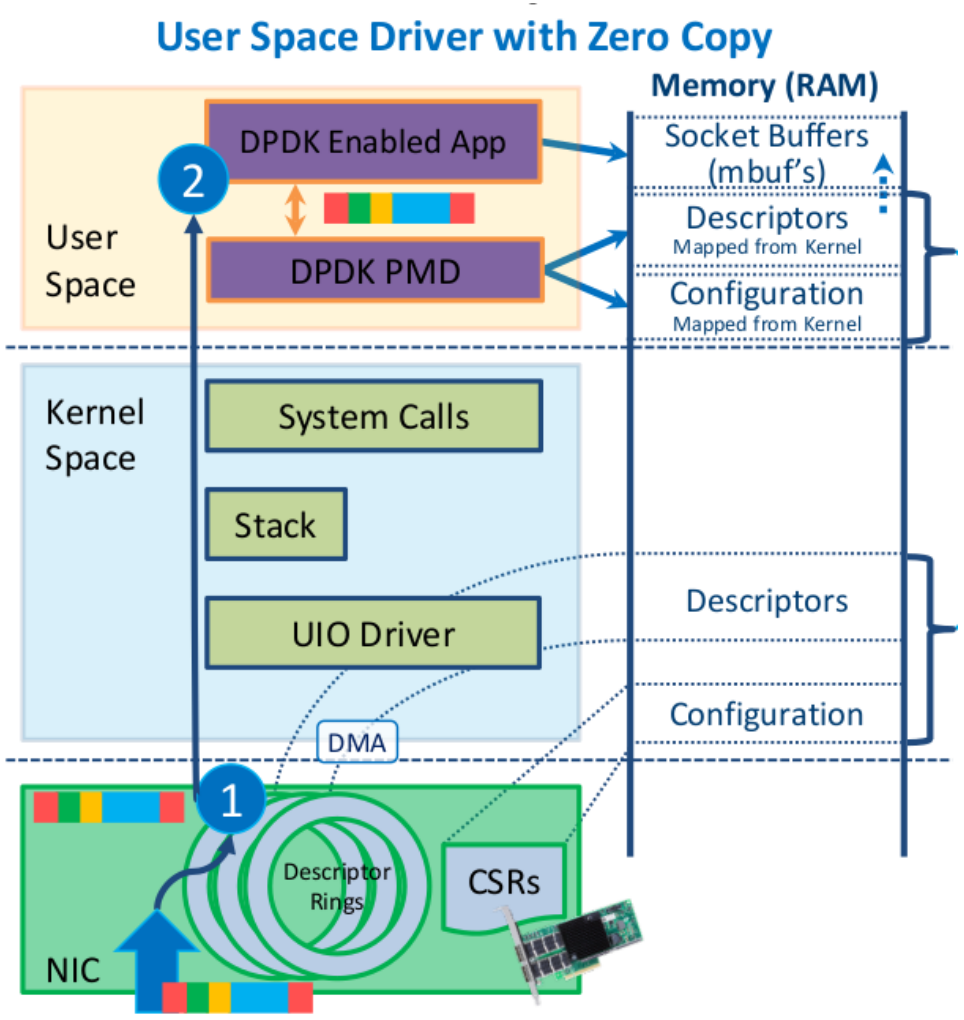
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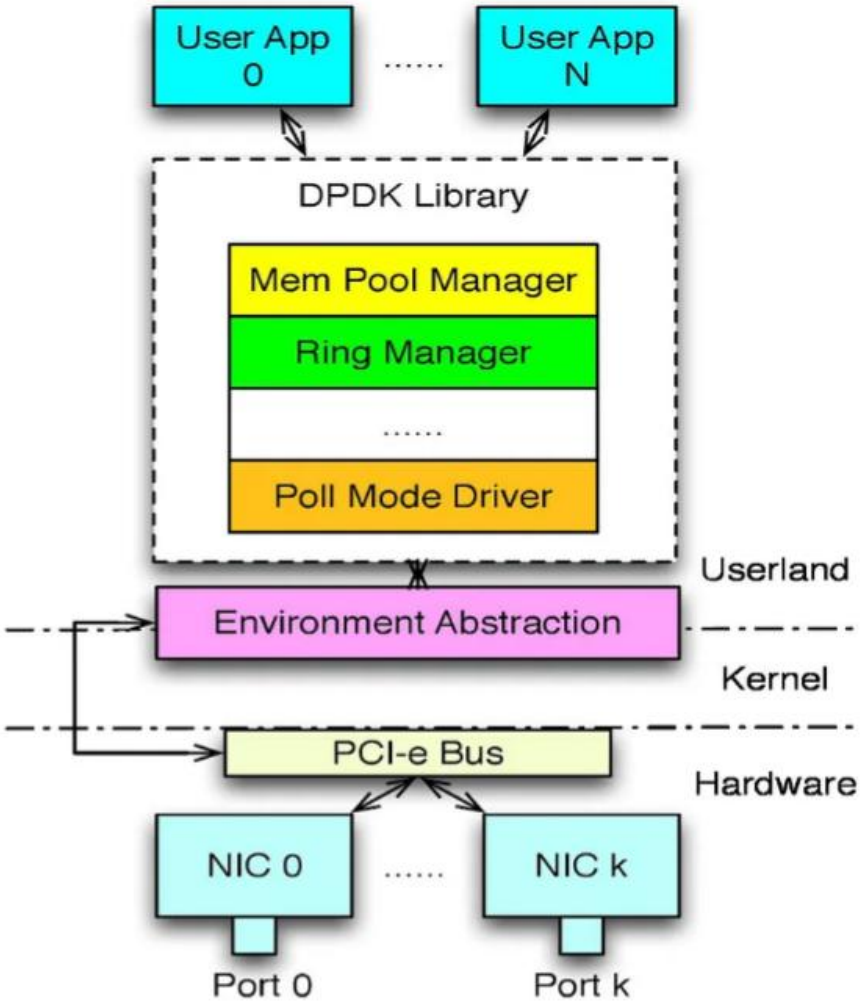




Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation, Intel

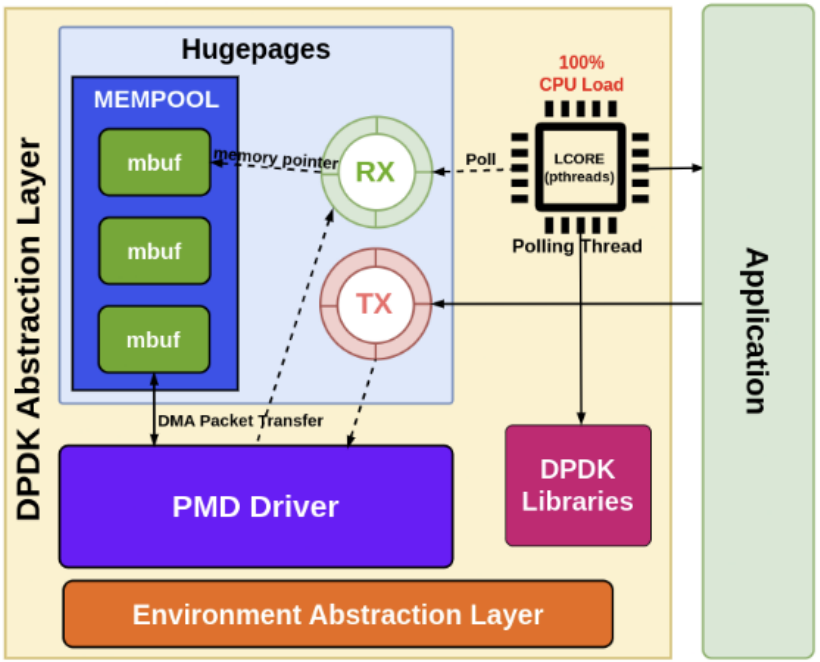
2.1 DPDK(3)

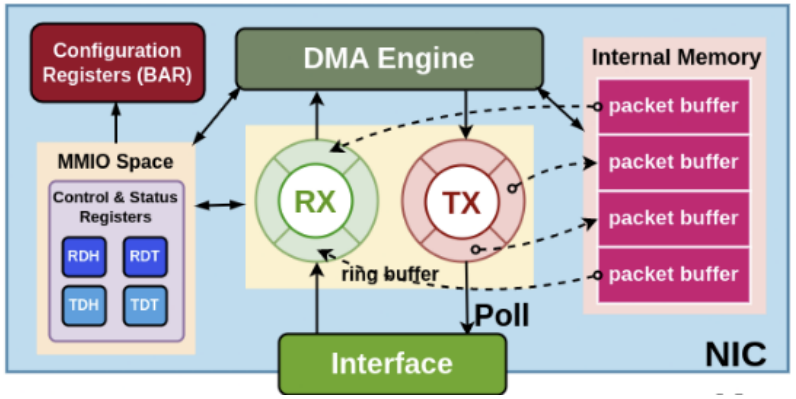


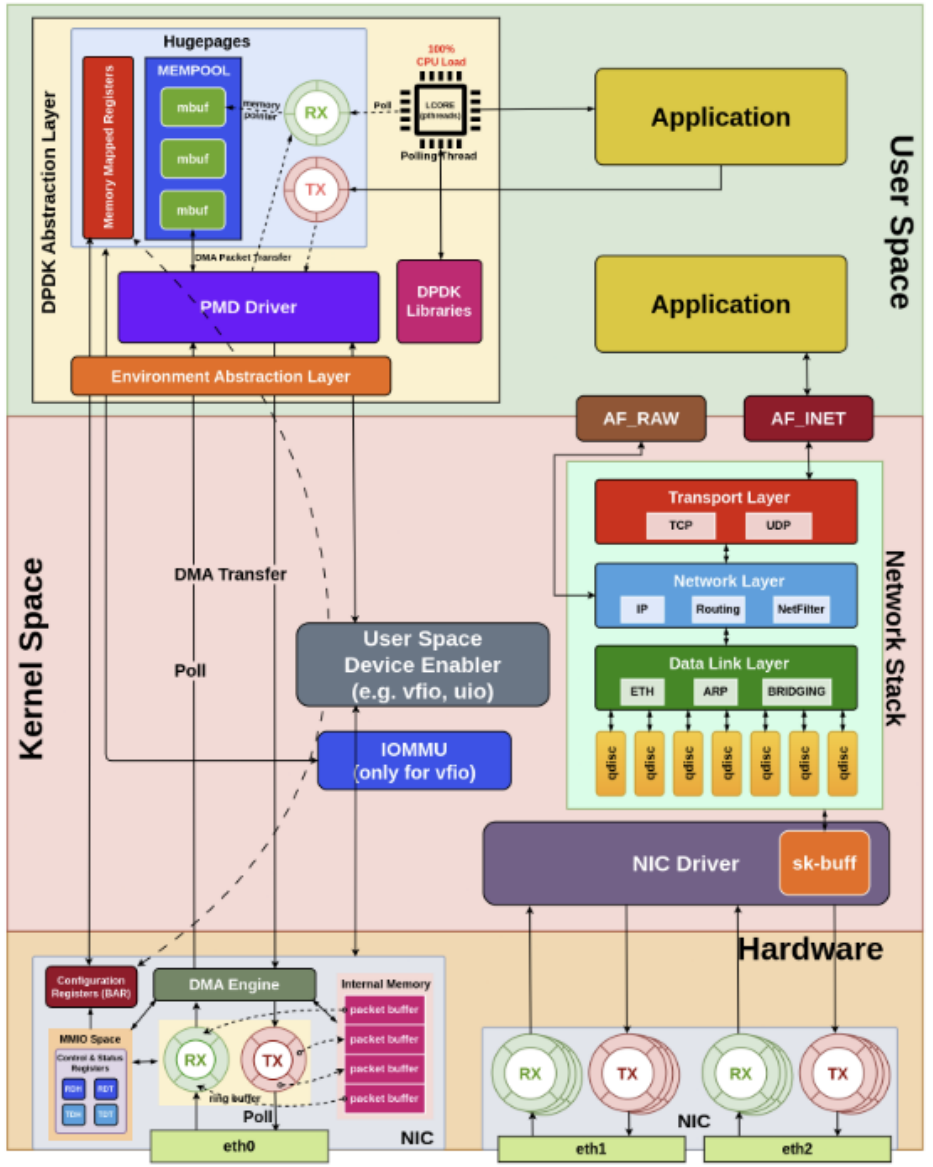


Ref) https://medium.com/google-cloud/forwarding-over-100-mpps-with-fd-io-vpp-on-x86-62b9447da554

2.1 DPDK(4)







Ref) https://docs.ngkore.com/blogs/dpdk/dpdk.html#hardware-requirements

2.2 FD.io VPP

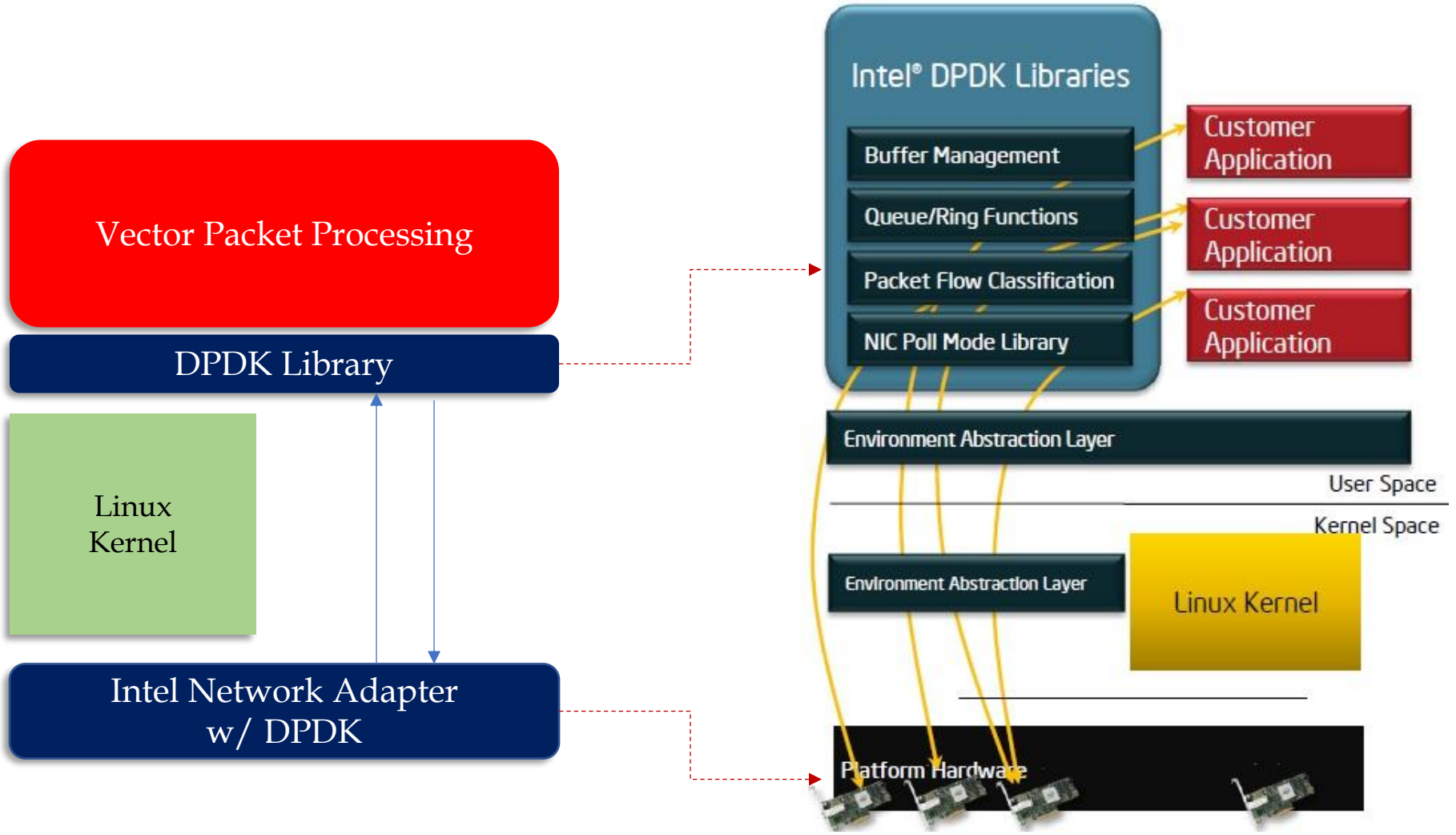
Vector Packet Processing



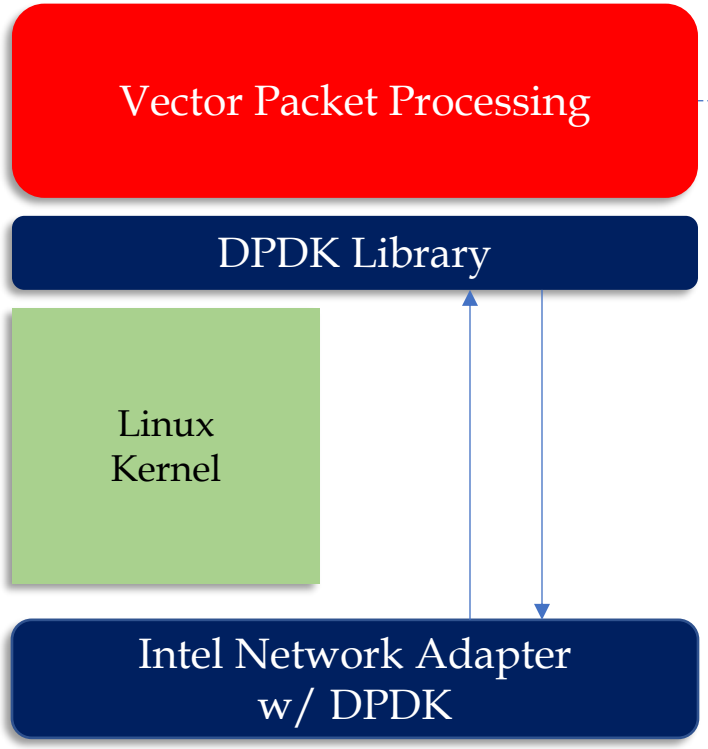
VPP

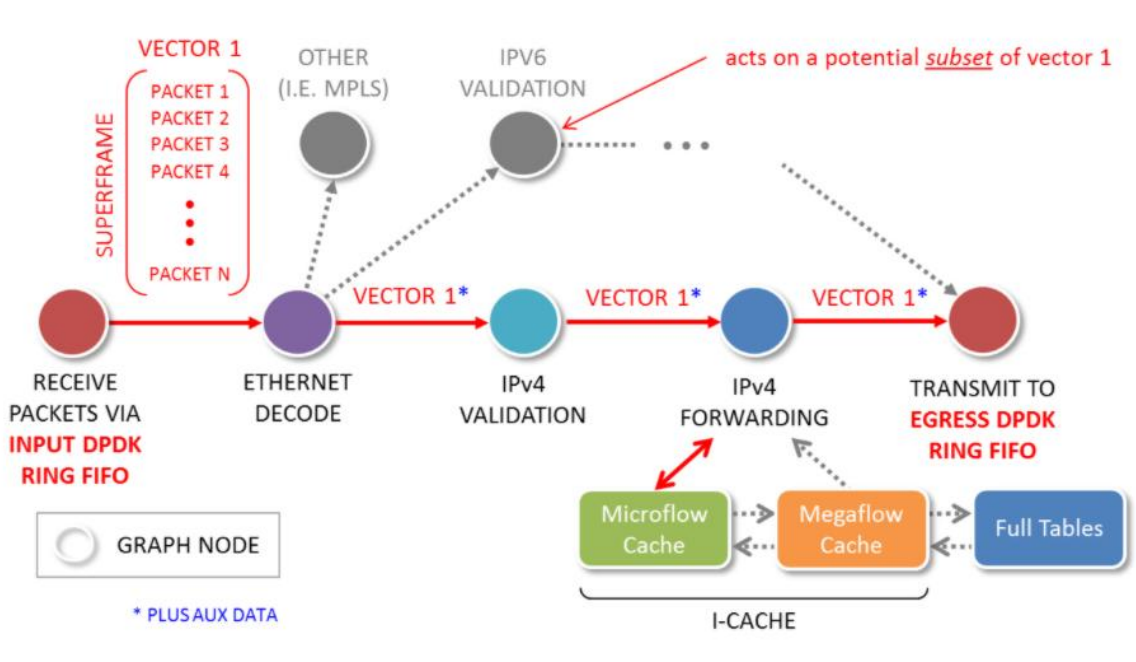
https://fd.io/

2.2 FD.io VPP(1)


Vector Packet Processing
DPDK Library
Linux Kernel
Intel Network Adapter w/ DPDK
Intel® DPDK Libraries
Buffer Management
Queue/Ring Functions
Packet Flow Classification
NIC Poll Mode Library
Customer Application
Customer Application
Customer Application
Environment Abstraction Layer
User Space
Kernel Space
Environment Abstraction Layer
Linux Kernel
Platform Hardware

2.2 FD.io VPP(2)



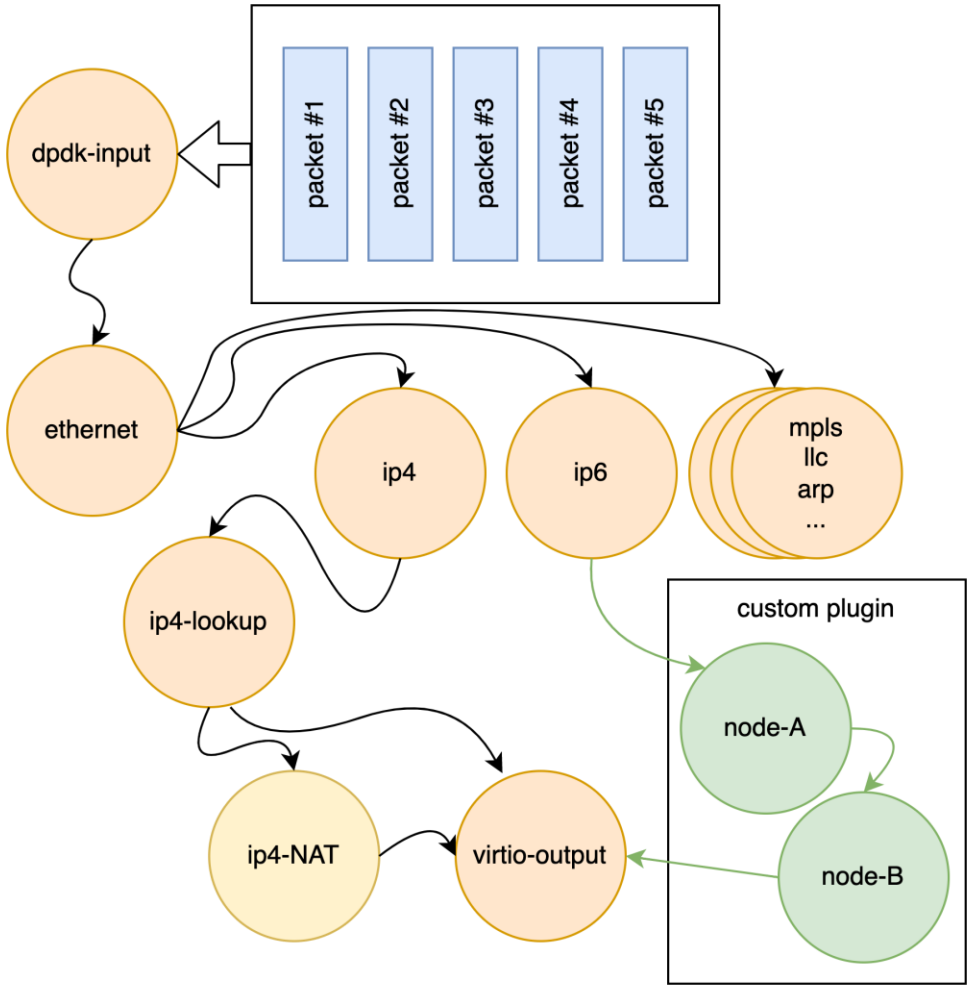


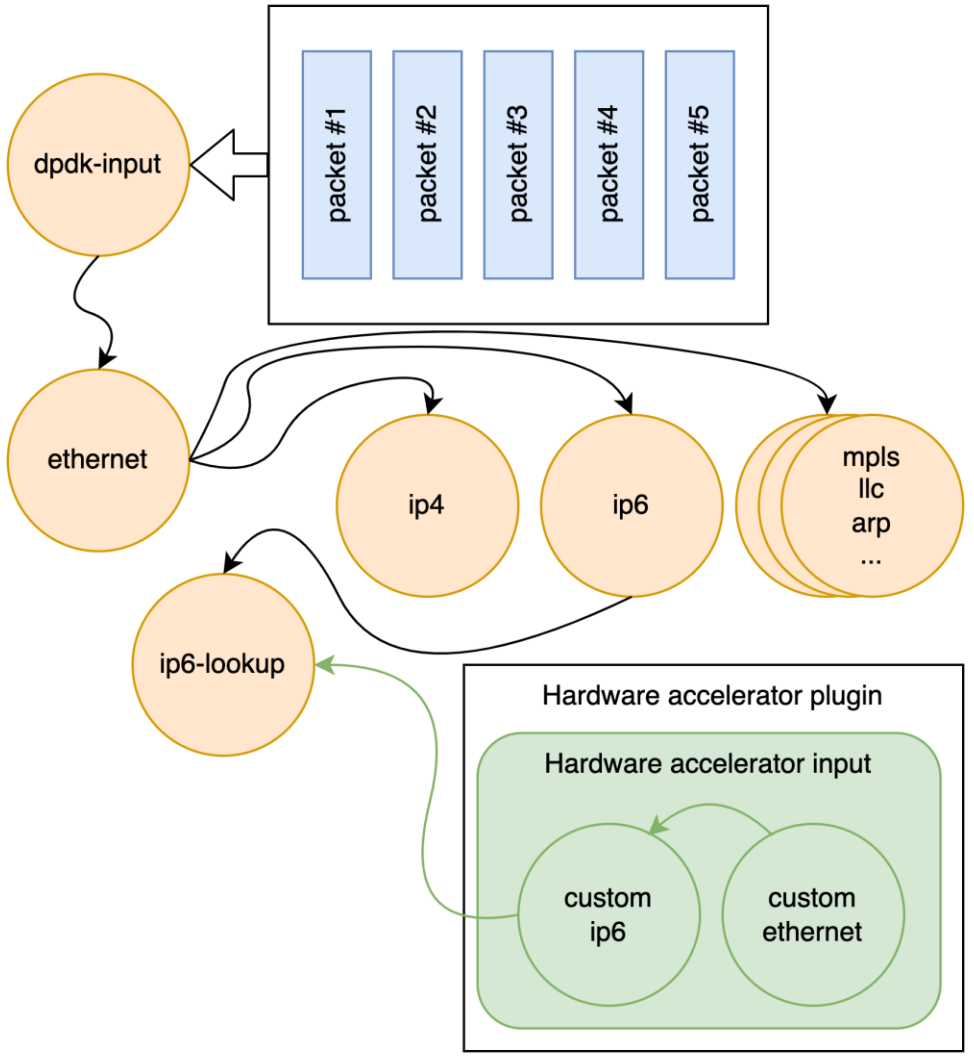


VPP

2.2 FD.io VPP(3)

Ref) https://fd.io/technology/





2.2 FD.io VPP(4) – Applications Area(1)

Routers/Universal CPE etc.

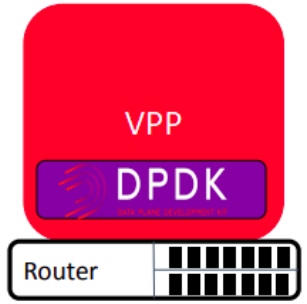
FD.io VPP supports entry hardware options from number of hardware vendors for building Customer Premise Equipmen FD.io based commercial options are available from vendors such as Netgate with TNSR, Cisco with the ASR 9000 and Carr Grade Services Engine, PANTHEON.tech with StoneWork Enterprise, and many more.

Commercial implementations integrate Strong Swan for IKE.

The implementations are accelerated with DPDK Cryptodev for whole platform crypto.

Featuring:

Data Plane	Control Plane
L2 Switch	Netconf/Yang
VLAN/ Q-inQ	BGP
L3 Router	IKEv2
NAT	
ACL (mac, ip, port)	



Broadband Network Gateway

FD.io VPP has a growing list of network traffic management and security features to support gateway uses cases such as Broadband Network Gateway.

Featuring:

Data Plane	Control Plane
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
hQoS	
ACL	
TM (Policing, Metering)	

Cloud Load Balancer

FD.io VPP has a rich set of plugin's to enhance its capabilities. Cloud load-balancing is just one of number of feature enhancin plugins available to the end user.

For example: Google Maglev Implementation, Consistent Hashing, Stateful and stateless load balancing, Kube-proxy integrat

Featuring:

Data Plane	Control Plane
Bonding	K8s Kube-Proxy
VLAN / Q-in-Q	
NAT	
ACL (blacklist)	
TM (policing, metering)	
L4 Load Balancer	

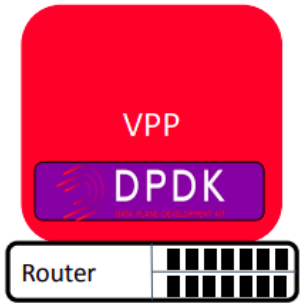
Intrusion Prevention System

FD.io VPP has four different Access Control List technologies; ranging from the simple IP-address whitelisting (called COP) to the sophisticated FD.io VPP Classifiers.

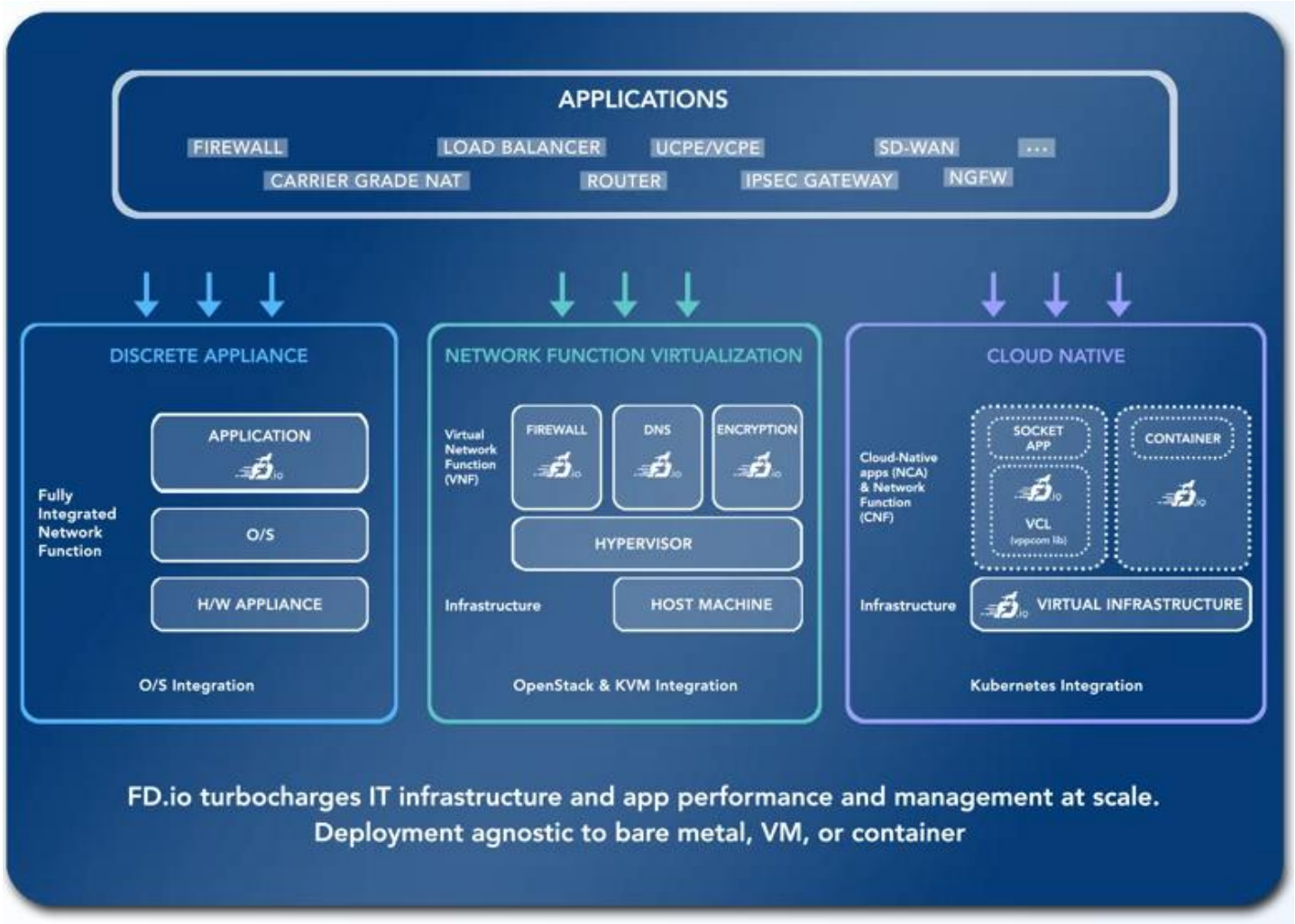
Featuring:

Data Plane	Control Plane
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
NAT (mac, ip, port)	
ACL (mac, ip, port)	

2.2 FD.io VPP(4) – Applications Area(2)



- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- …



Ref) https://www.netgate.com/resources/articles-vector-packet-processing

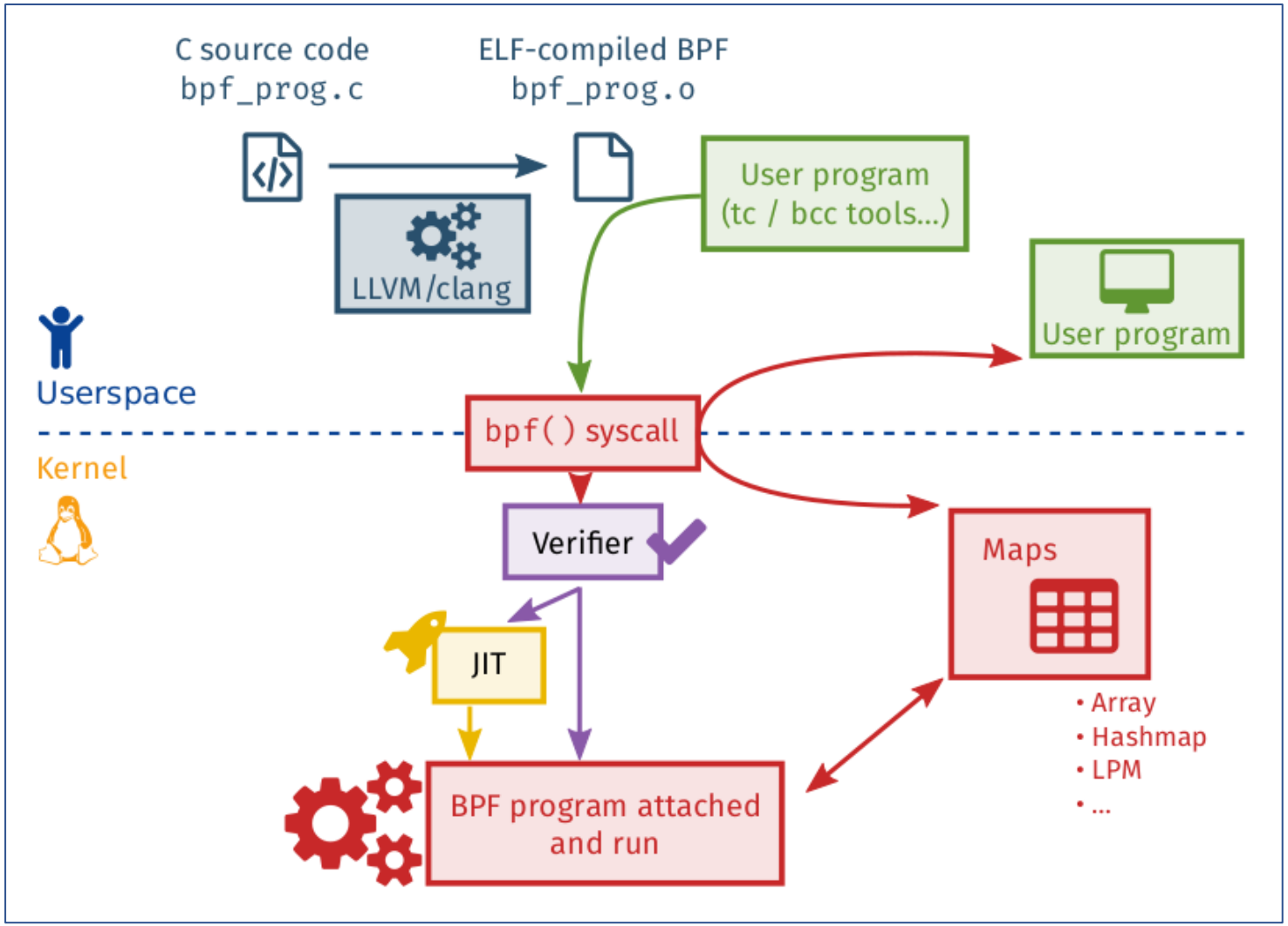
2.3 eBPF XDP

eBPF XDP



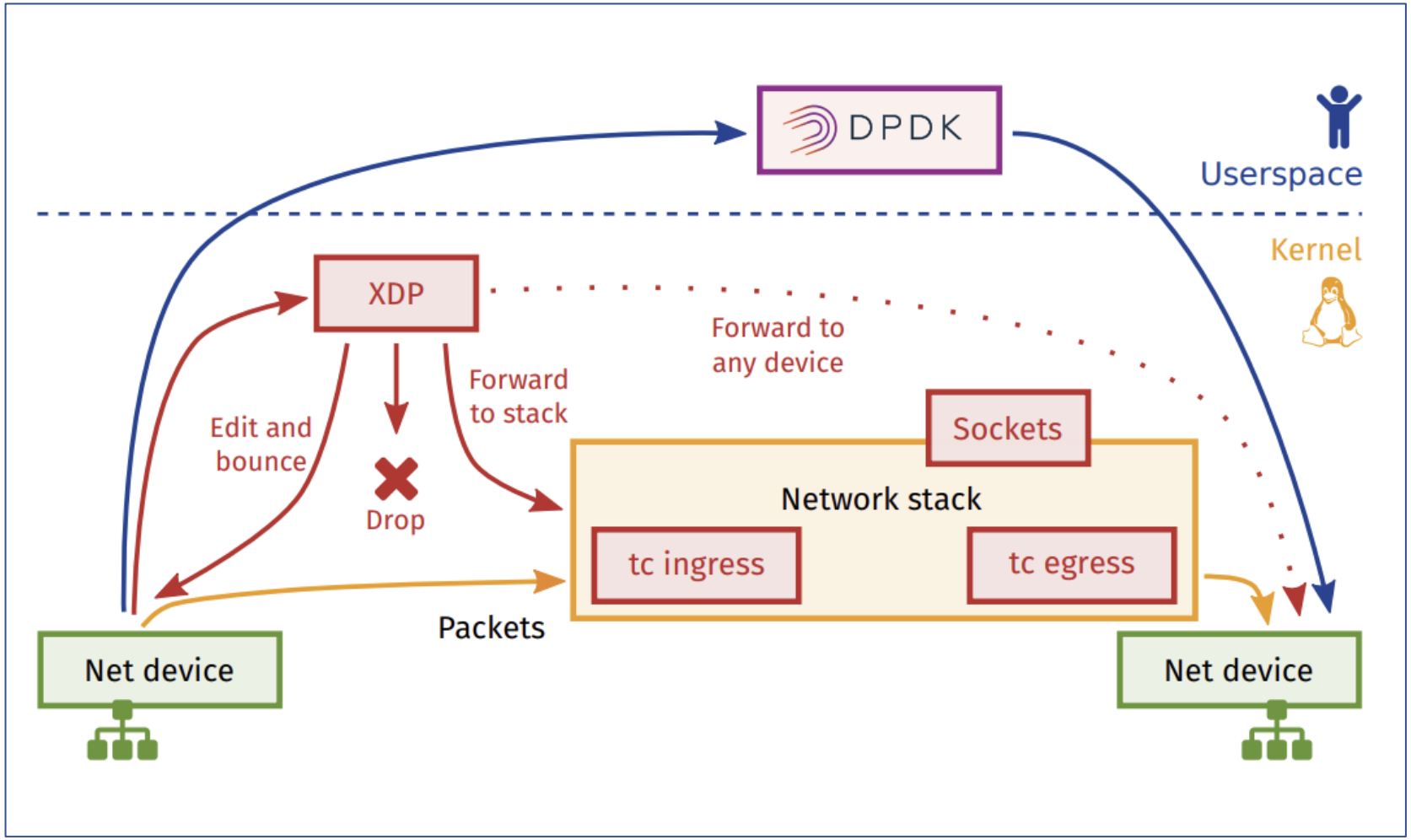
https://ebpf.io/

2.3 eBPF XDP(1) – eBPF Concept



Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

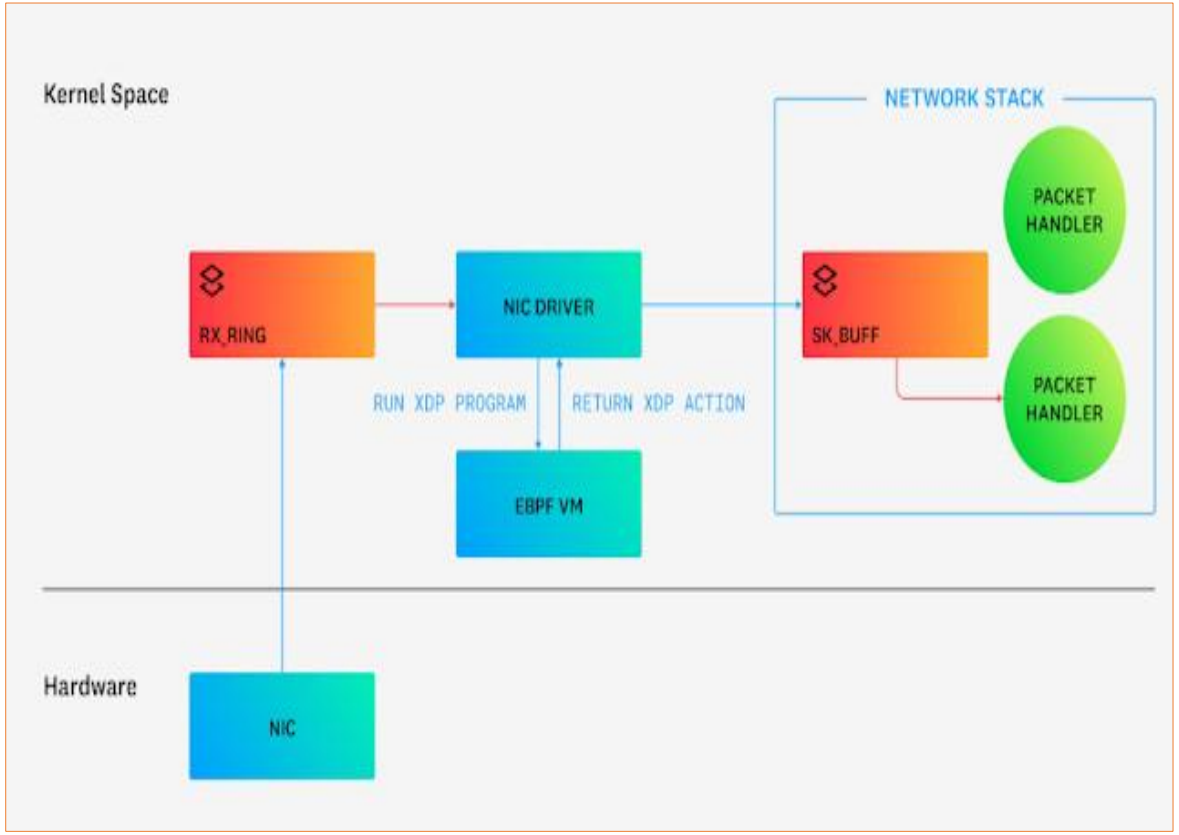
2.3 eBPF XDP(2) – XDP Concept(1)

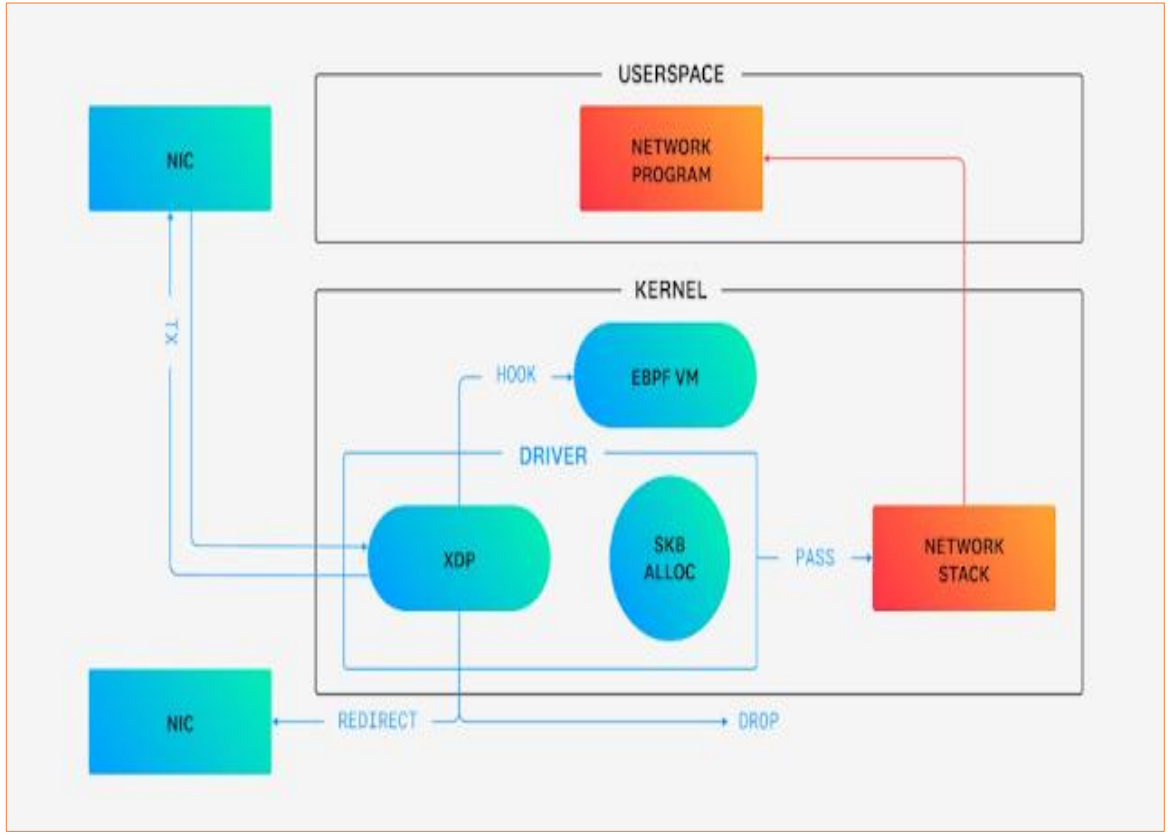


Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

2.3 eBPF XDP(2) – XDP Concept(2)

Ref) https://www.datadoghq.com/blog/xdp-intro/

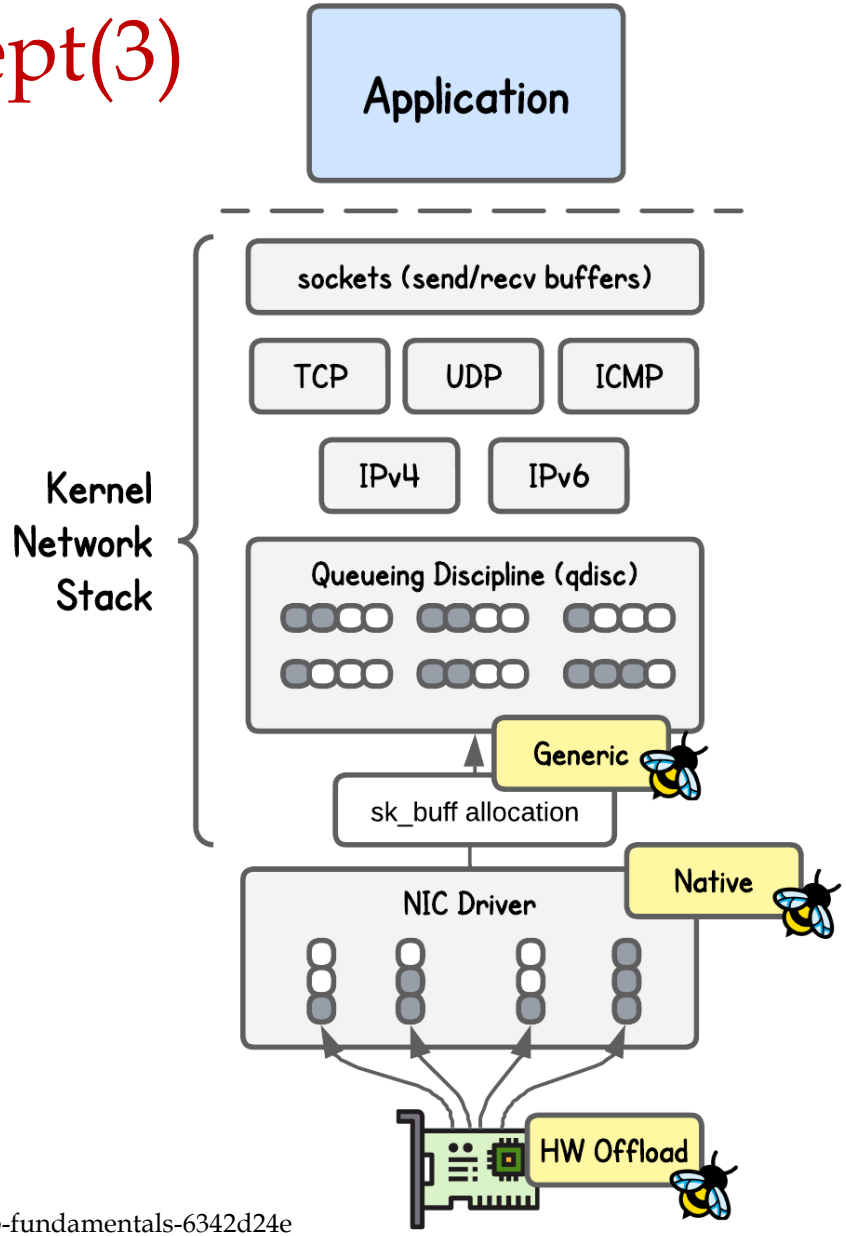




eBPF XDP

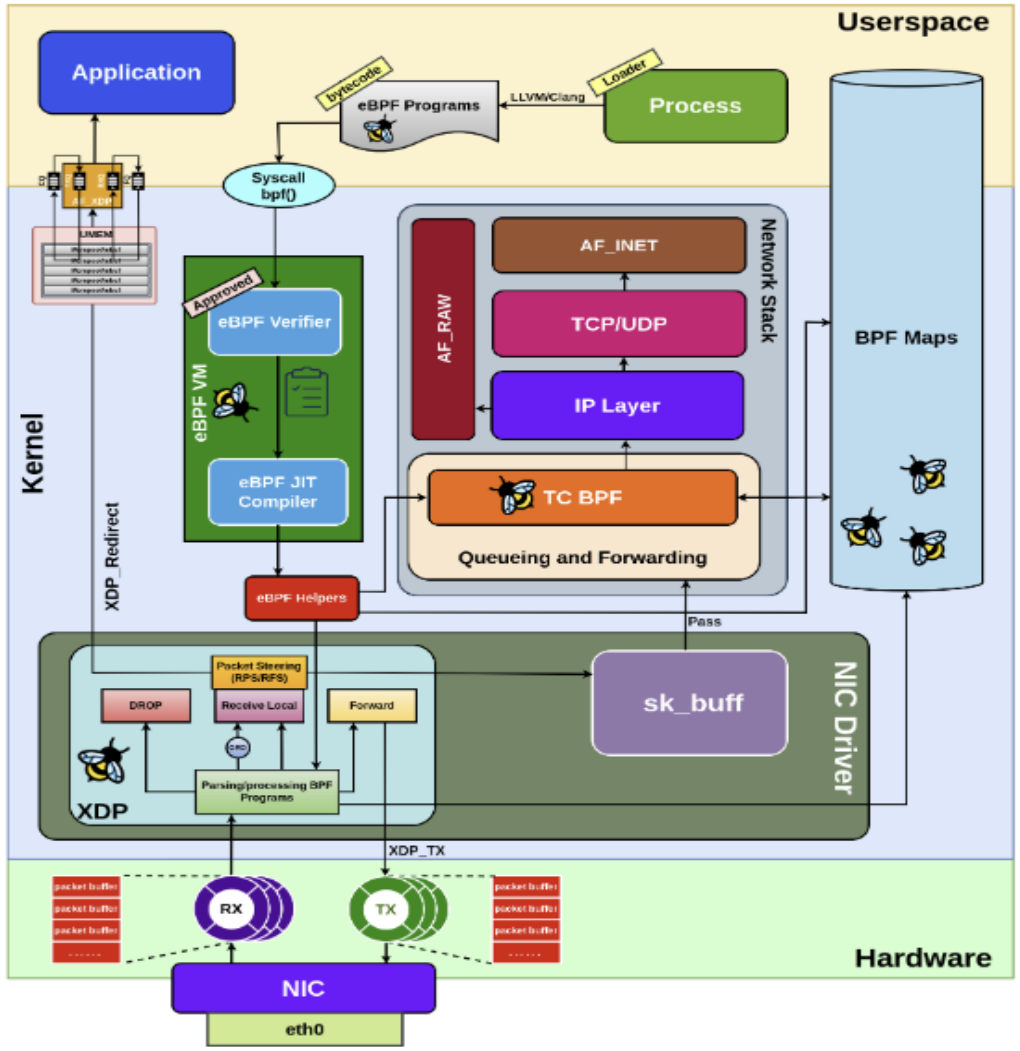
2.3 eBPF XDP(2) – XDP Concept(3)

- Generic Mode
- Native Mode
- HW Offload Mode



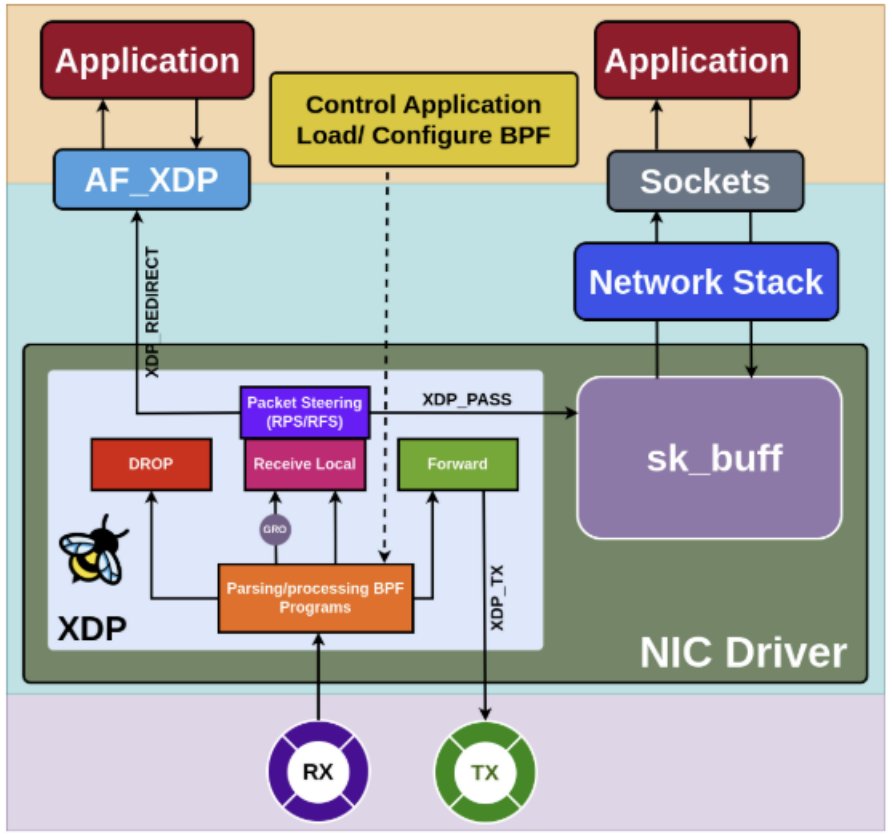
Ref) https://labs.iximiuz.com/tutorials/ebpf-xdp-fundamentals-6342d24e

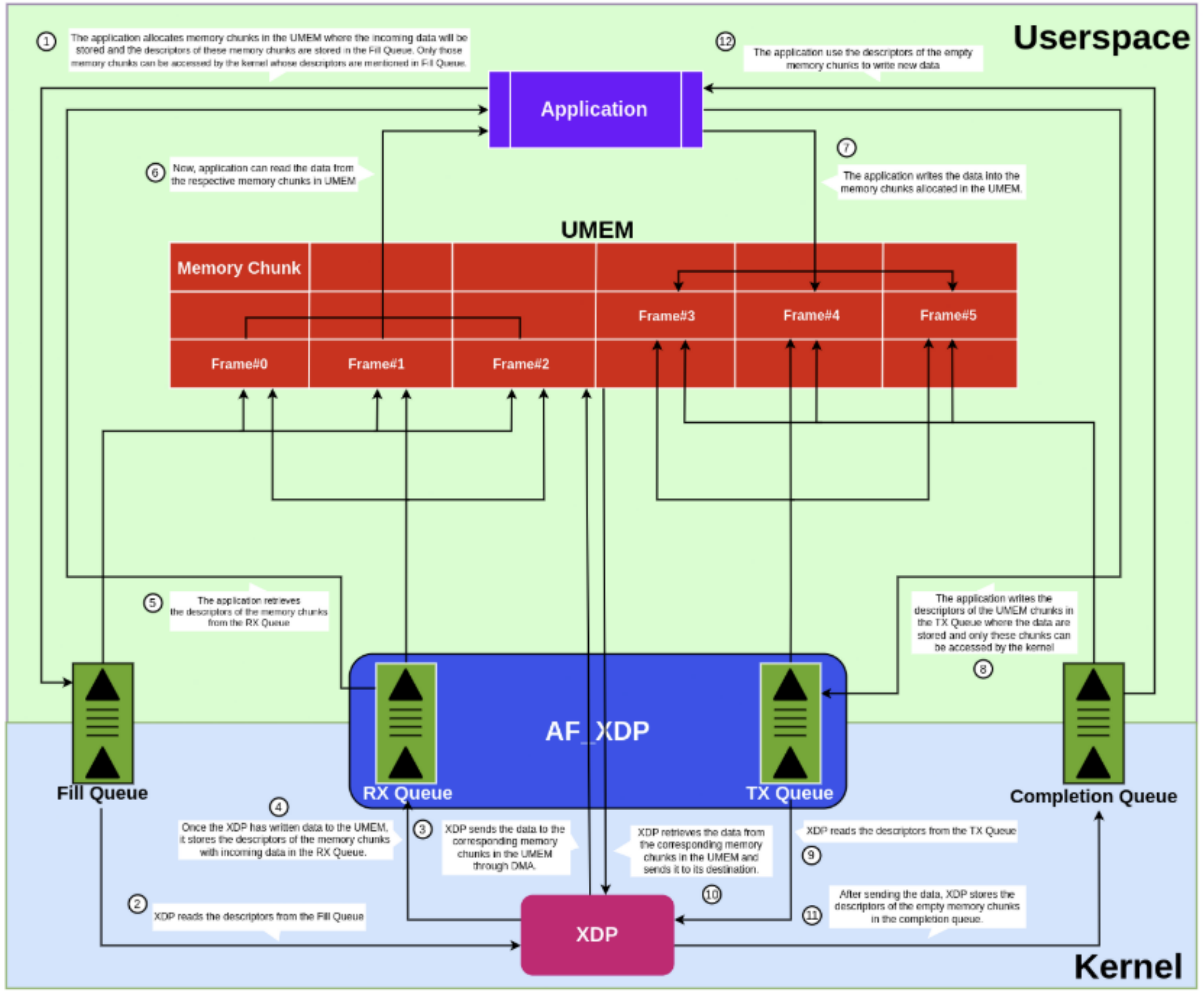
2.3 eBPF XDP(3) – XDP & AF_XDP(1)



Ref) https://docs.ngkore.com/blogs/afxdp/afxdp.html#hardware-requirements

2.3 eBPF XDP(3) - XDP & AF_XDP(2)





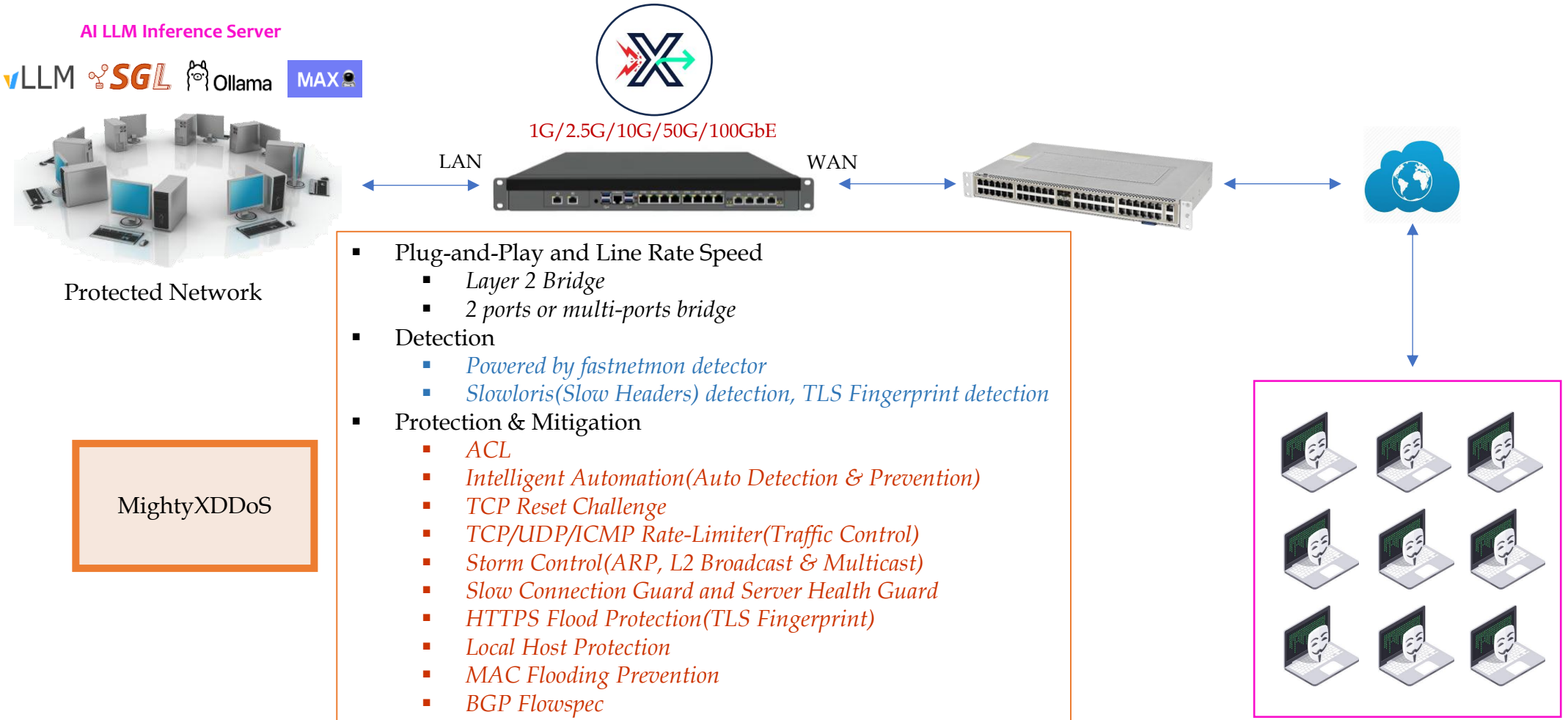
Ref) https://docs.ngkore.com/blogs/afxdp/afxdp.html#hardware-requirements

3. MightyXDDoS™



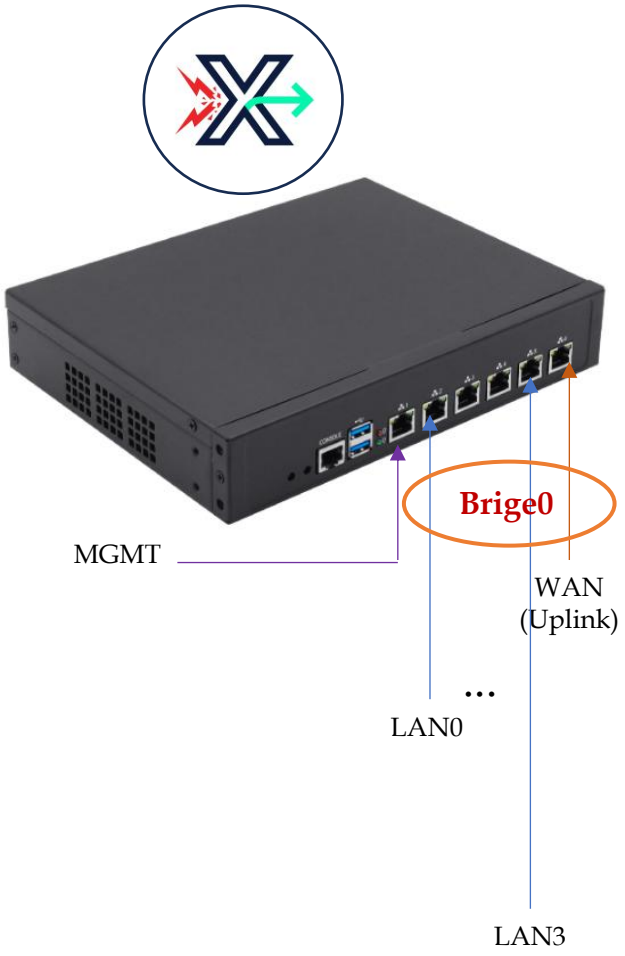
XDDoS Mitigator

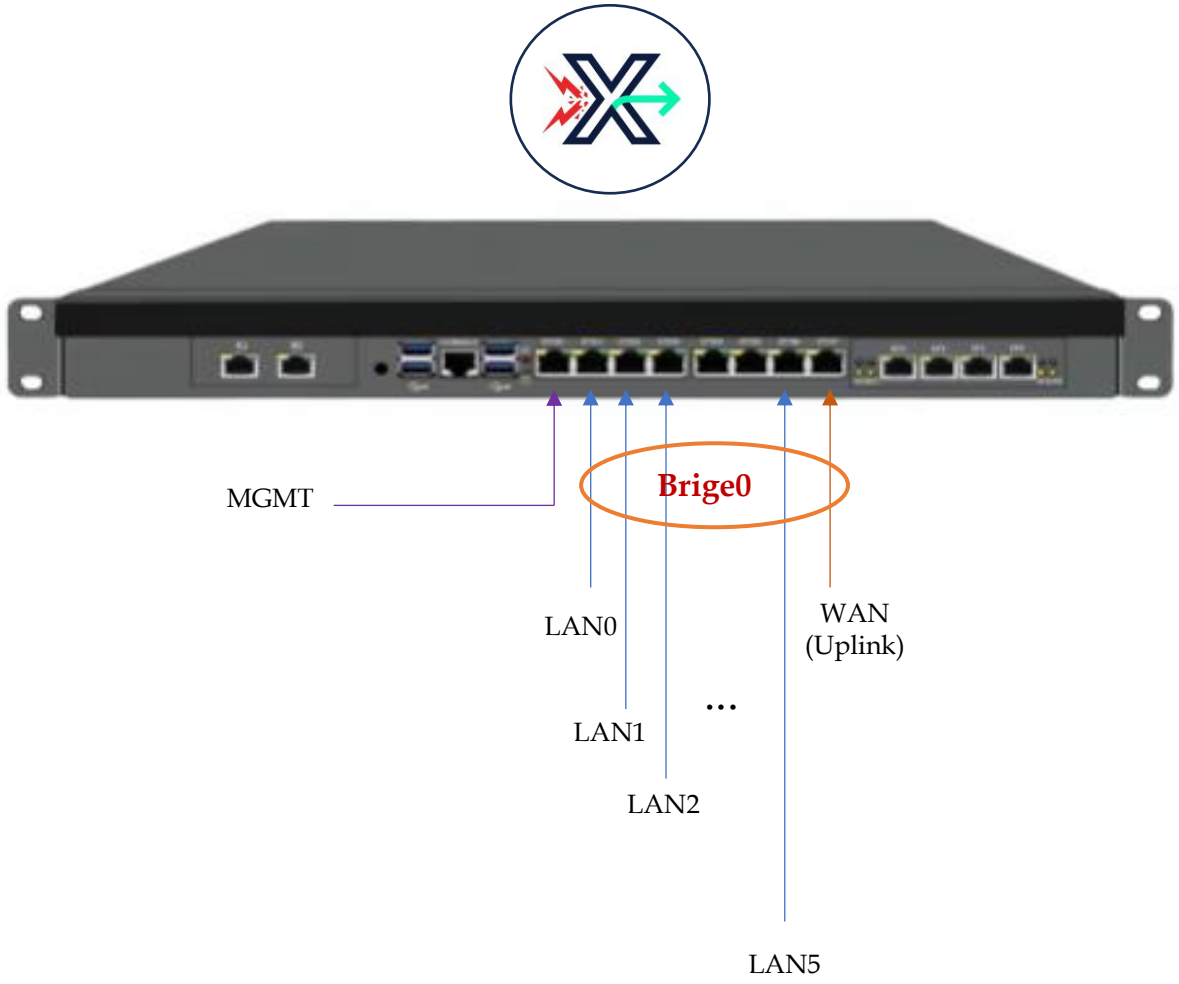
3. MightyXDDoS(1) – Inline Mitigator Box(1)



3. MightyXDDoS(1) – Inline Mitigator Box(2)

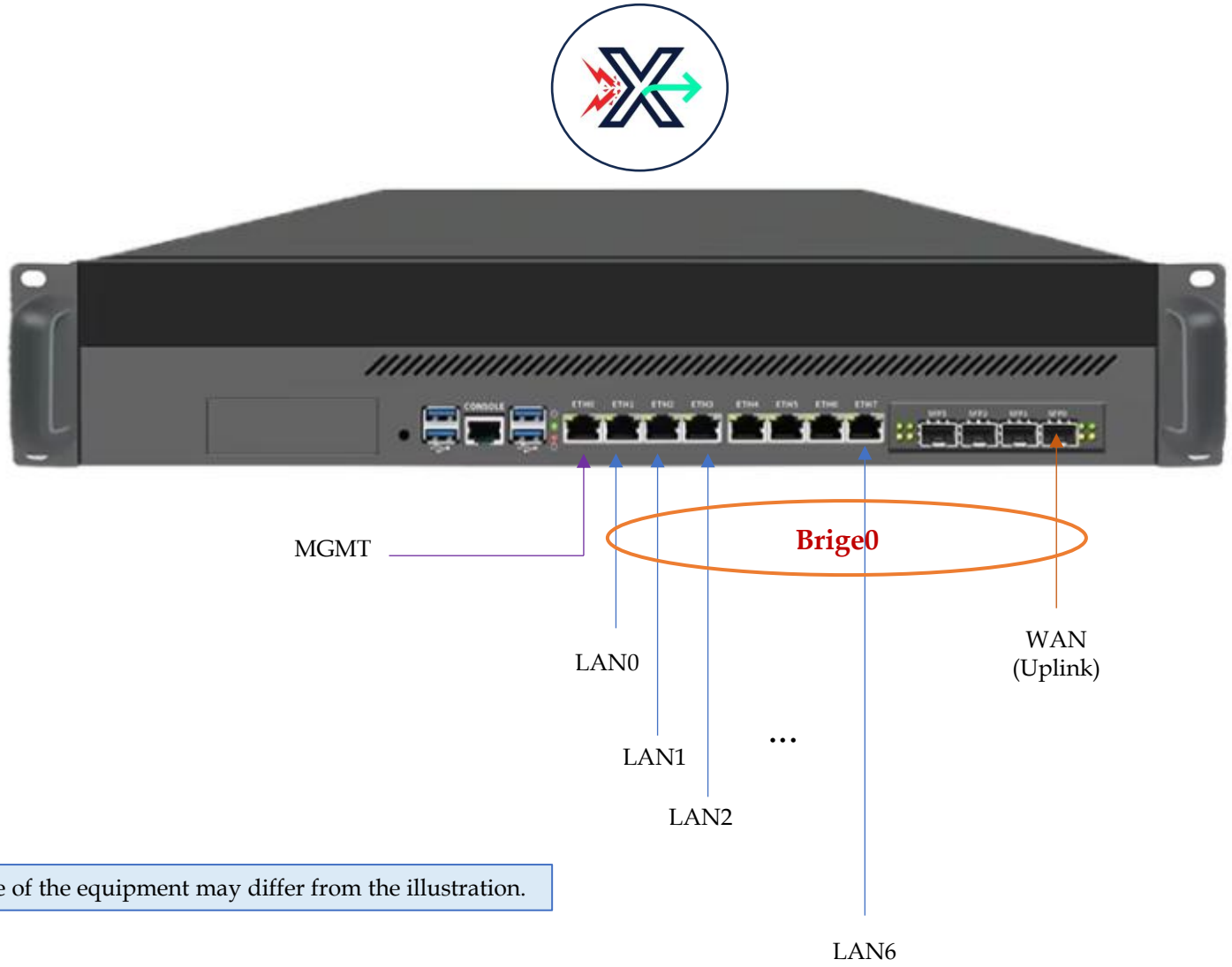
DDoS Mitigator for 1 Server





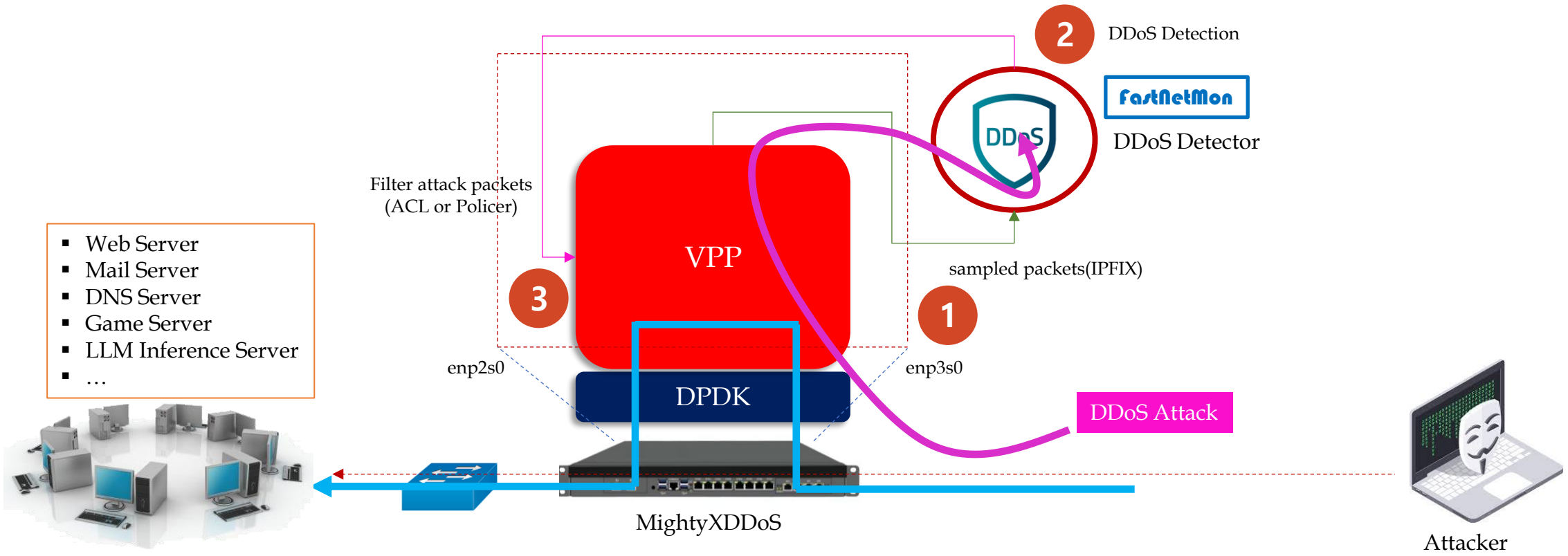
Note: The appearance of the equipment may differ from the illustration.

3. MightyXDDoS(1) – Inline Mitigator Box(3)

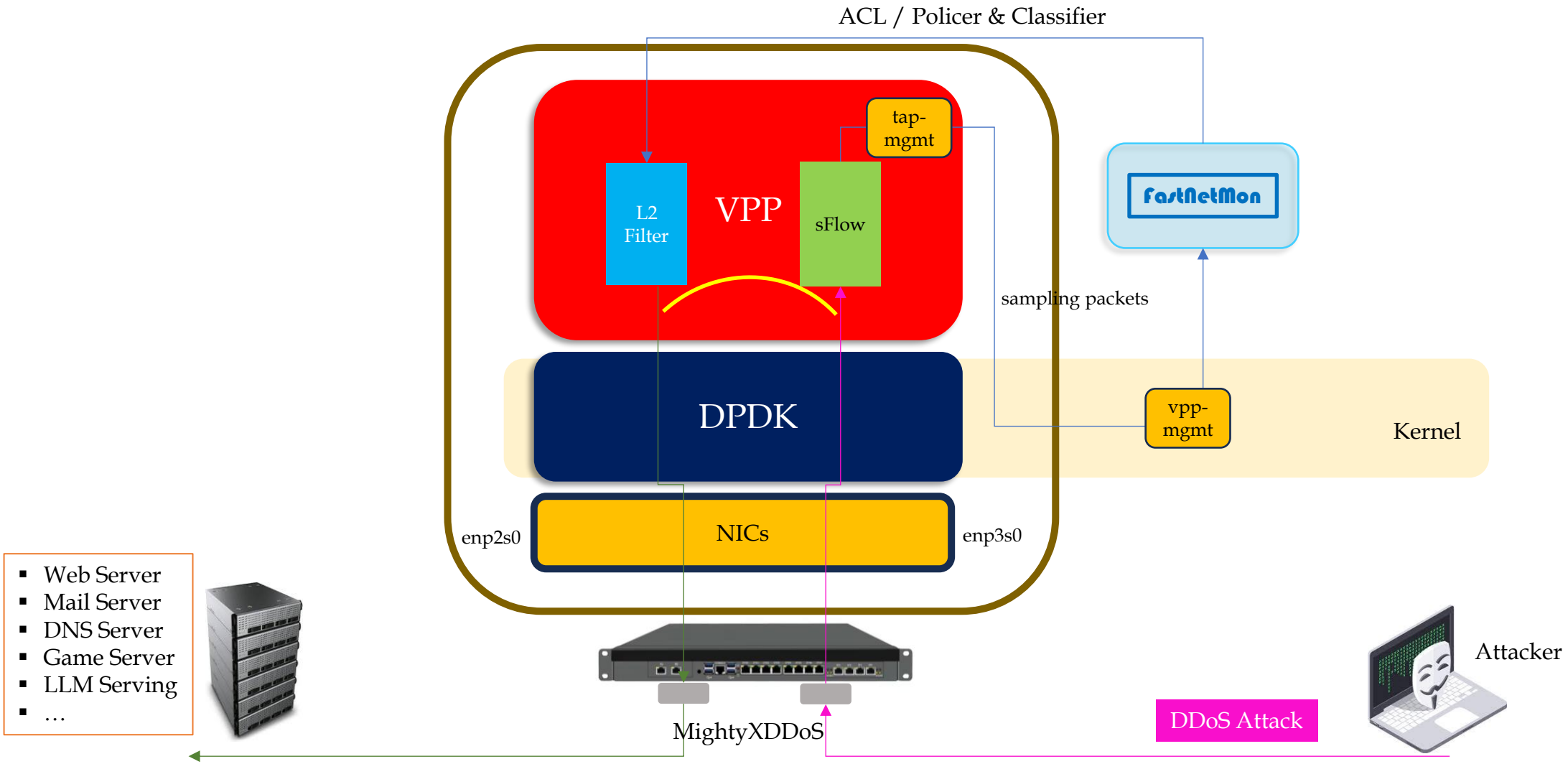


Note: The appearance of the equipment may differ from the illustration.

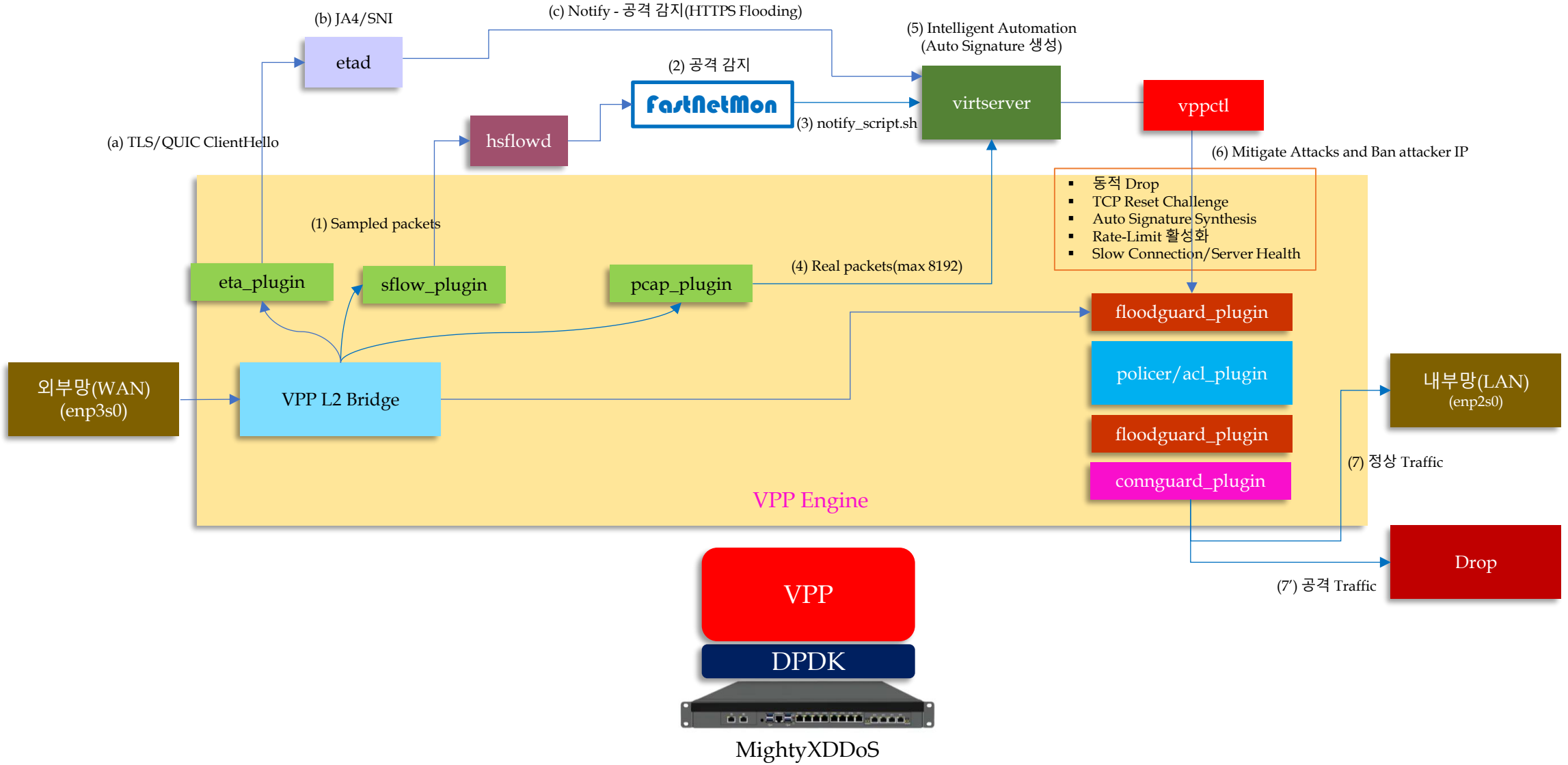
3. MightyXDDoS(2) – VPP Mitigator(1)



3. MightyXDDoS(2) – VPP Mitigator(2)



3. MightyXDDoS(2) – VPP Mitigator(3)

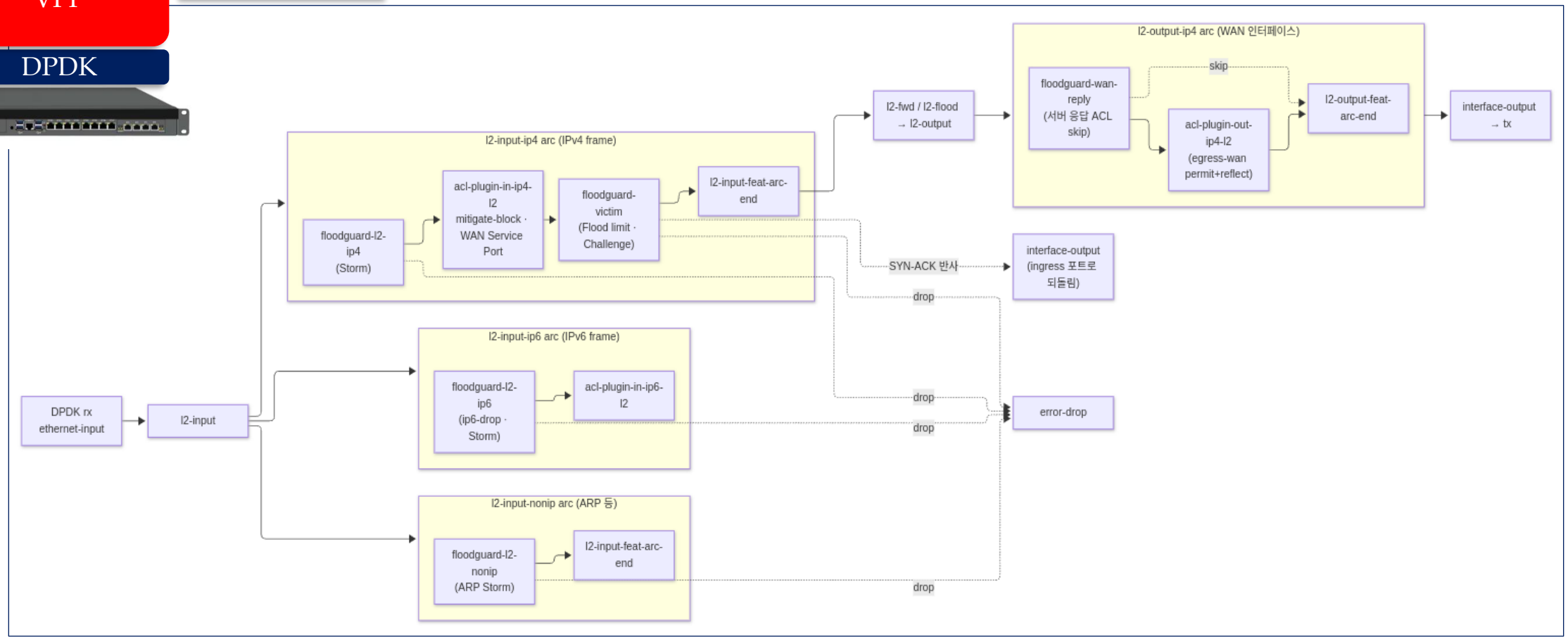


MightyXDDoS

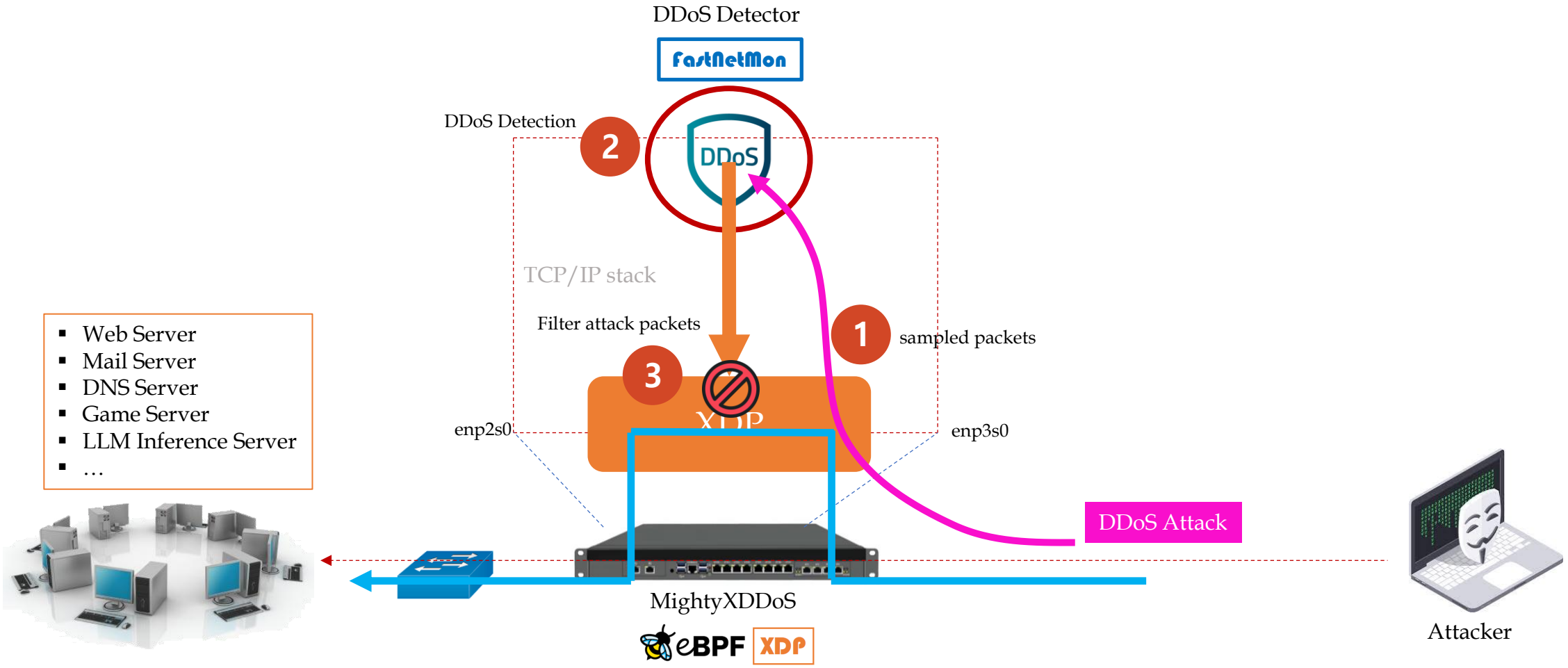
3. MightyXDDoS(2) – VPP Mitigator(4)



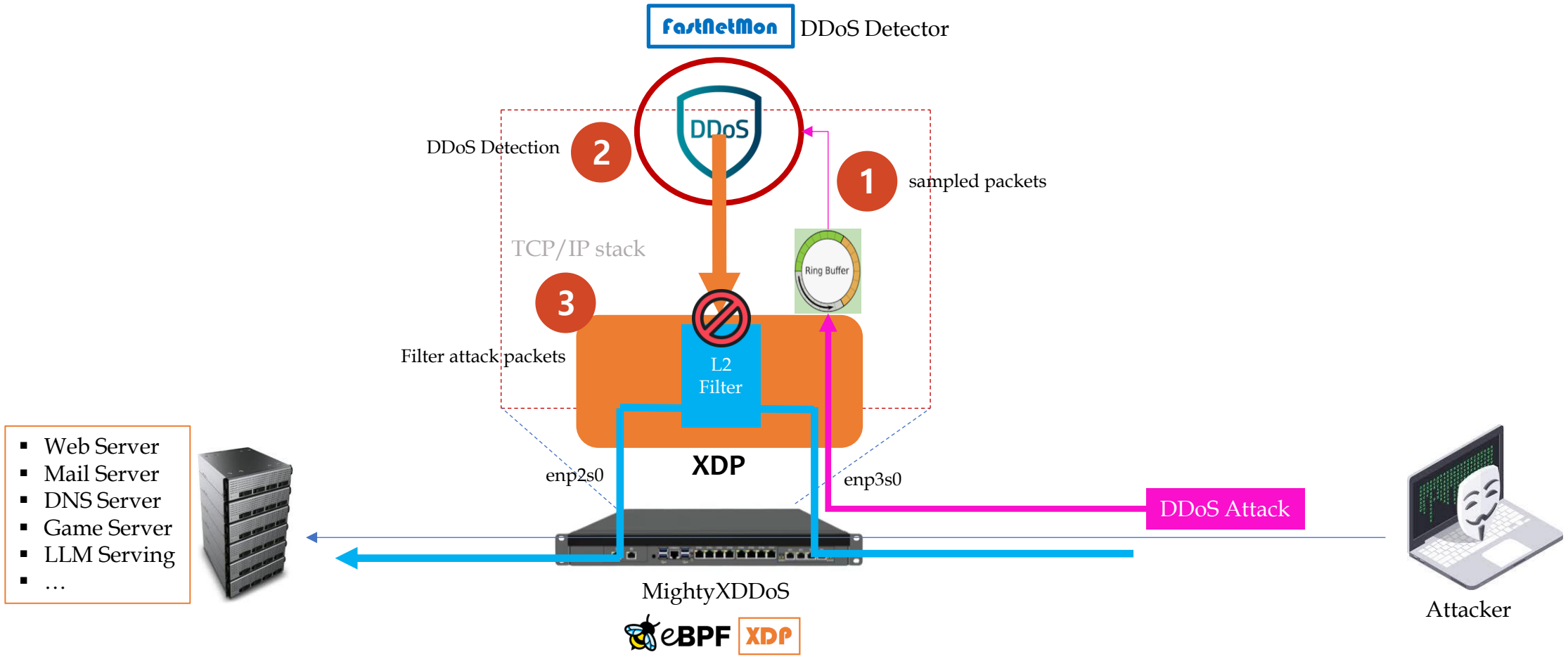
floodguard_plugin



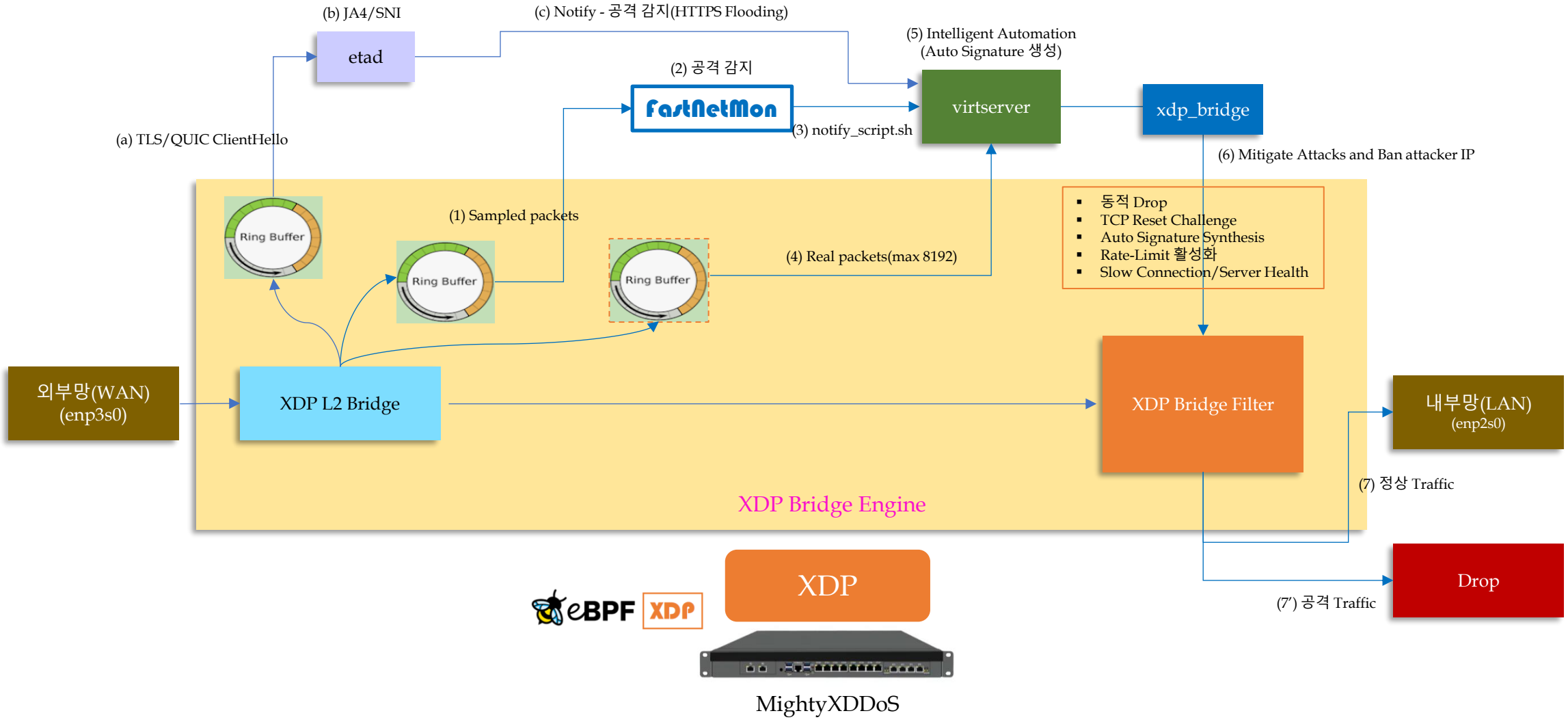
3. MightyXDDoS(3) – XDP Mitigator(1)



3. MightyXDDoS(3) – XDP Mitigator(2)

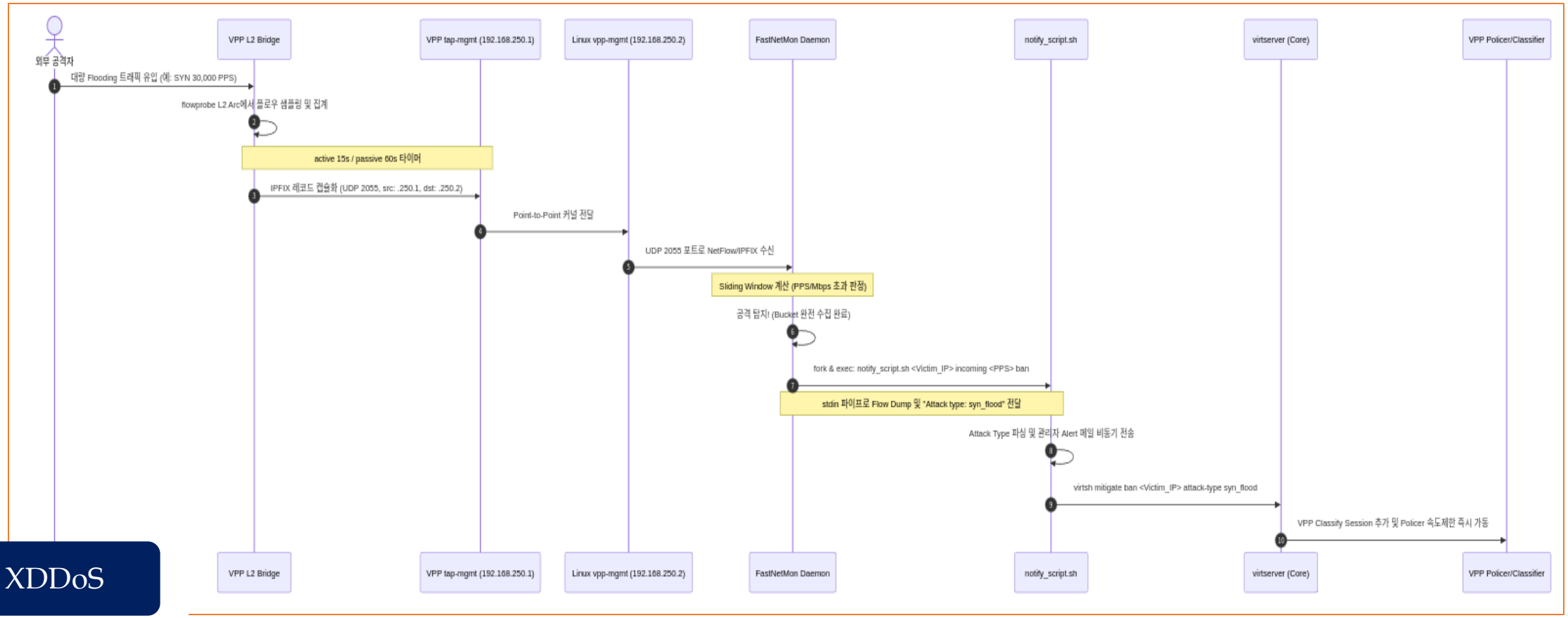


3. MightyXDDoS(3) – XDP Mitigator(3)



3. MightyXDDoS(4) – w/ FastNetMon Detector(1)

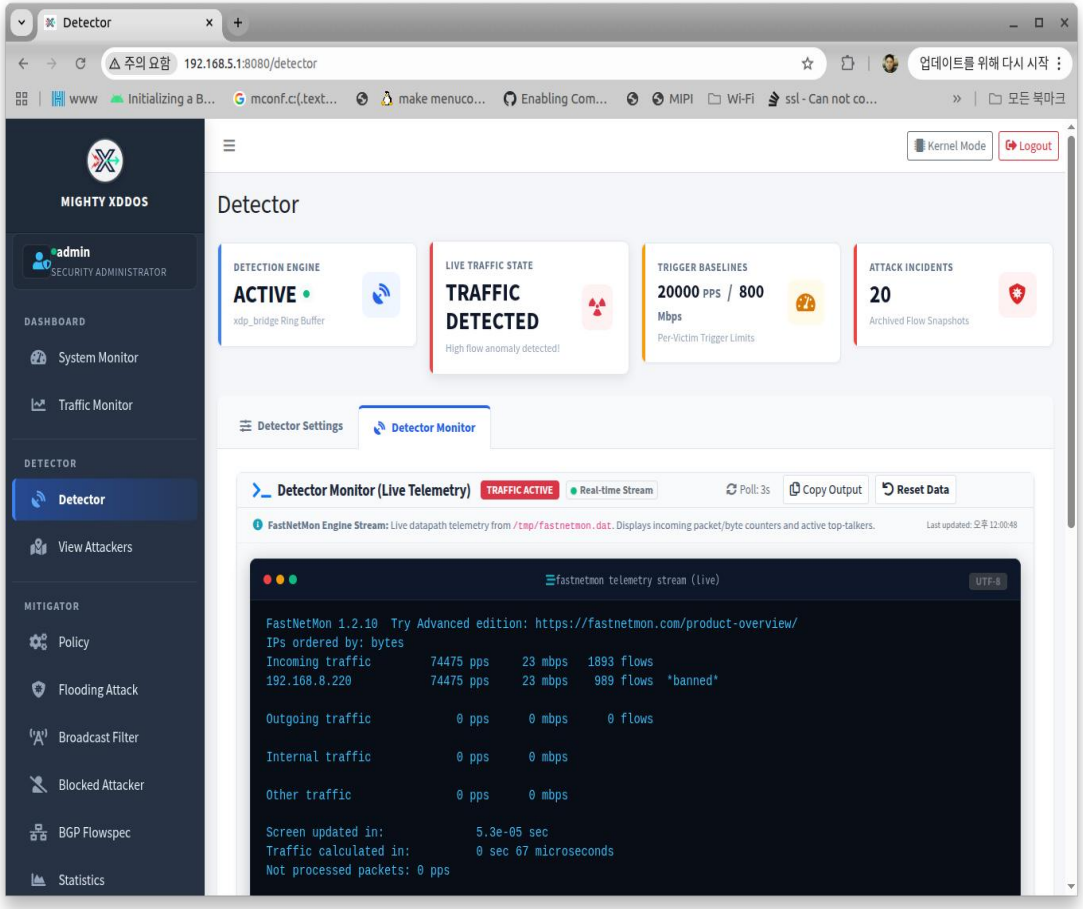
FastNetMon

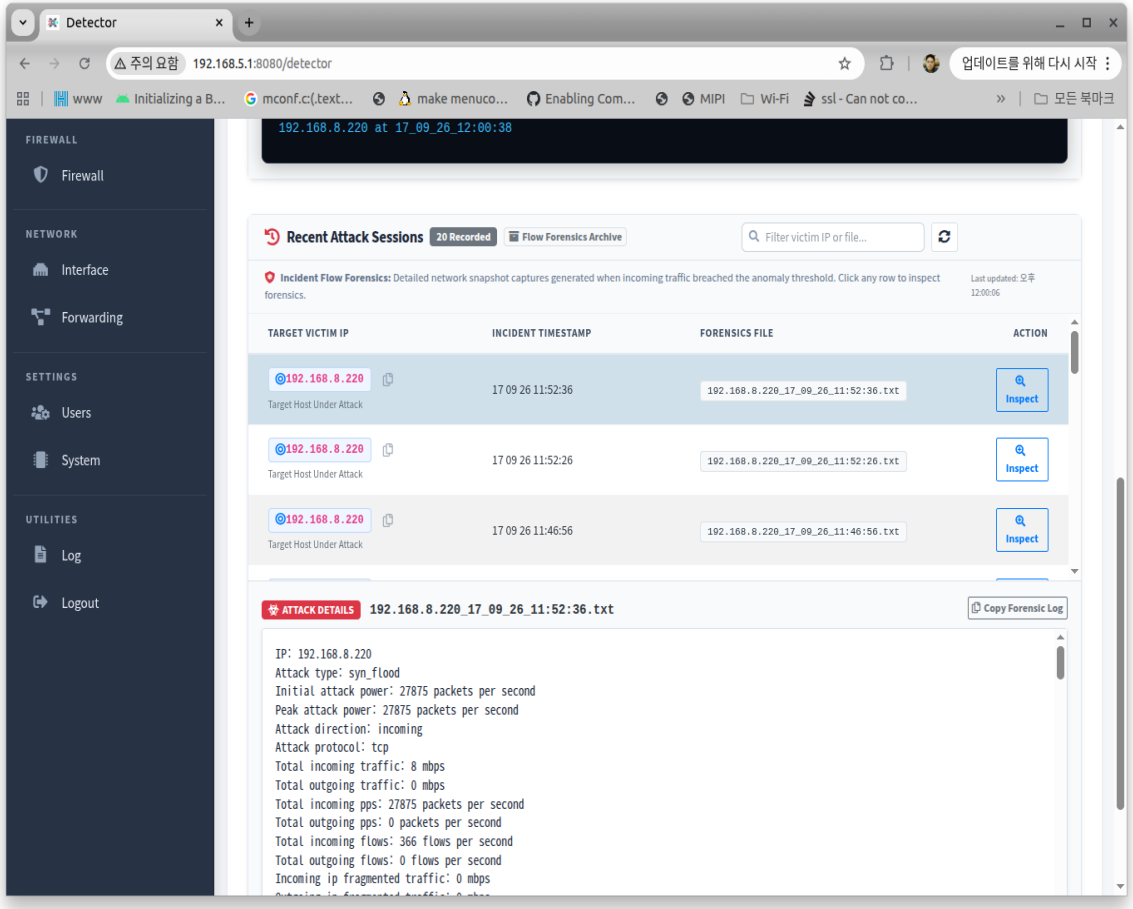


XDDoS



3. MightyXDDoS(4) – w/ FastNetMon Detector(2)





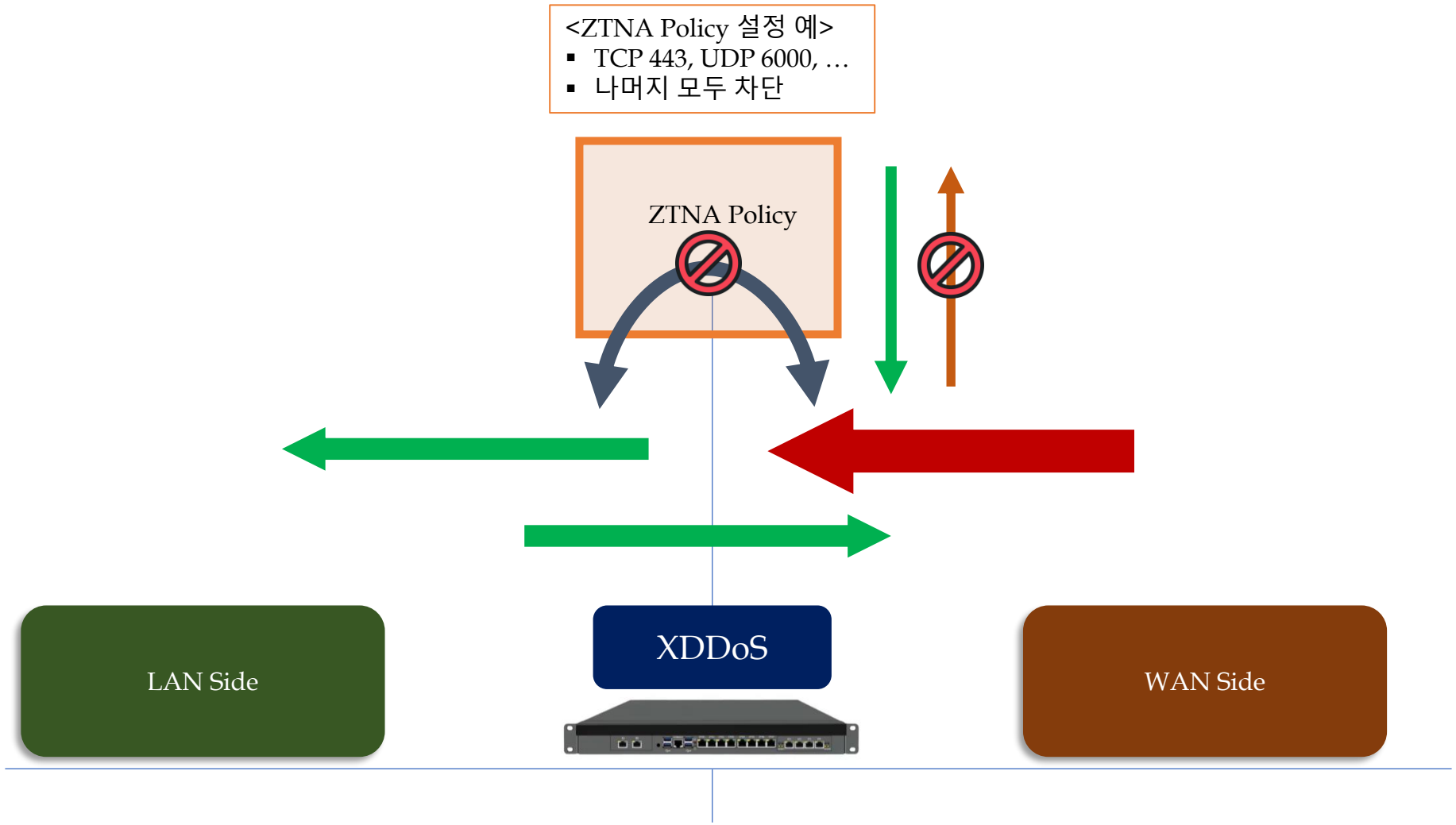
3. MightyXDDoS(5) – Mitigation Types

- **[1] ZTNA Policy**
  - 허용할 Port를 제외하고는 모두 차단
  - Line speed packet forwarding
- **[2] TCP DDoS Attack Protection**
  - TCP Reset Challenge(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
  - TCP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
  - TCP Reset Challenge Only
  - TCP PPS Threshold 기반의 Rate-Limiting
- **[3] UDP DDoS Attack Protection**
  - UDP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
  - UDP PPS Threshold 기반의 Rate-Limiting
- **[4] ICMP DDoS Attack Protection**
  - ICMP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
  - ICMP PPS Threshold 기반의 Rate-Limiting
  - 기본적으로 ZTNA Policy에서 ICMP는 허용 안함으로 설정

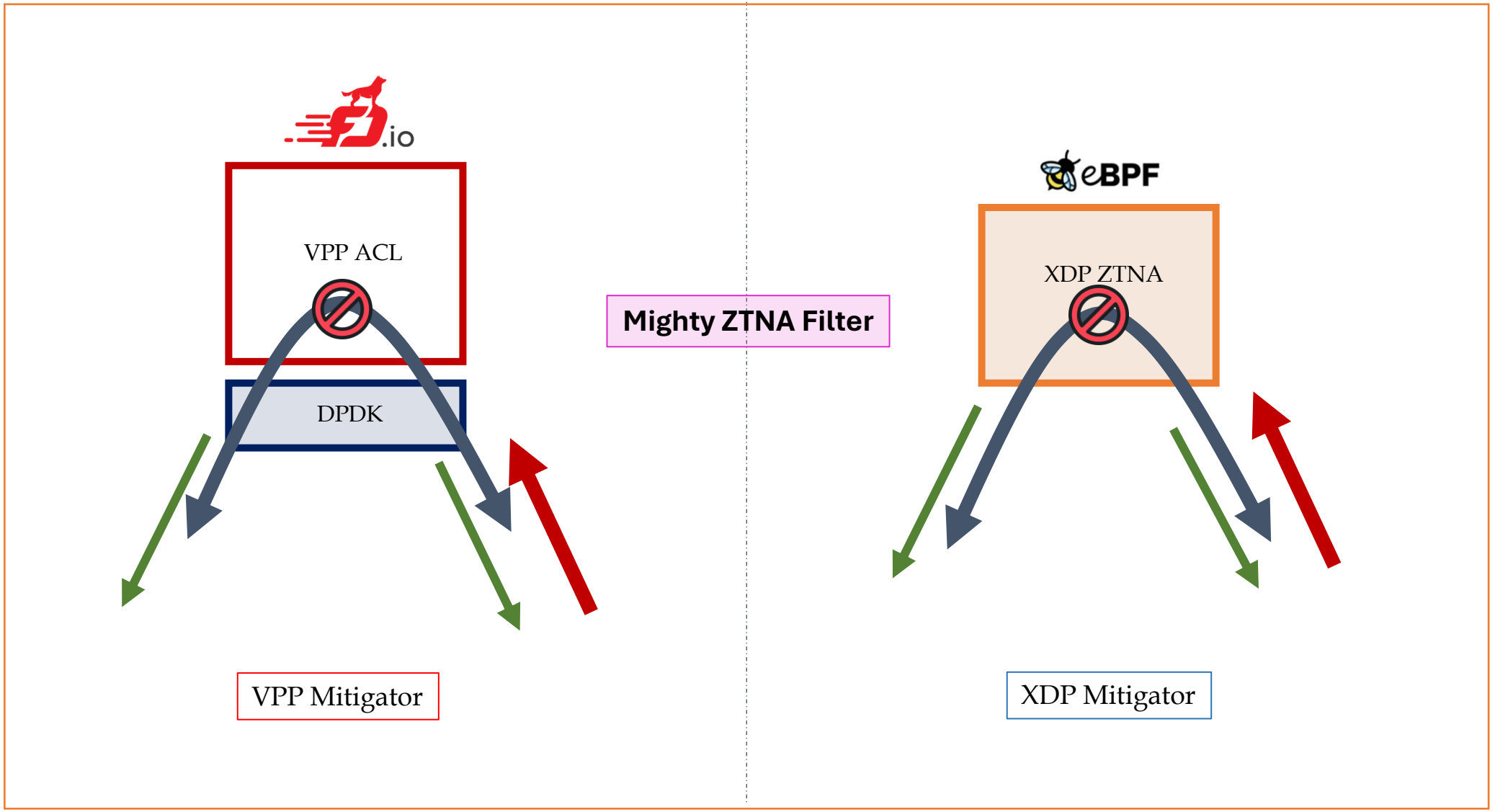
- **[5] Box Local Attack Protection**
  - Box -> Outbound Session 허용, 나머지 모두 차단
  - Rate-Limit
- **[6] Storm Control**
  - L2 Broadcast/Multicast, ARP
- **[7] Slow Connection Guard**
  - Slowloris Attack Protection
  - Slow Request/Hold/Read
- **[8] Server Health Guard**
  - Server 과부하 상태 체크(SYN+ACK 응답률 하락 체크, RST 비율 체크 등)
  - 서버 응답 데이터량 대비 요청 수 비율 변화 체크
- **[9] HTTPS Flood(TLS Fingerprint) Protection**
  - HTTPS Auto Signature & JA4 Fingerprint
- **[10] BGP Flowspec Protection**
  - 회선의 85%를 초과하는 트래픽 유입 시, BGP Flowspec 기능 동작



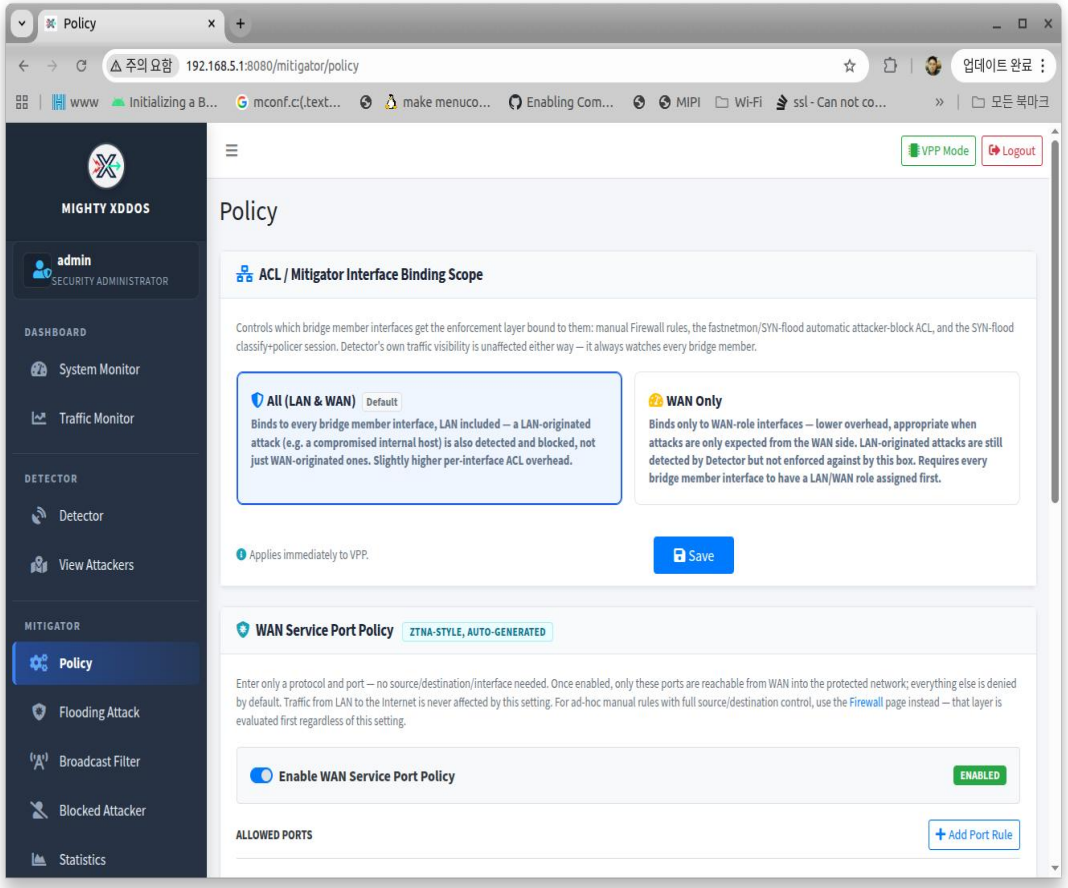
3. MightyXDDoS(6) – ACL Policy(1)

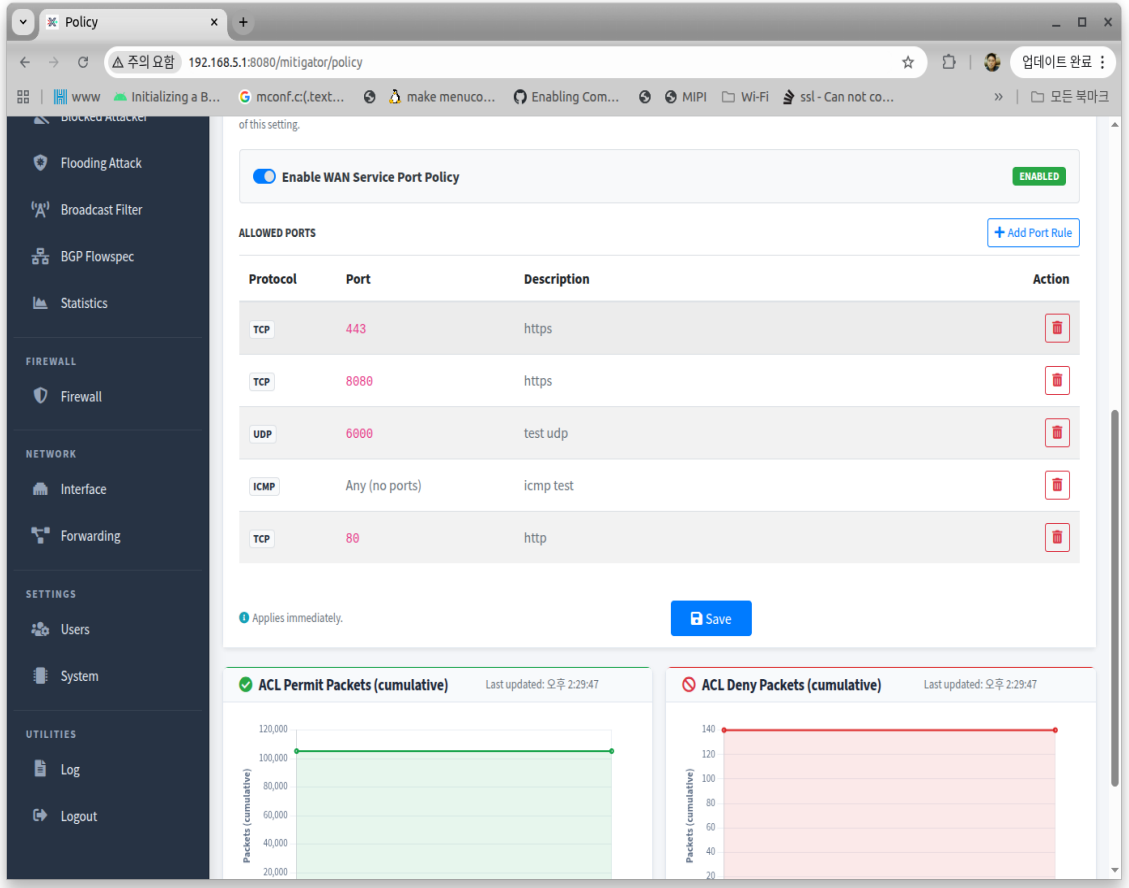


3. MightyXDDoS(6) – ACL Policy(2)



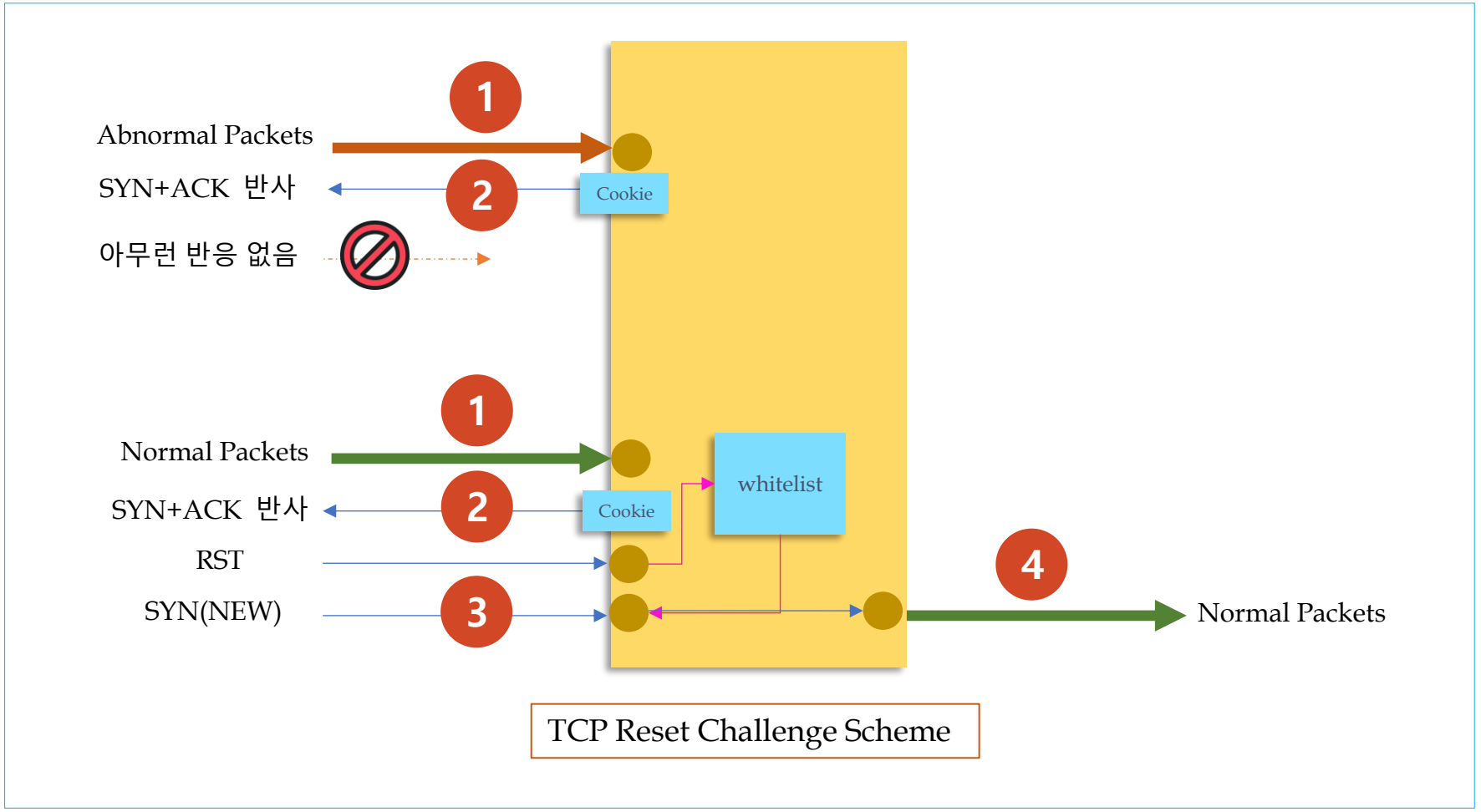
3. MightyXDDoS(6) – ACL Policy(3)



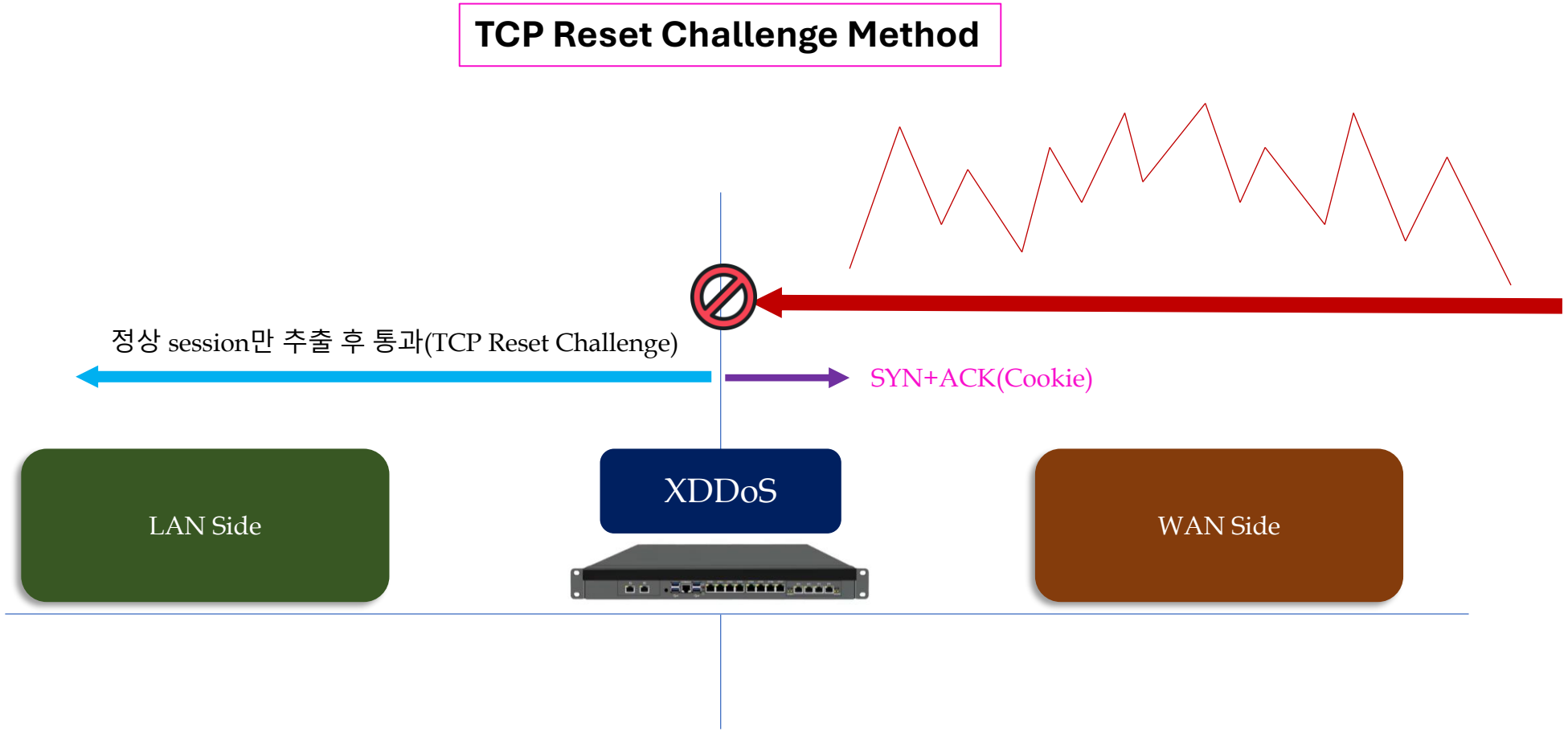


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(1)

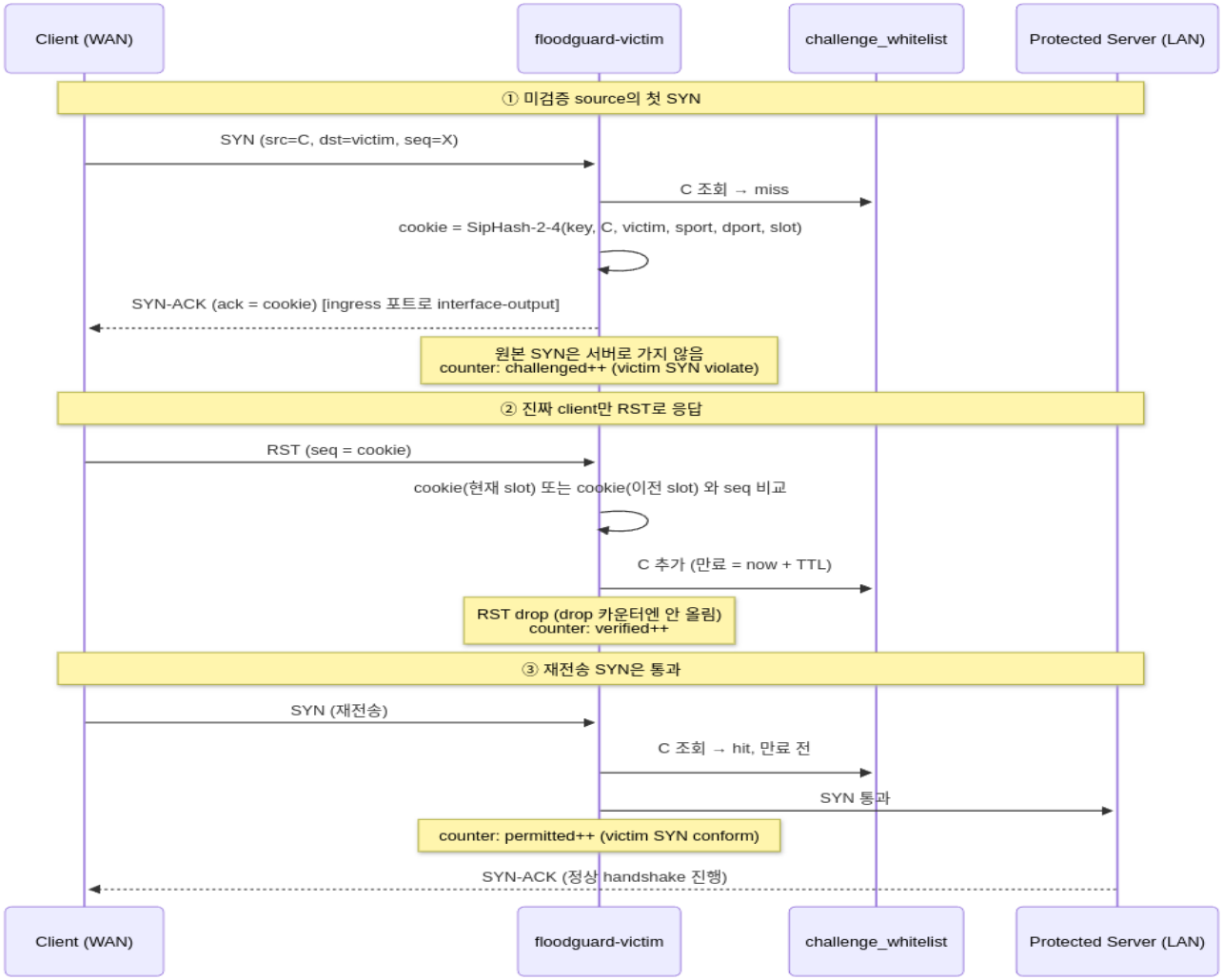
공격 패킷 차단, 정상 TCP 접속만 선별 후 통과 기법



3. MightyXDDoS(7) – TCP Reset Challenge Scheme(2)

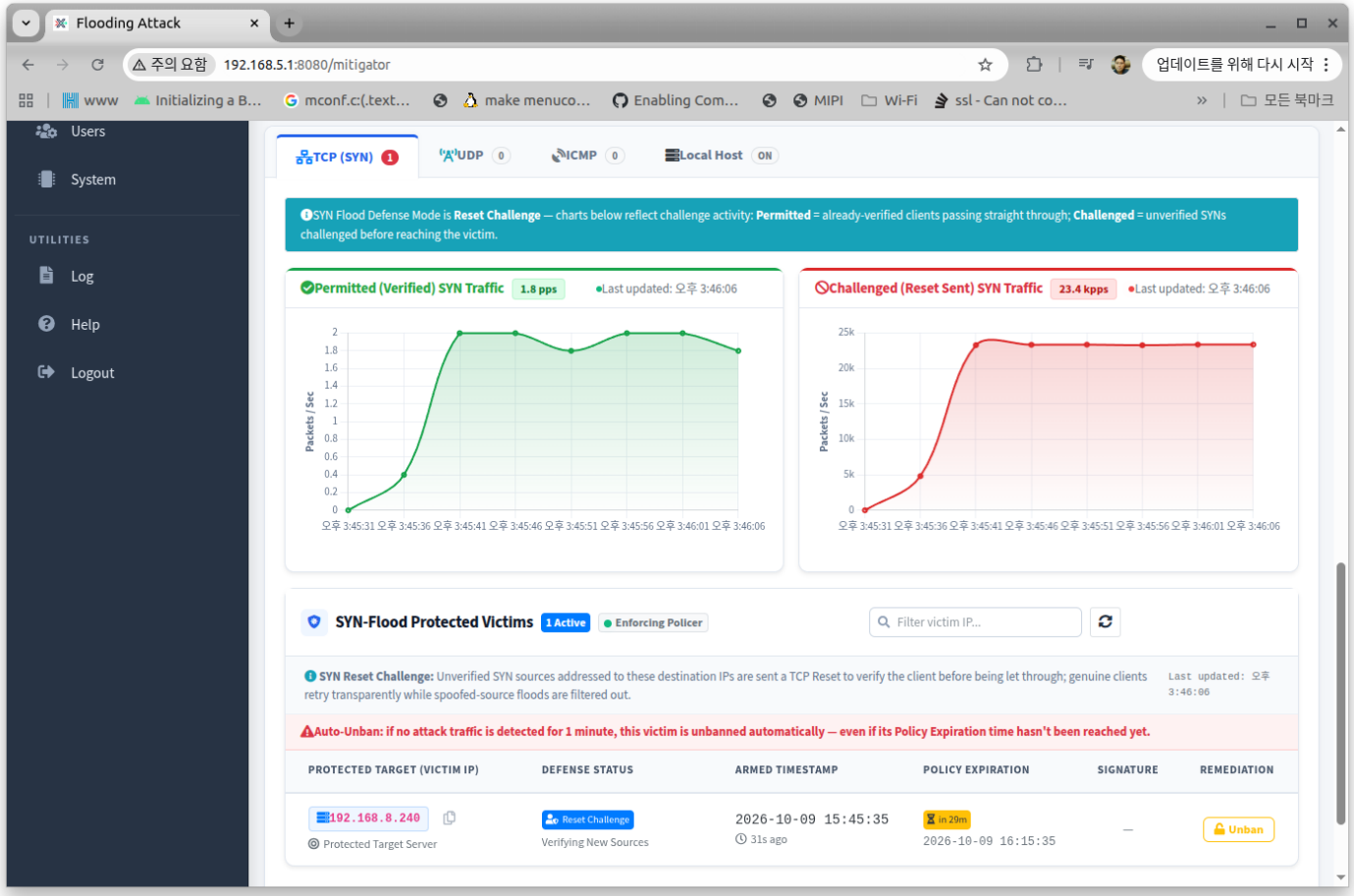


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(3)



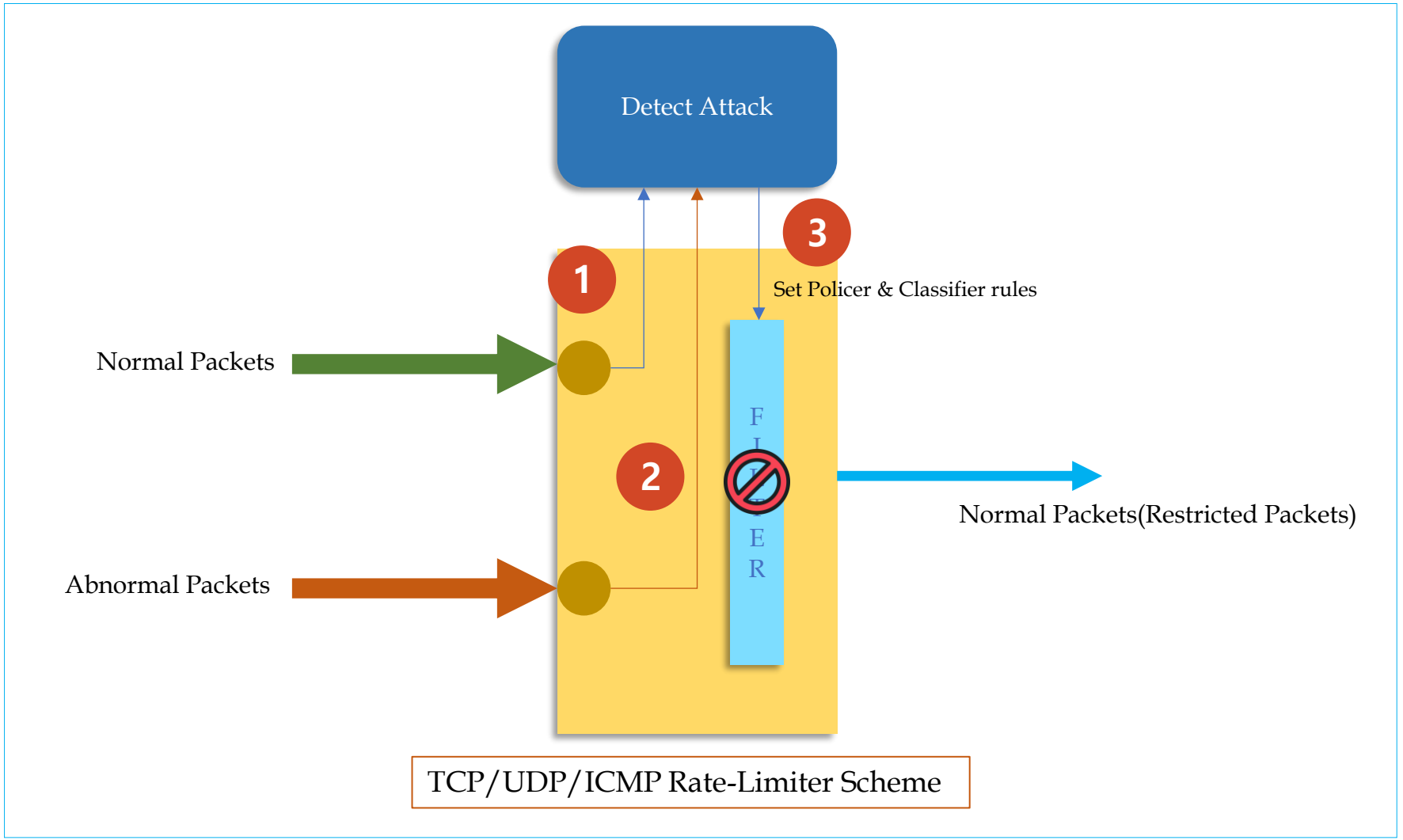


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(4)

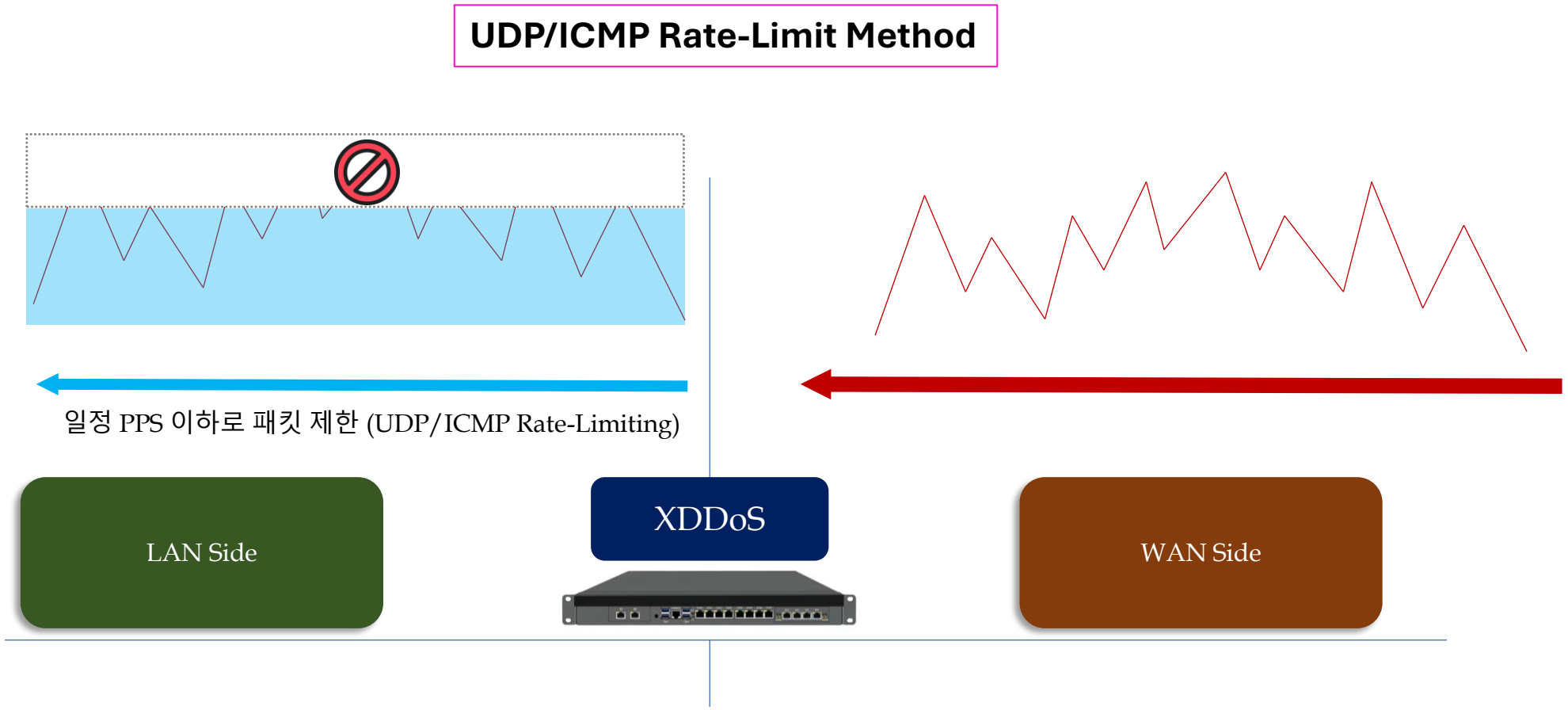


3. MightyXDDoS(8) – Rate-Limiting Scheme(1)

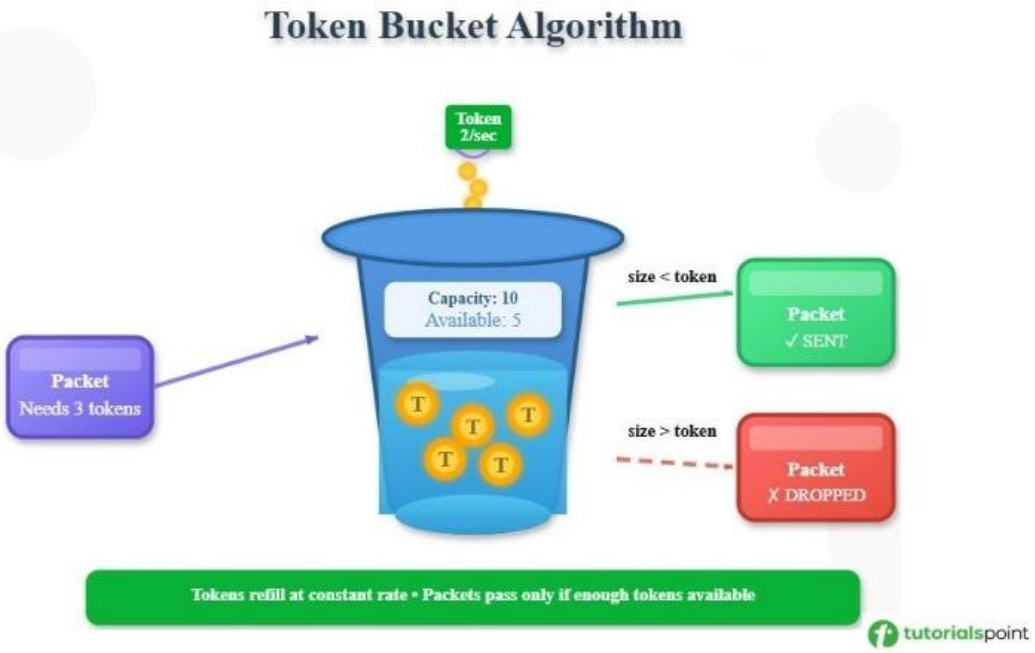
Traffic Control (Policer & Classifier) 기법

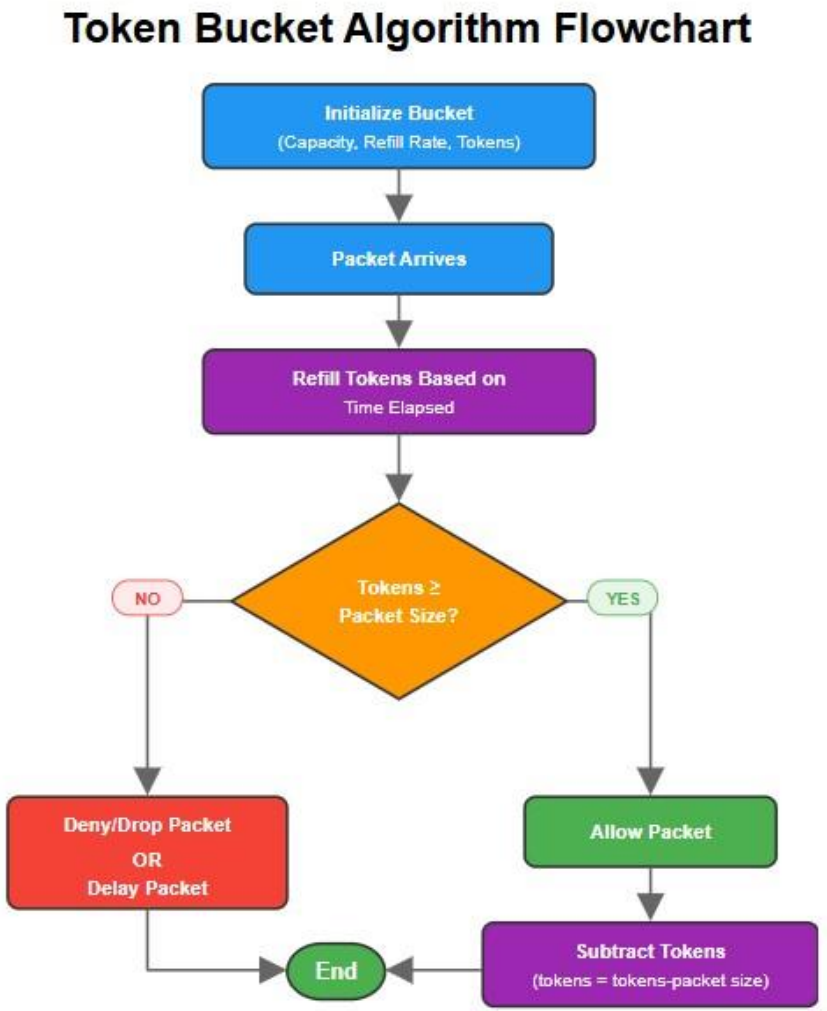


3. MightyXDDoS(8) – Rate-Limiting Scheme(2)



3. MightyXDDoS(8) – Rate-Limiting Scheme(3)



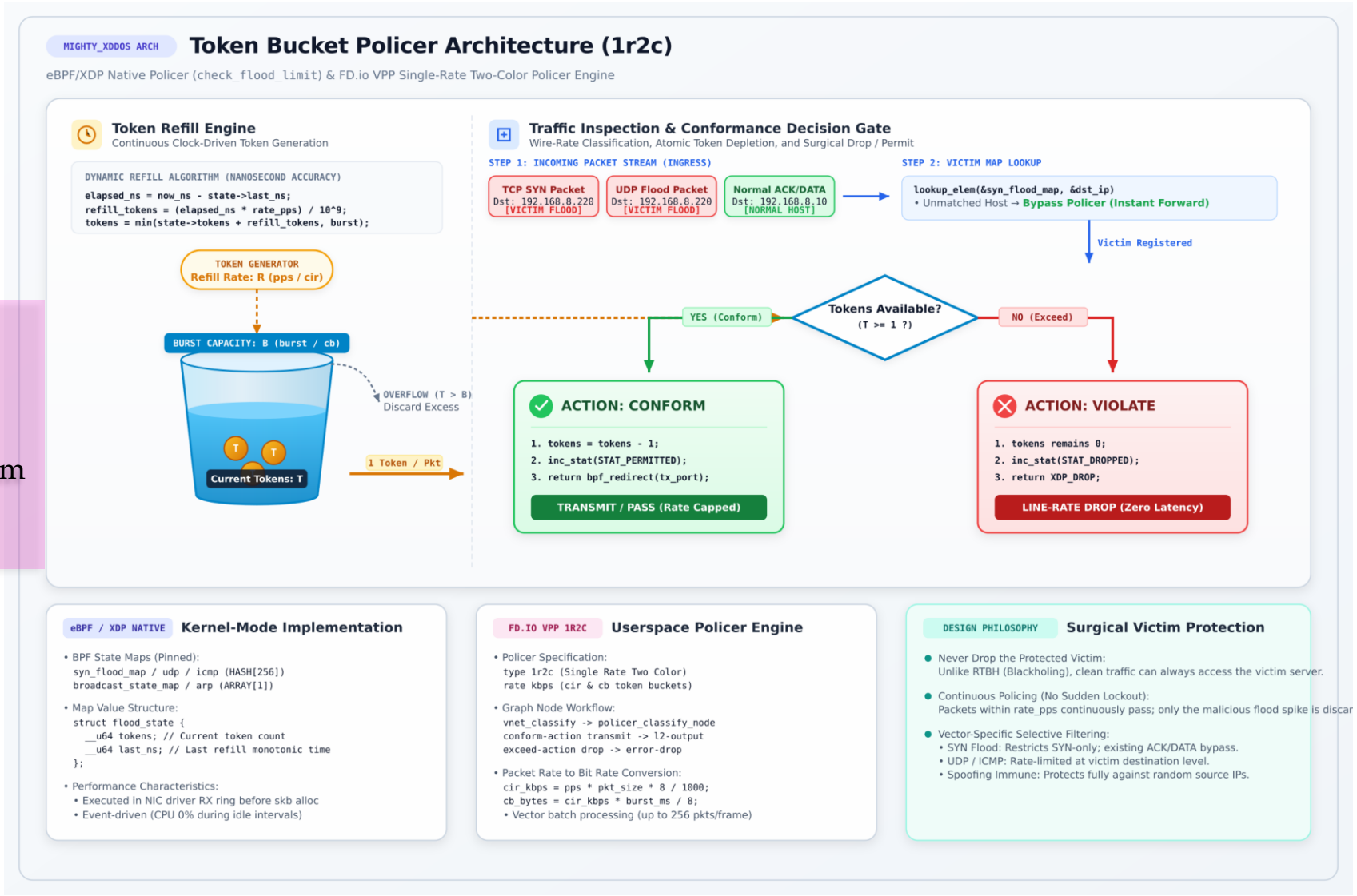


Ref) https://www.tutorialspoint.com/data_communication_computer_network/token_bucket_algorithm.htm

3. MightyXDDoS(8) – Rate-Limiting Scheme(4)

Rate-Limiter

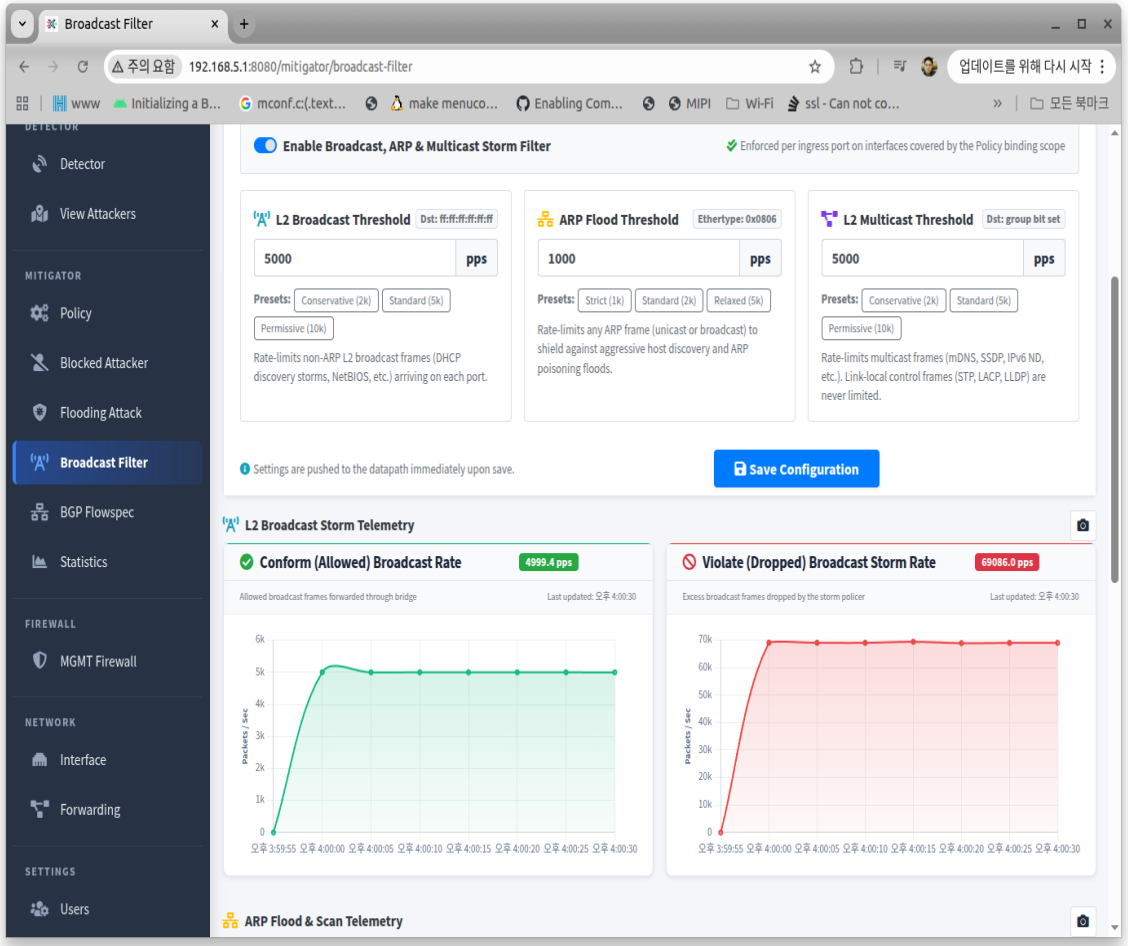
- TCP Syn Flooding
- UDP Flooding
- ICMP Flooding
- ARP Broadcast Storm
- Broadcast/Multicast Storm
- MAC Flooding
- …



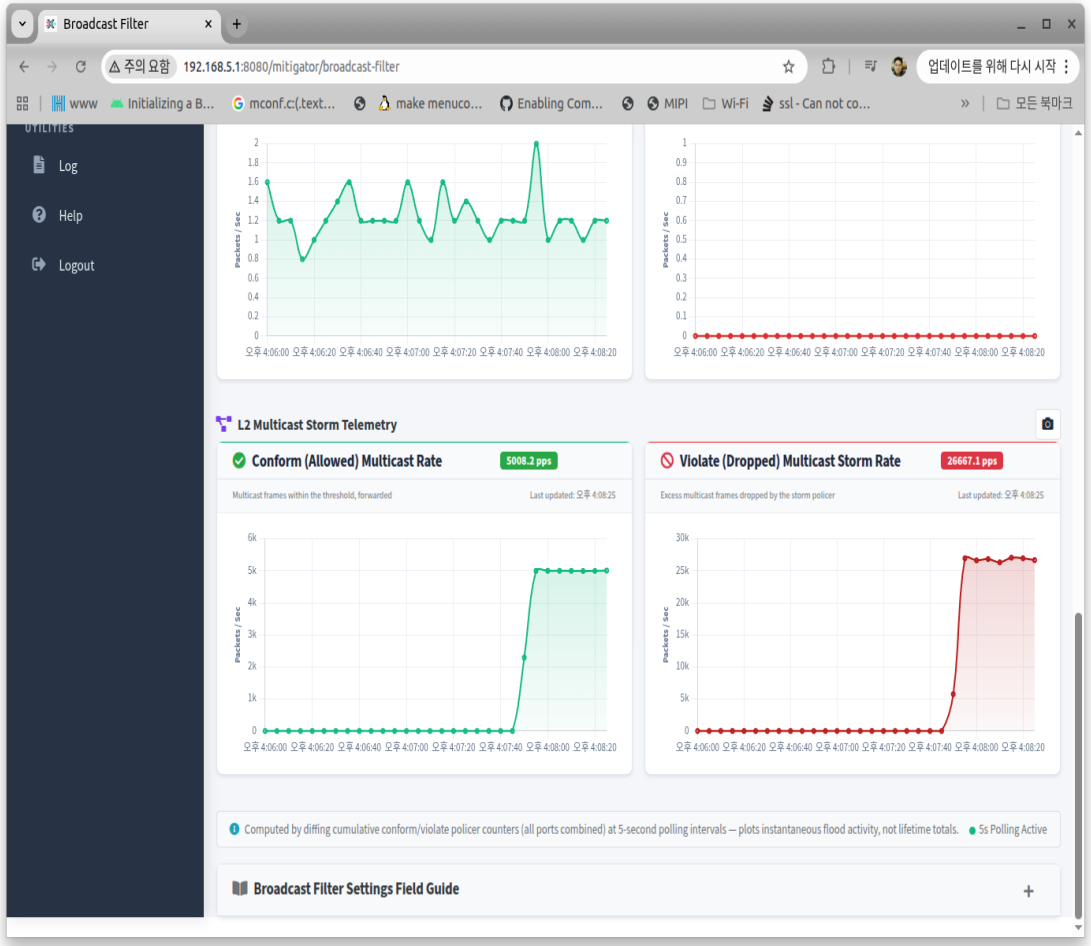
3. MightyXDDoS(8) – Rate-Limiting Scheme(5)

L2 Broadcast/Multicast, ARP Storm Filter

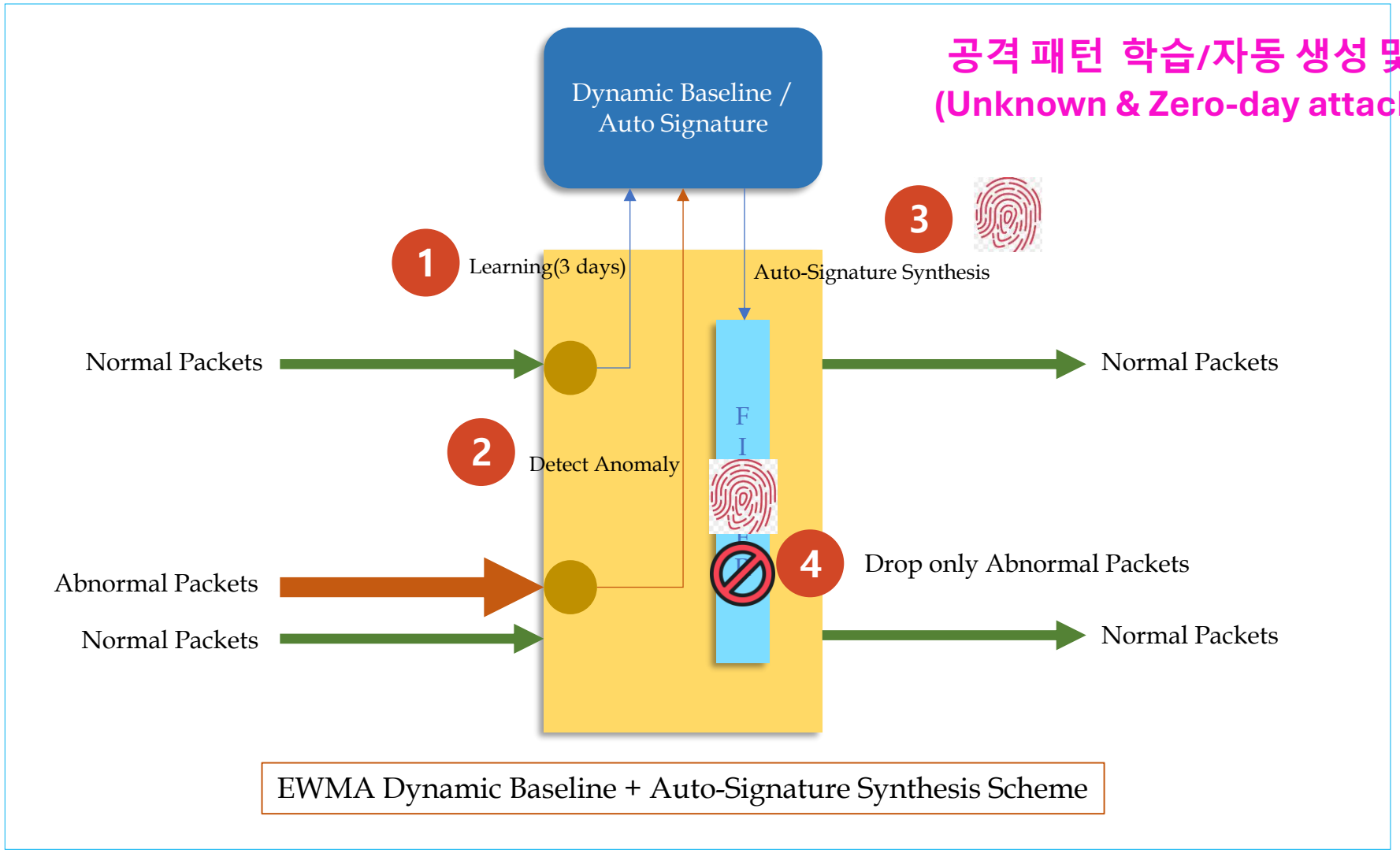
L2 Broadcast Storm Filter



L2 Multicast Storm Filter

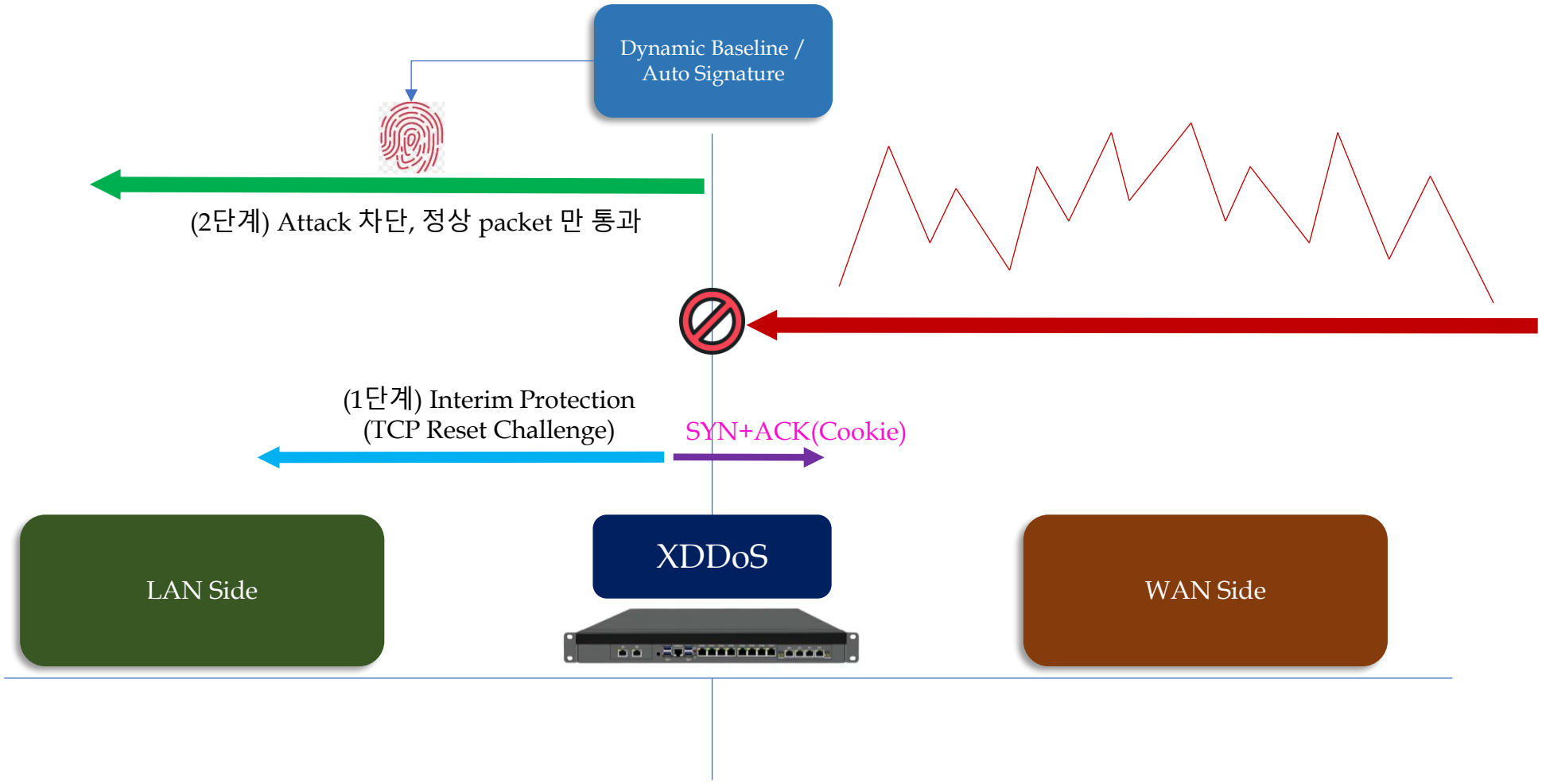


3. MightyXDDoS(9) – Intelligent Automation(1)



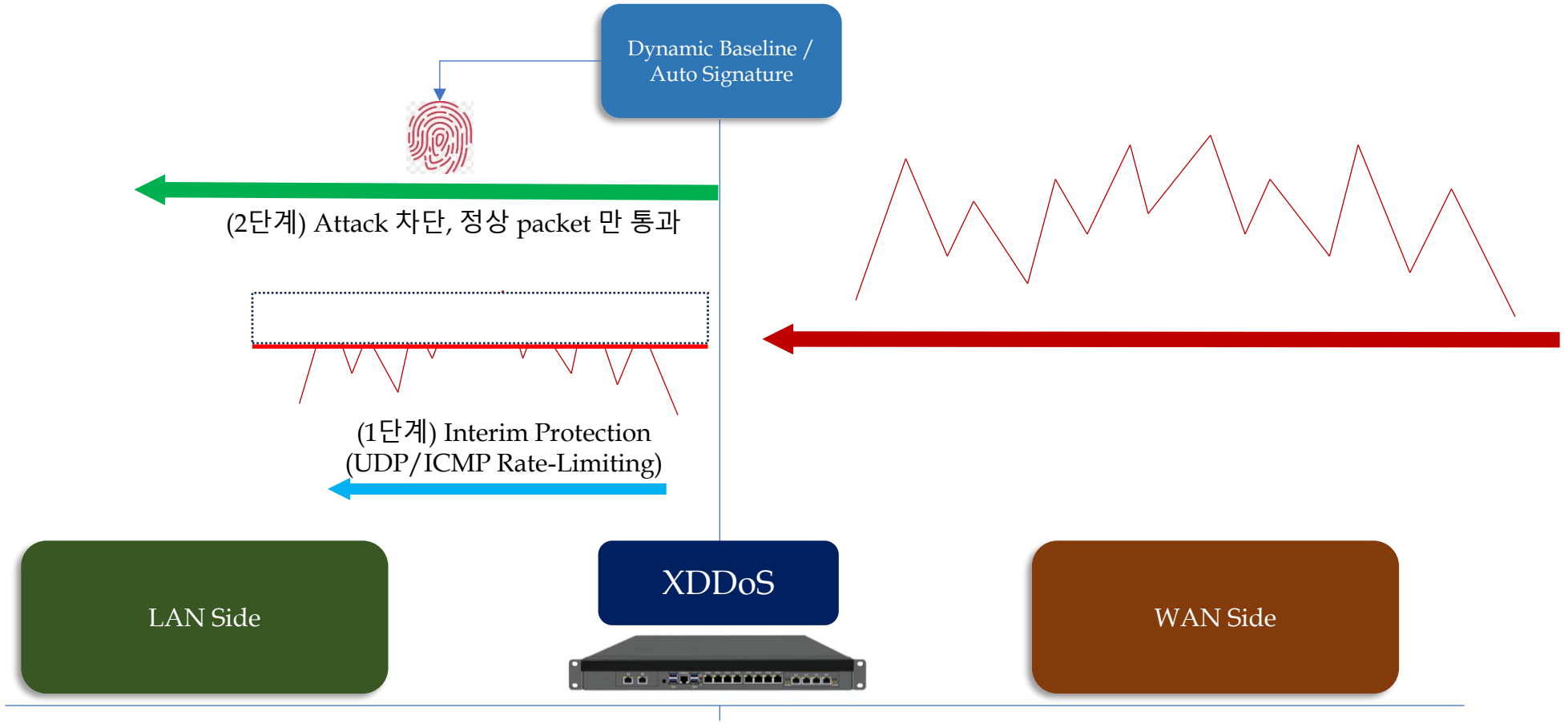
3. MightyXDDoS(9) – Intelligent Automation(2)

Mode1) Interim(Reset Challenge) + Auto-Signature Synthesis Method



3. MightyXDDoS(9) – Intelligent Automation(3)

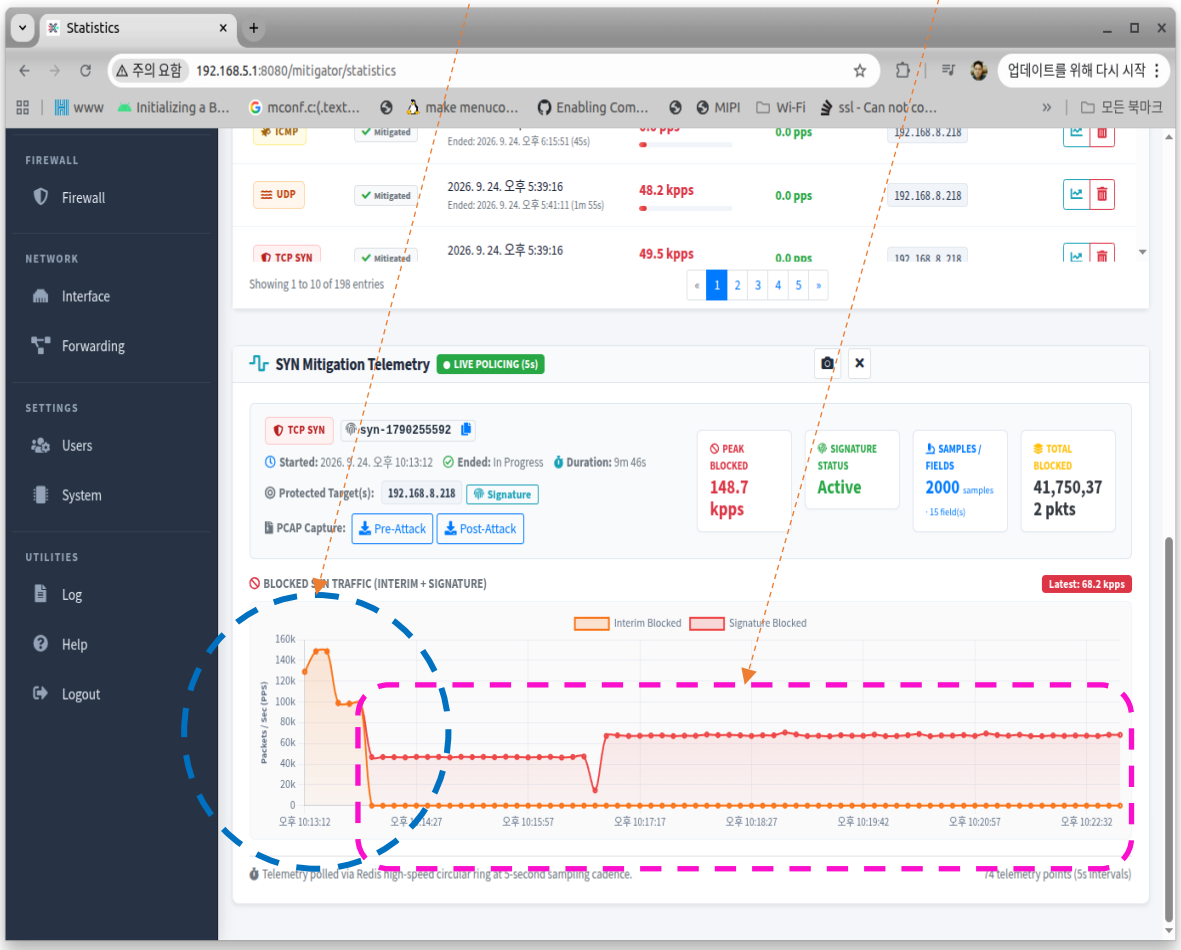
Mode2) Interim(Rate-Limit) + Auto-Signature Synthesis Method

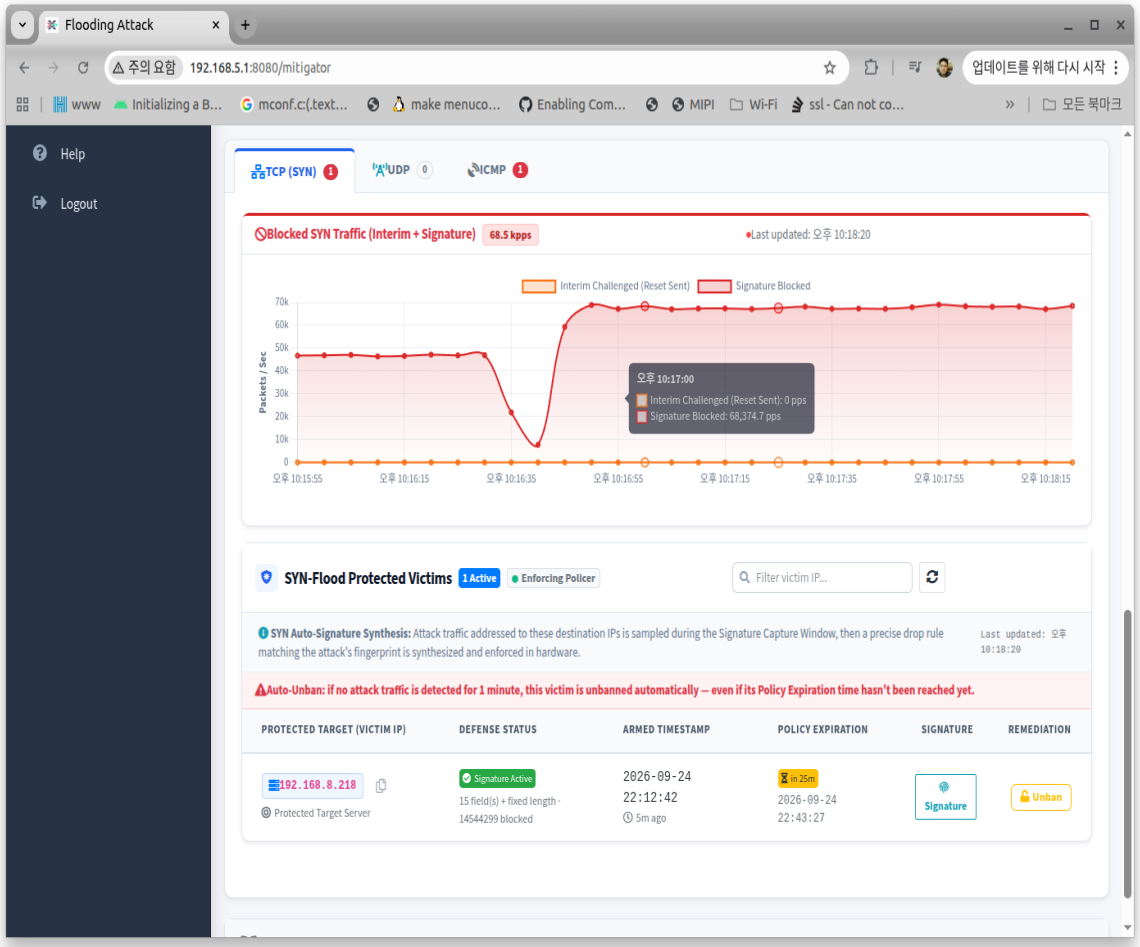


3. MightyXDDoS(9) – Intelligent Automation(4)

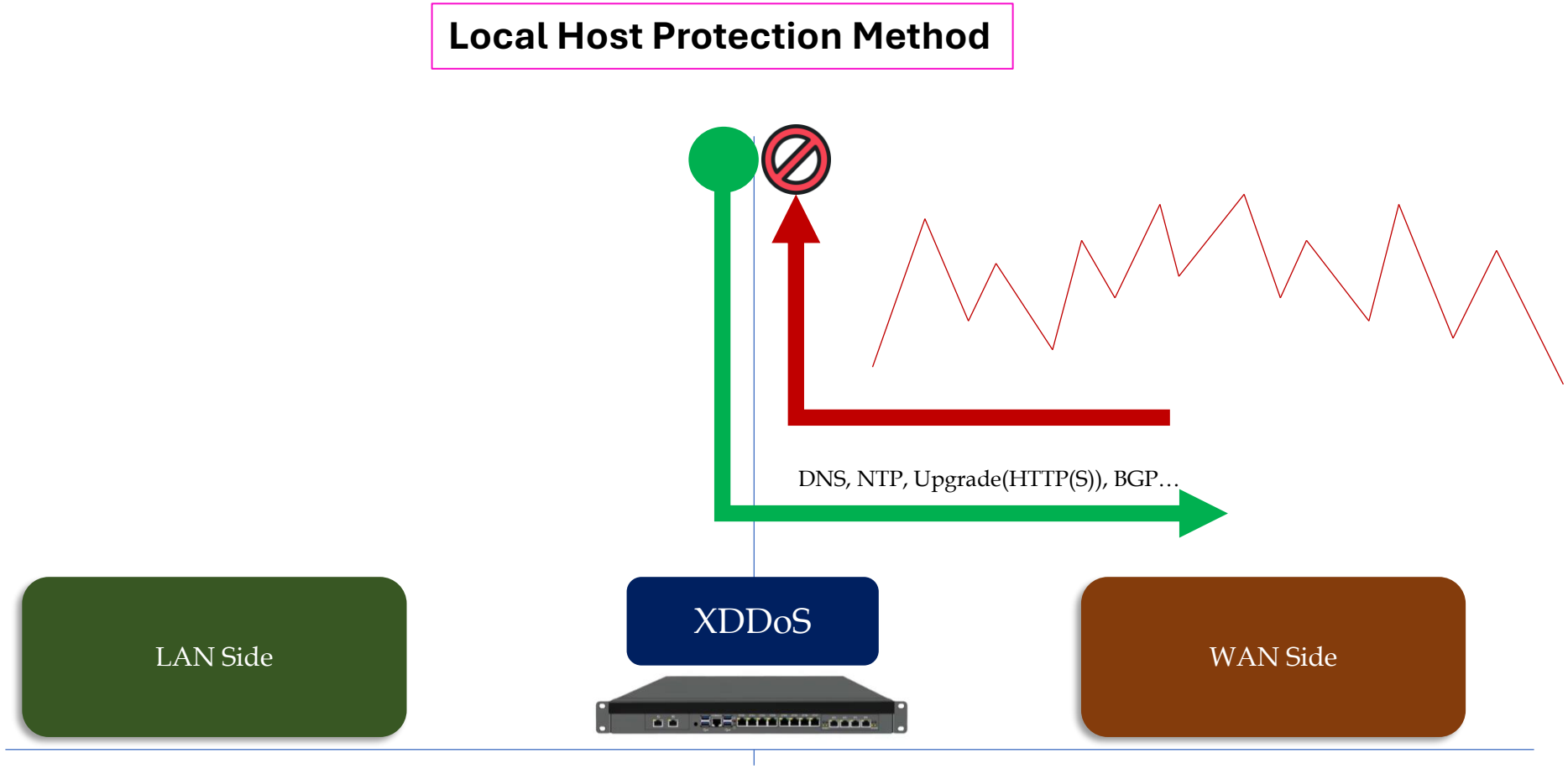
(1단계) Interim Protection
(TCP Reset Challenge)

(2단계) Auto-Signature Protection

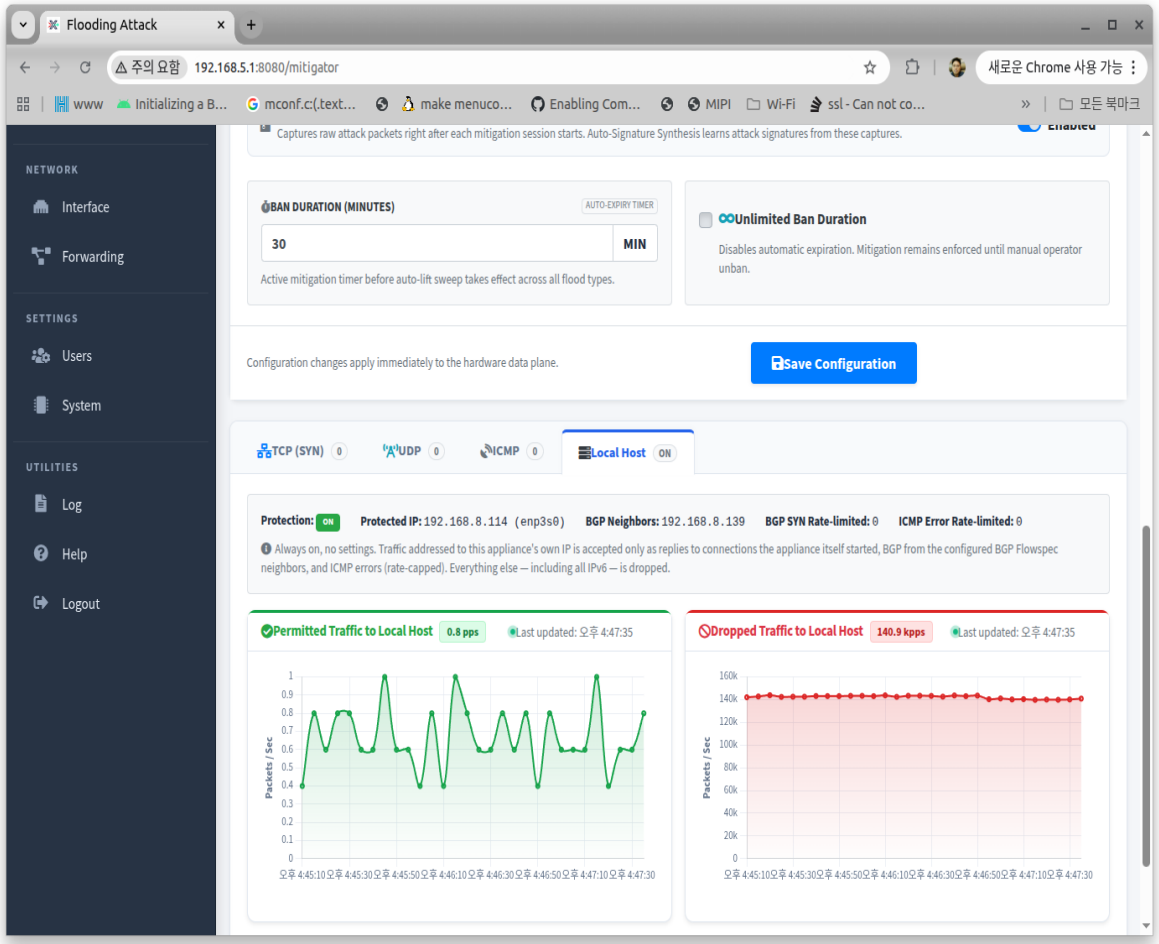


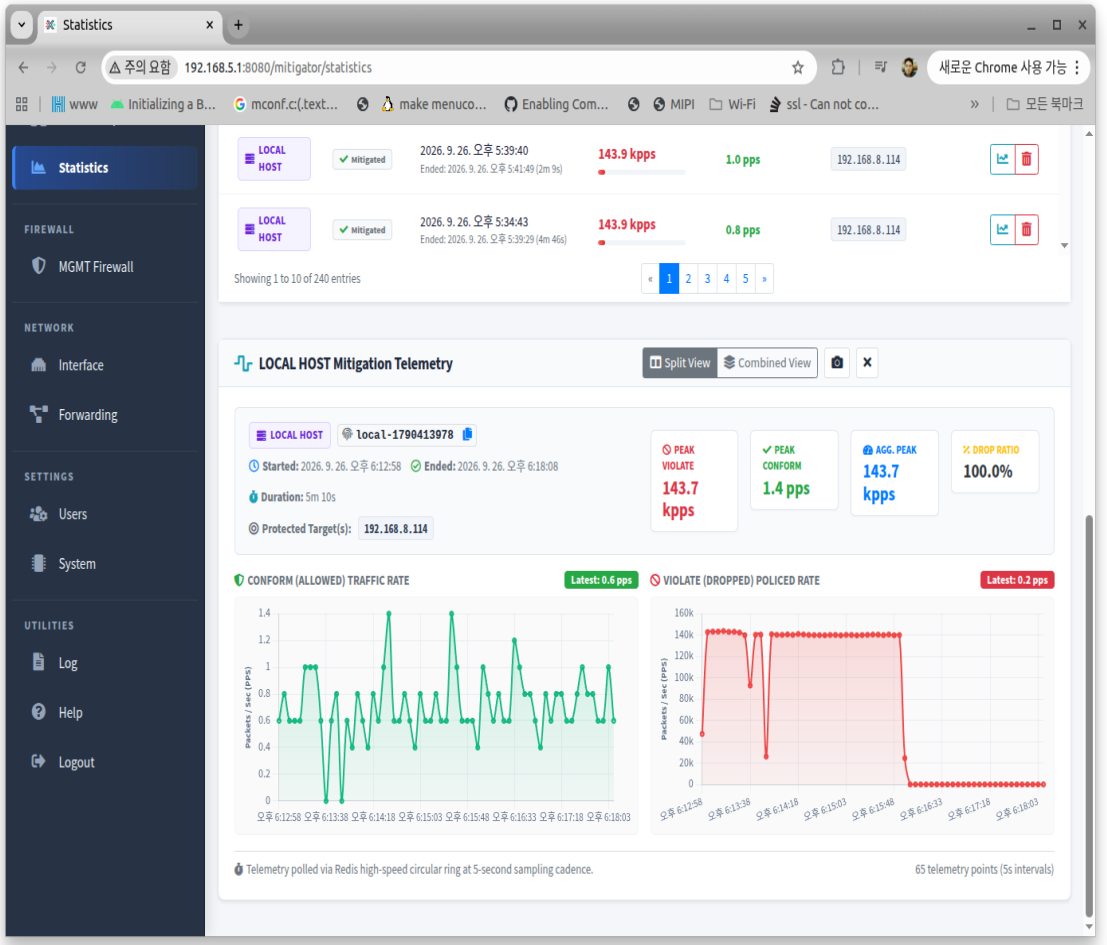


3. MightyXDDoS(10) – Local Host Mitigation(1)

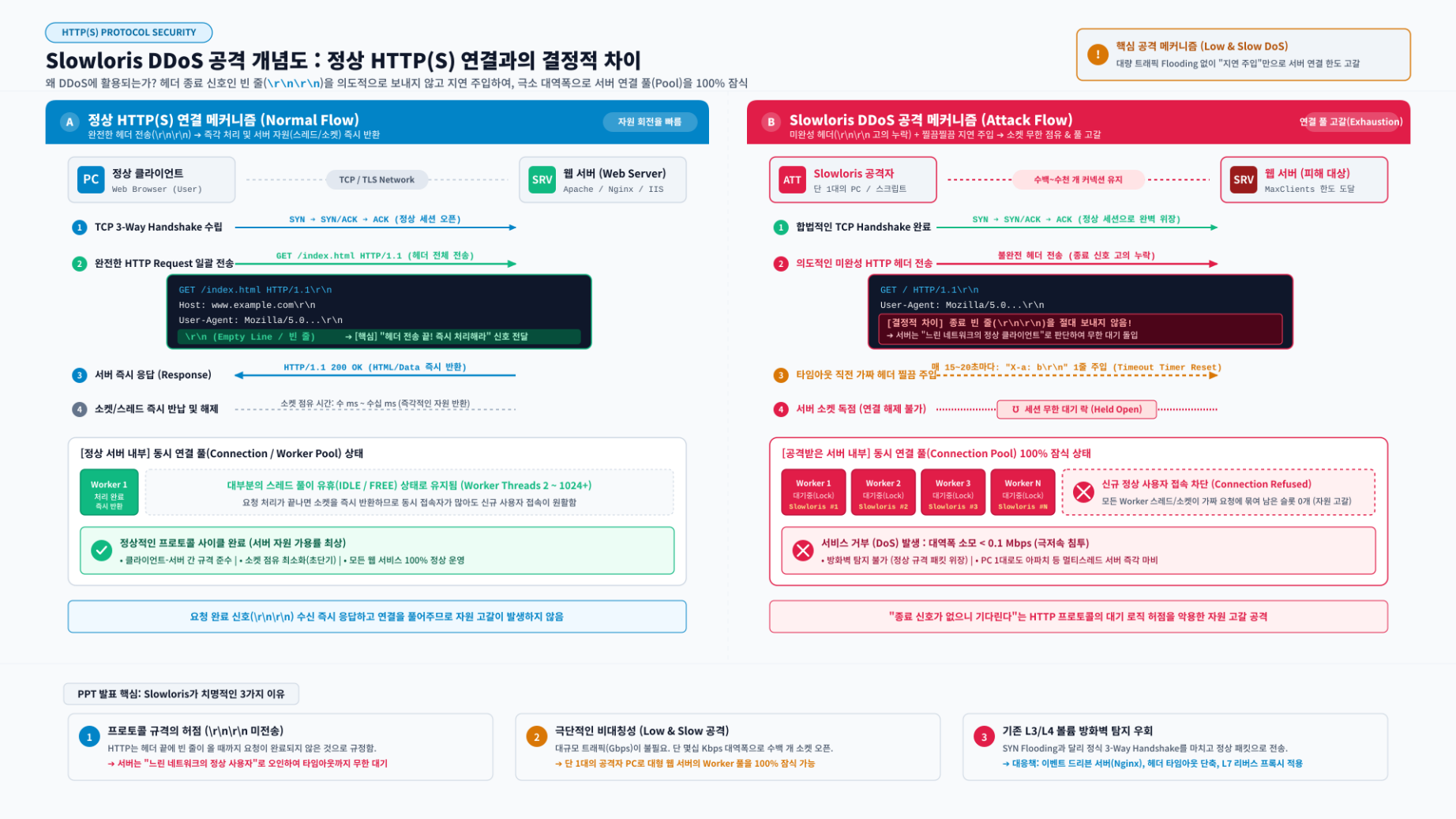


3. MightyXDDoS(10) – Local Host Mitigation(2)

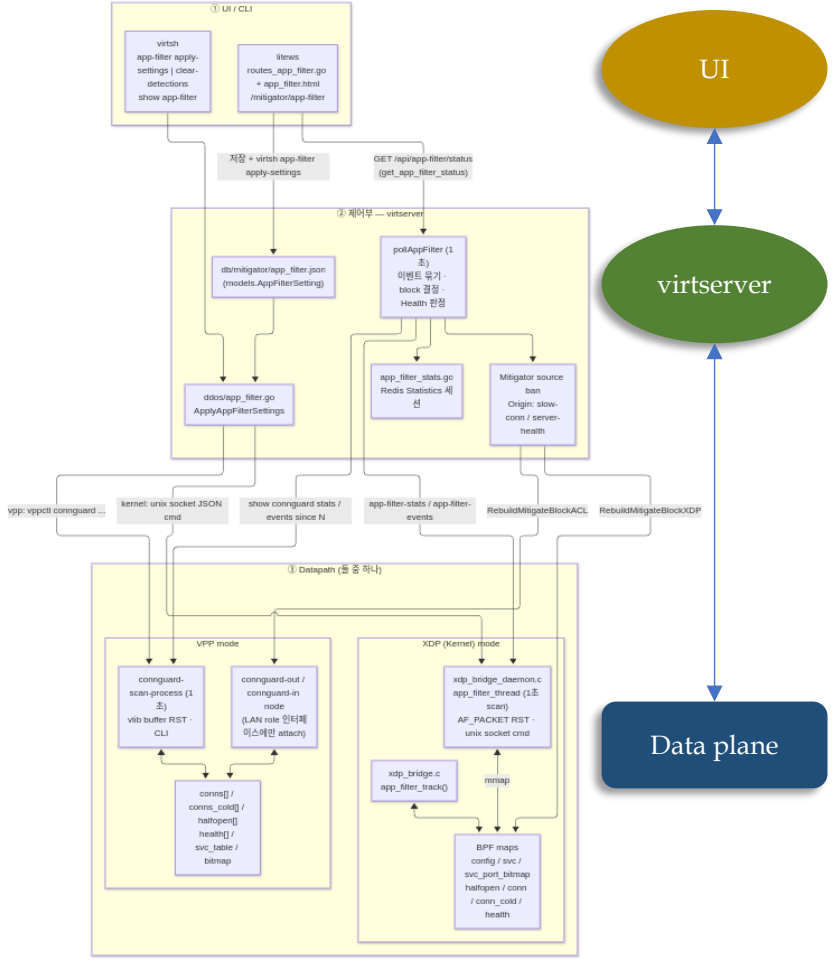


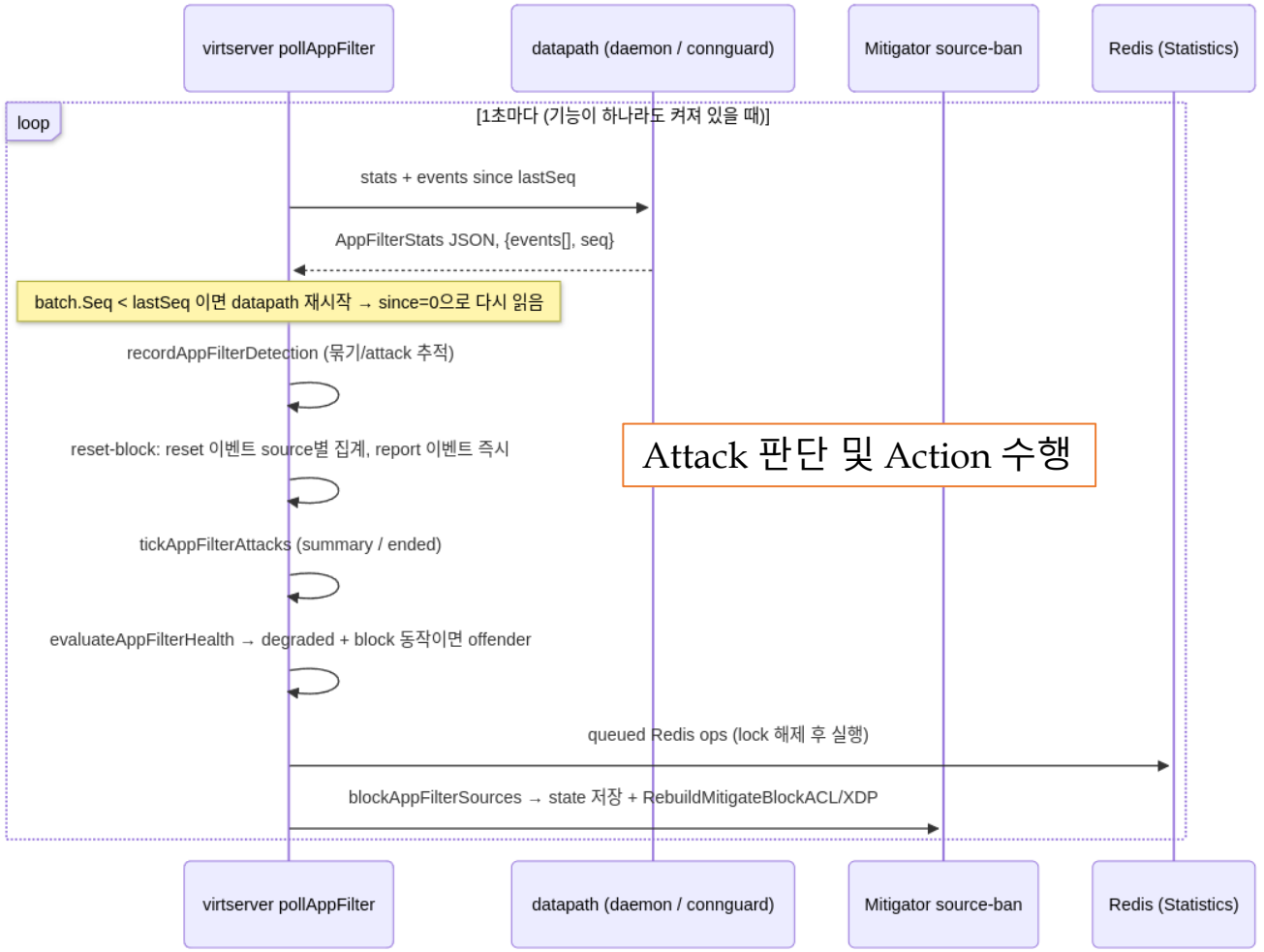


3. MightyXDDoS(11) – Slow Connection Guard(1)



3. MightyXDDoS(11) – Slow Connection Guard(2)

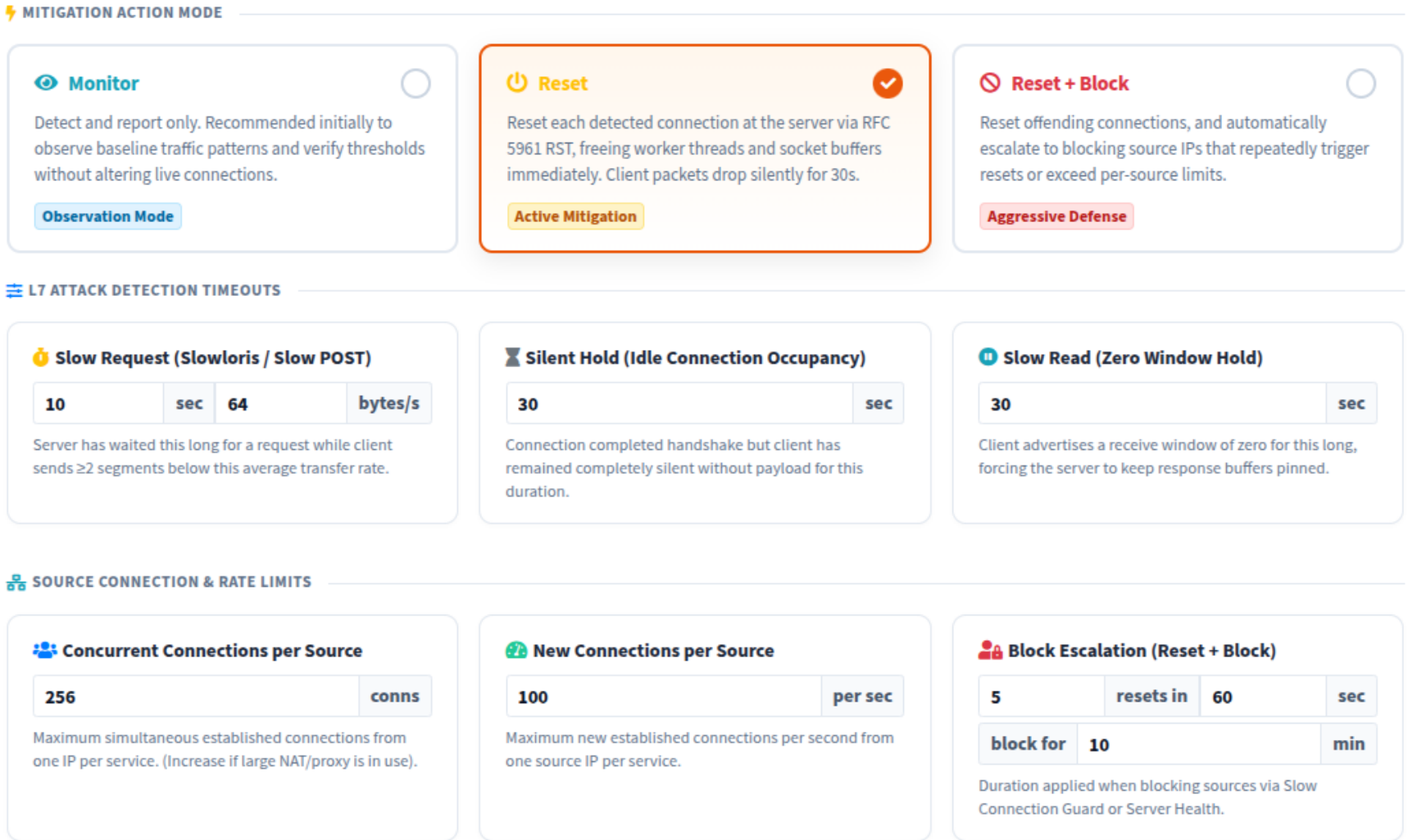




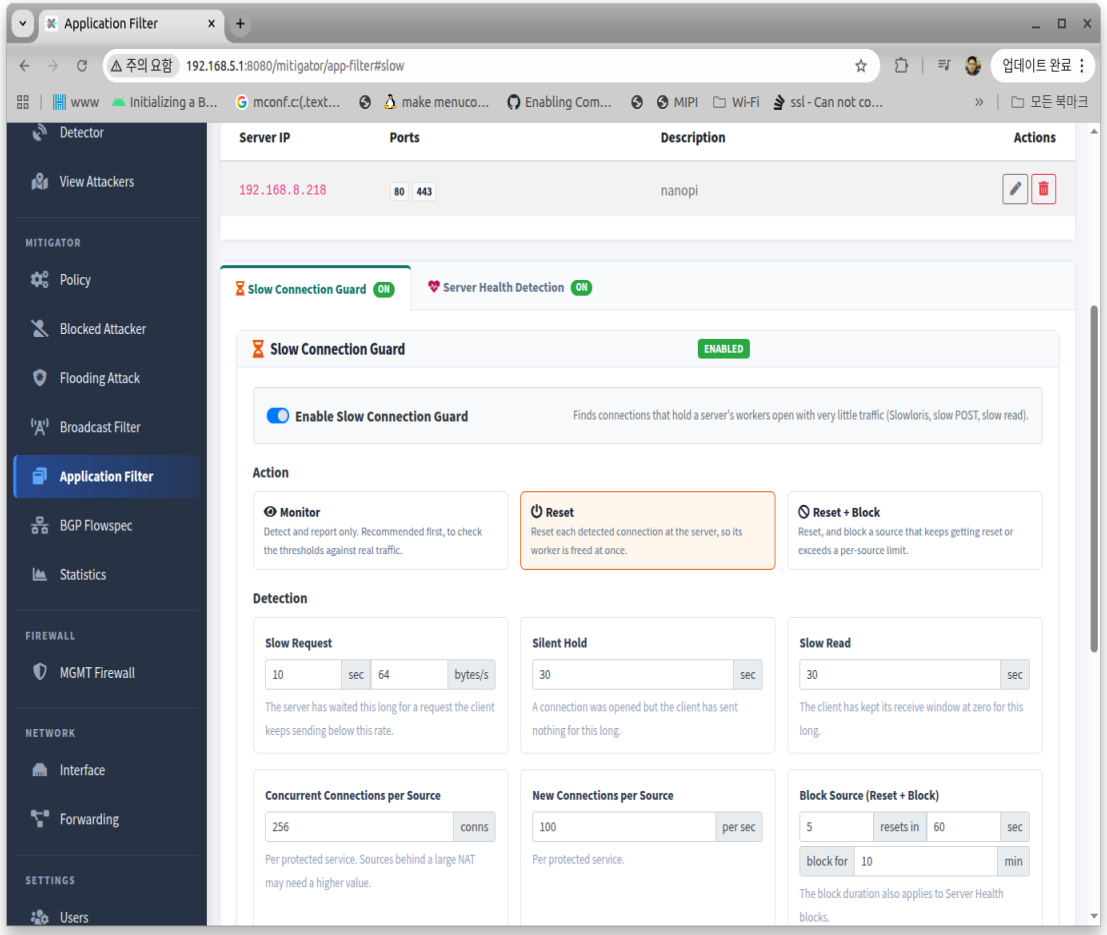
3. MightyXDDoS(11) – Slow Connection Guard(3)

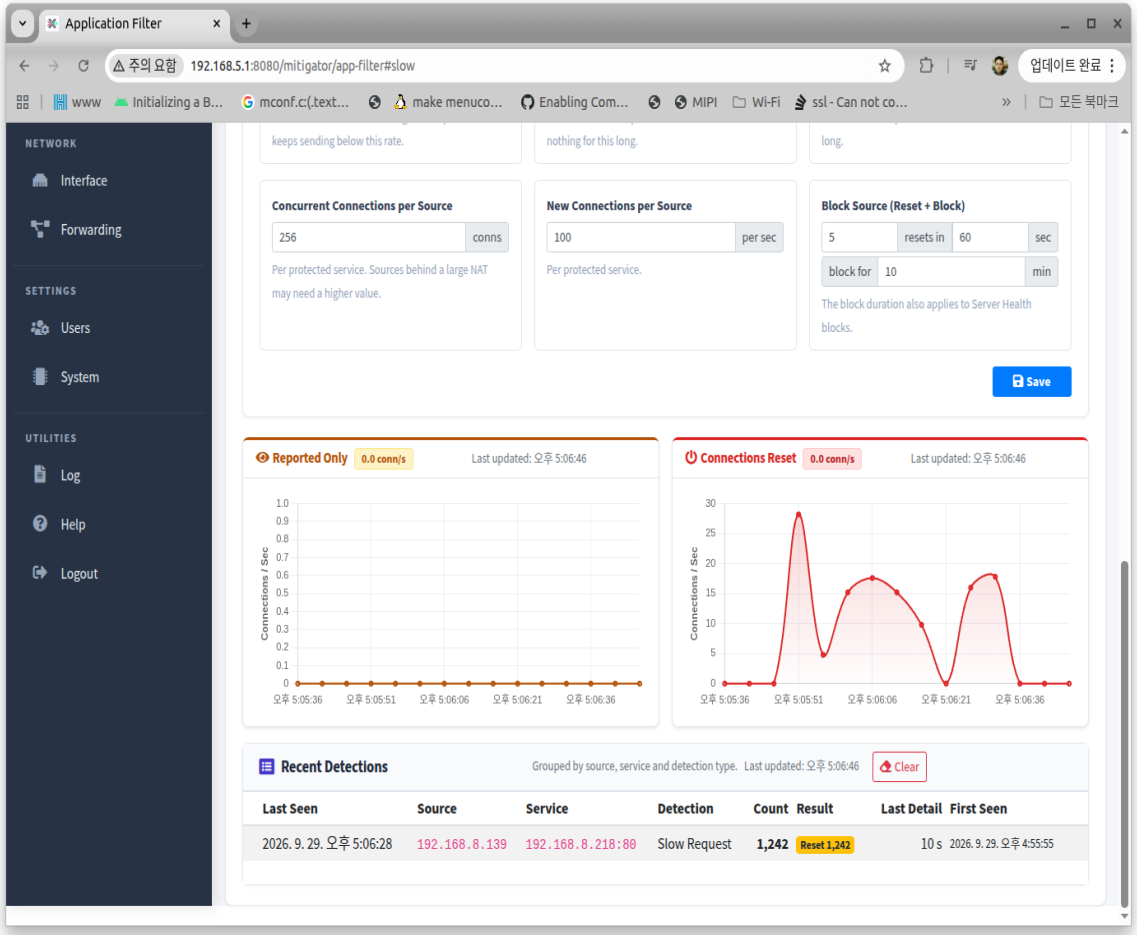
Action

Attack Detection 조건

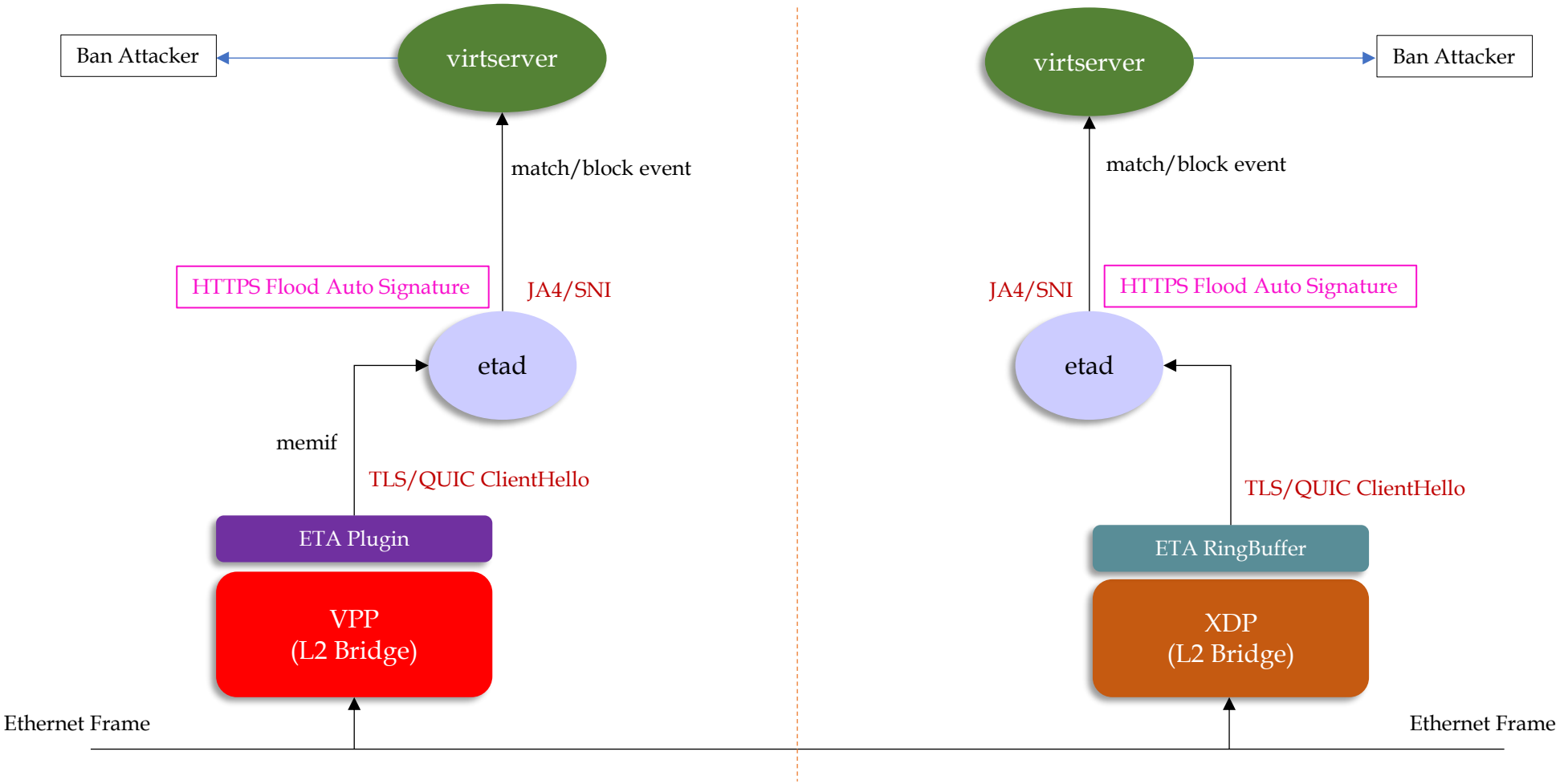


3. MightyXDDoS(11) – Slow Connection Guard(4)

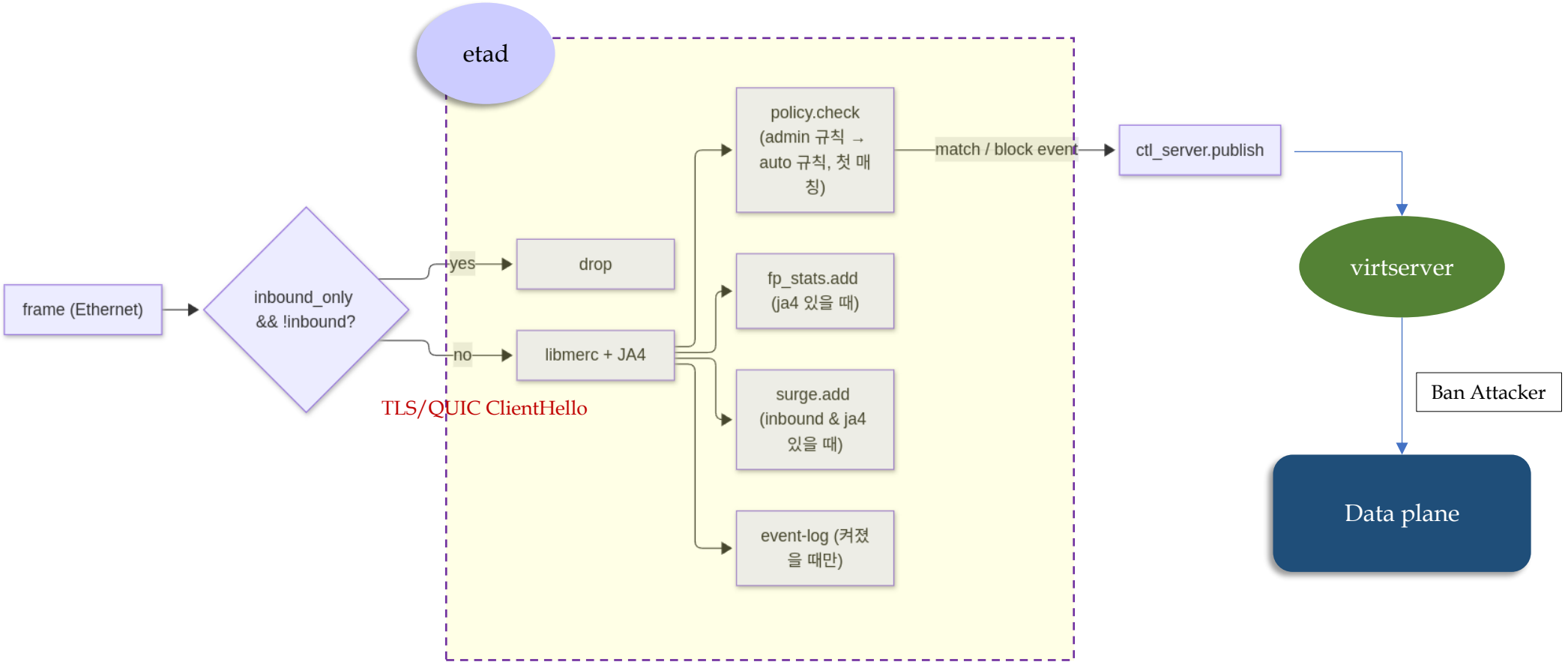




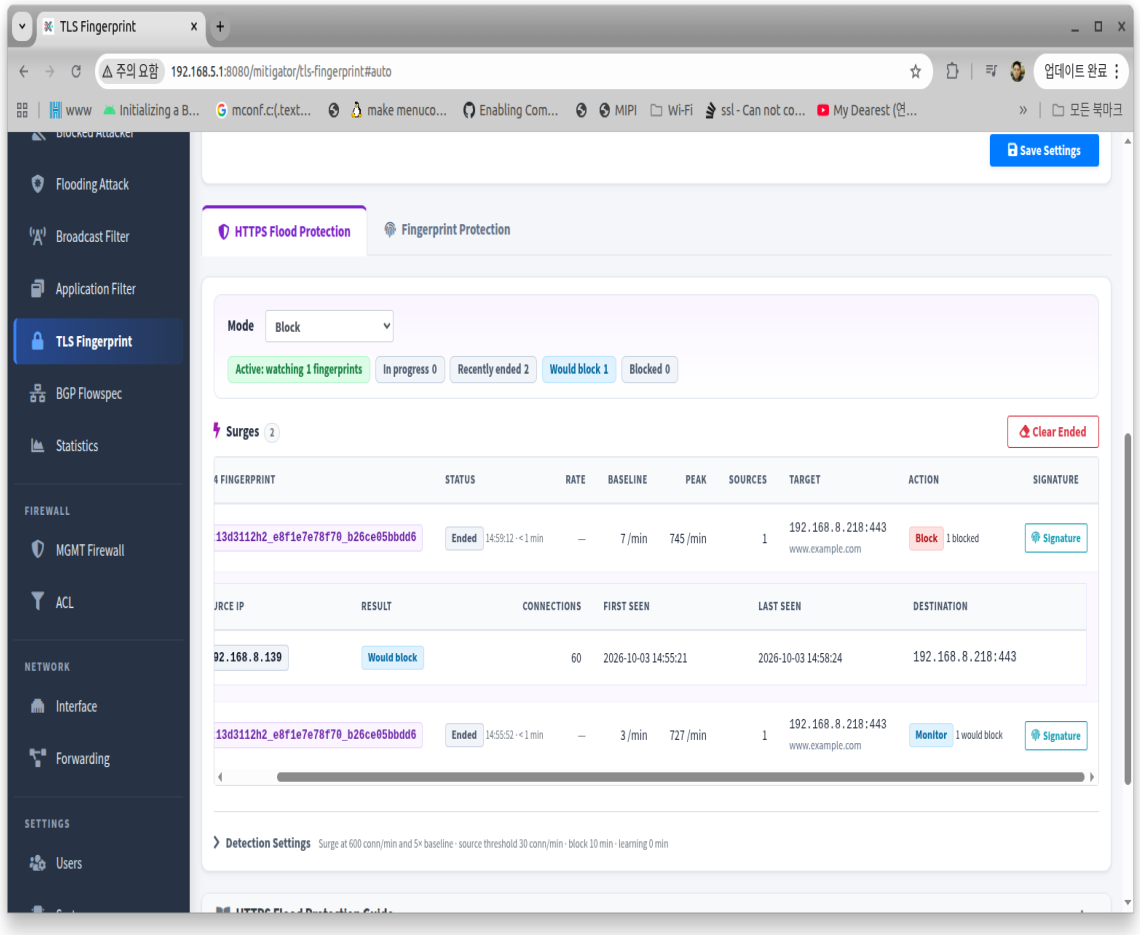
3. MightyXDDoS(12) – HTTPS Flood Protection(1)

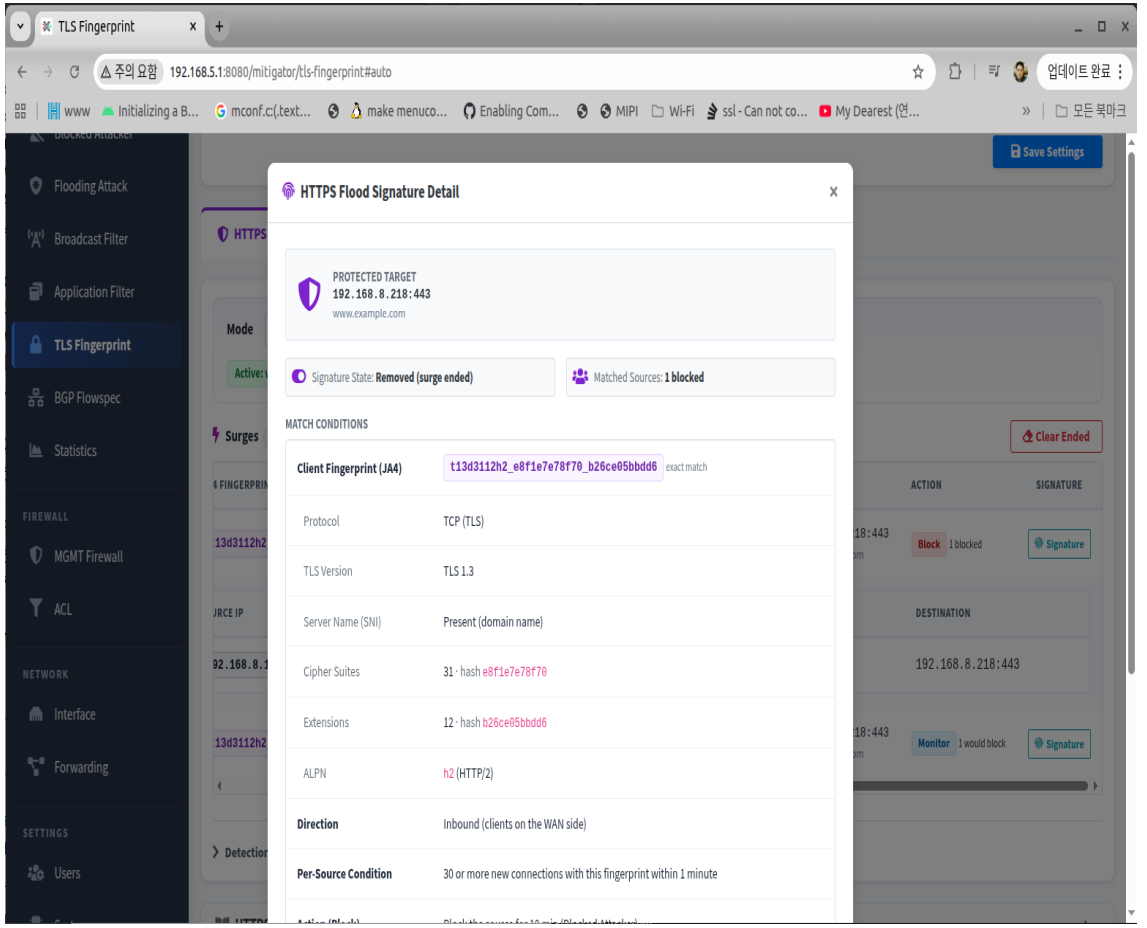


3. MightyXDDoS(12) – HTTPS Flood Protection(2)

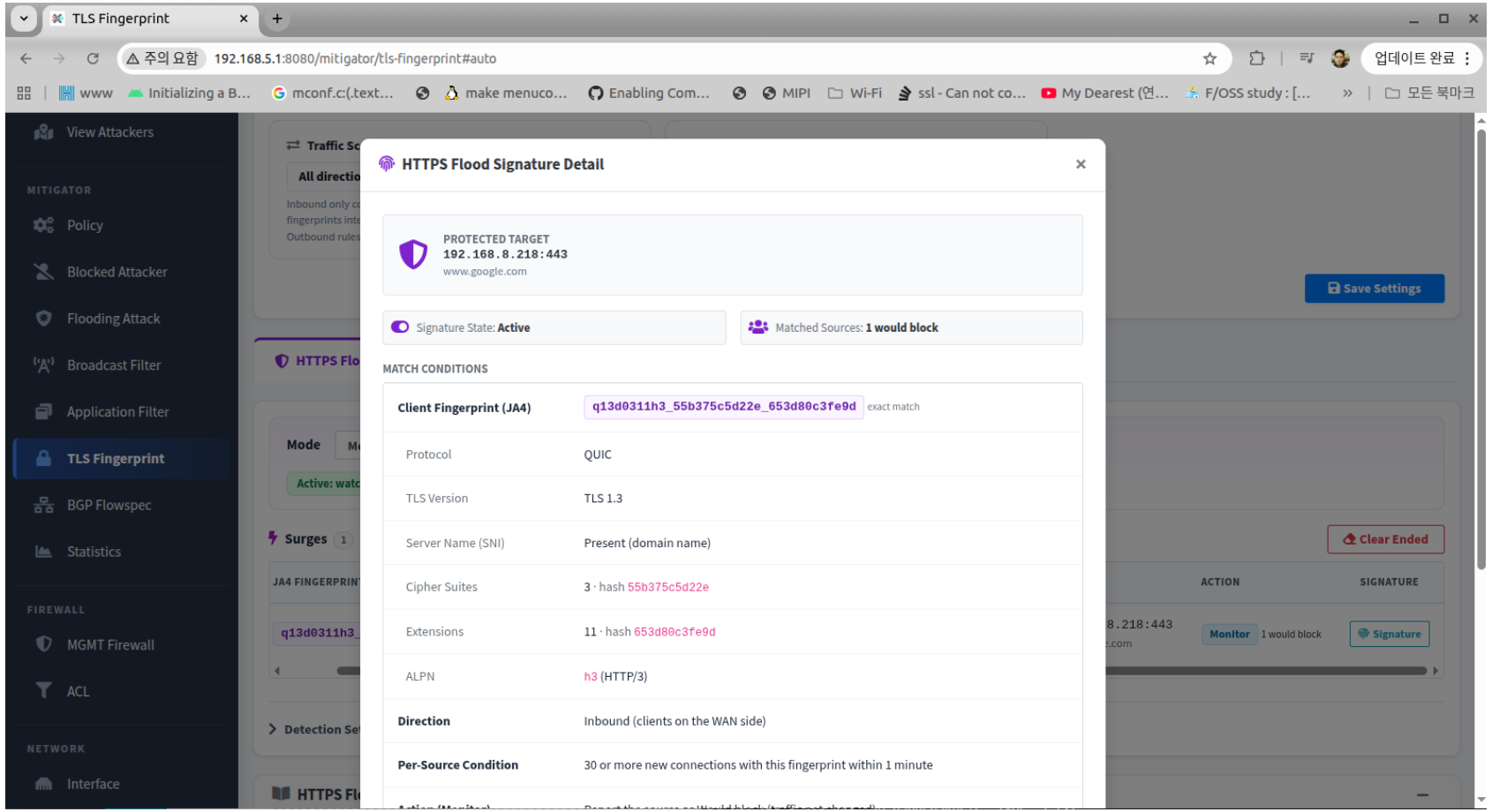


3. MightyXDDoS(12) – HTTPS Flood Protection(3)



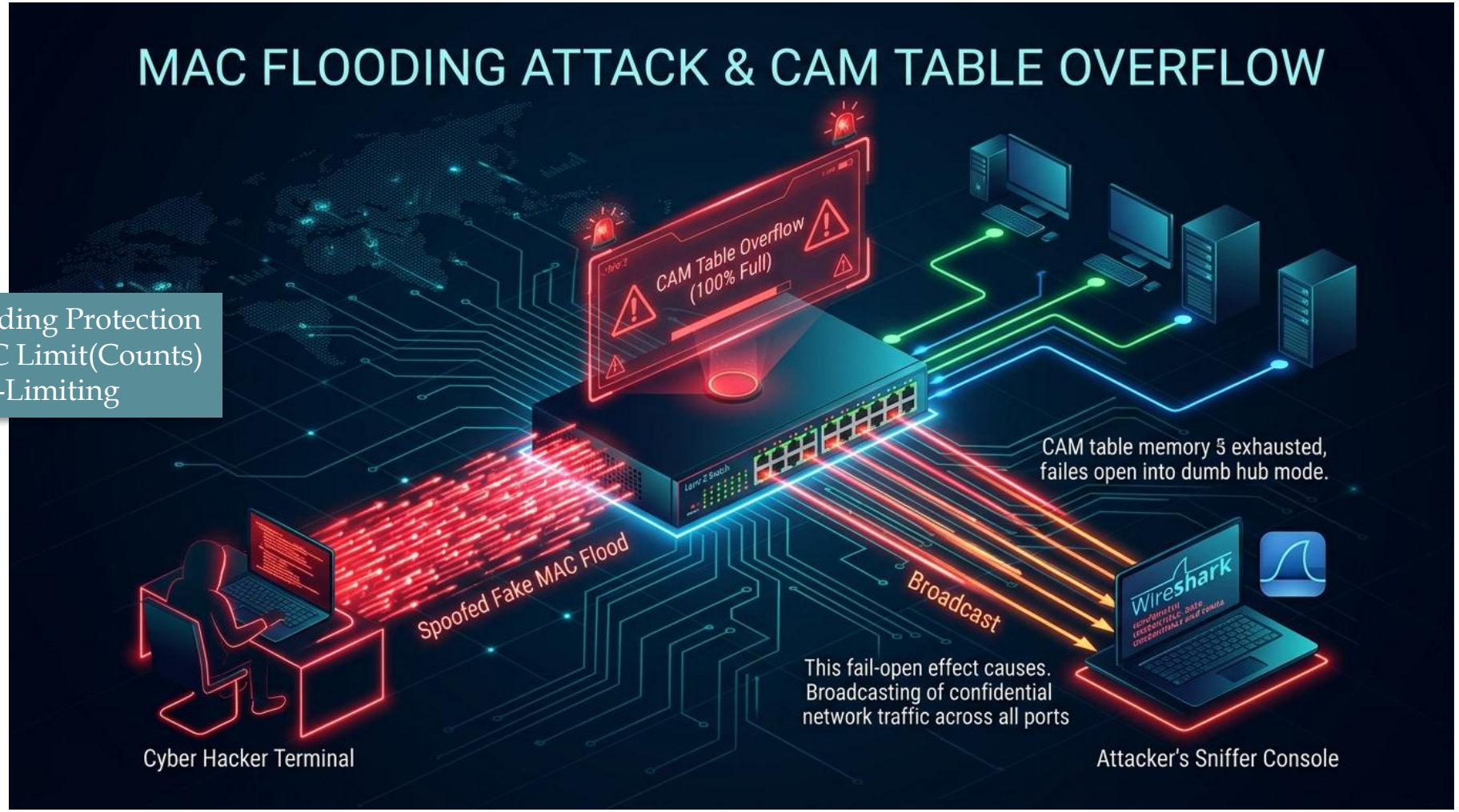


3. MightyXDDoS(12) – HTTPS Flood Protection(4)



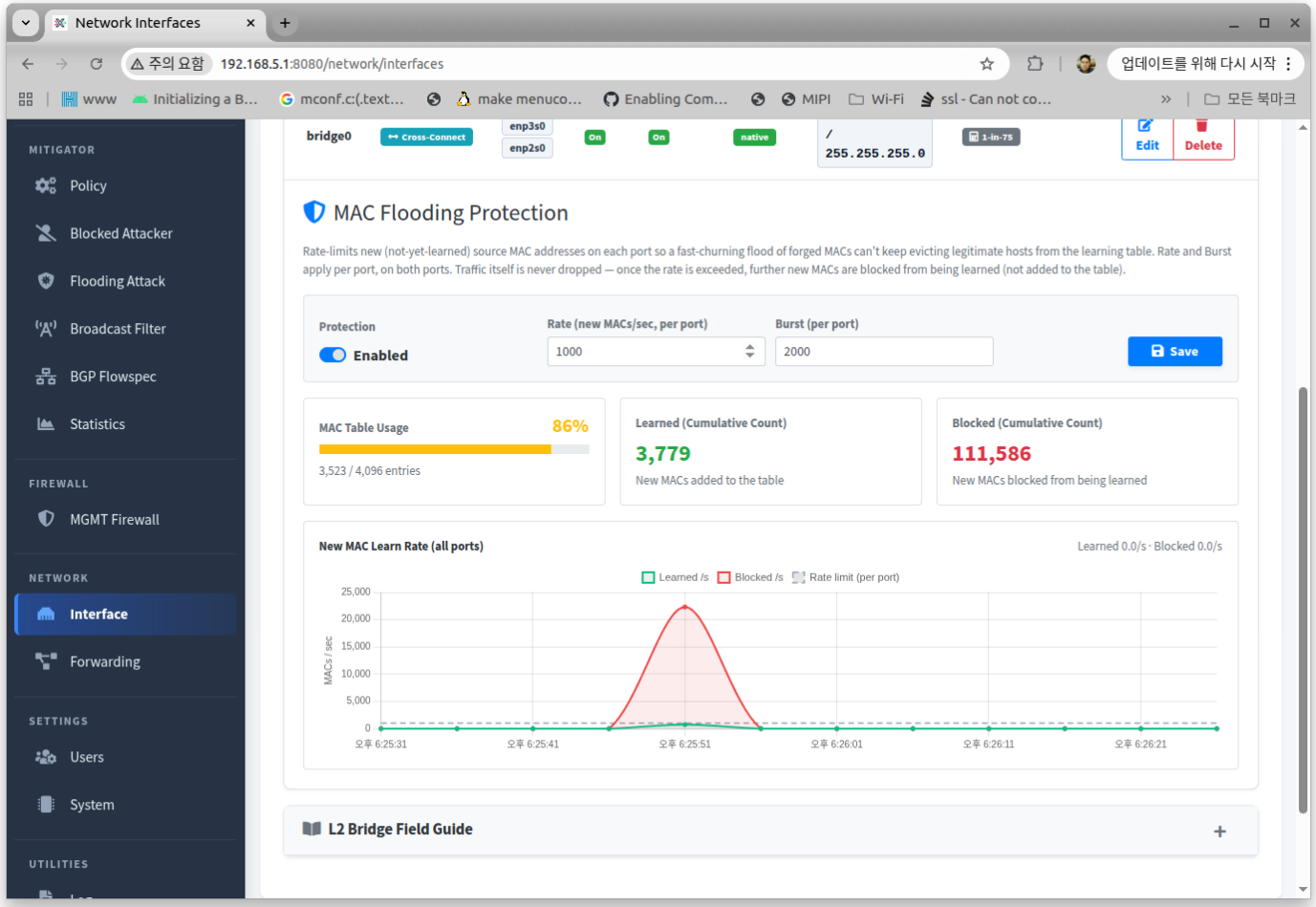
3. MightyXDDoS(13) – MAC Flooding Protection(1)

- MAC Flooding Protection
  - ✓ MAC Limit(Counts)
  - ✓ Rate-Limiting

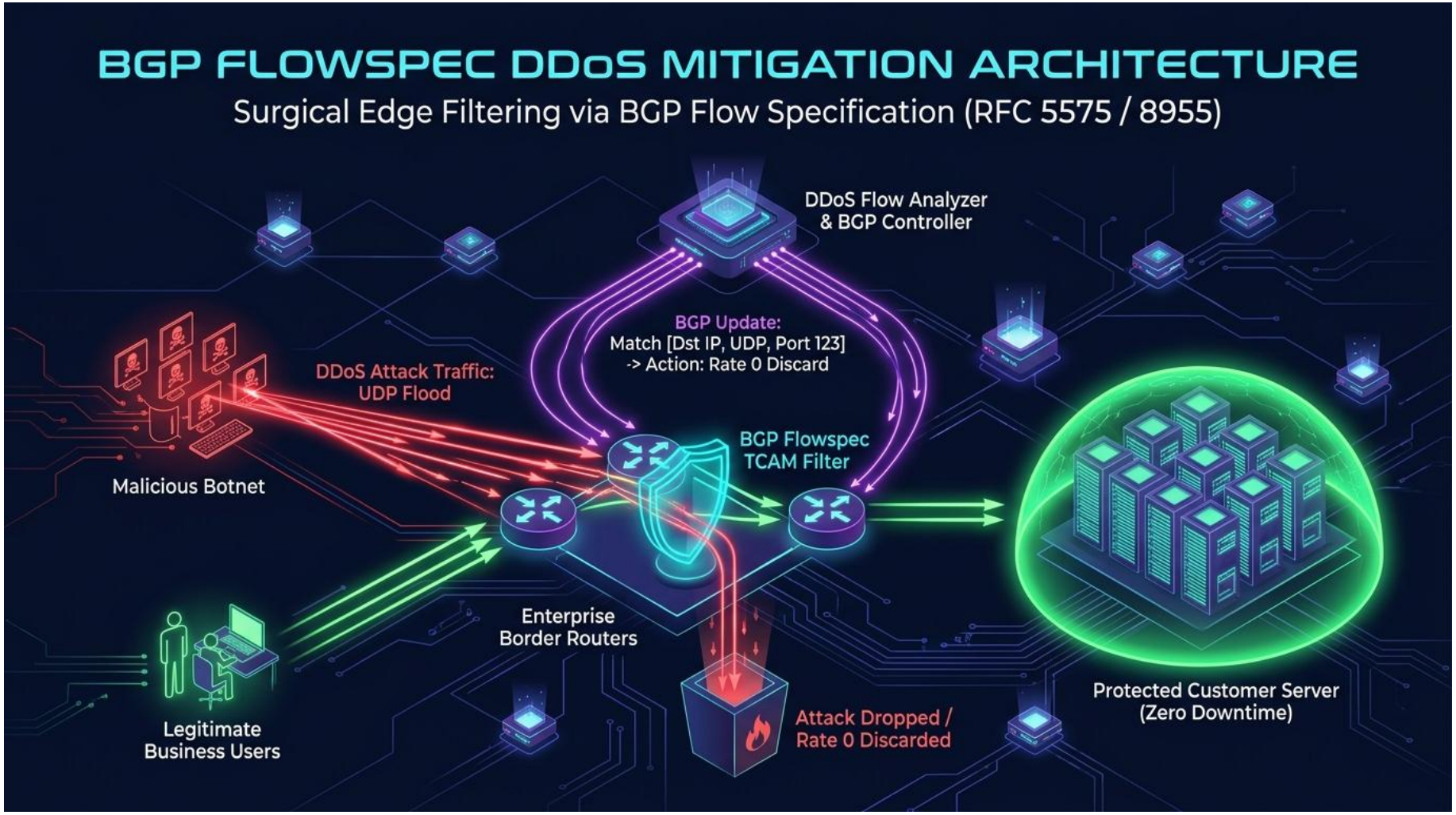


Generated by Gemini

3. MightyXDDoS(13) – MAC Flooding Protection(2)

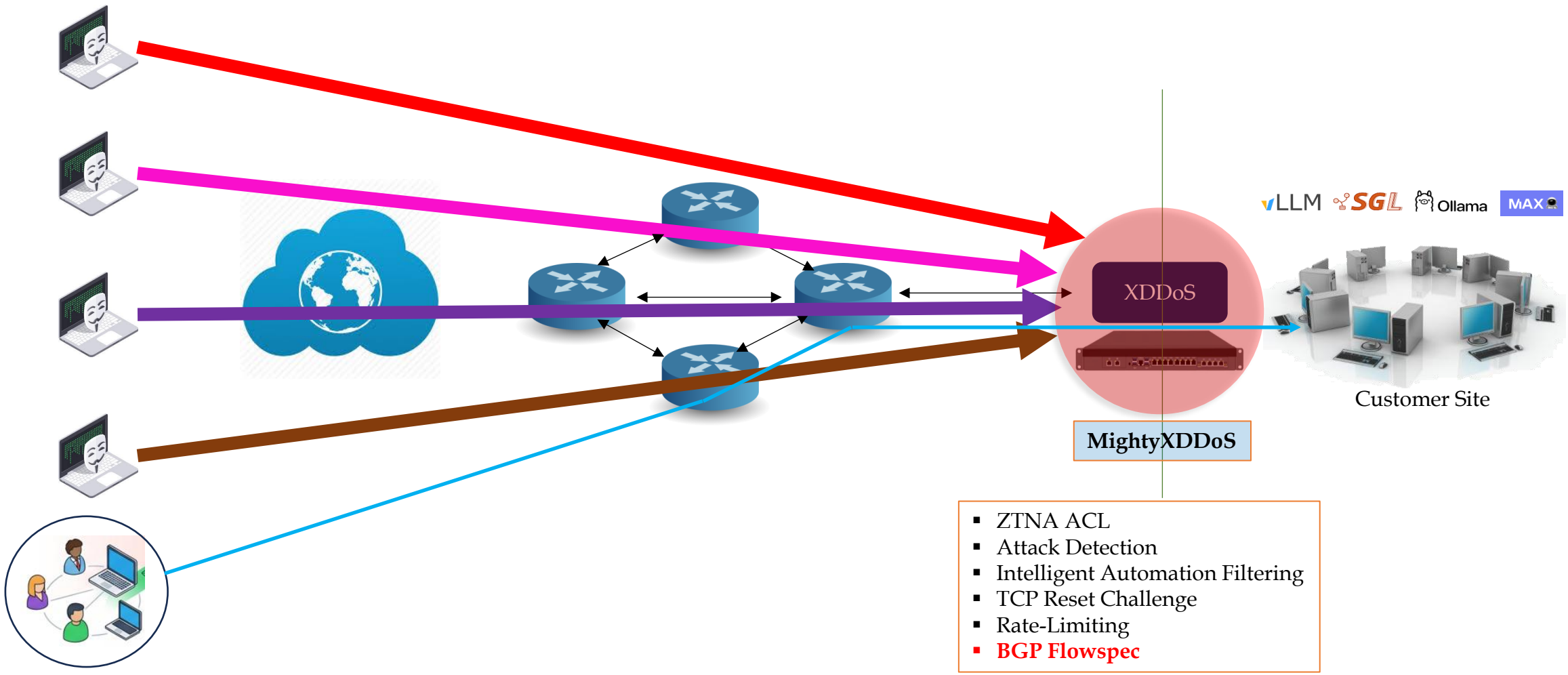


3. MightyXDDoS(14) – BGP Flowspec(1)



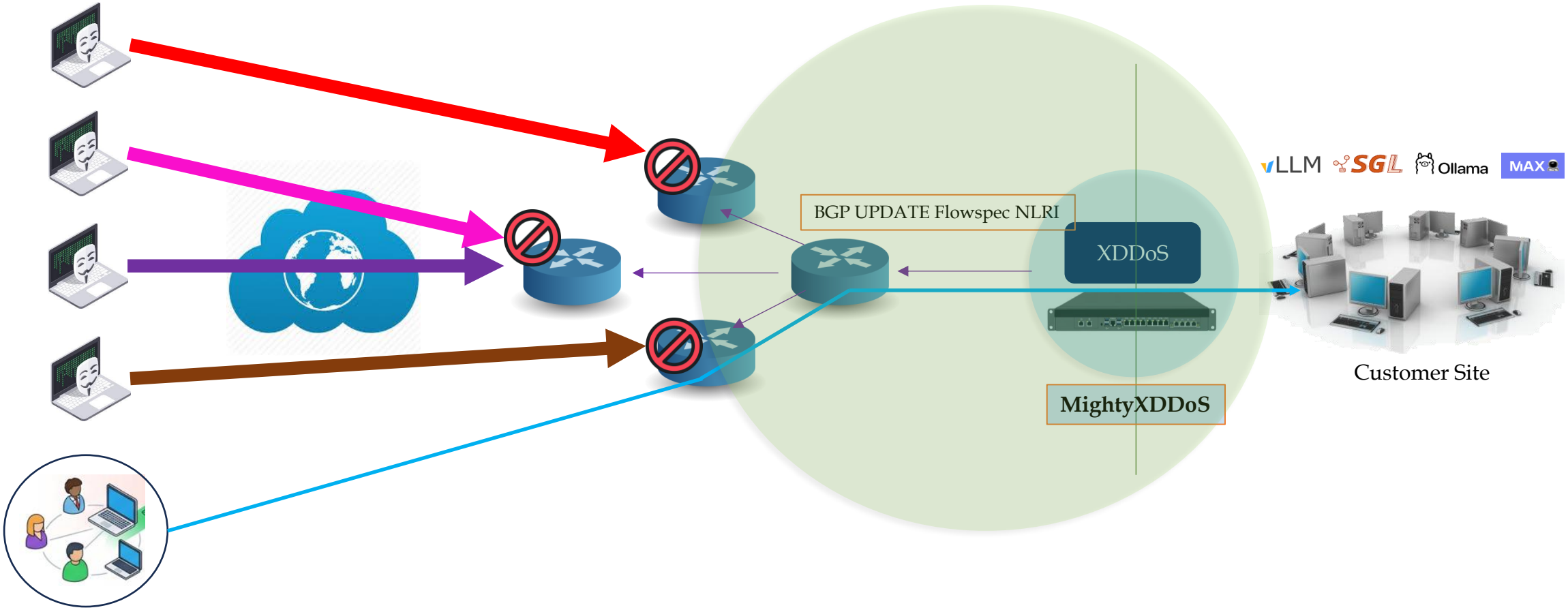
Generated by Gemini

3. MightyXDDoS(14) – BGP Flowspec(2)

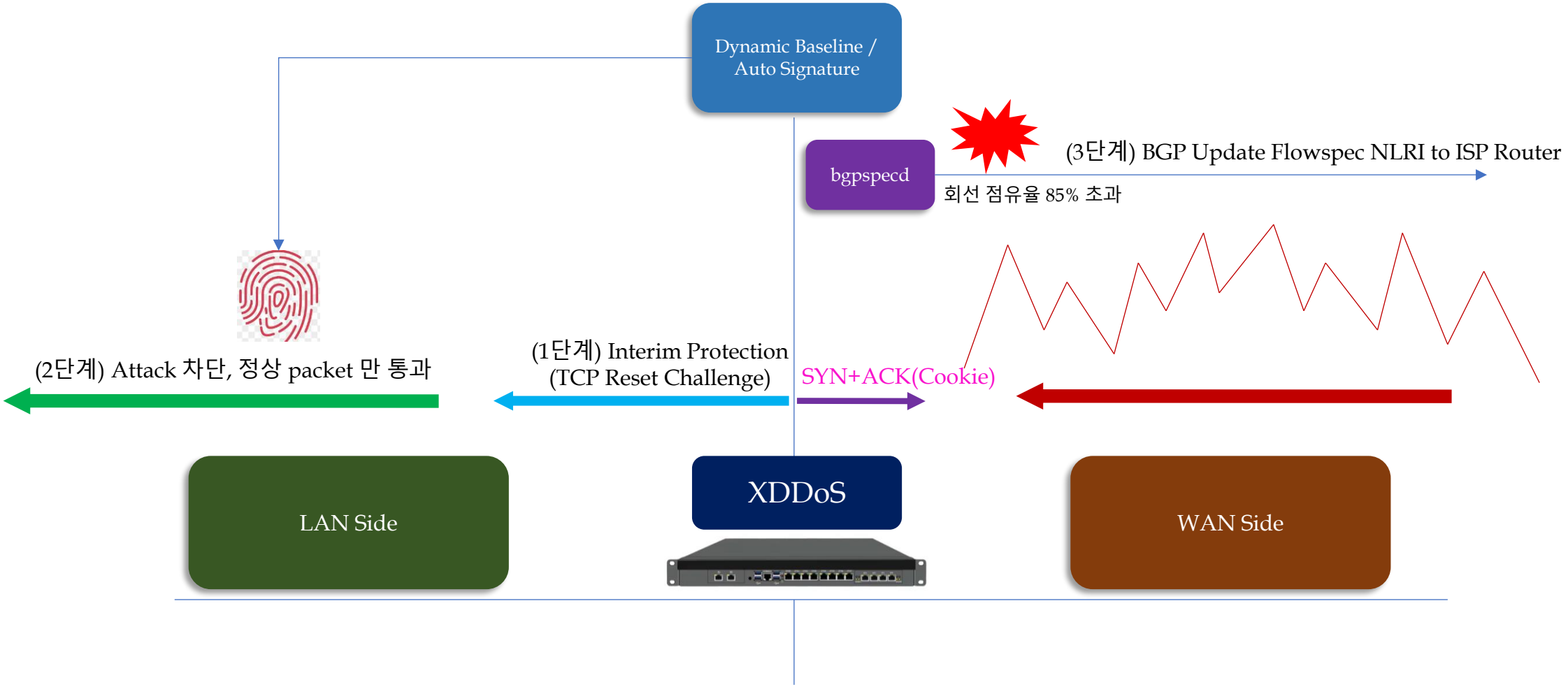


- ZTNA ACL
- Attack Detection
- Intelligent Automation Filtering
- TCP Reset Challenge
- Rate-Limiting
- **BGP Flowspec**

3. MightyXDDoS(14) – BGP Flowspec(3)

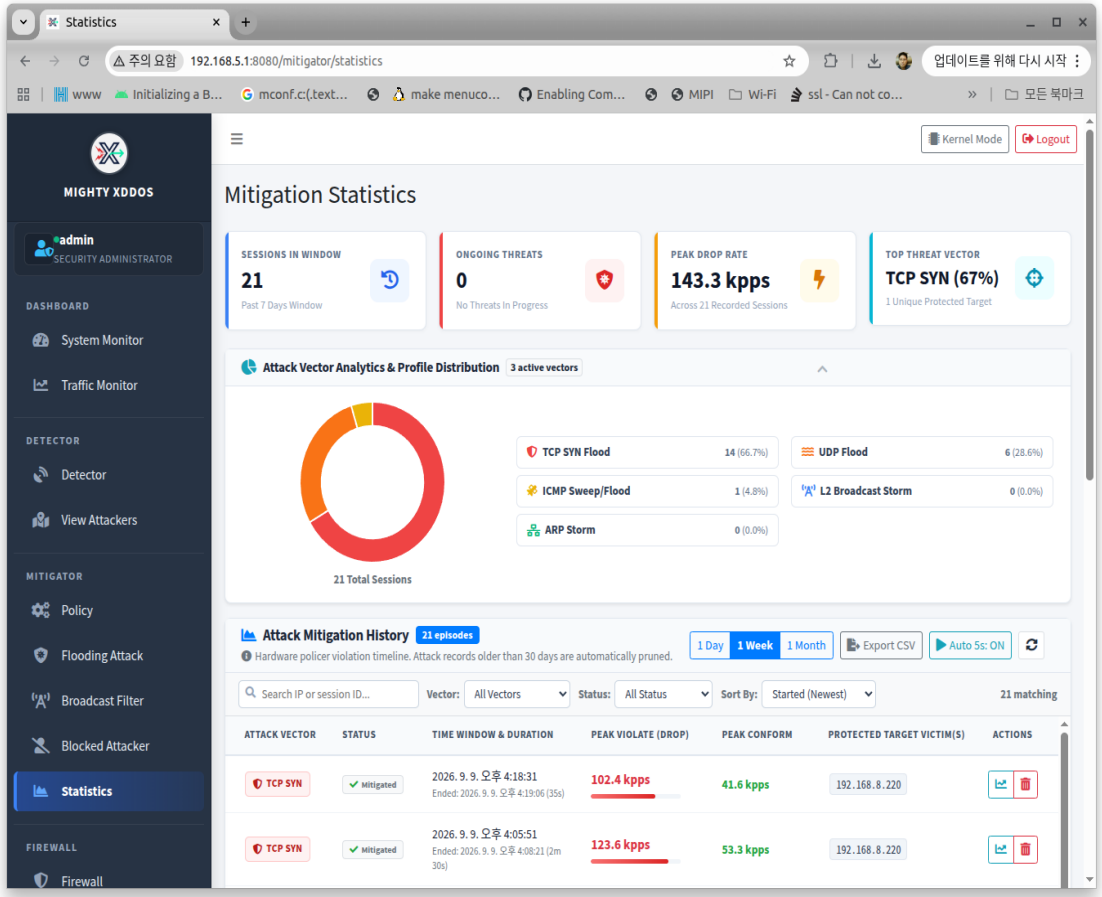


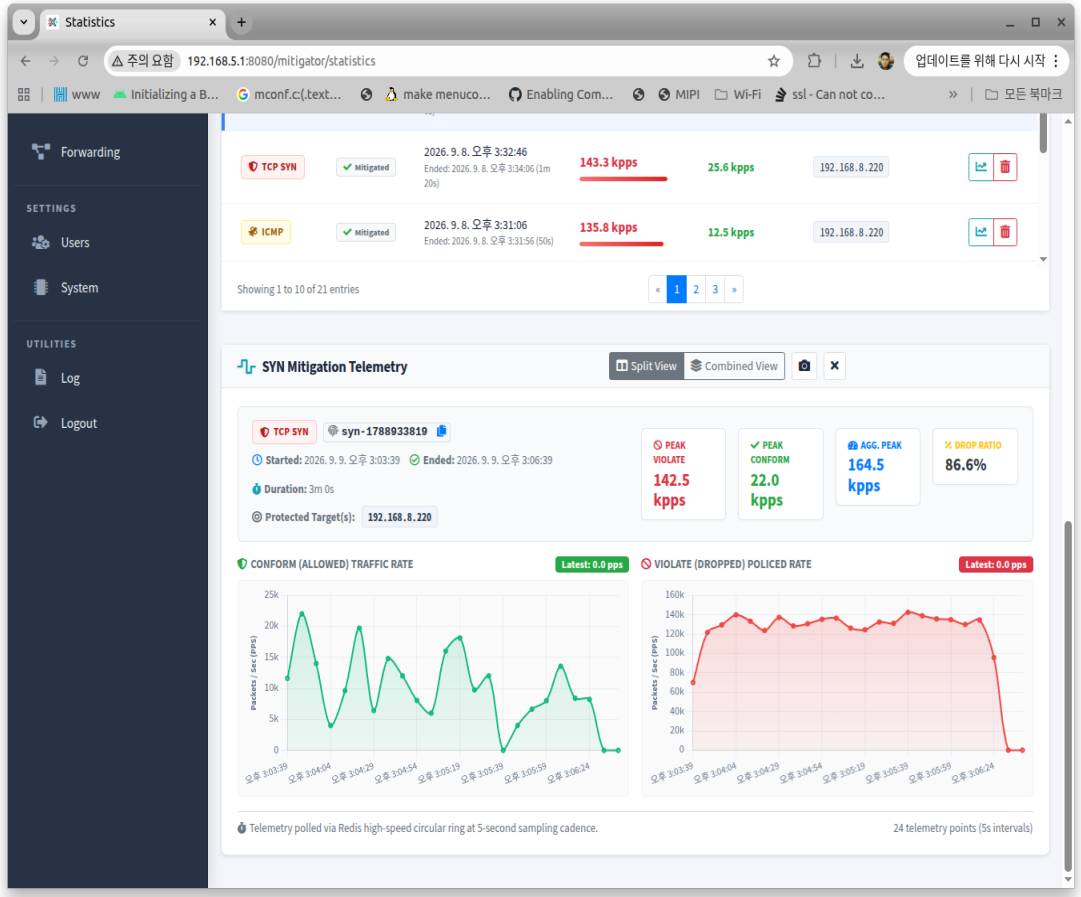
3. MightyXDDoS(14) – BGP Flowspec(4)



3. MightyXDDoS(15) – Statistics(1)

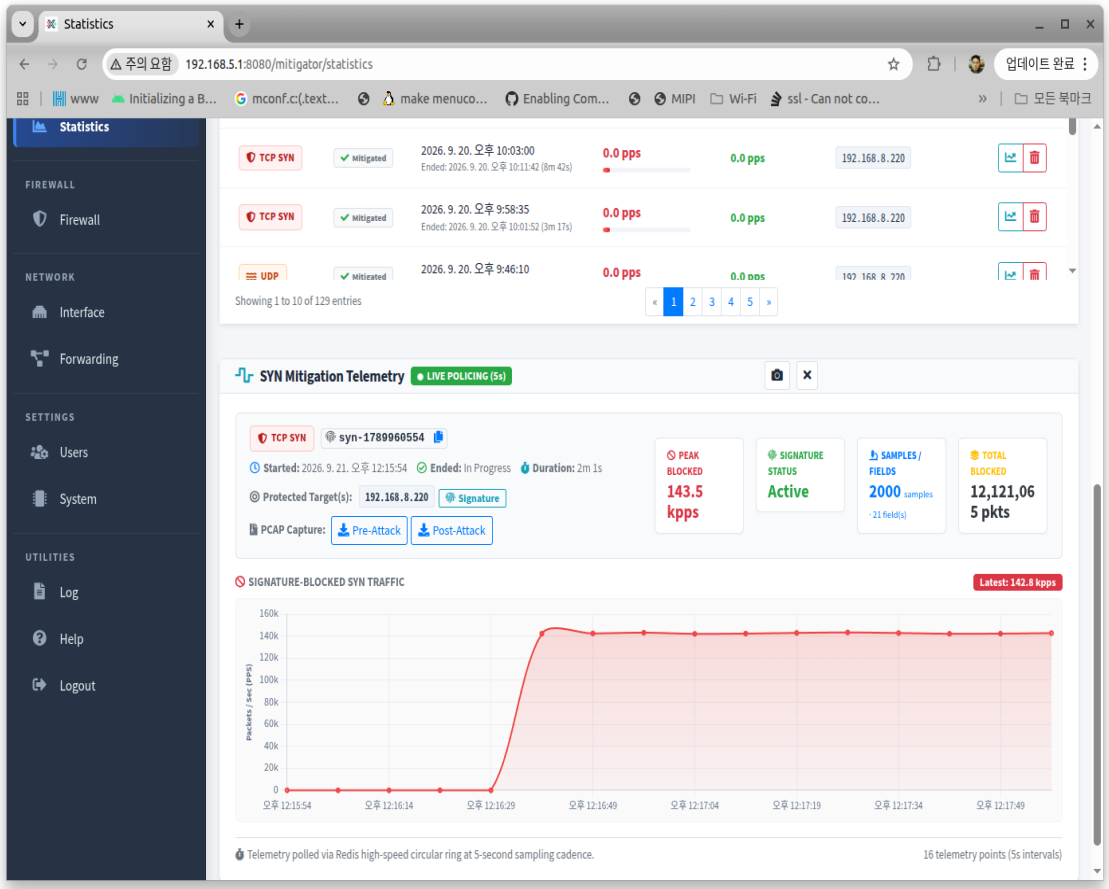
Defense Mode: TCP Reset Challenge Mode

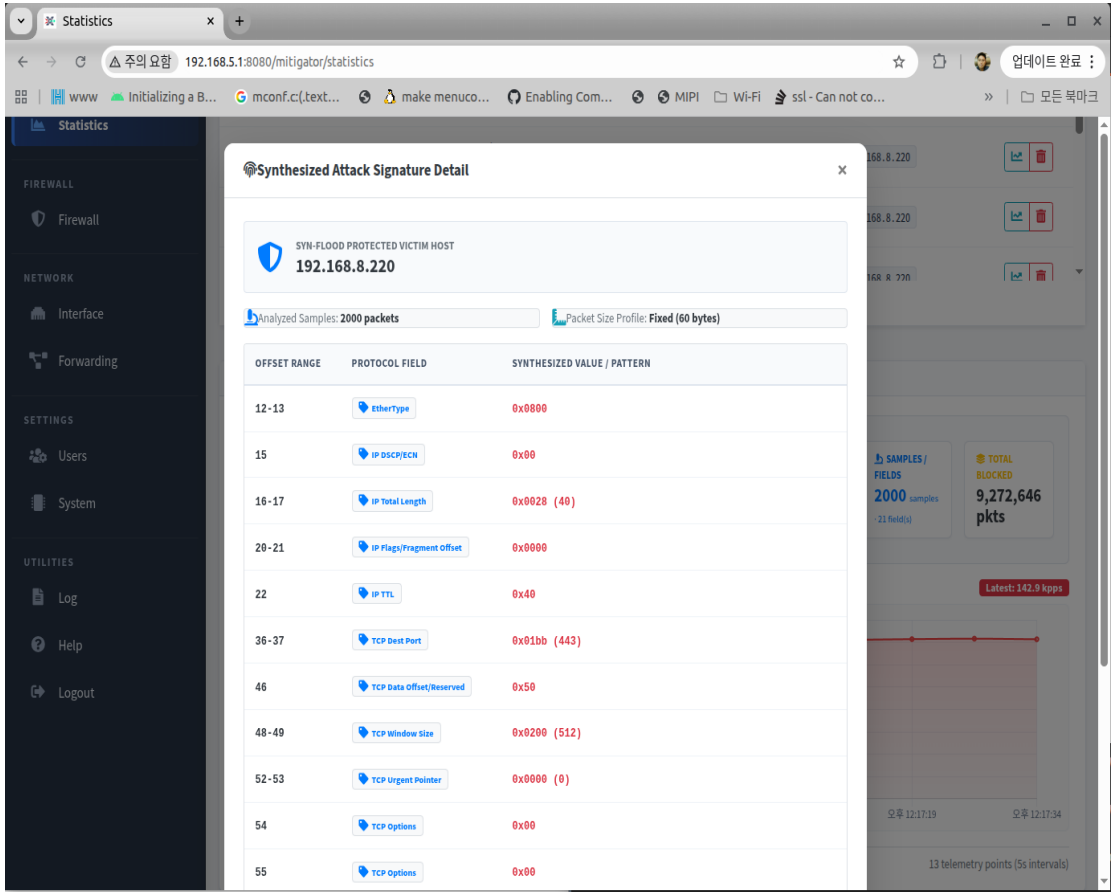




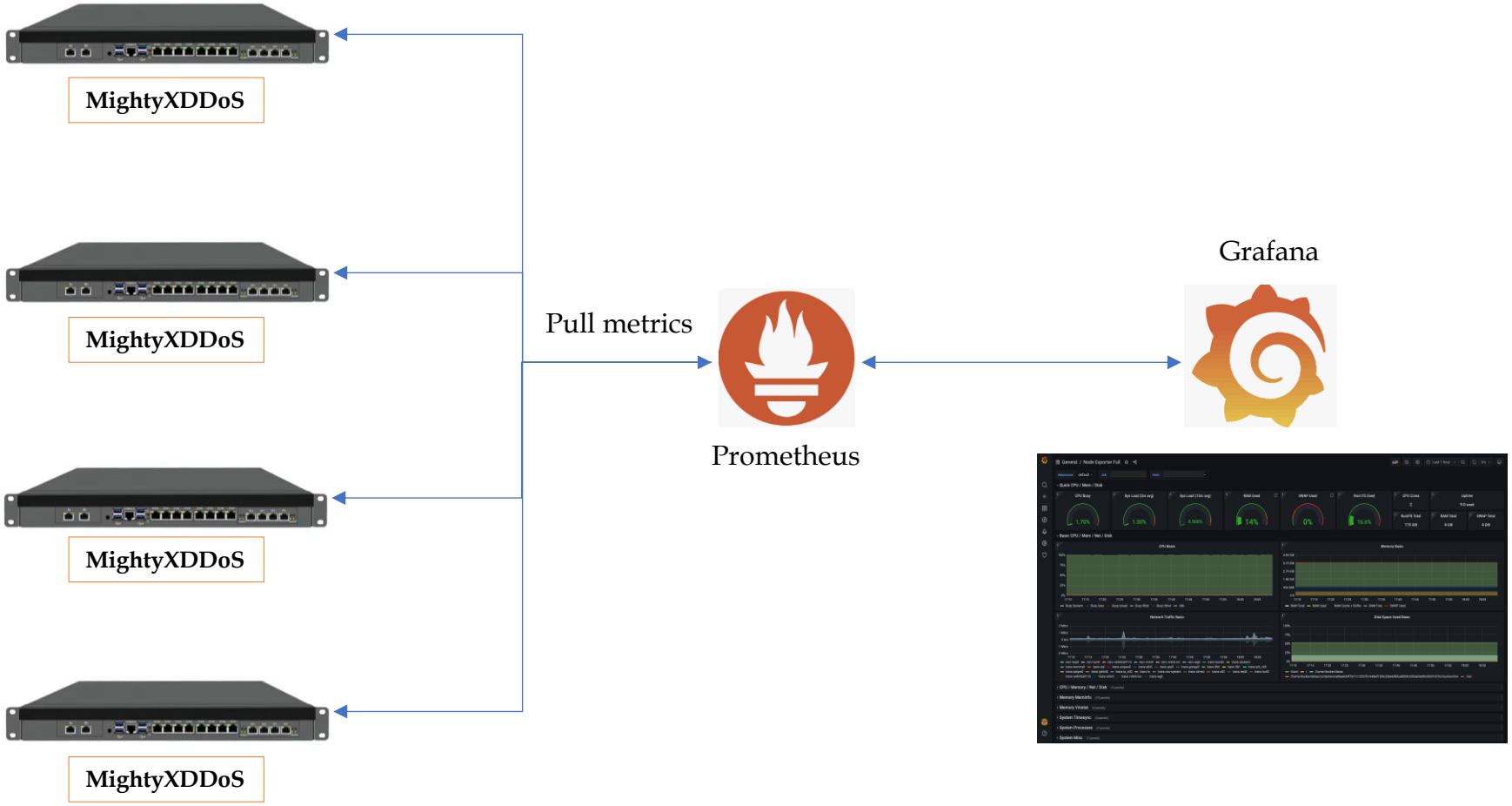
3. MightyXDDoS(15) – Statistics(2)

Defense Mode: Auto-Signature Synthesis Mode

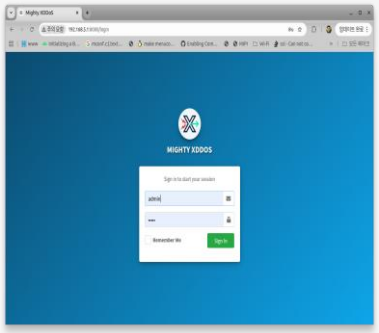




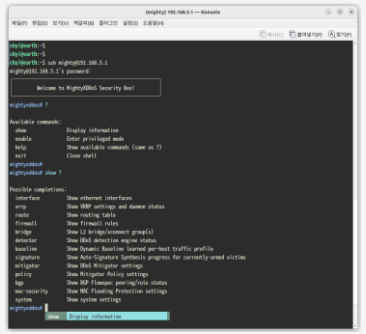
3. MightyXDDoS(16) – Prometheus Metrics Exporter



4. XDDoS User Interface

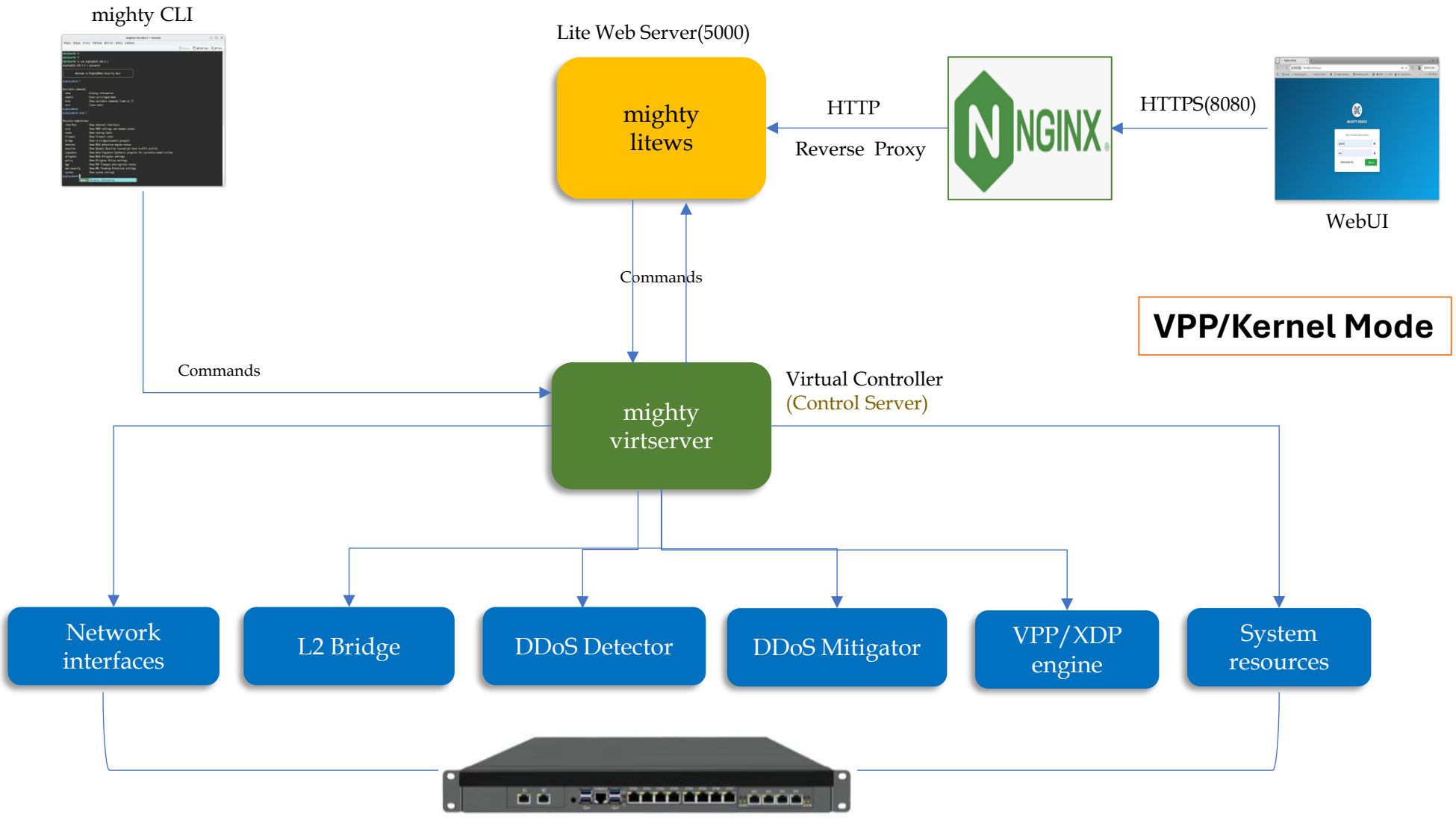


WebUI

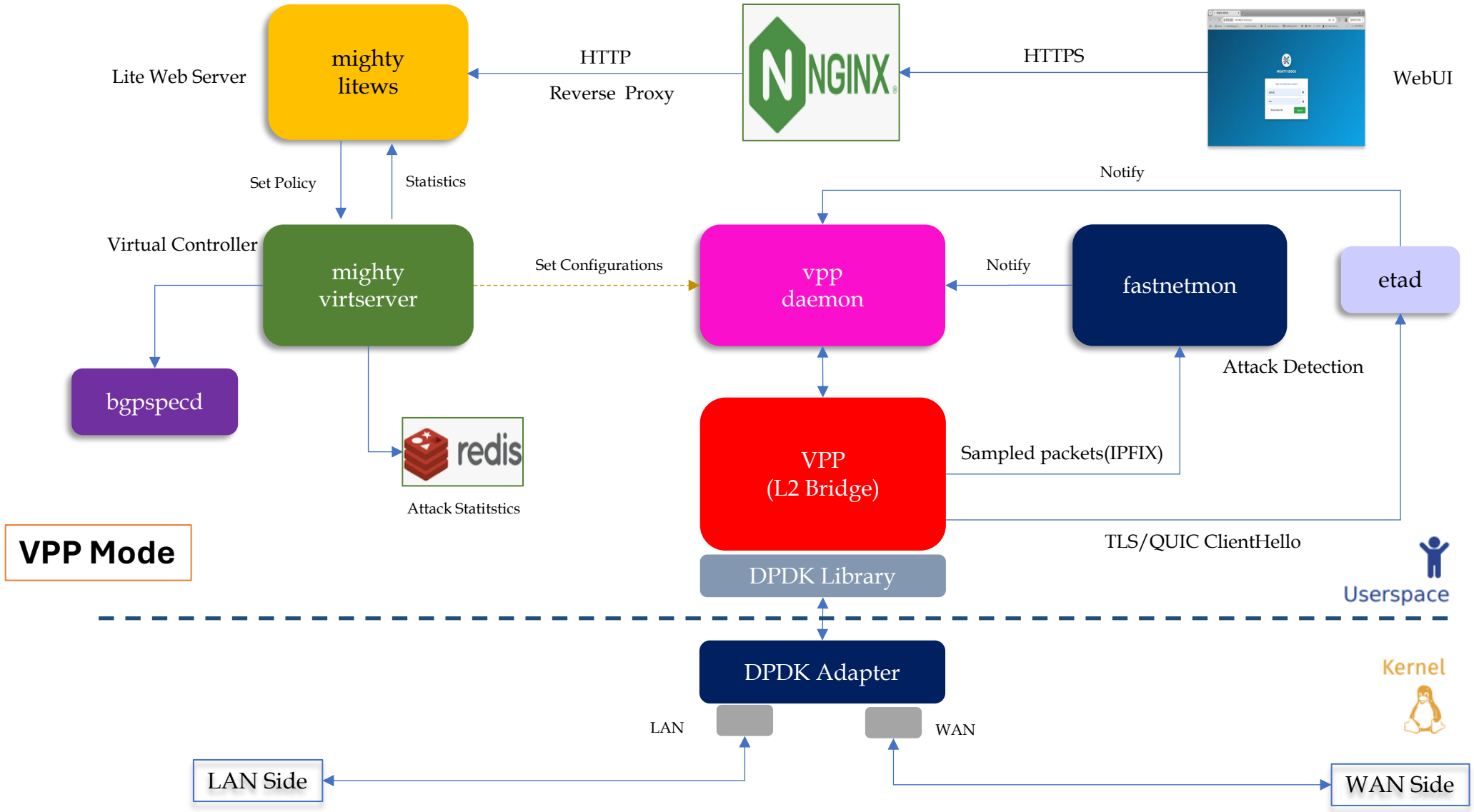


CLI

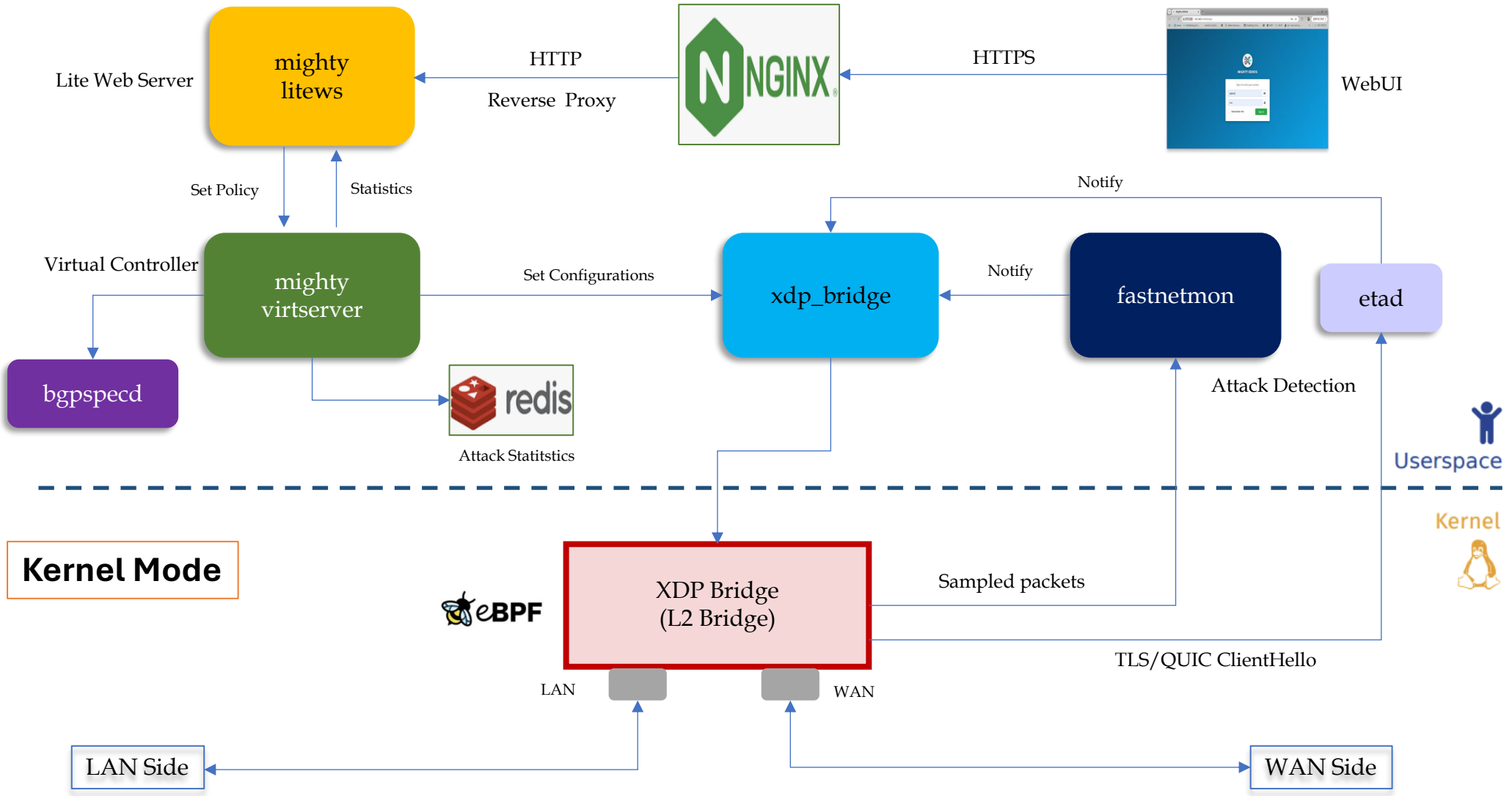
4. XDDoS User Interface(1) – Mighty Engine(1)



4. XDDoS User Interface(1) – Mighty Engine(2)

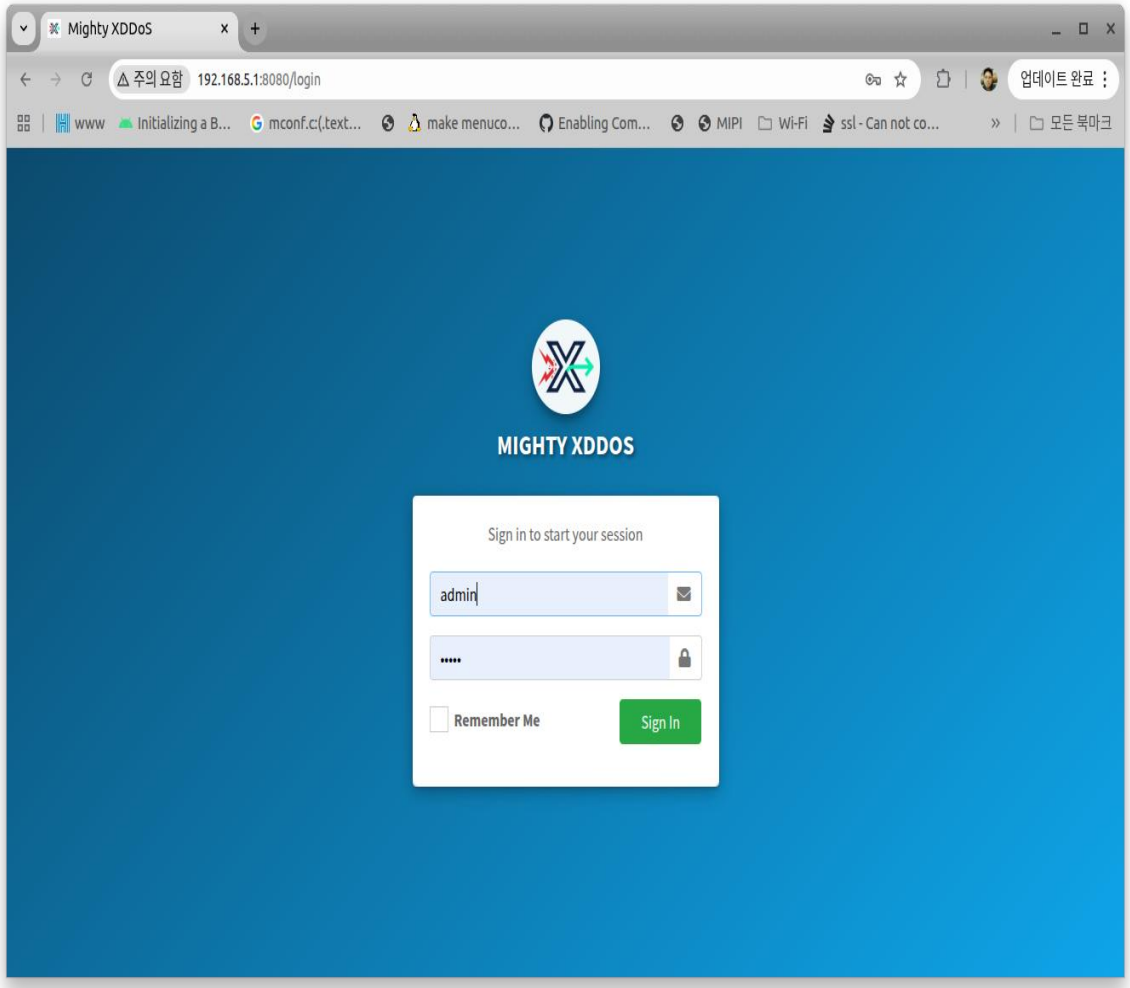


4. XDDoS User Interface(1) – Mighty Engine(3)

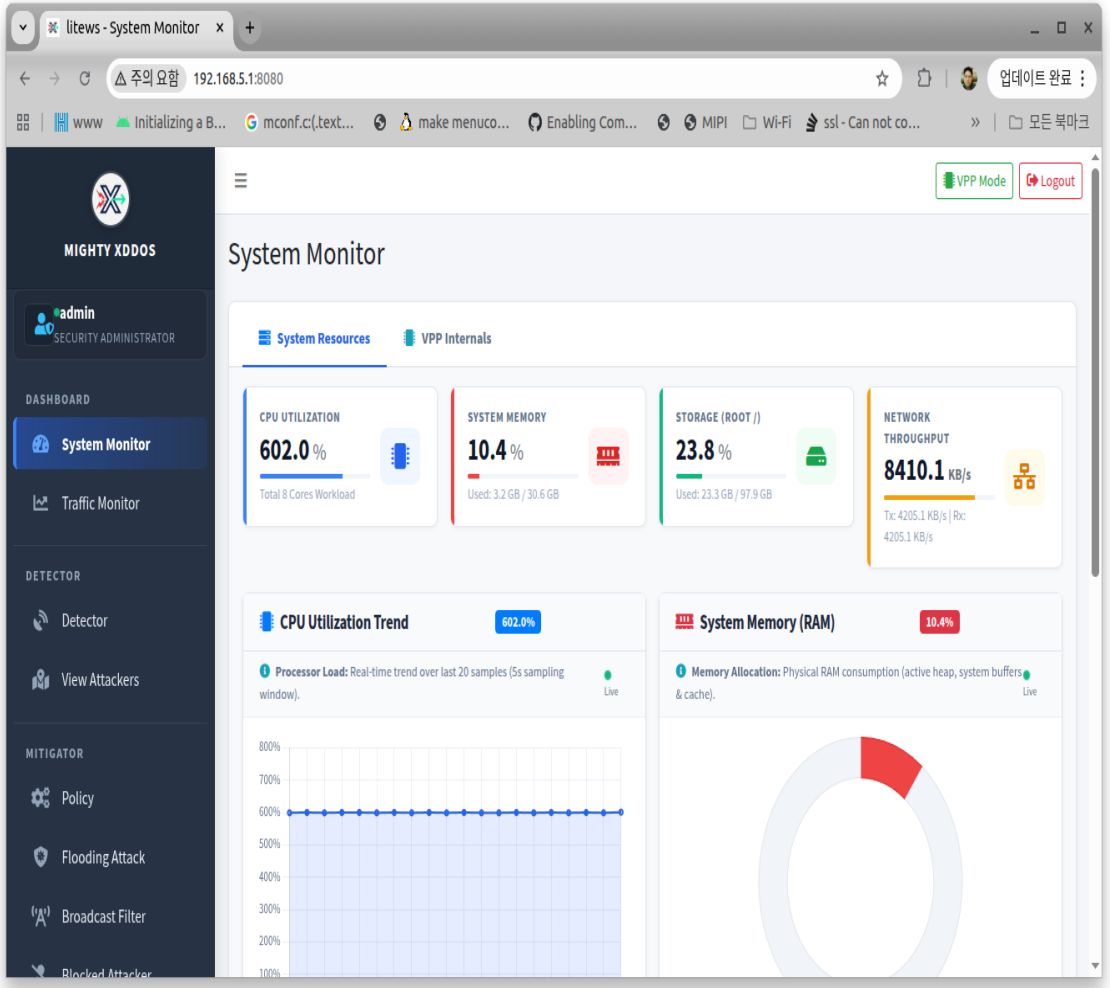


4. XDDoS User Interface(2) – WebUI(1)

Login page

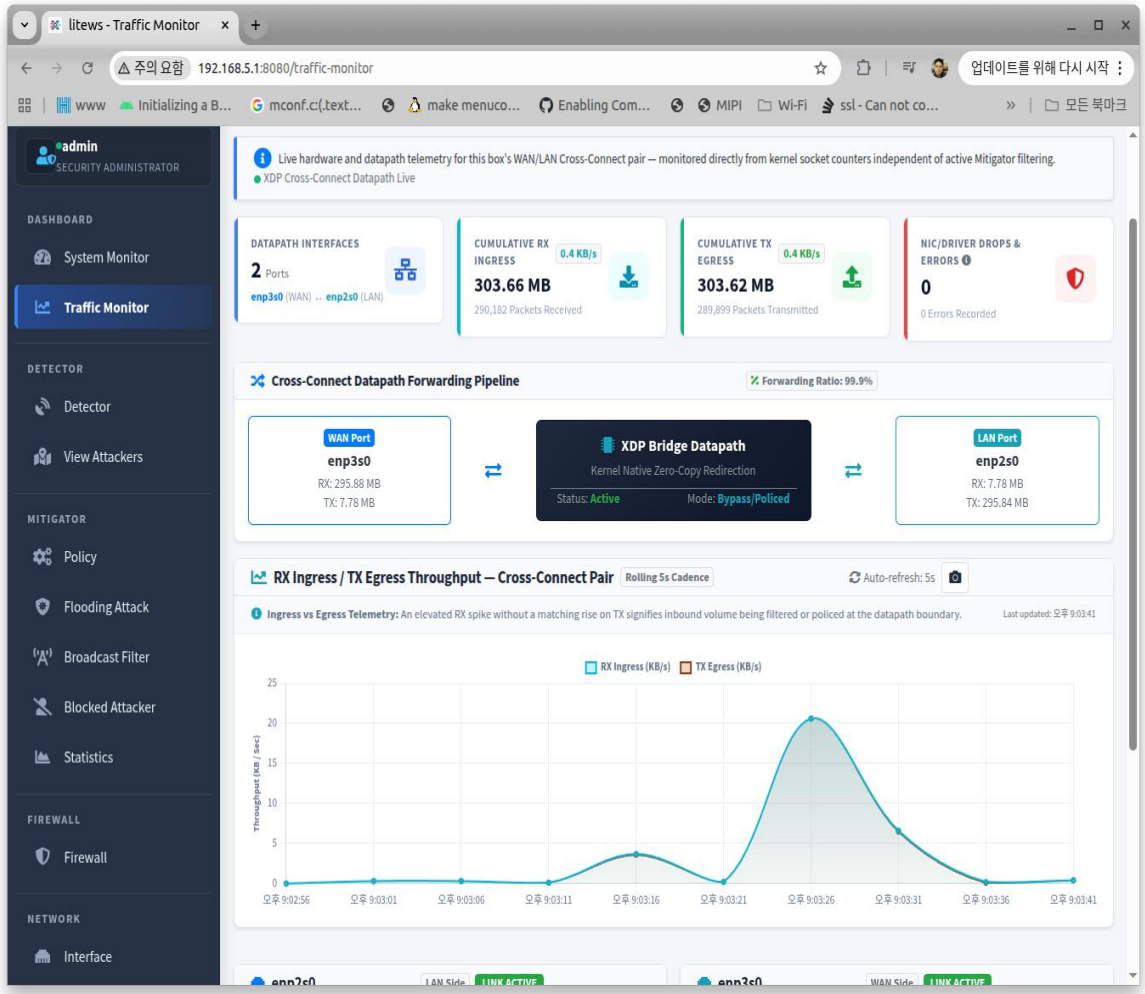


DashBoard – System Monitor Page

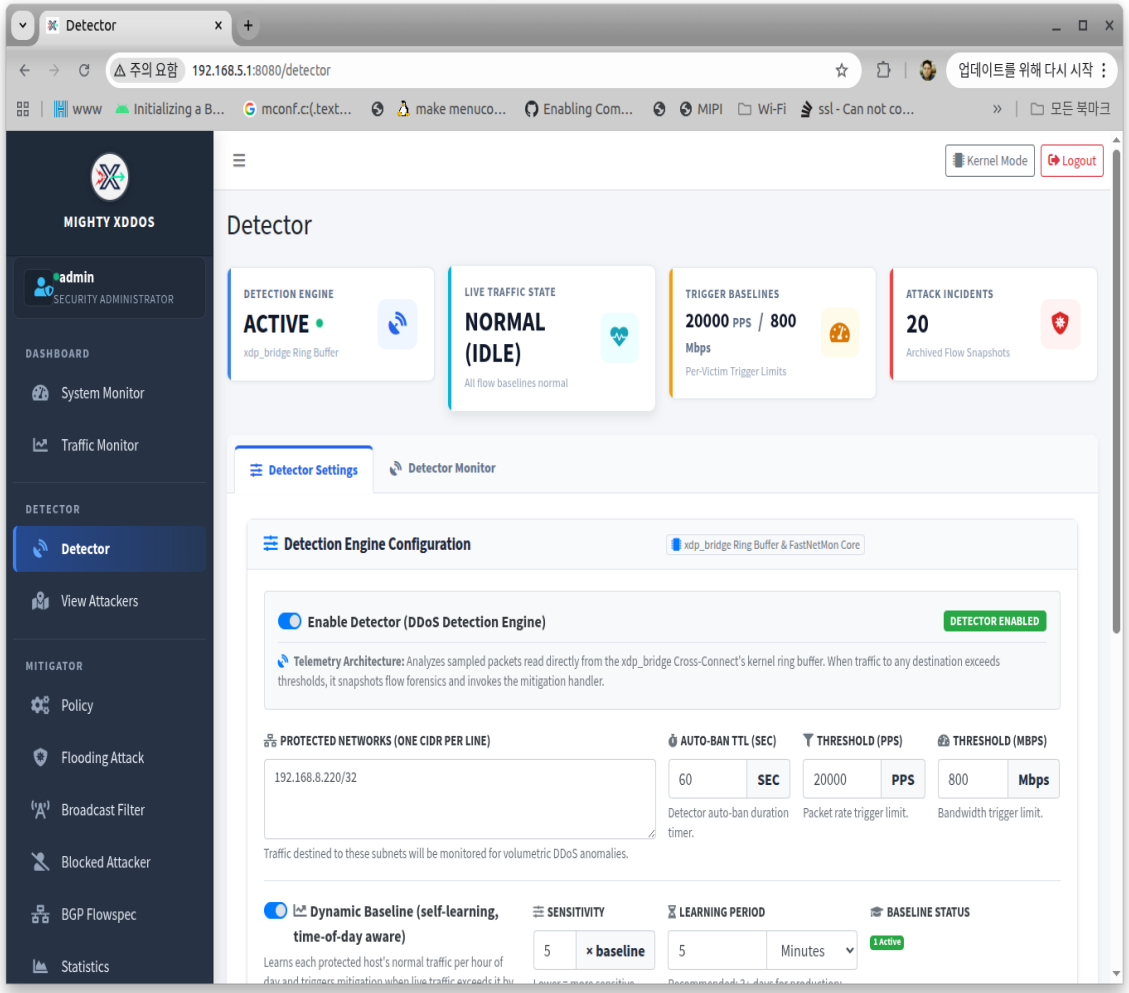


4. XDDoS User Interface(2) – WebUI(2)

DashBoard – Traffic Monitor Page

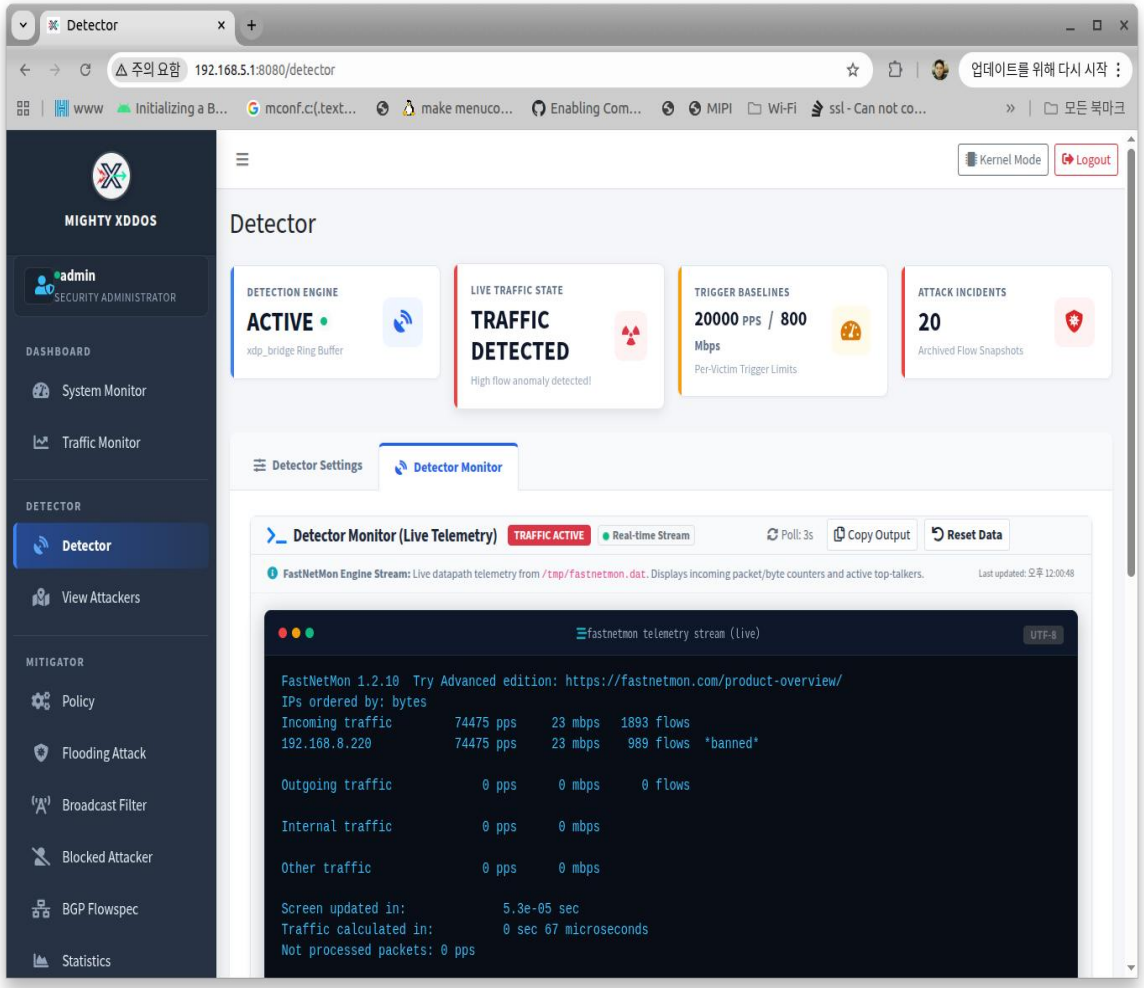


Detector Page#1

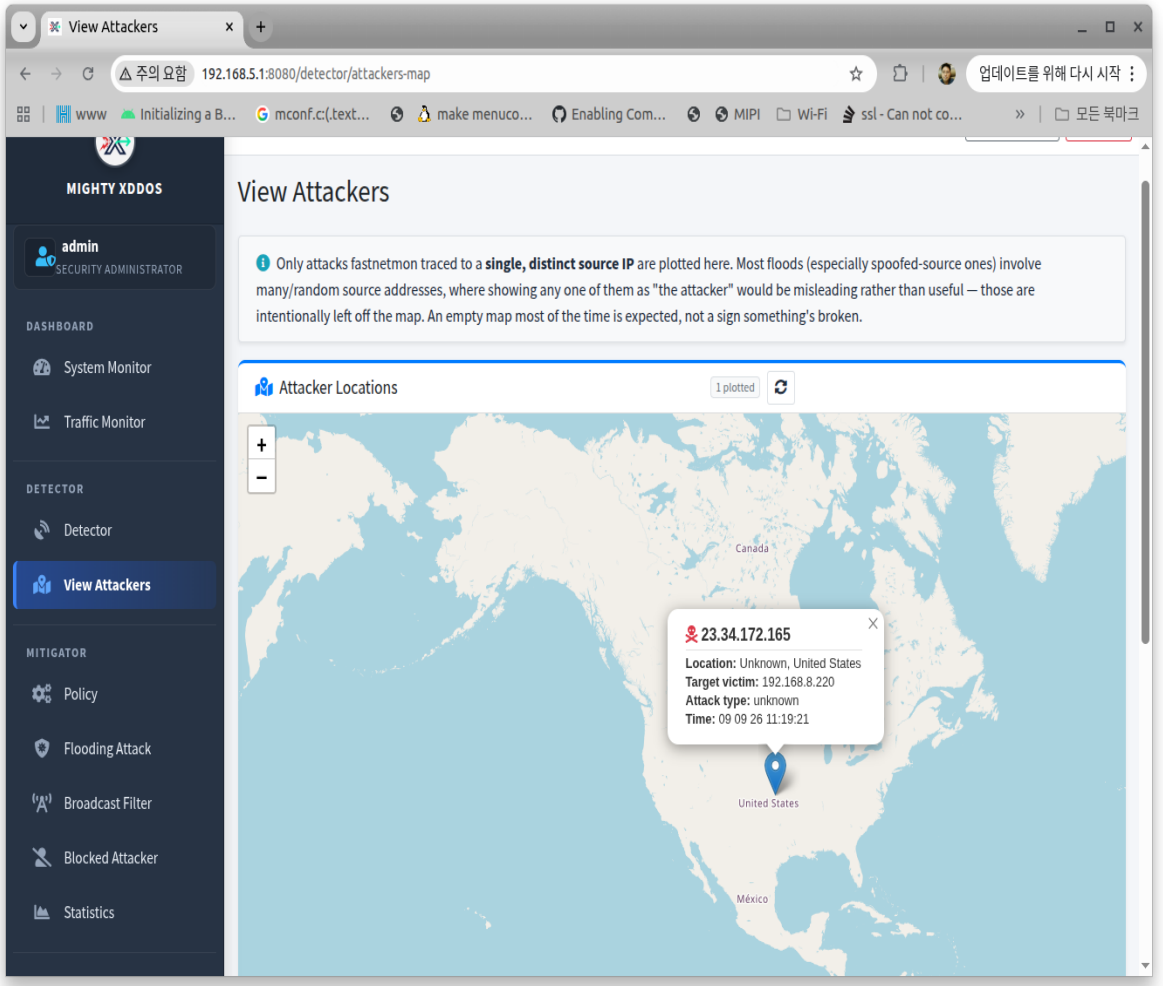


4. XDDoS User Interface(2) – WebUI(3)

Detector Page#2

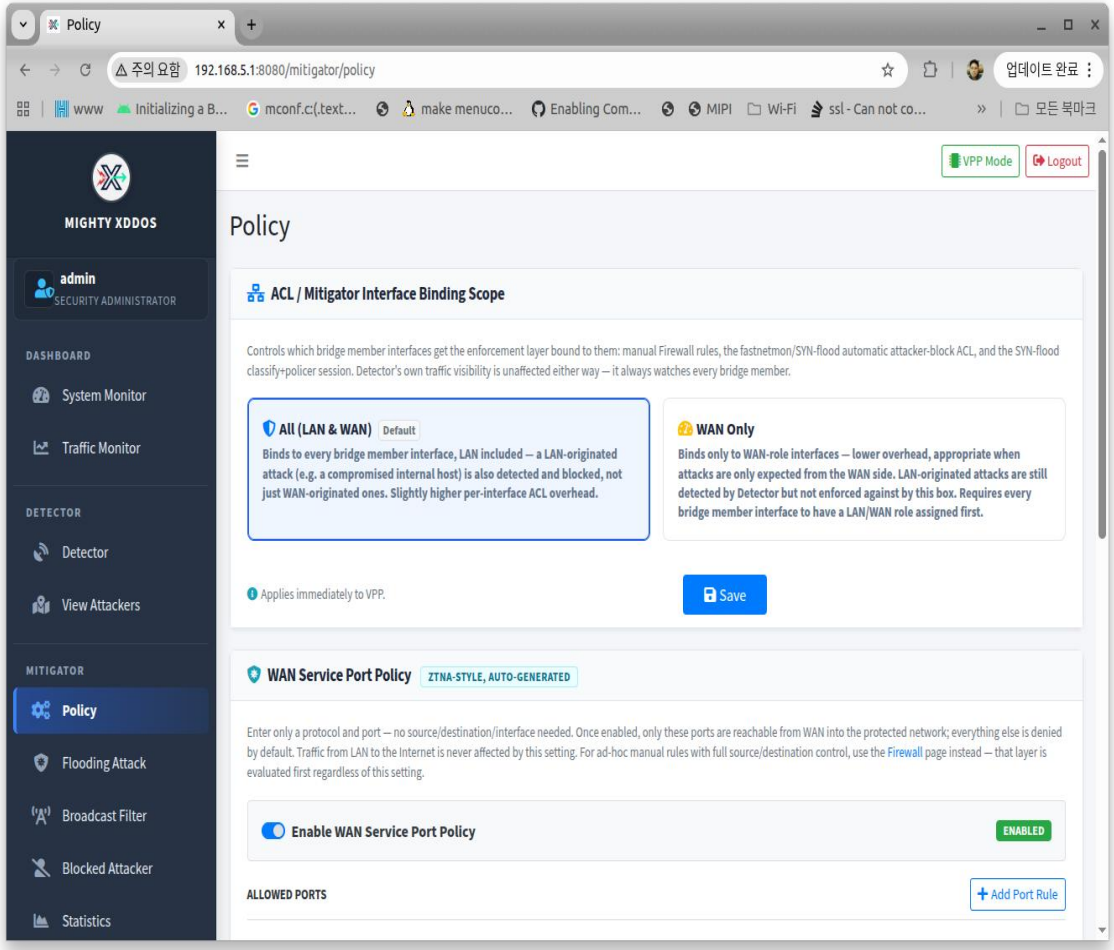


Detector - View Attackers page

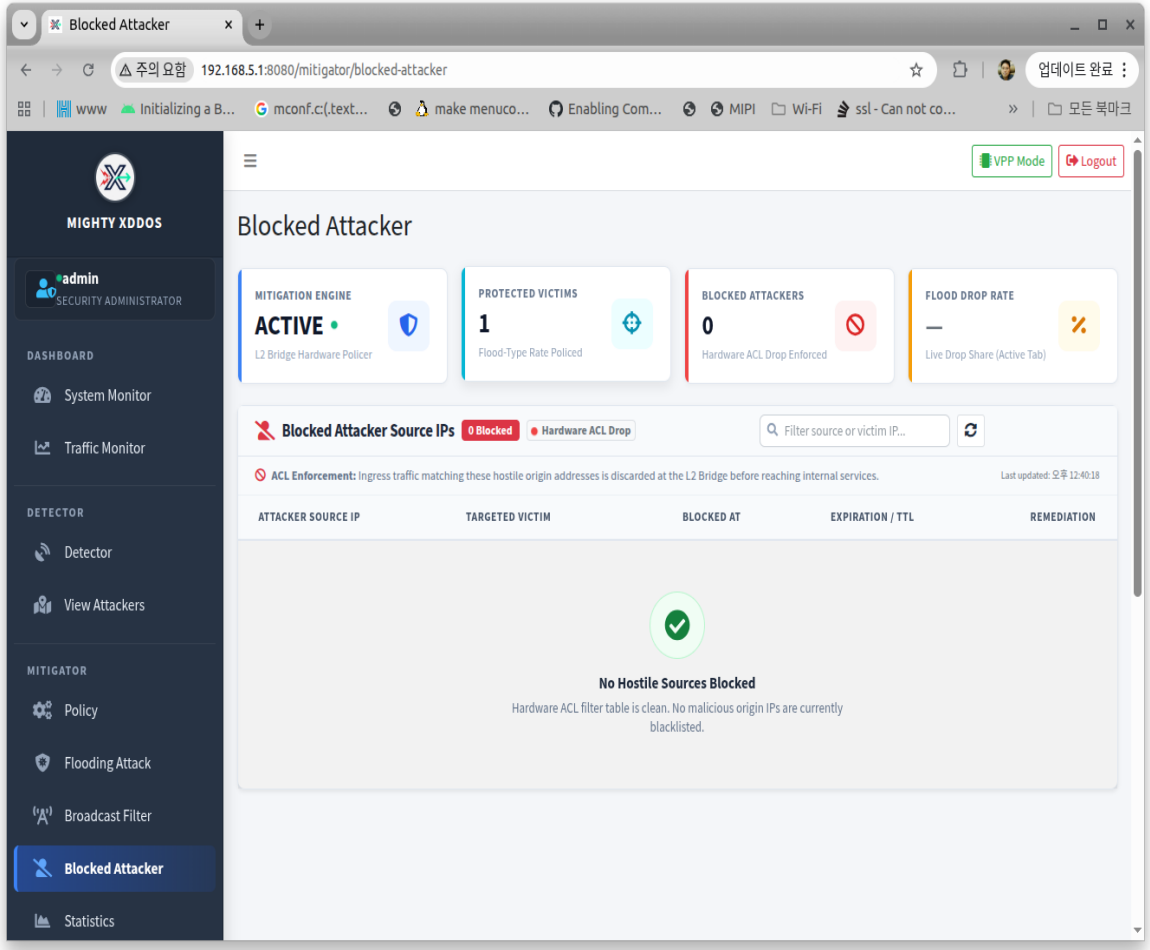


4. XDDoS User Interface(2) – WebUI(4)

Mitigator - Policy page

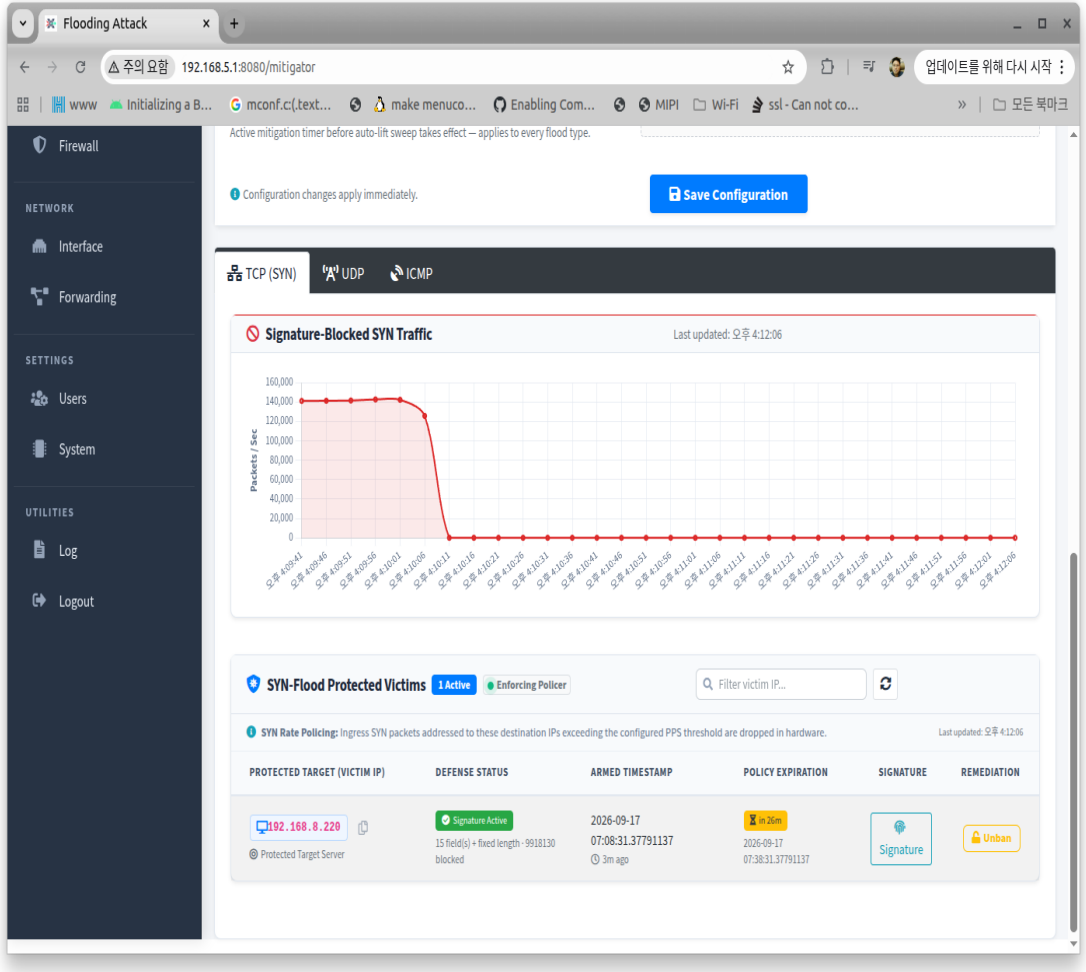


Mitigator - Blocked Attacker page

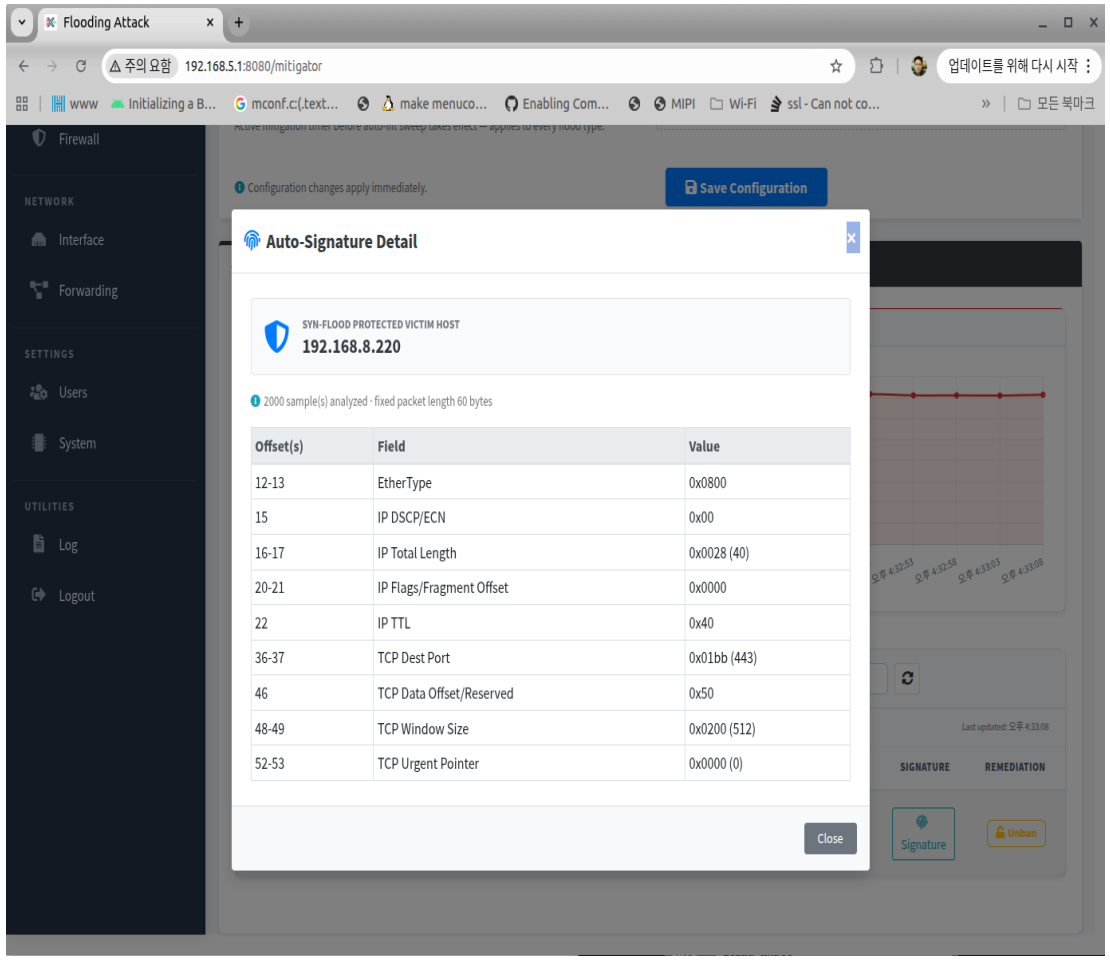


4. XDDoS User Interface(2) – WebUI(5)

Mitigator - Flooding Attack/TCP(SYN) page

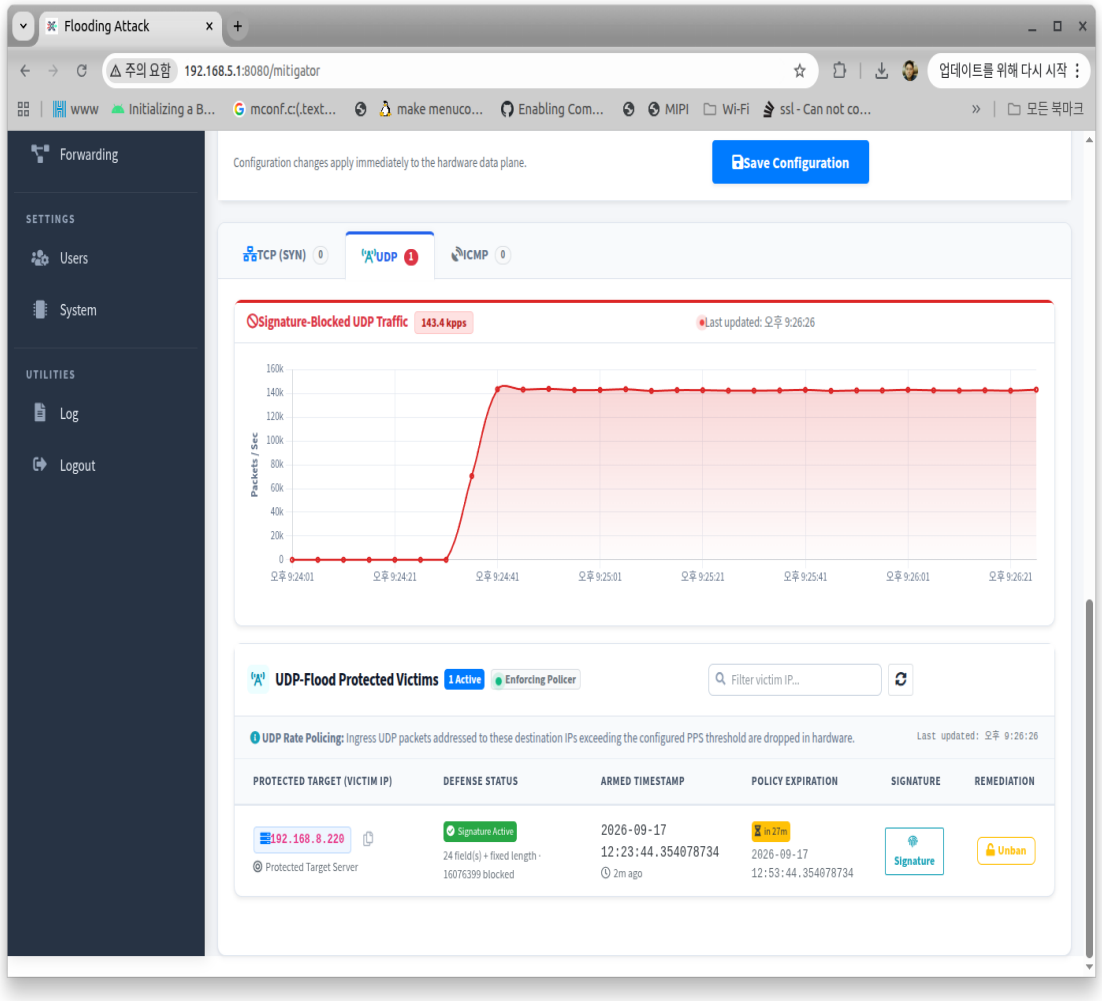


Mitigator - Flooding Attack/TCP(SYN) page - Signature

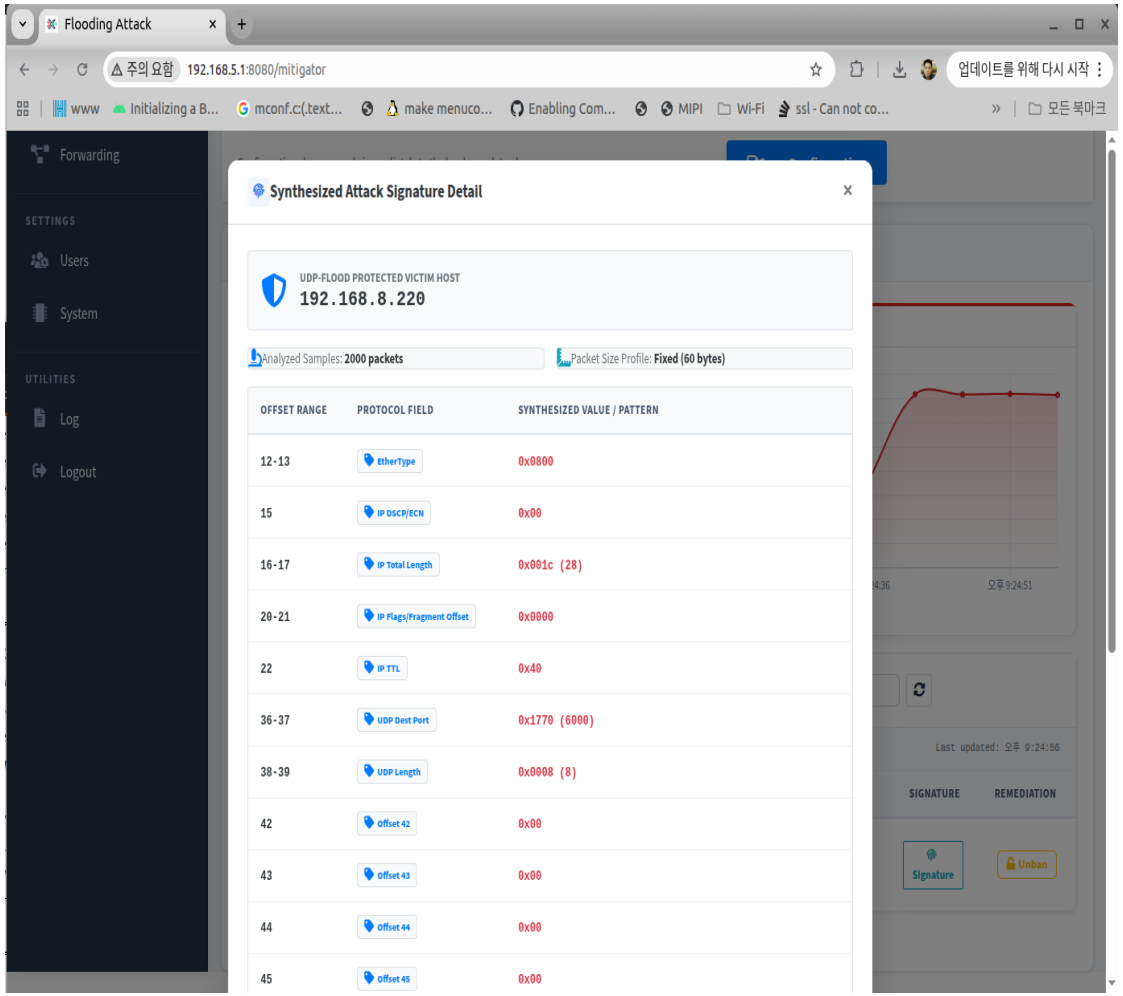


4. XDDoS User Interface(2) – WebUI(6)

Mitigator - Flooding Attack/UDP page

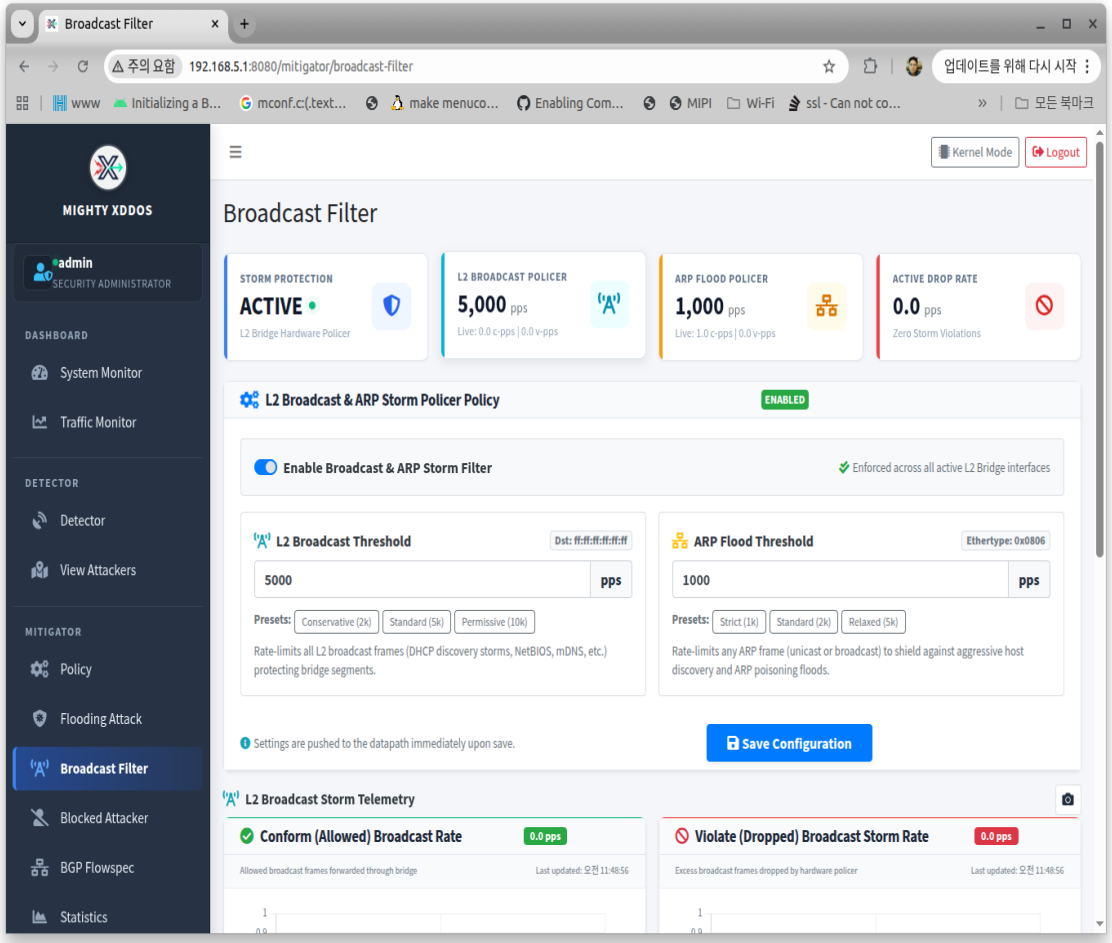


Mitigator - Flooding Attack/UDP - Signature page

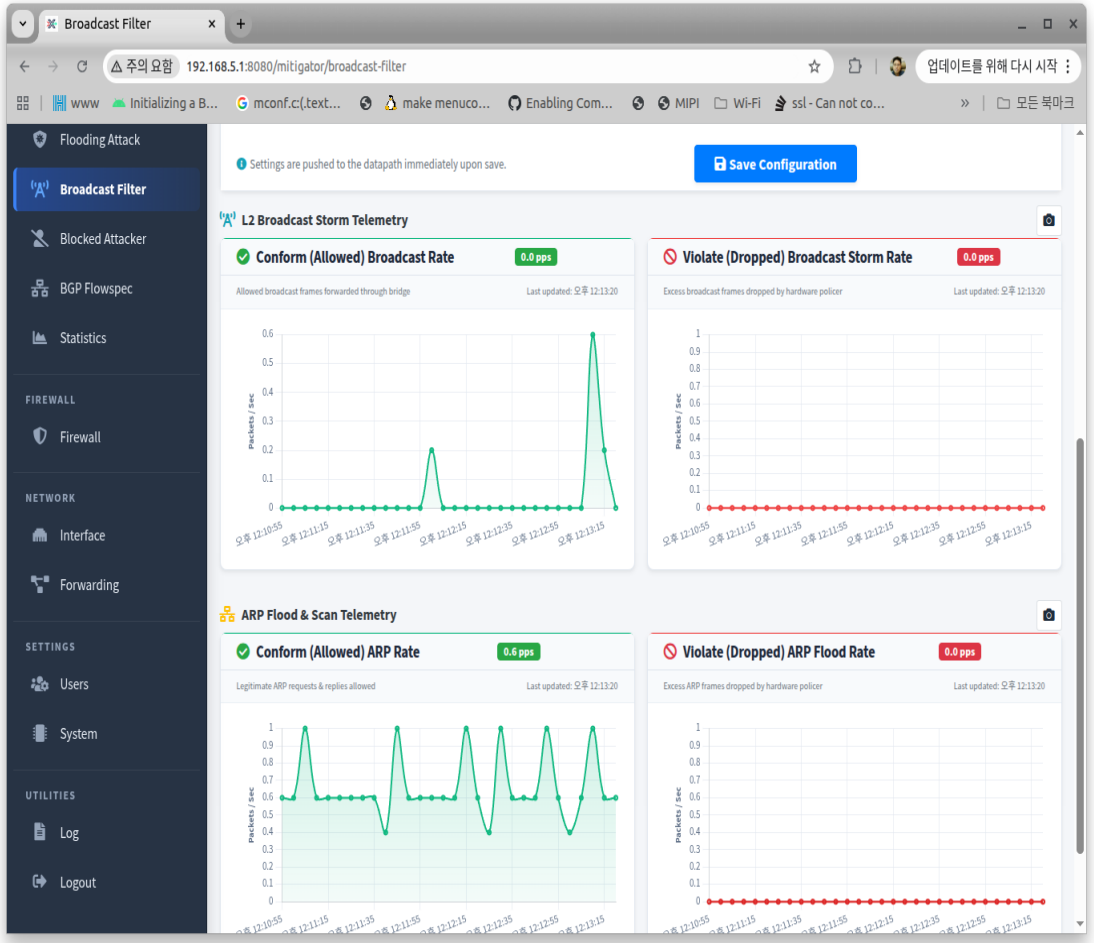


4. XDDoS User Interface(2) – WebUI(7)

Mitigator - Broadcast Filter Page#1

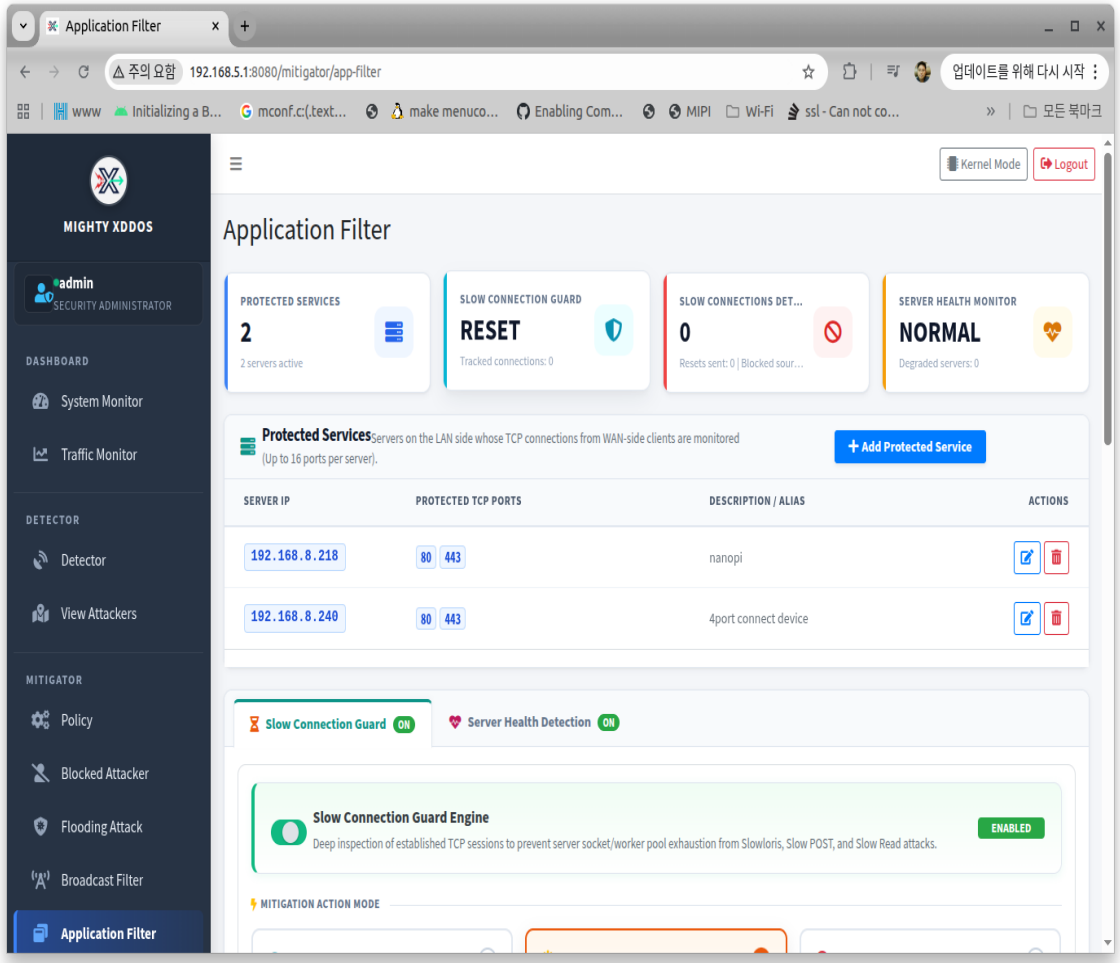


Mitigator - Broadcast Filter Page#2

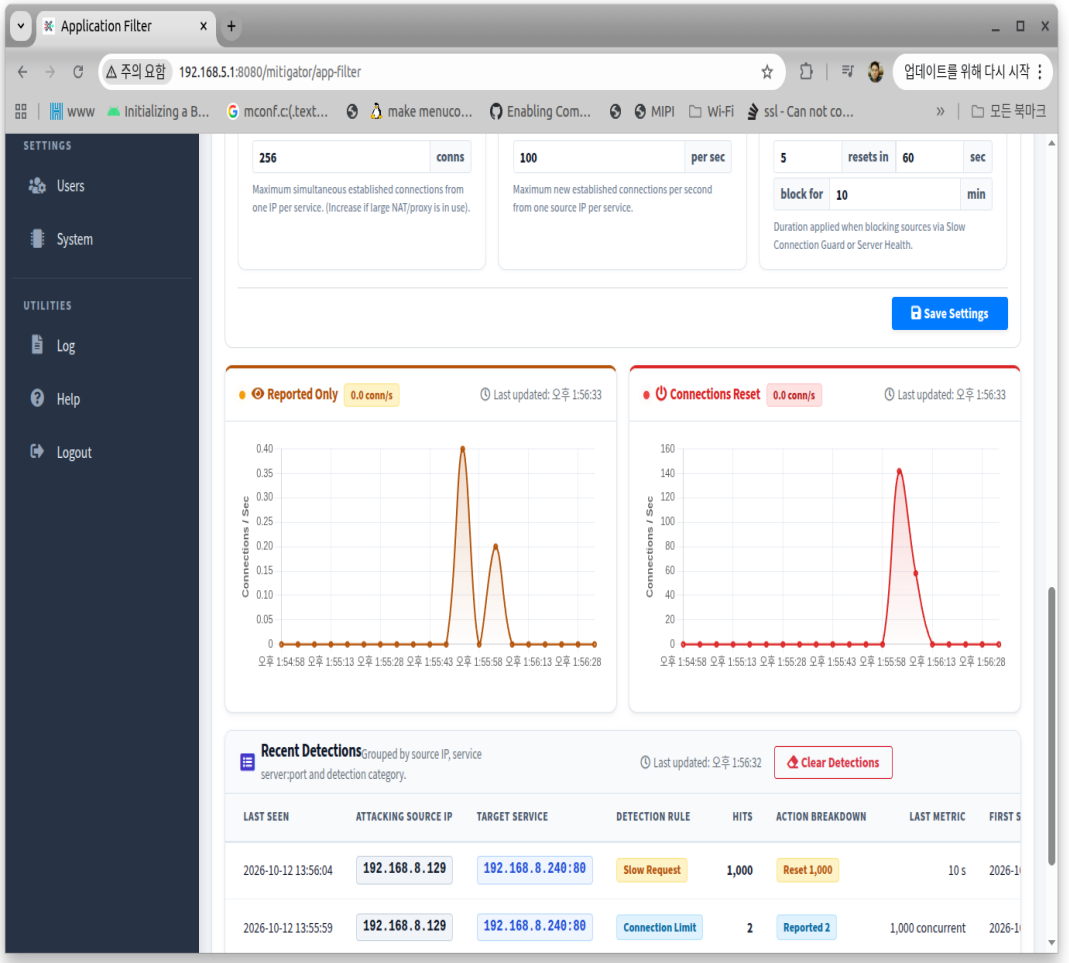


4. XDDoS User Interface(2) – WebUI(8)

Mitigator - Application Filter Page#1

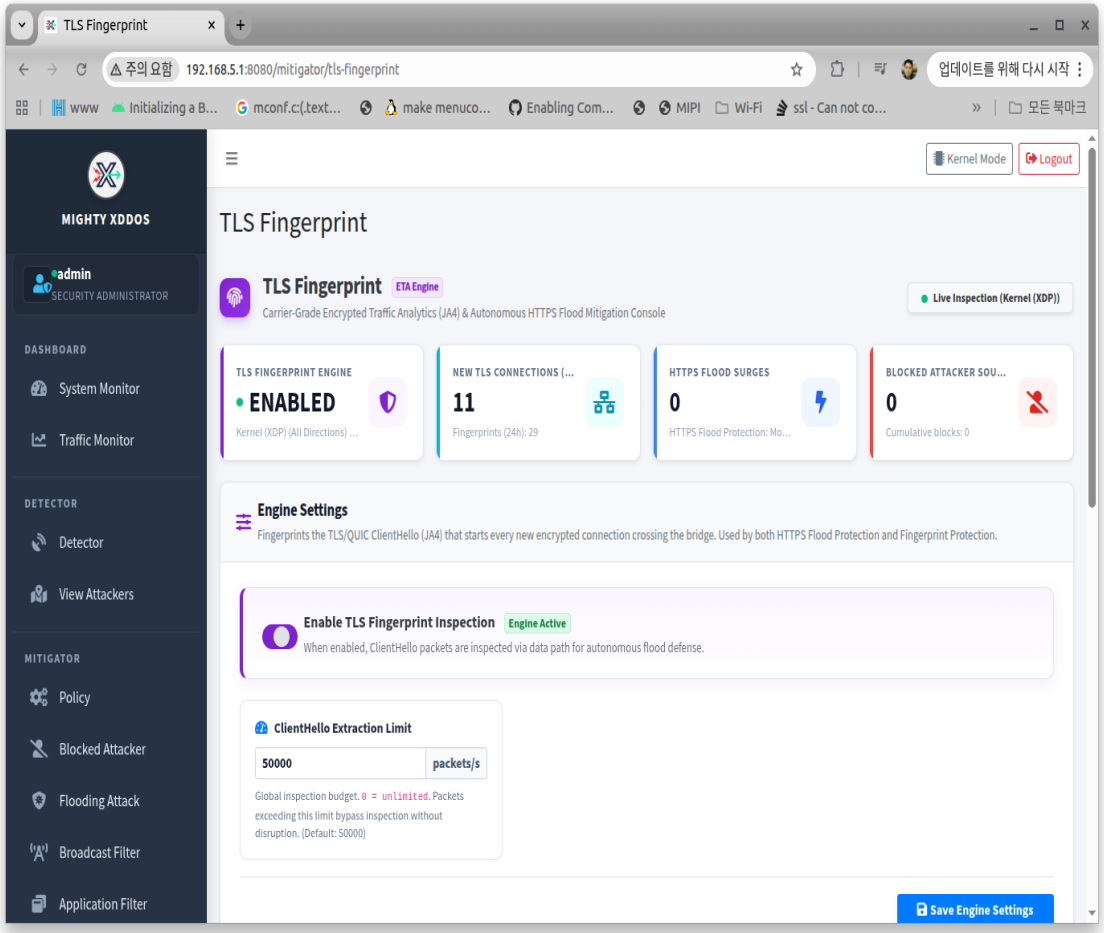


Mitigator - Application Filter Page#2

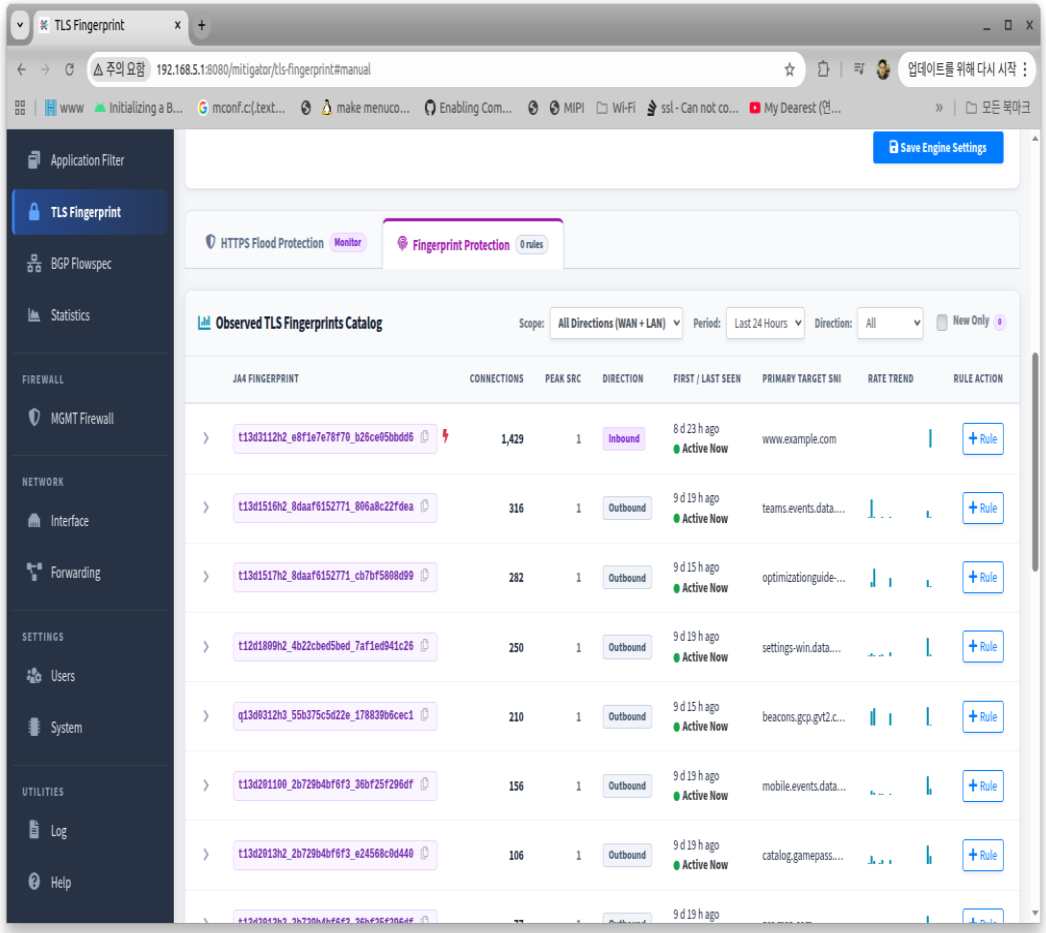


4. XDDoS User Interface(2) – WebUI(9)

Mitigator - TLS Fingerprint Page#1

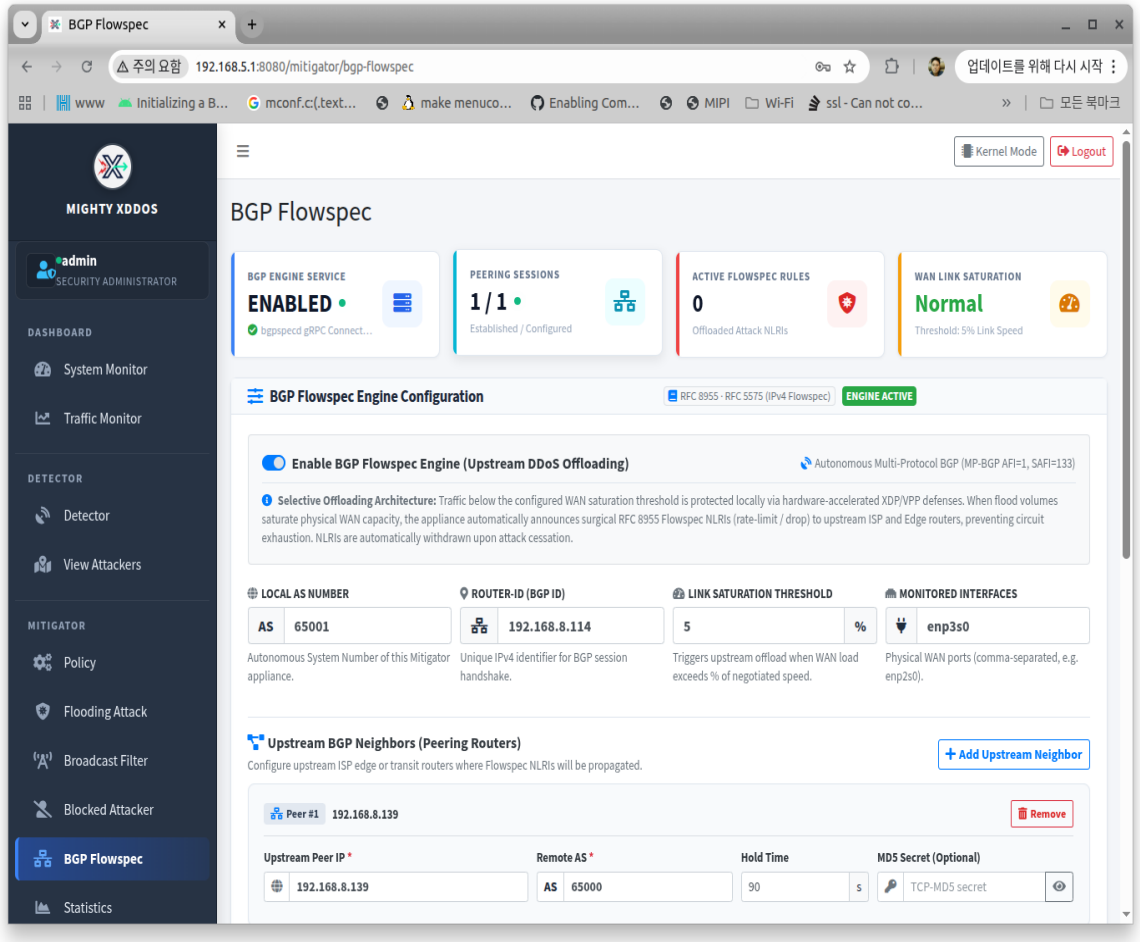


Mitigator - TLS Fingerprint Page#2

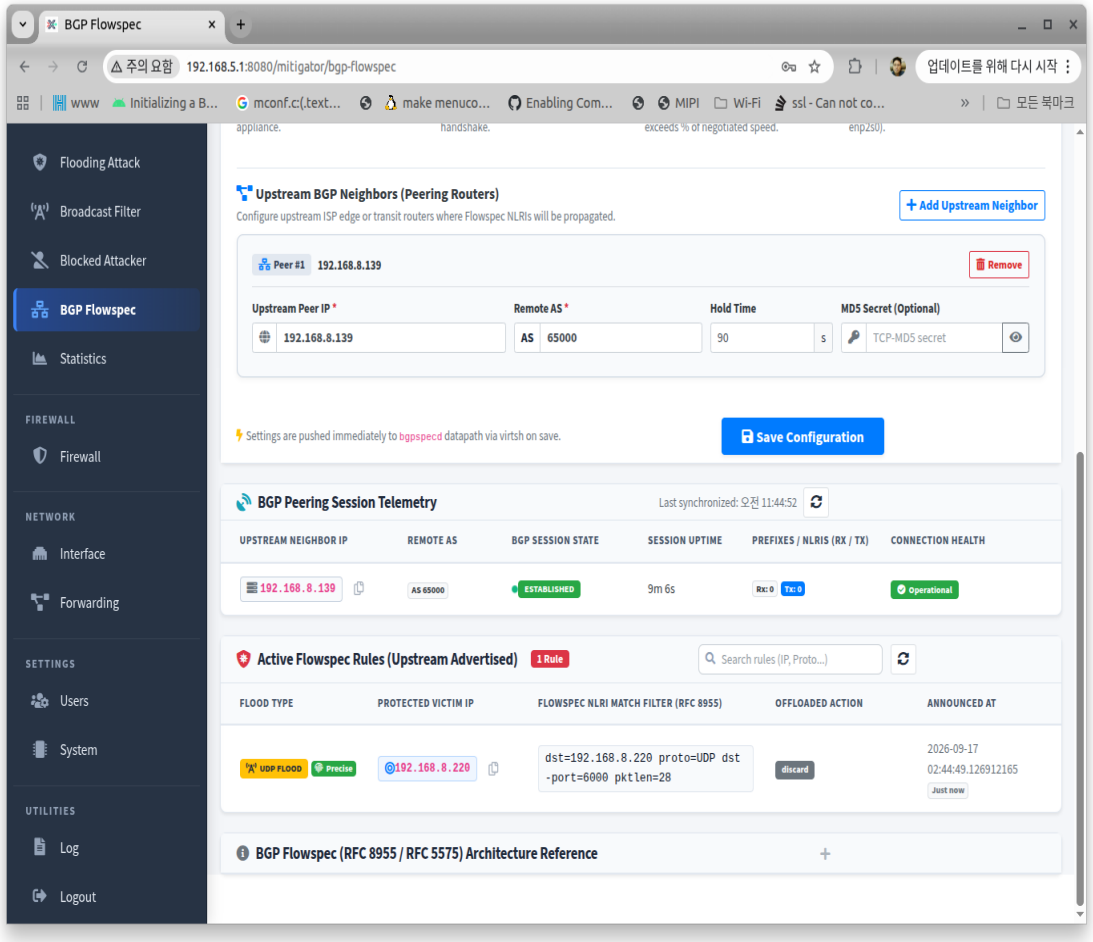


4. XDDoS User Interface(2) – WebUI(10)

Mitigator - BGP Flowspec Page#1

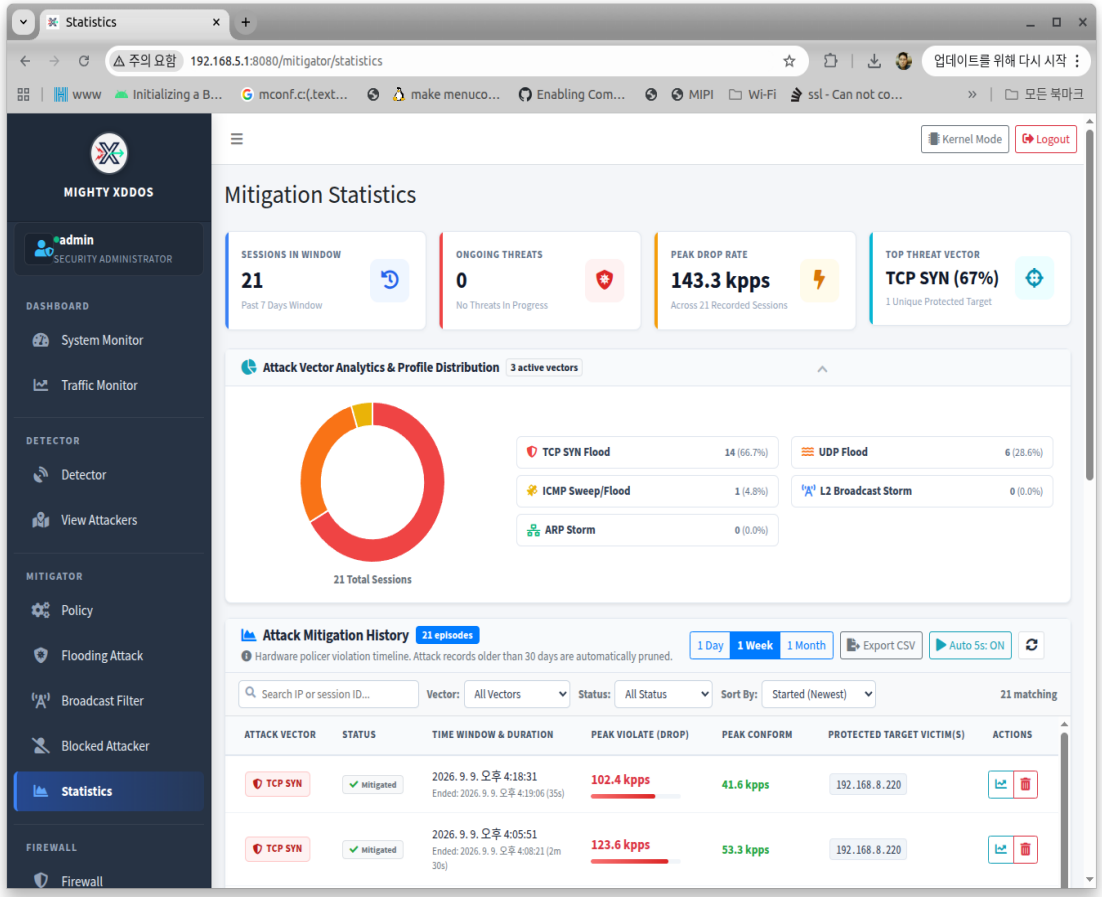


Mitigator - BGP Flowspec Page#2

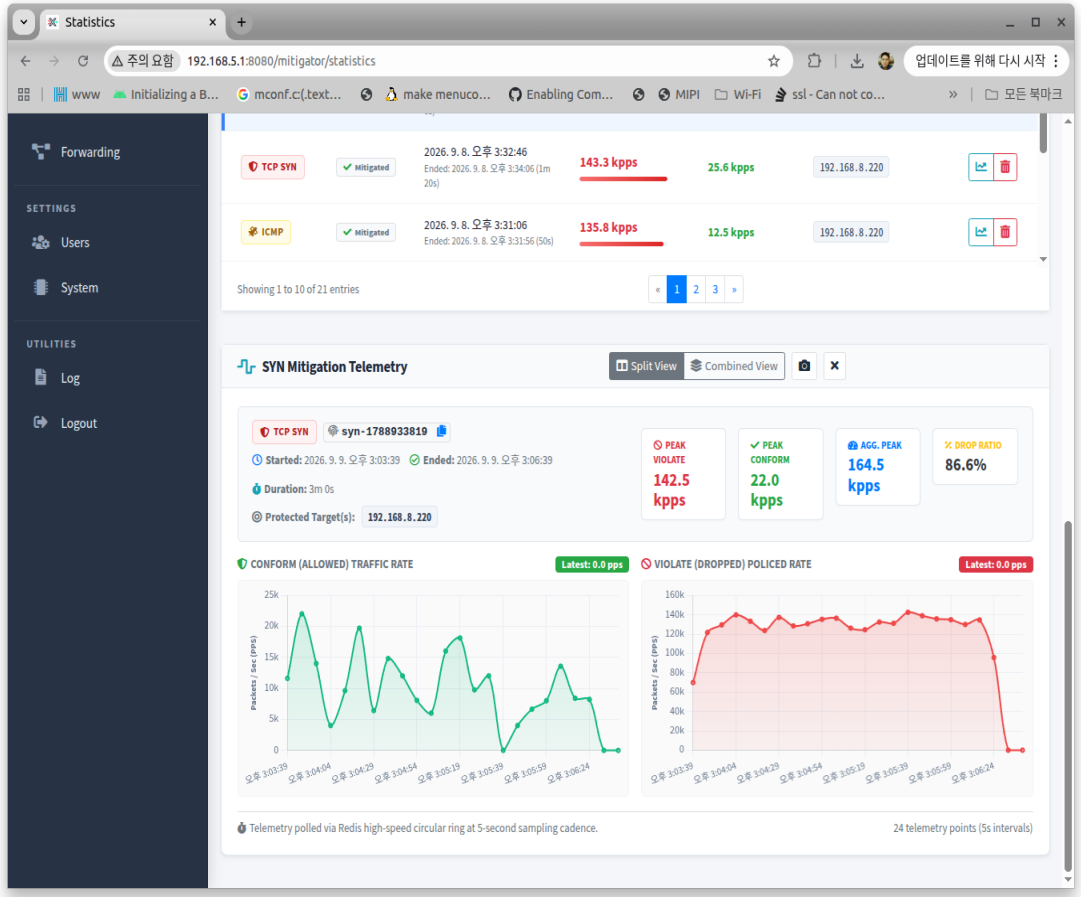


4. XDDoS User Interface(2) – WebUI(11)

Mitigator - Statistics Page#1

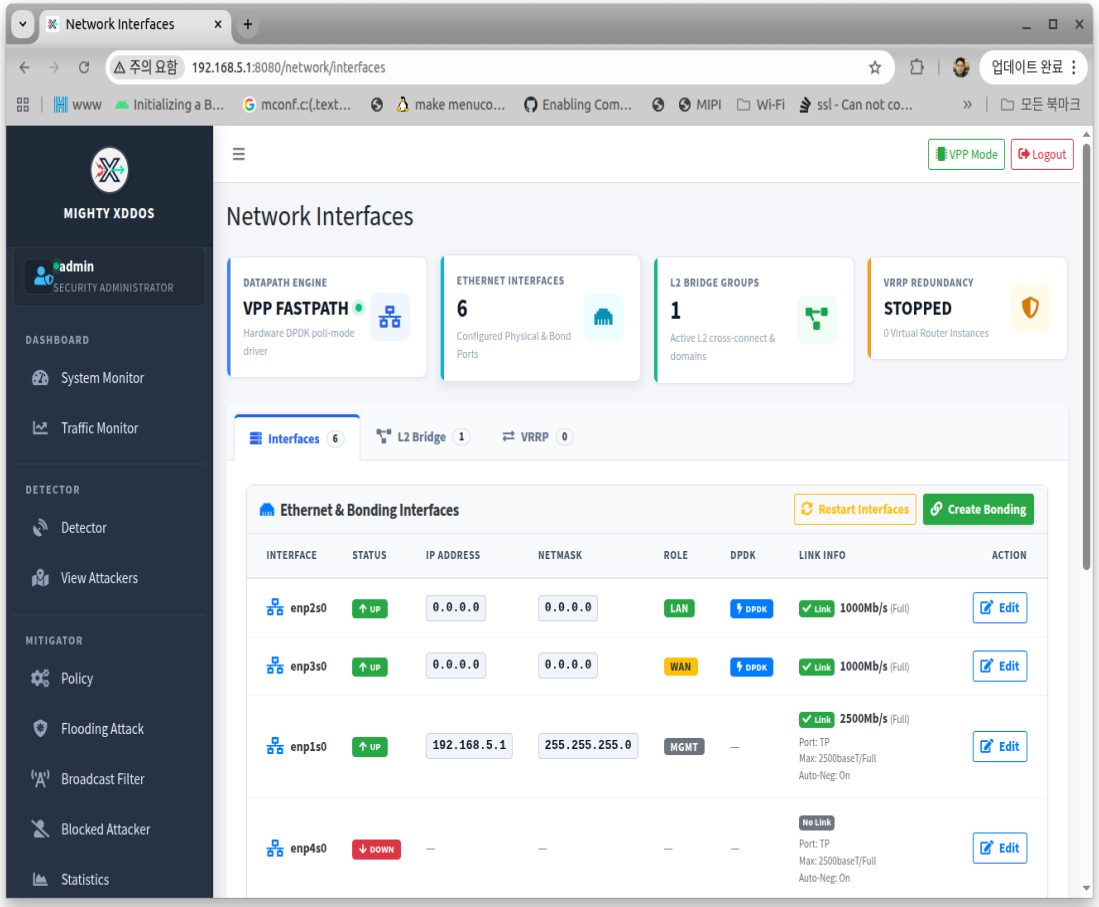


Mitigator - Statistics Page#2

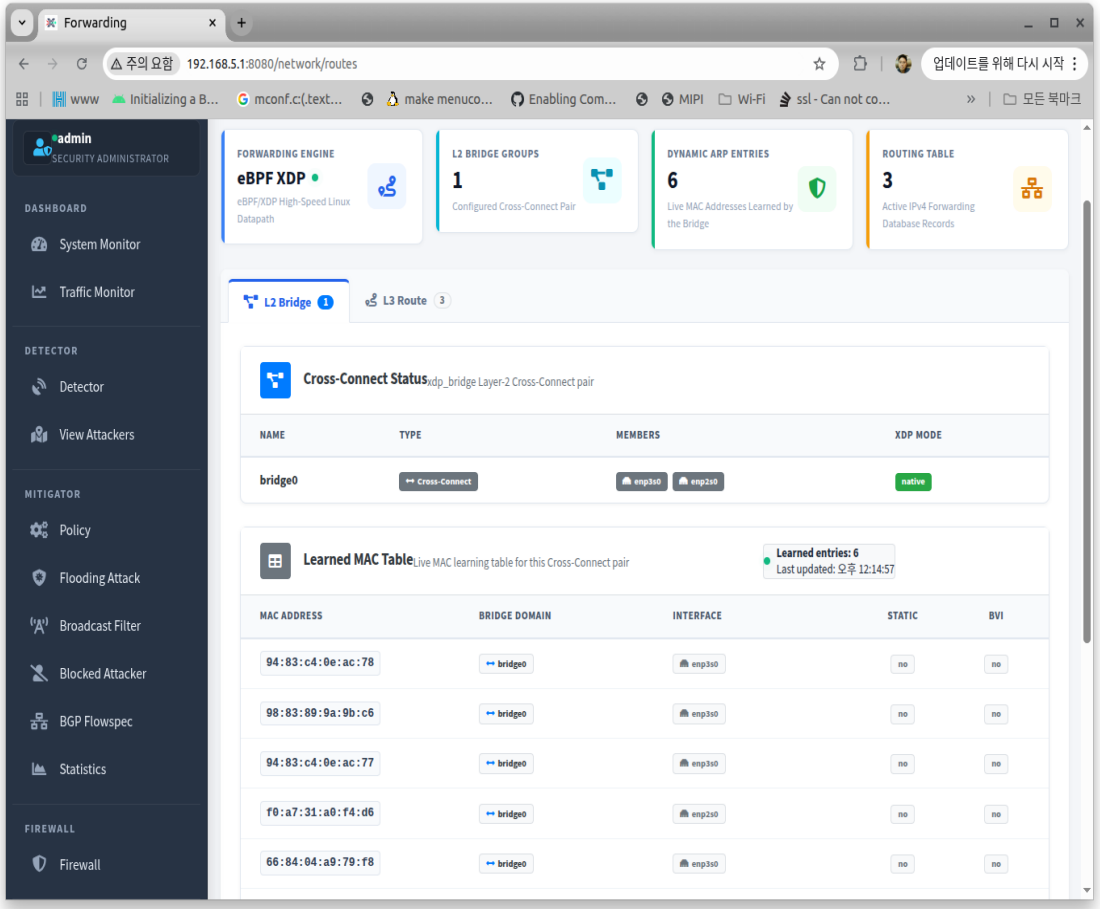


4. XDDoS User Interface(2) – WebUI(12)

Network - Interfaces Page

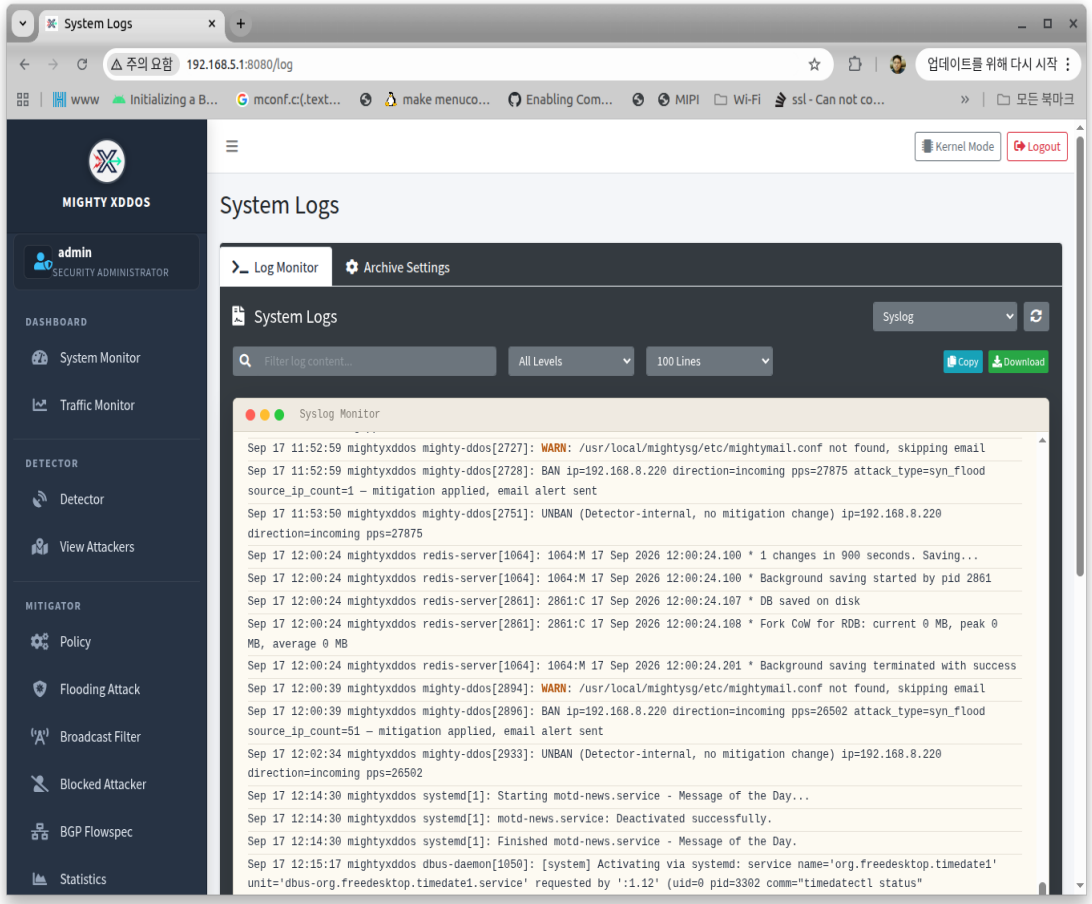


Network - Forwarding Page

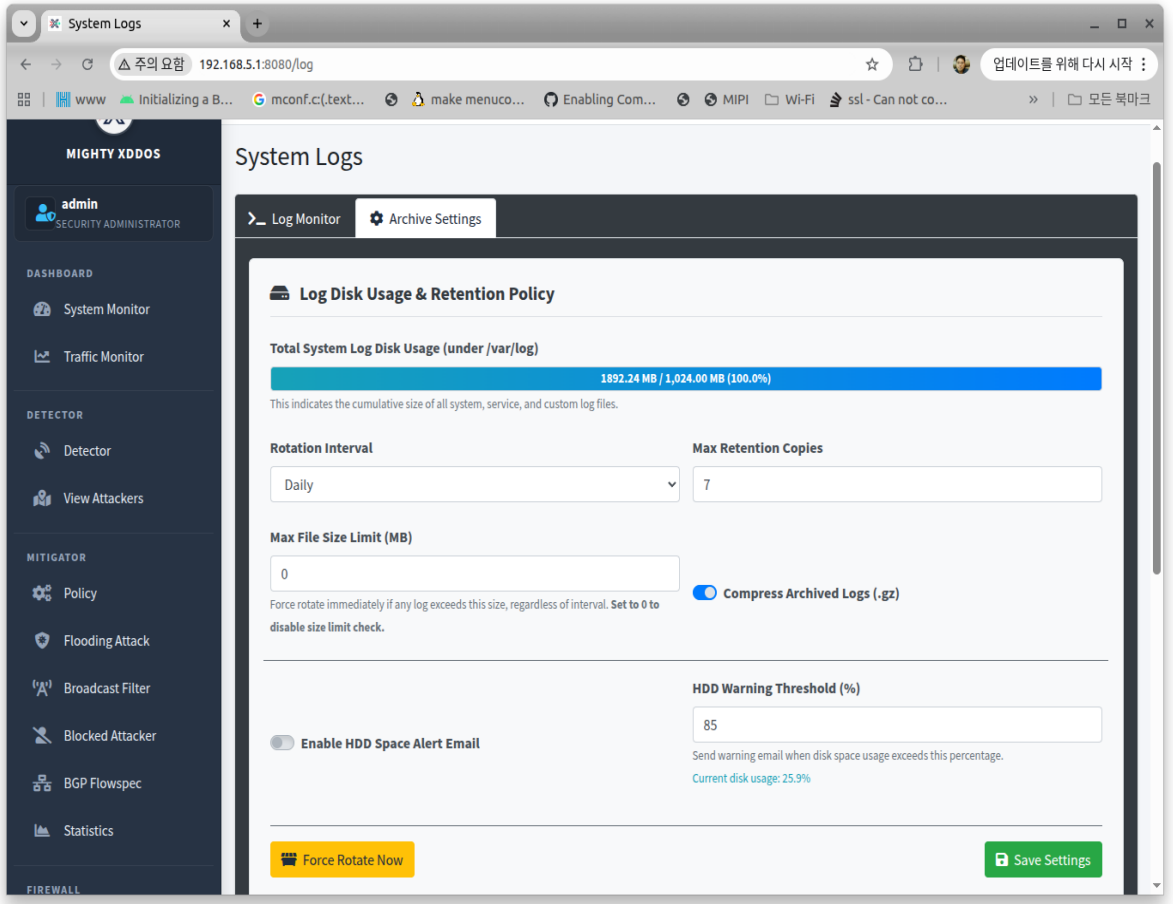


4. XDDoS User Interface(2) – WebUI(13)

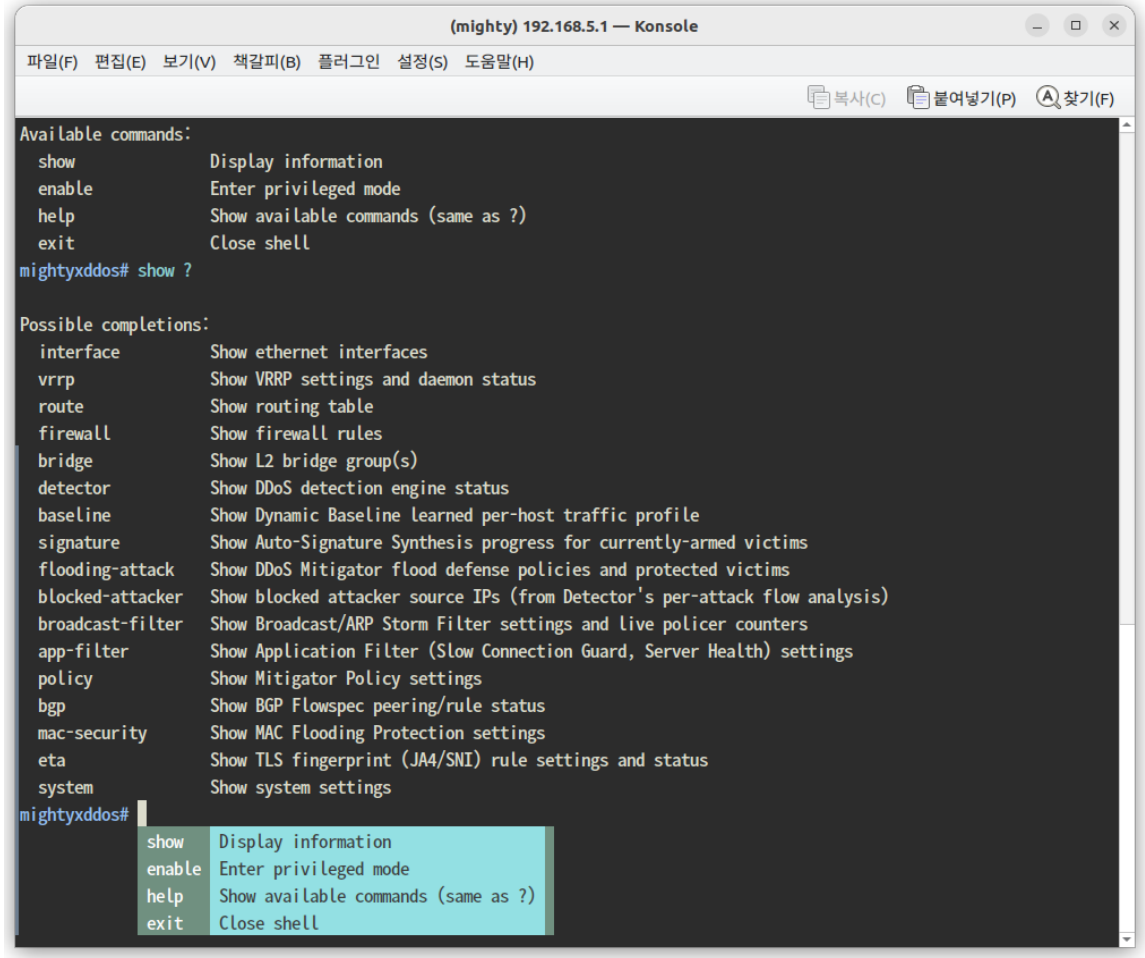
Utilities - Log Page#1

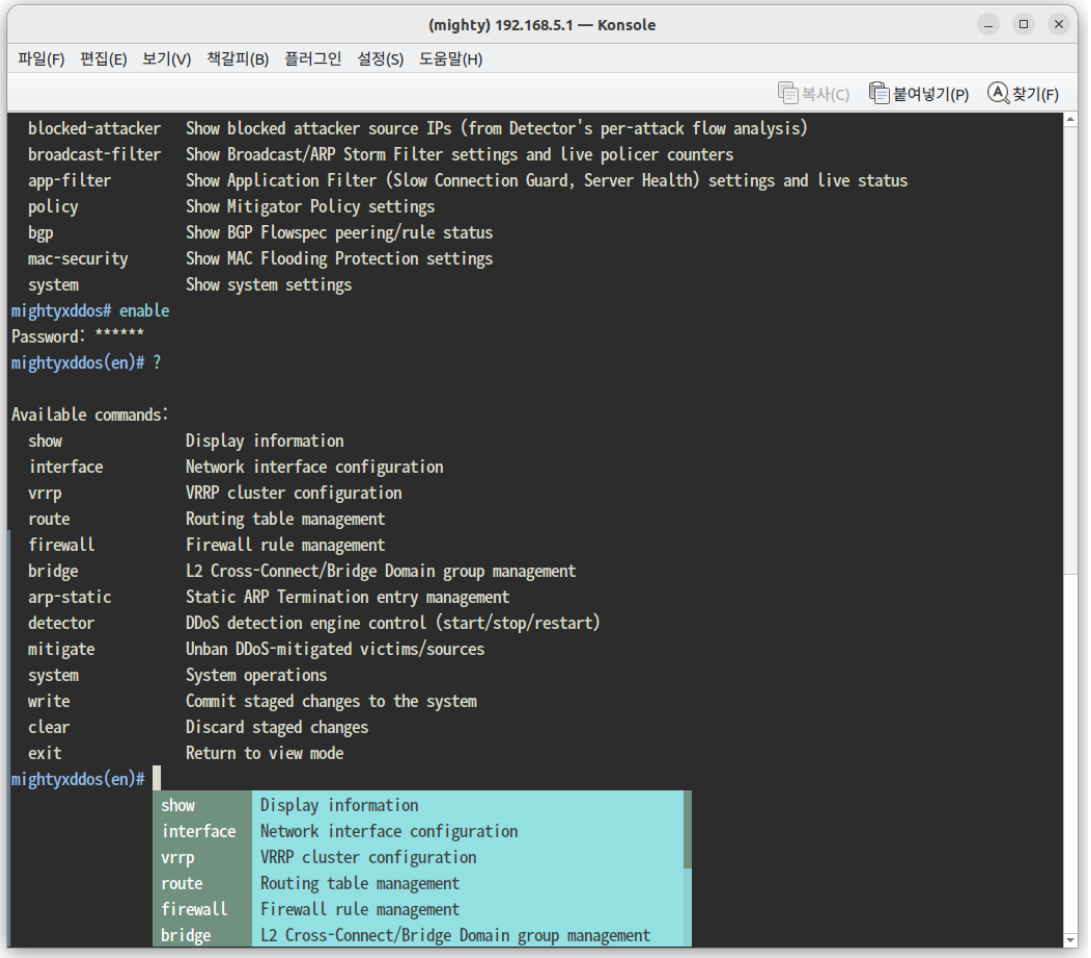


Utilities - Log Page#2

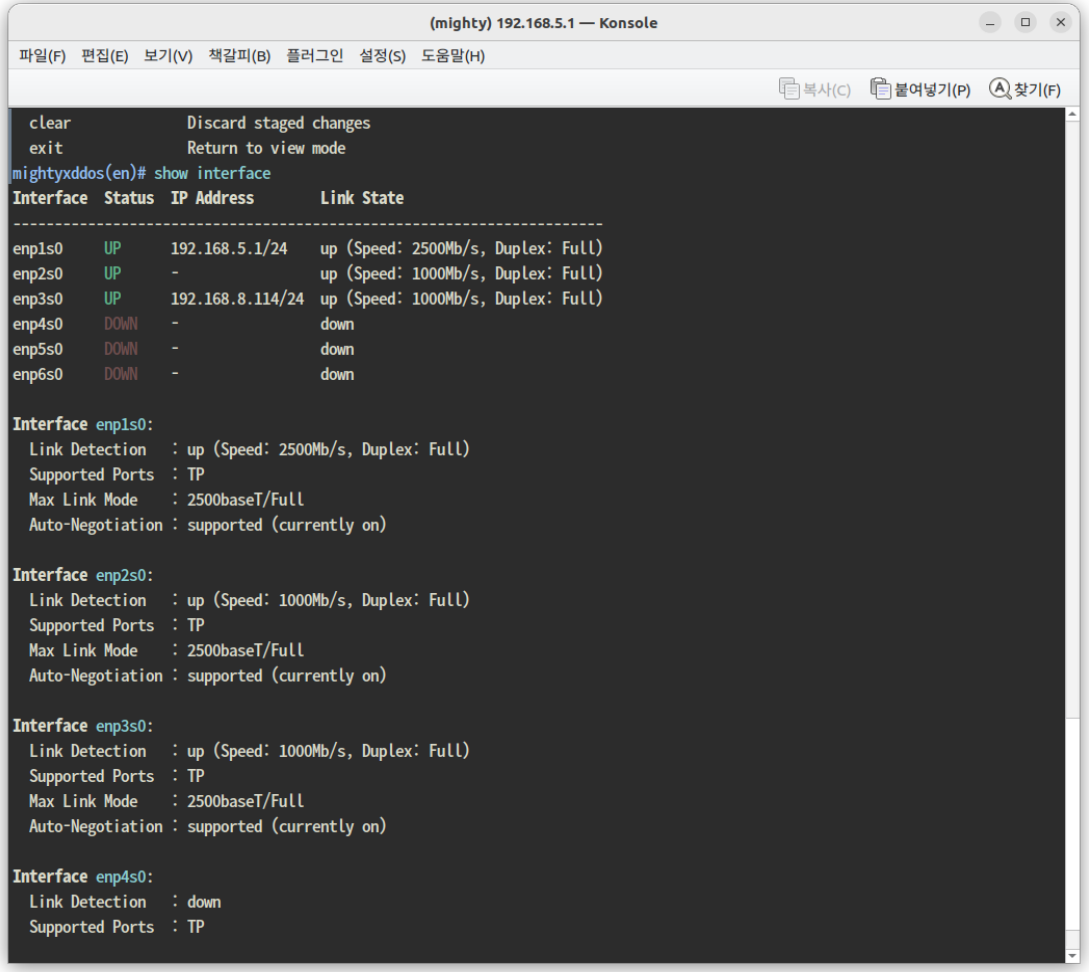


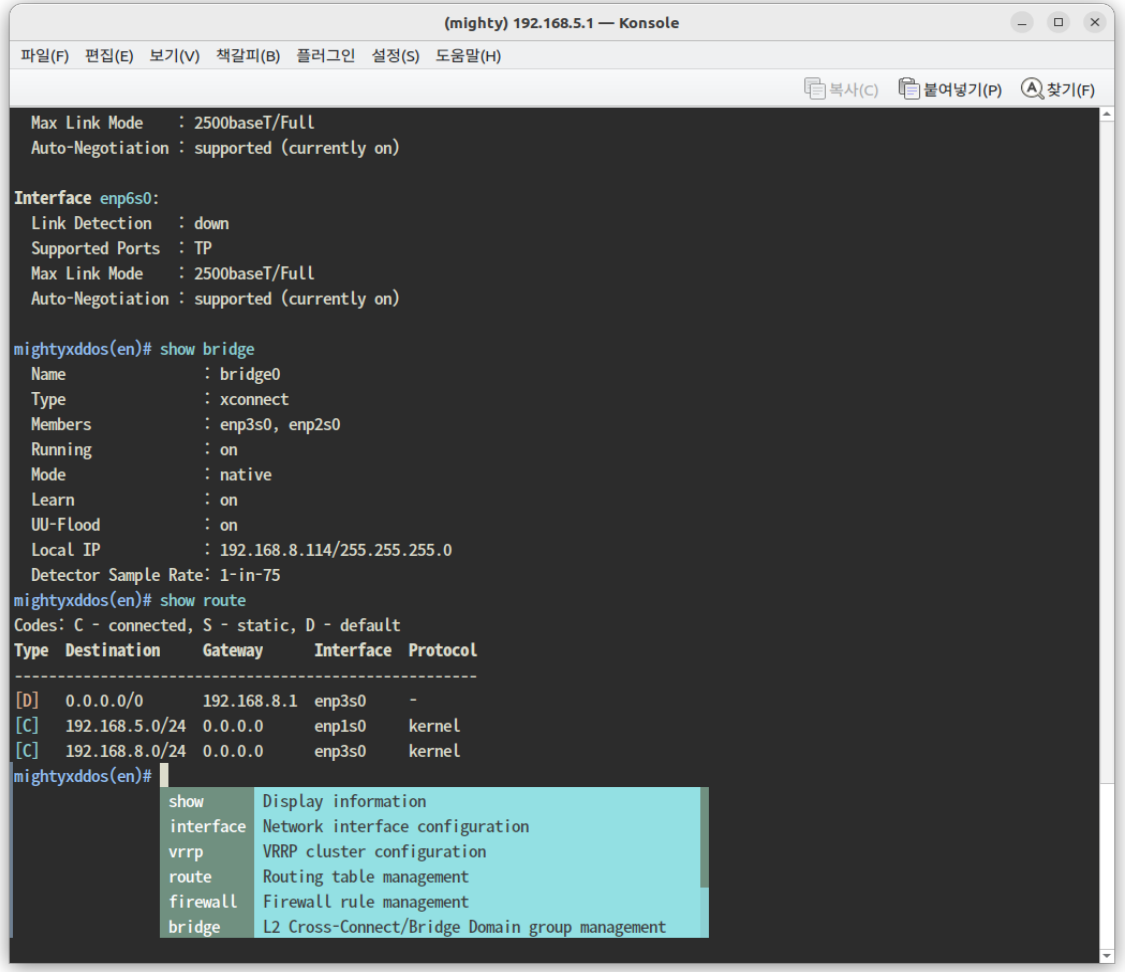
4. XDDoS User Interface(3) – CLI(1)



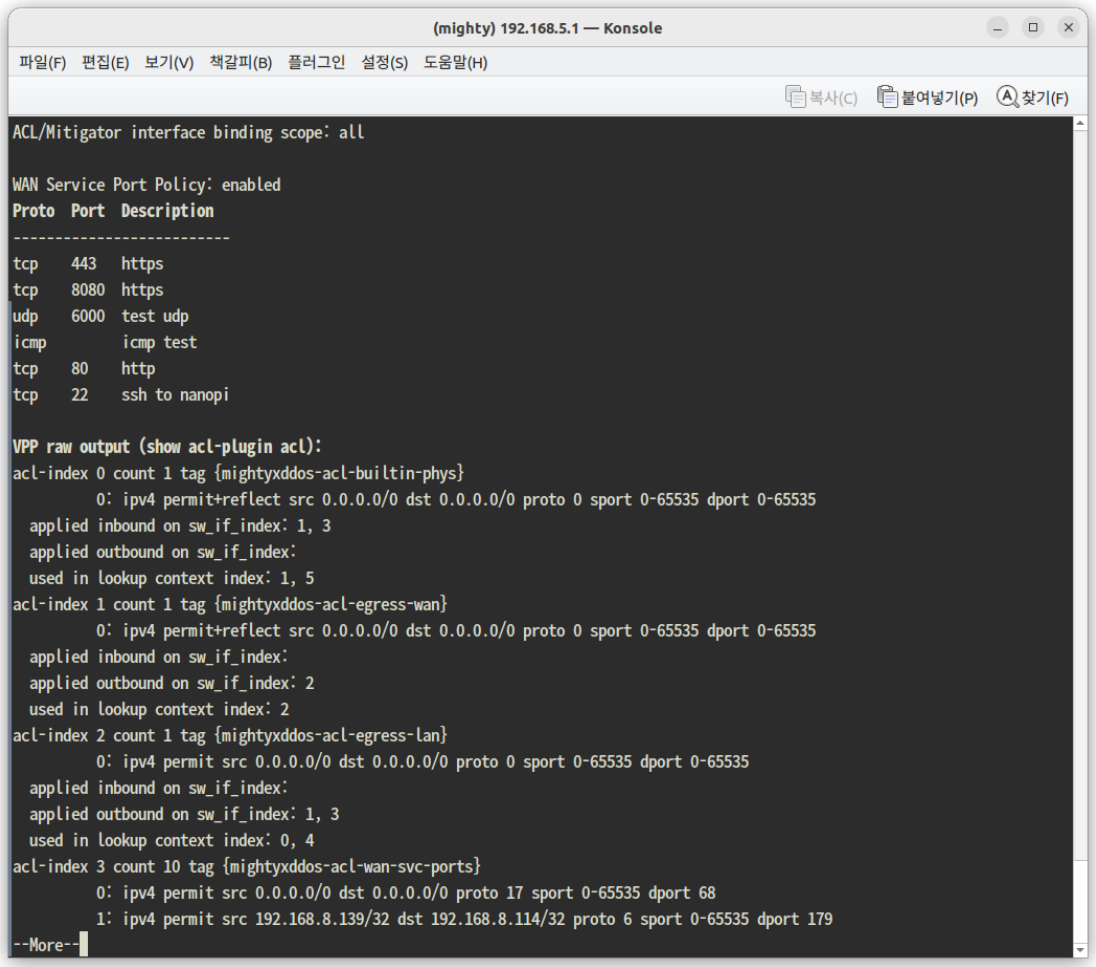


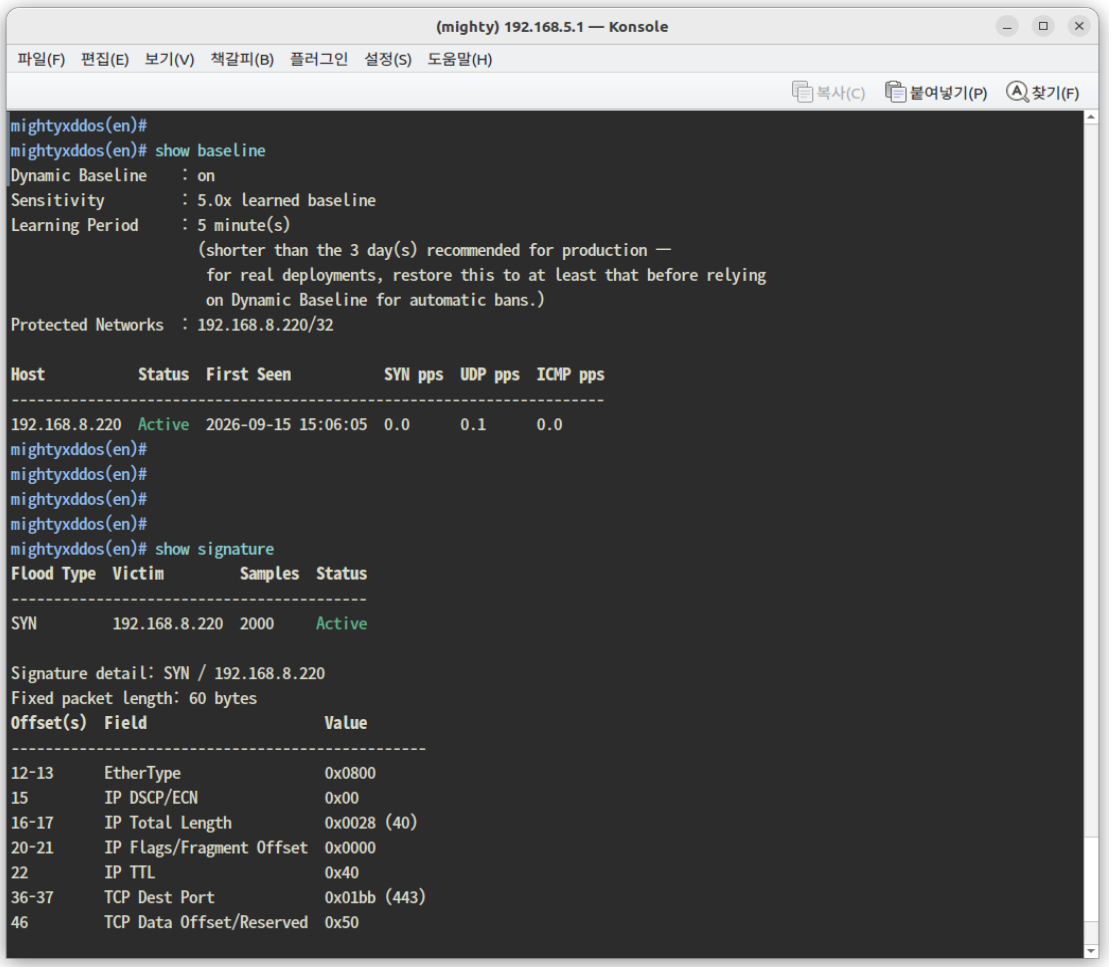
4. XDDoS User Interface(3) – CLI(2)



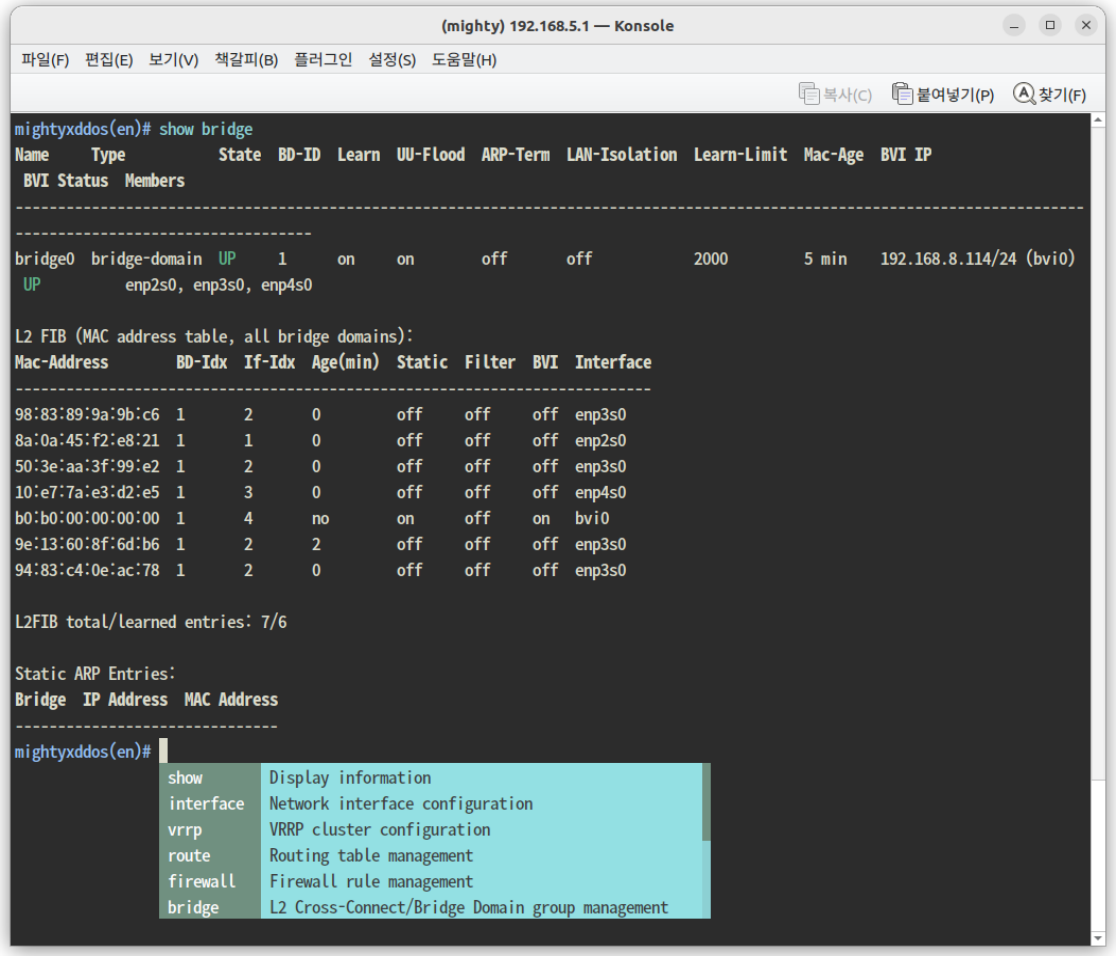


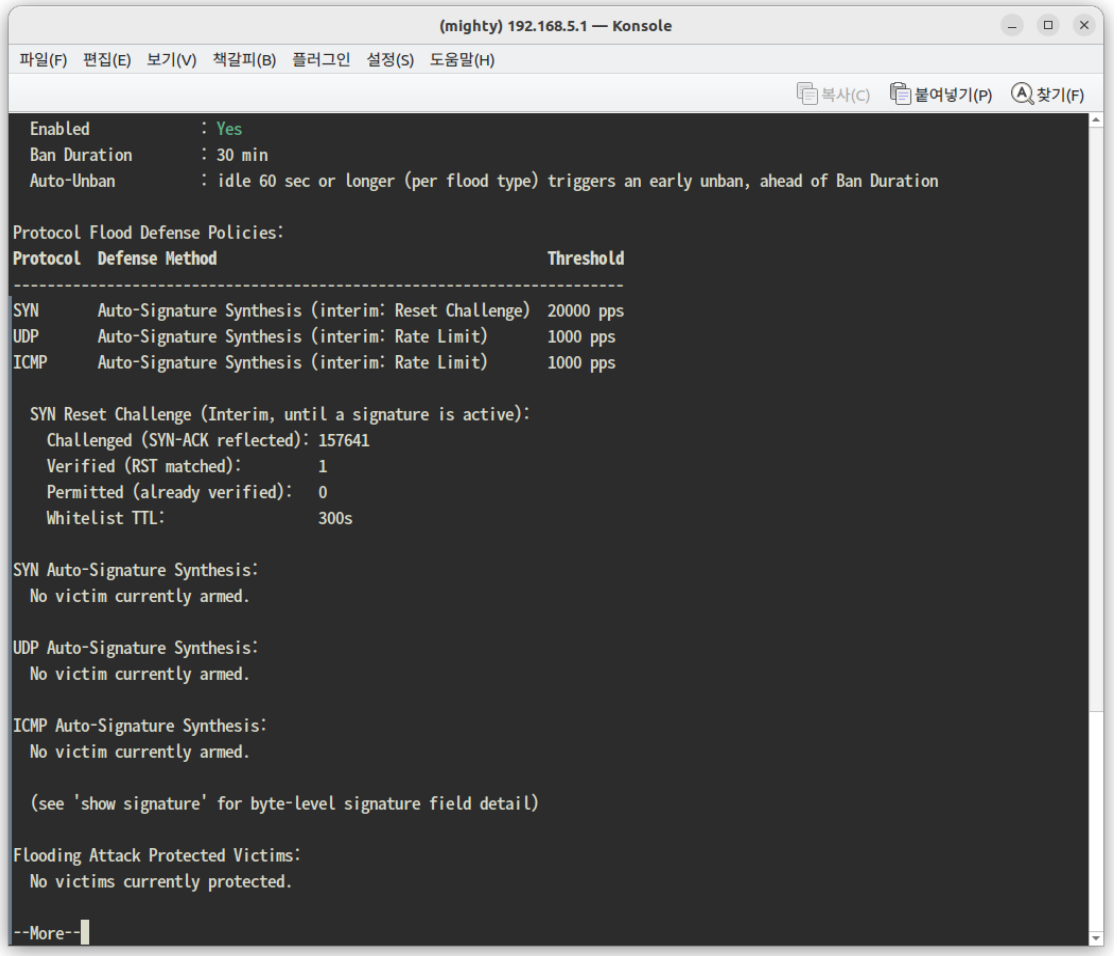
4. XDDoS User Interface(3) – CLI(3)



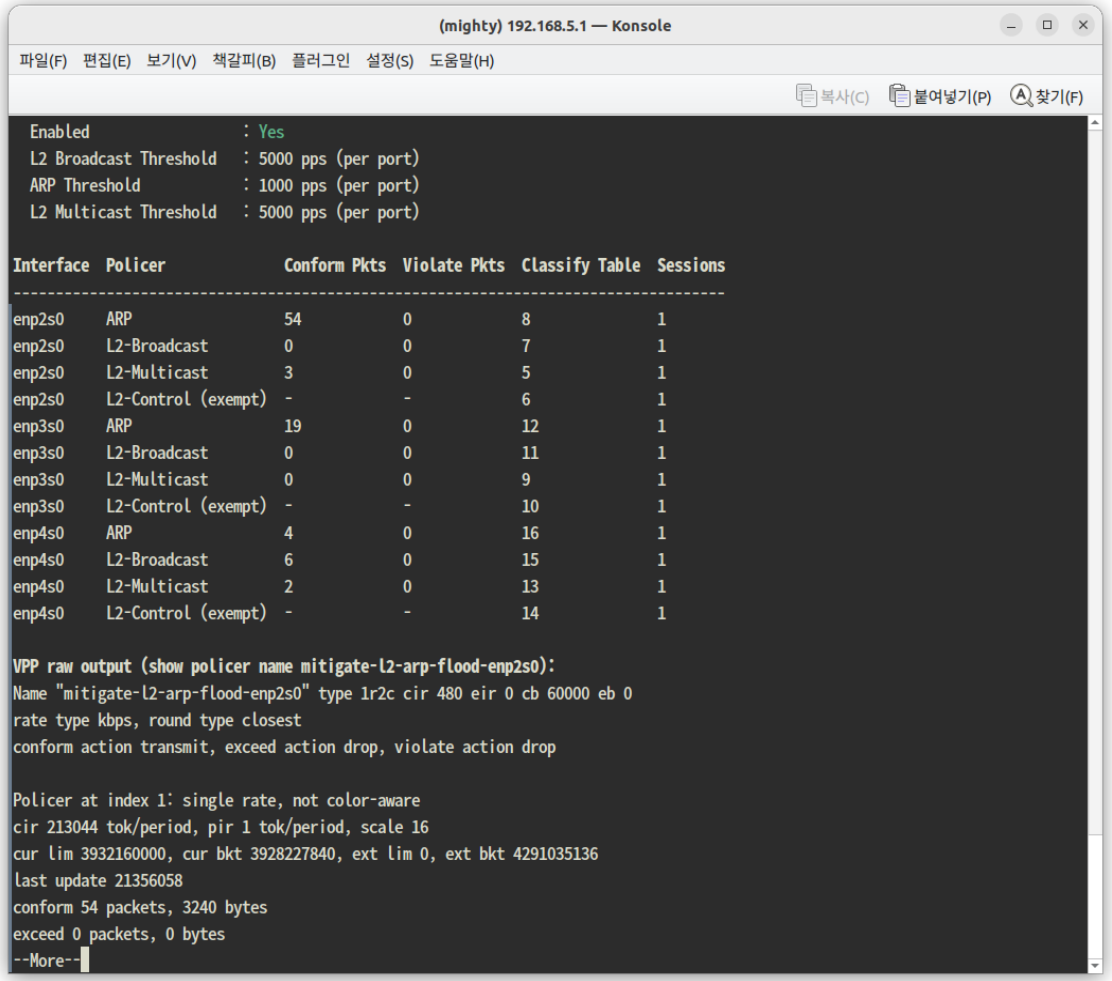


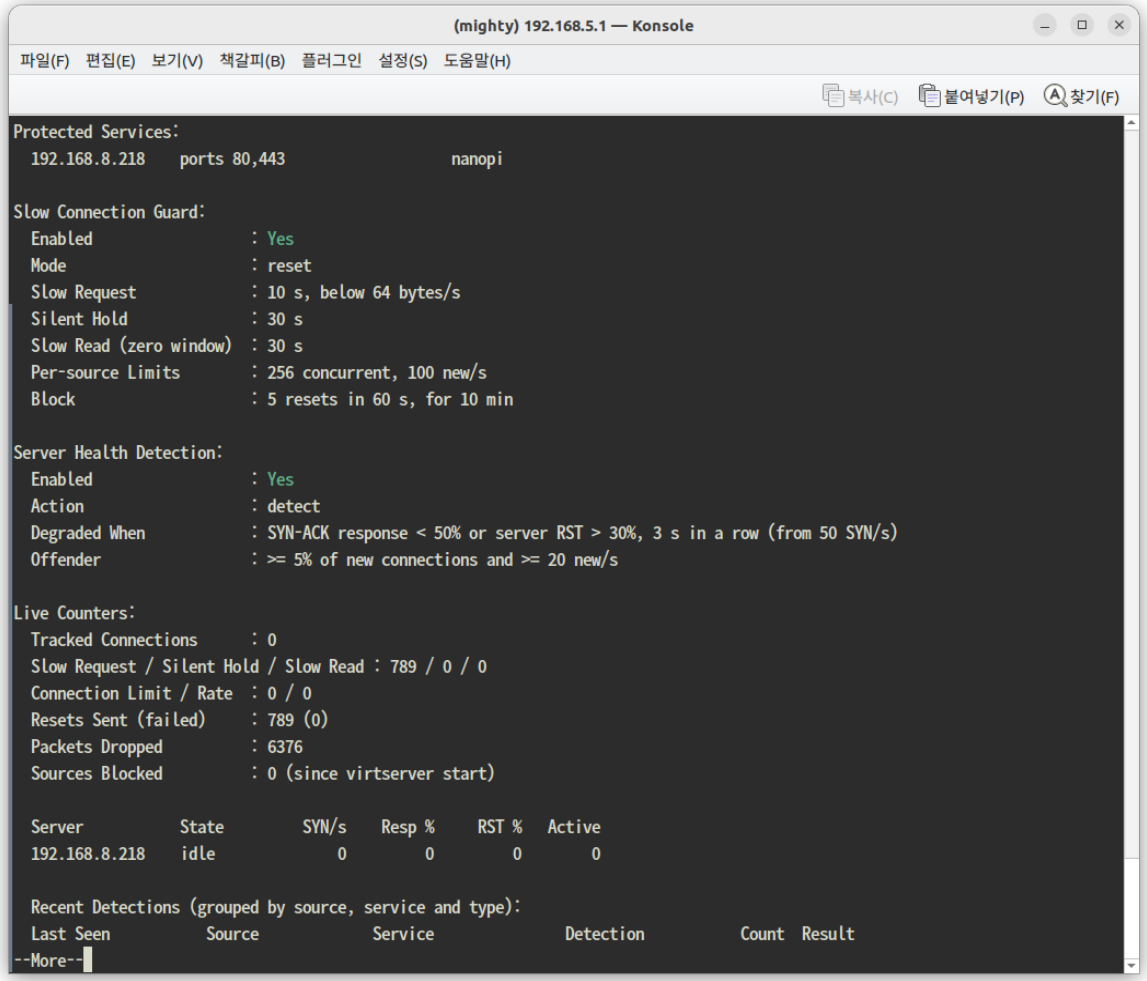
4. XDDoS User Interface(3) – CLI(4)



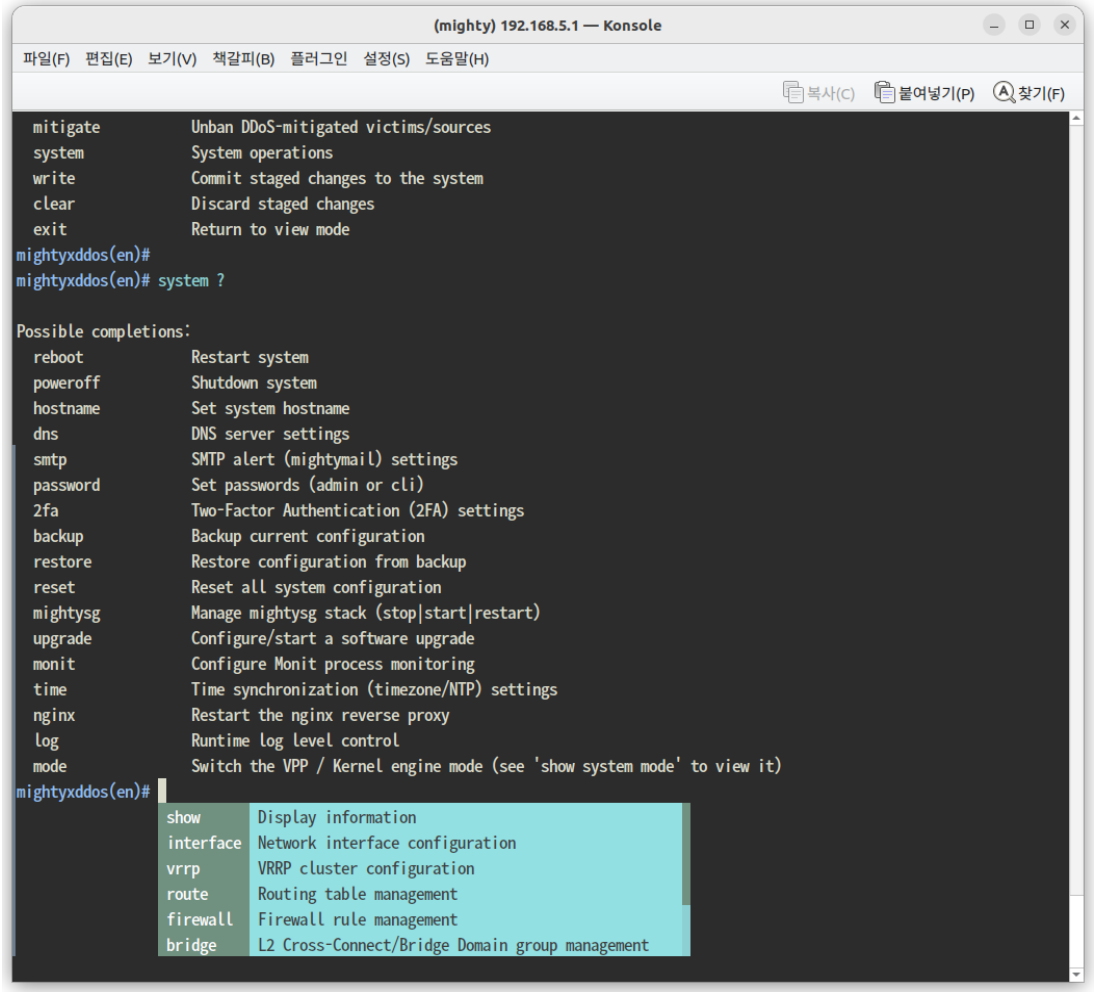


4. XDDoS User Interface(3) – CLI(5)





4. XDDoS User Interface(3) – CLI(6)



Appendix



A. Testbed(1) – TCP/UDP/ICMP & MAC Flooding Test

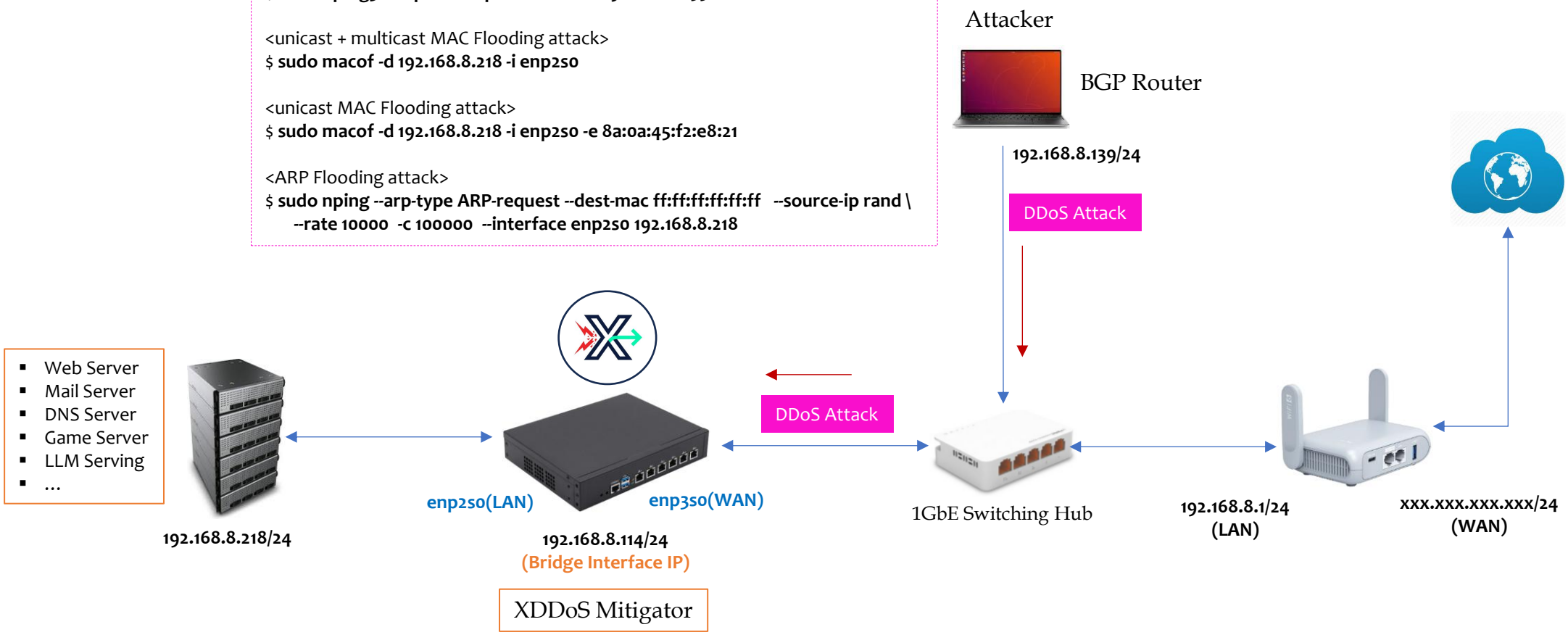
<TCP/UDP/ICMP Flooding attacks – random source IPs>
$ **sudo hping3 -S -p 443 --flood --rand-source 192.168.8.218**
$ **sudo hping3 -udp --flood -p 6000 --rand-source 192.168.8.218**
$ **sudo hping3 --icmp --flood --rand-source 192.168.8.218**

<L2 broadcast attack>
$ **sudo hping3 --udp -d 100 -p 6000 --flood 192.168.8.255**

<unicast + multicast MAC Flooding attack>
$ **sudo macof -d 192.168.8.218 -i enp2s0**

<unicast MAC Flooding attack>
$ **sudo macof -d 192.168.8.218 -i enp2s0 -e 8a:0a:45:f2:e8:21**

<ARP Flooding attack>
$ **sudo nping --arp-type ARP-request --dest-mac ff:ff:ff:ff:ff:ff --source-ip rand **
--rate 10000 -c 100000 --interface enp2s0 192.168.8.218

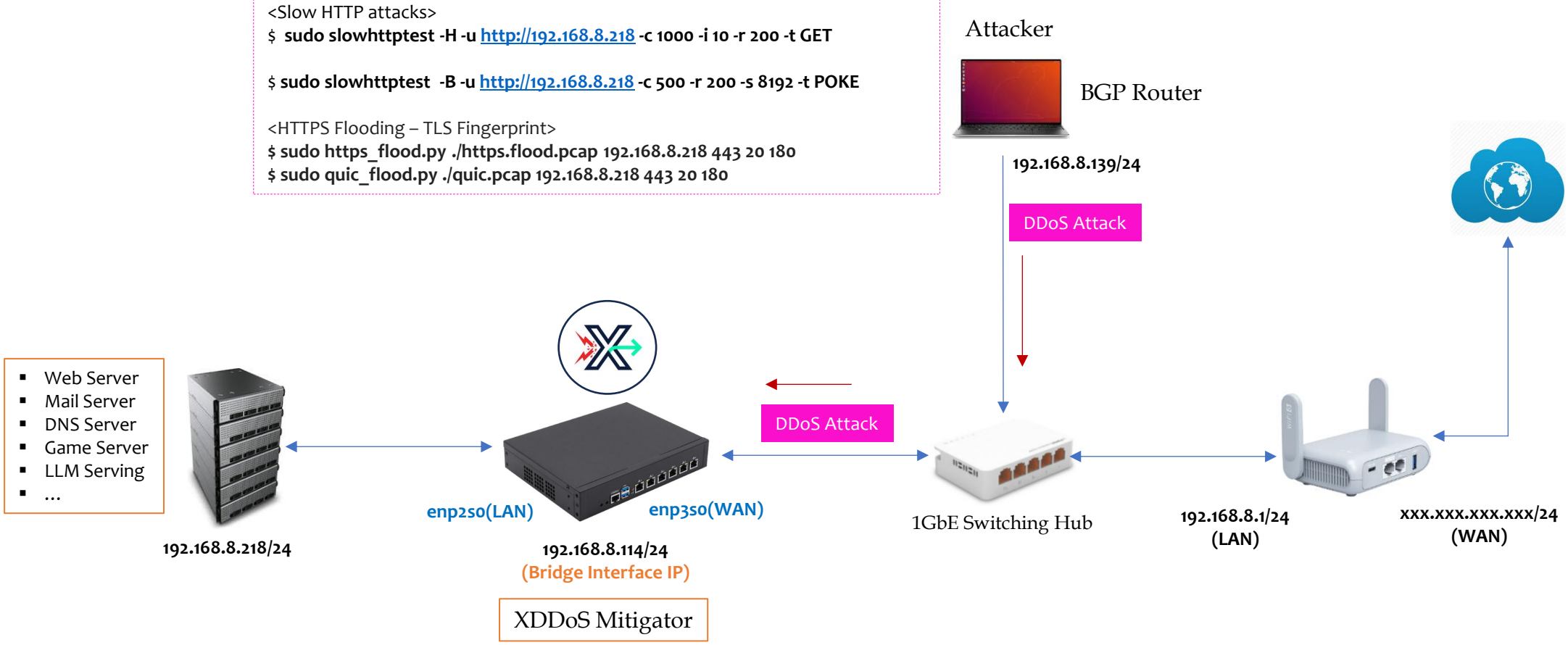


A. Testbed(2) – Slowloris and HTTPS Flooding Test

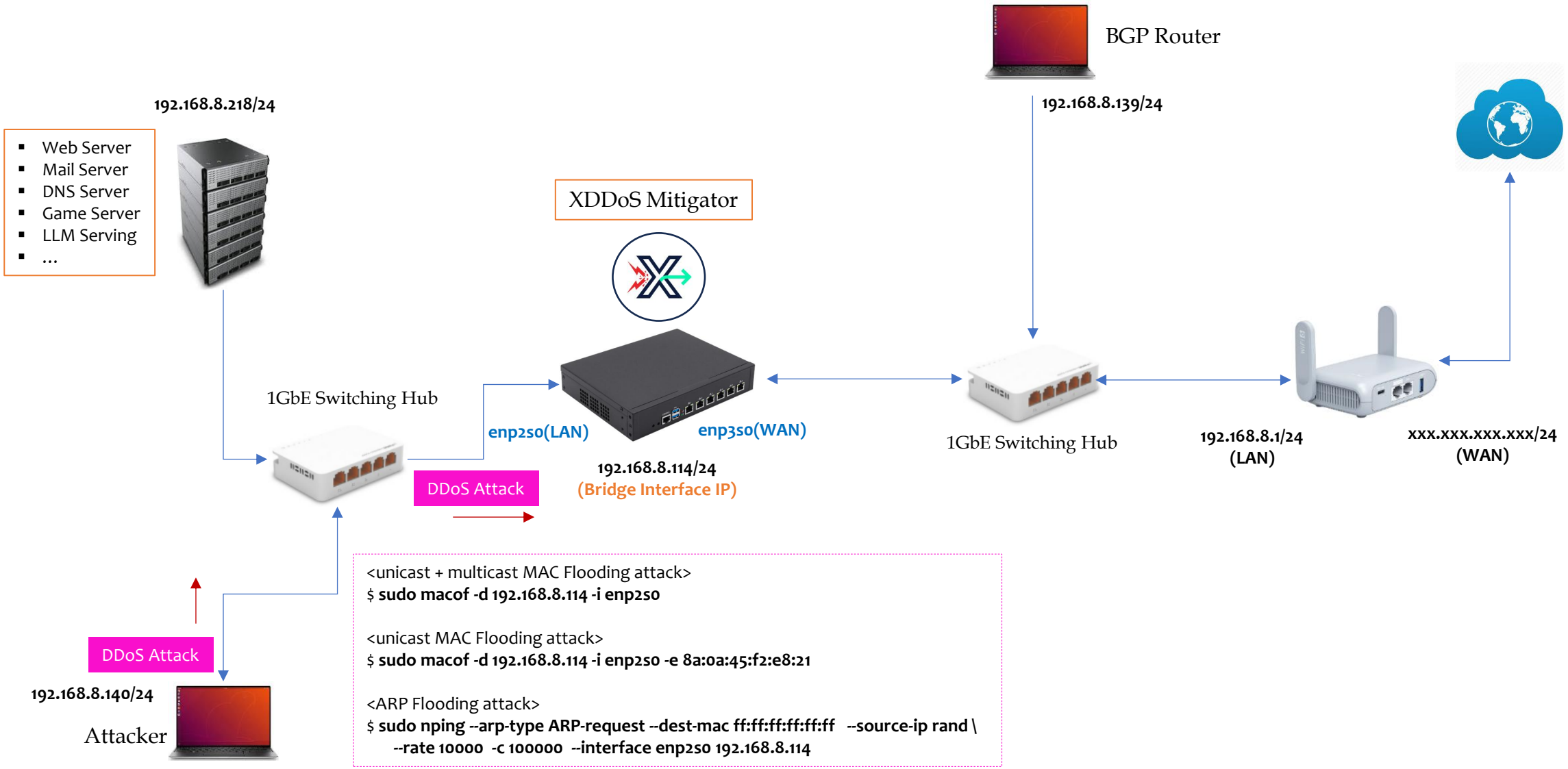
<Slow HTTP attacks>
$ **sudo slowhttptest -H -u http://192.168.8.218 -c 1000 -i 10 -r 200 -t GET**

$ **sudo slowhttptest -B -u http://192.168.8.218 -c 500 -r 200 -s 8192 -t POKE**

<HTTPS Flooding – TLS Fingerprint>
$ sudo https_flood.py ./https.flood.pcap 192.168.8.218 443 20 180
$ sudo quic_flood.py ./quic.pcap 192.168.8.218 443 20 180



A. Testbed(3) – MAC Flooding Test



<unicast + multicast MAC Flooding attack>
$ **sudo macof -d 192.168.8.114 -i enp2s0**

<unicast MAC Flooding attack>
$ **sudo macof -d 192.168.8.114 -i enp2s0 -e 8a:0a:45:f2:e8:21**

<ARP Flooding attack>
$ **sudo nping --arp-type ARP-request --dest-mac ff:ff:ff:ff:ff:ff --source-ip rand **
--rate 10000 -c 100000 --interface enp2s0 192.168.8.114

B. References

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